R101 pretrained B vs random A | Fixed target36 vs R101/A

Native/FP64 winners; true rescues/losses and false-pick cost, not selection-frequency accuracy.

Split	True winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	919	639	127	535	1.1035
VAL	4625	52	42	5	29	1.1243
TEST	4598	50	38	7	38	1.0874

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.116324	24168	22472	-0.000724919	-0.006093
VAL	0.049383	1308	1268	-0.000272309	-0.002608
TEST	0.112346	1326	1235	-0.000388076	-0.002009

Split	New-false cost contribution	New-false cost delta	New-false regret pp
TRAIN	0.003040981	0.000079047	0.001686
VAL	0.003171510	0.000128414	0.002138
TEST	0.003721341	0.000180927	0.002915

Cost contributions: sum over newly false-picked instances / task n; ALL averages18 tasks.

R89B/R84A are historical, not12800+57600 budget-matched; zero-count movement is not success.

R101 B vs A | Original student latency

First128 GPU-resident original TEST inputs/task; three synchronized repetitions.

Problem	A student ms/row	B student ms/row
TSP	0.243893	0.242471
CVRP	0.291117	0.291154
ATSP	0.065964	0.061758
OVRP	0.064915	0.061512
VRPB	0.065113	0.061487
VRPL	0.065013	0.061102
VRPTW	0.064890	0.061006
OVRPTW	0.065490	0.060956
OVRPB	0.064941	0.061135
OVRPL	0.064911	0.060924
VRPBL	0.065333	0.060681
VRPBTW	0.065099	0.061029
VRPLTW	0.064982	0.060763
OVRPBL	0.065081	0.061138
OVRPBTW	0.065137	0.061167
OVRPLTW	0.064811	0.061032
VRPBLTW	0.065158	0.060978
OVRPBLTW	0.065094	0.060994

Original1842953-parameter student only; no online donor, teacher feature I/O or solver decoder.

Excludes dataset I/O, host transfer and Stage-I pretraining; not single-instance/E2E latency.

No latency or speedup claim is inferred from R94 cached-selector numbers.

Hardware is disclosed, not bitwise matched; no model reselection based on timing.

R101/A | Two-stage provenance

Fixed TRAIN hint endpoint, then independently reset original full18 learning.

Teacher: random. Stage I:12800 updates, logical batch1024, seed2; CVRP+15MVRP TRAIN81000/task only.

Separate non-affine per-node LayerNorm MSE; one dropout forward. No outcome labels, learned adapter or teacher VAL/TEST. Fixed endpoint, not validation selected.

Stage II: fresh original R84/R88B student; ONLY instance_encoder transferred. All non-encoder parameters remain identical fresh R84. Optimizer, scaler, RNG and task/sample streams reset.

Original1842953 parameters; no added heads or features. R77 CE .35 / comparison .35 / risk .02 / R-Drop .35. No hint loss, teacher arrays or online donor in Stage II or deployment.

Both A/B have12800+57600 updates; R89B/R84A are NOT total-budget-matched. B teacher has additional external solver pretraining history. The paper memo provides conditional motivation only.

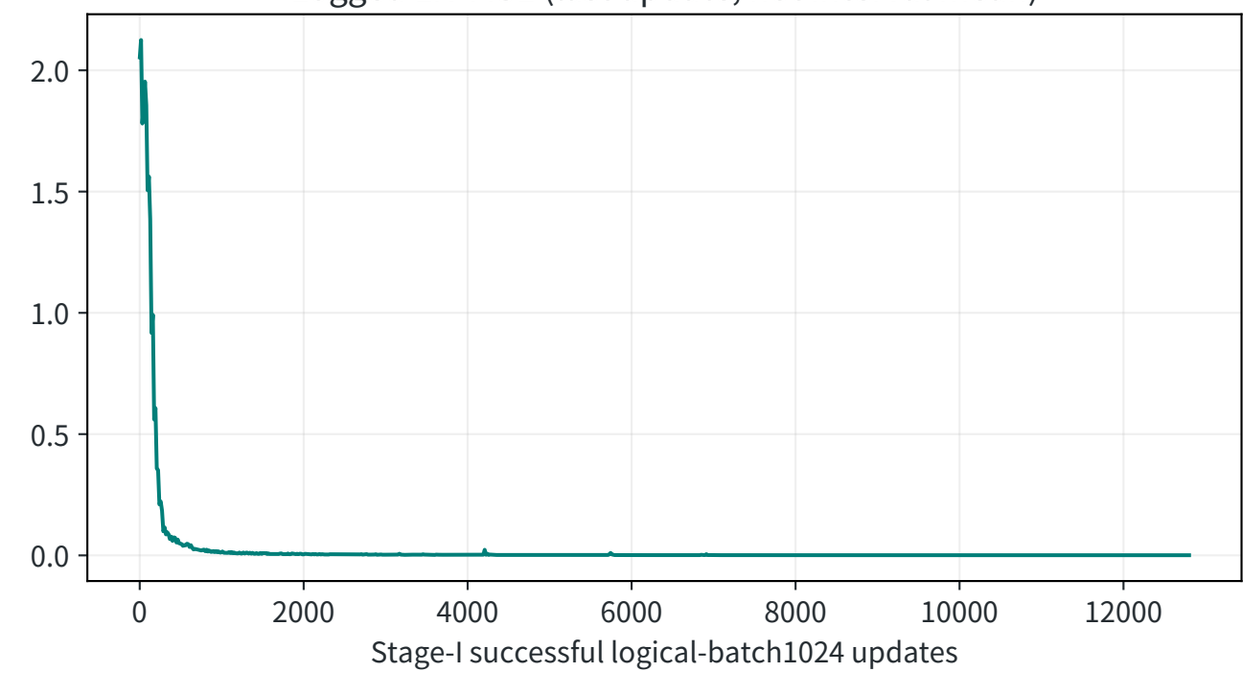
Native/FP64 ind winners; legal-pool Oracle and TRAIN-fitted SBS; ALL is task-equal. Development-exposed TEST; no TEST/LIB checkpoint selection.

Stage-I encoder SHA256: 1267da285be94fe4ff642cffcf4bac3b84d5bfb2ce6f6b5b255af3308dab02bb

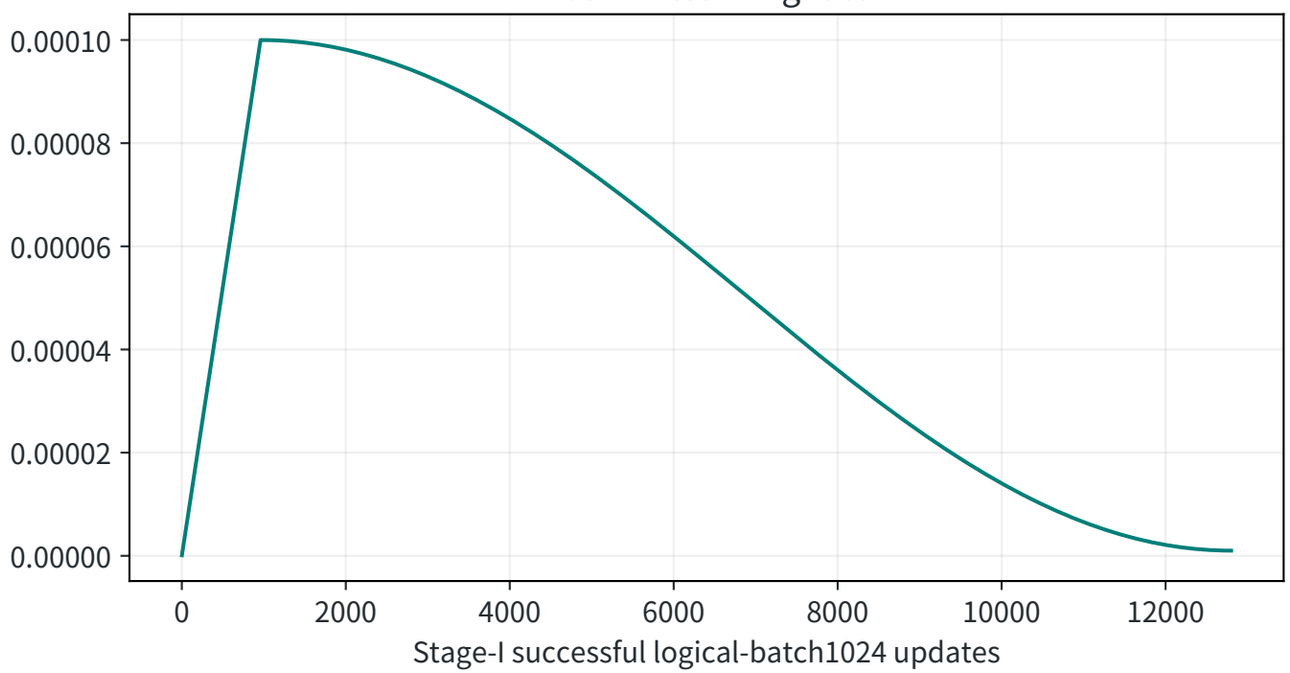
Unchanged non-encoder SHA256: 970a8aa2a548f9034f34b73d060662f71808e46359335185ce07bfc5b1bb2fa6

R101/A | TRAIN-only random node hints; fixed12800 endpoint

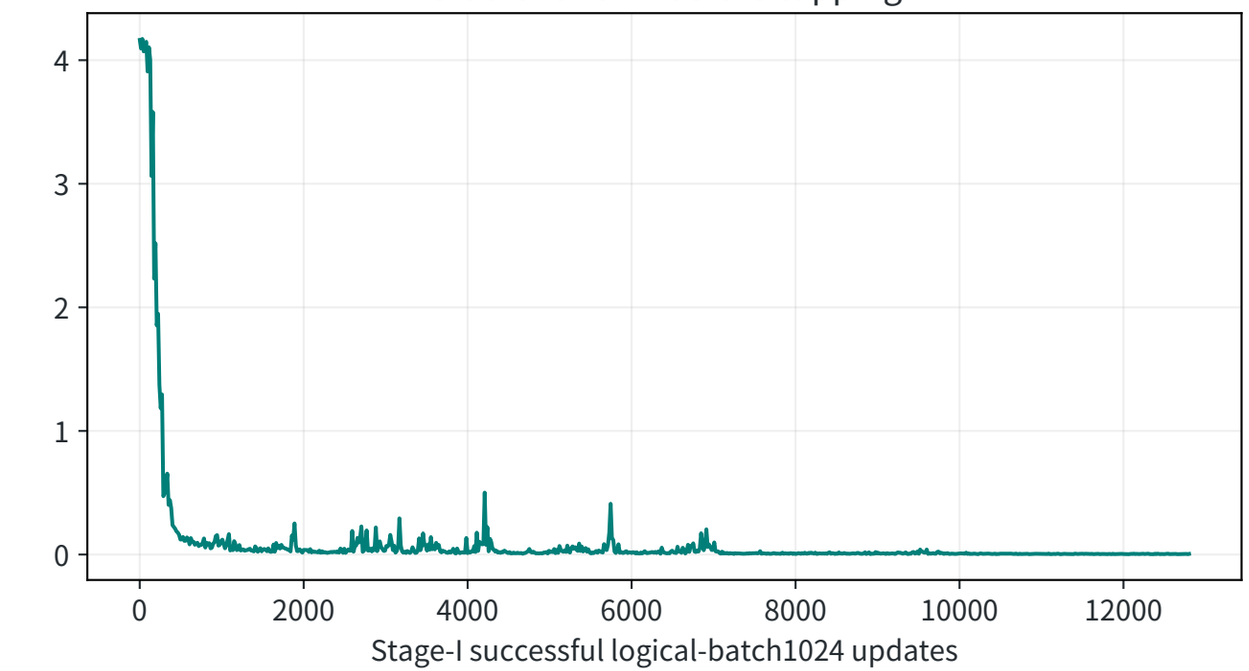
Logged LN-MSE (last update, not interval mean)



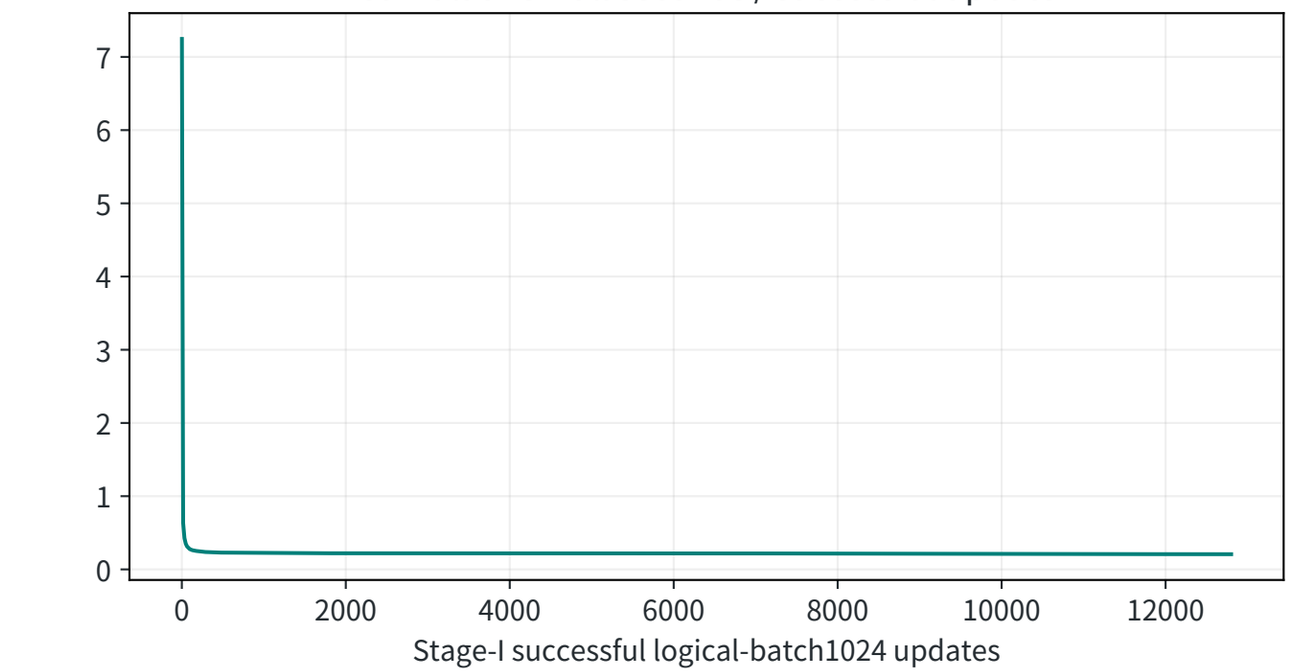
AdamW learning rate



Gradient norm before clipping



Cumulative wall seconds / successful update



R101/A | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 29 | update 41760 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.1519	66.2037	11.416572	11.427978	11.371692	-0.1949	+0.3927	0.4026
TEST ALL	71.3222	66.1802	11.373839	11.385299	11.329932	-0.1979	+0.3853	0.3930

Dataset	n	NSS Top1 (%)	R101 Top1 (%)	NSS cost	R101 cost
TSP TEST	4500	45.888889	47.000000	7.970400719	7.970917341
CVRP TEST	4500	45.977778	48.200000	18.004802368	18.009998522
TSPLIB	49	20.408163	28.571429	8.217907074	8.210717059
CVRPLIB	100	70.000000	62.000000	67.101111717	67.254707232

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original R77 native CE

R101/A | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.1519	66.2037	11.416572	11.427978	11.371692	-0.1949	+0.3927	0.4026	per-problem TRAIN SBS
TSP	46.7333	33.1111	8.024500	8.045354	7.990533	-0.2592	+0.4251	0.4144	T2T
CVRP	46.7556	18.7111	18.159052	18.269377	18.064087	-0.6039	+0.5257	0.7022	RELD_CVRP
ATSP	77.2222	47.4889	1.638651	1.676905	1.632471	-2.2812	+0.3786	0.3745	UNICO_MatPOENet
OVRP	41.1556	35.7556	7.462603	7.469579	7.415777	-0.0934	+0.6314	0.6447	MoSES_RF
VRPB	81.1111	80.8444	9.354106	9.354876	9.318750	-0.0082	+0.3794	0.3873	reld_mt
VRPL	38.2000	33.0889	12.432087	12.437959	12.383444	-0.0472	+0.3928	0.3996	MoSES_RF
VRPTW	79.9333	79.9556	18.819353	18.819431	18.766139	-0.0004	+0.2836	0.2986	reld_mt
OVRPTW	83.8444	83.7778	10.142885	10.143730	10.116859	-0.0083	+0.2573	0.2546	reld_mt
OVRPB	88.5333	88.5111	6.357159	6.357254	6.341078	-0.0015	+0.2536	0.2637	reld_mt
OVRPL	41.7333	35.9111	7.437128	7.445709	7.390416	-0.1153	+0.6321	0.6473	MoSES_RF
VRPBL	78.9111	78.5111	9.368180	9.371523	9.328366	-0.0357	+0.4268	0.4291	reld_mt
VRPBTW	78.8000	78.4222	19.944422	19.950430	19.851084	-0.0301	+0.4702	0.4490	reld_mt
VRPLTW	78.8889	78.7556	18.763599	18.765871	18.707843	-0.0121	+0.2980	0.3145	reld_mt
OVRPBL	89.2444	89.2000	6.369733	6.370093	6.352824	-0.0057	+0.2662	0.2774	reld_mt
OVRPBTW	83.6000	83.4444	10.660230	10.661098	10.623245	-0.0081	+0.3482	0.3204	reld_mt
OVRPLTW	83.5778	83.4889	10.156922	10.157423	10.129818	-0.0049	+0.2676	0.2707	reld_mt
VRPBLTW	79.3111	79.4000	19.797034	19.797044	19.706420	-0.0001	+0.4598	0.4473	reld_mt
OVRPBLTW	83.1778	83.2889	10.610650	10.609955	10.571302	+0.0065	+0.3722	0.3517	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R101/A | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.3222	66.1802	11.373839	11.385299	11.329932	-0.1979	+0.3853	0.3930	per-problem TRAIN SBS
TSP	47.0000	34.5333	7.970917	7.989621	7.937593	-0.2341	+0.4198	0.4108	T2T
CVRP	48.2000	17.9333	18.009999	18.124983	17.918509	-0.6344	+0.5106	0.6687	RELD_CVRP
ATSP	76.9333	47.2444	1.635289	1.674207	1.629441	-2.3246	+0.3589	0.3580	UNICO_MatPOENet
OVRP	39.9778	33.6444	7.447082	7.455750	7.400487	-0.1163	+0.6296	0.6388	MoSES_RF
VRPB	80.6222	80.4000	9.350278	9.351810	9.315298	-0.0164	+0.3755	0.3797	reld_mt
VRPL	37.7111	32.2667	12.349928	12.356138	12.300958	-0.0503	+0.3981	0.4060	MoSES_RF
VRPTW	81.1556	80.8444	18.688759	18.690545	18.637408	-0.0096	+0.2755	0.2865	reld_mt
OVRPTW	83.9333	83.9333	10.092823	10.093063	10.068135	-0.0024	+0.2452	0.2363	reld_mt
OVRPB	88.1556	88.1333	6.370594	6.370826	6.352225	-0.0036	+0.2892	0.2953	reld_mt
OVRPL	41.8889	34.6000	7.462971	7.472542	7.418033	-0.1281	+0.6058	0.6150	MoSES_RF
VRPBL	79.5111	79.4222	9.395304	9.396379	9.355967	-0.0114	+0.4205	0.4177	reld_mt
VRPBTW	78.7333	78.7556	19.667108	19.665725	19.577666	+0.0070	+0.4569	0.4527	reld_mt
VRPLTW	79.4000	79.2444	18.801316	18.803136	18.745143	-0.0097	+0.2997	0.3167	reld_mt
OVRPBL	89.2889	89.2889	6.375710	6.375741	6.359310	-0.0005	+0.2579	0.2693	reld_mt
OVRPBTW	84.9111	84.8222	10.659029	10.659591	10.624748	-0.0053	+0.3226	0.2945	reld_mt
OVRPLTW	82.8889	82.8667	10.085770	10.086112	10.059001	-0.0034	+0.2661	0.2668	reld_mt
VRPBLTW	79.7778	79.7333	19.704833	19.707068	19.613284	-0.0113	+0.4668	0.4486	reld_mt
OVRPBLTW	83.7111	83.5778	10.661385	10.662149	10.625574	-0.0072	+0.3370	0.3132	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R101/A | LIB and locked provenance

Current R101 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	28.5714	10.2041	8.210717	8.240435	8.142167	-0.3606	+0.8419	0.8808
CVRPLIB	62.0000	45.0000	67.254707	67.245364	66.945093	+0.0139	+0.4625	0.5744

SHA256: e96d42b96ec03155d8ae60e698459605bcf382e0d9f5d488a224fd809d0e23d9

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, original R88B physical TSP node0 slot3; original R77 hand34/methodID, joint2, score MLP/pair differences/maximin; stable ascending global-solver-ID ties; original inputs only; no teacher, hint, adapter or calibration

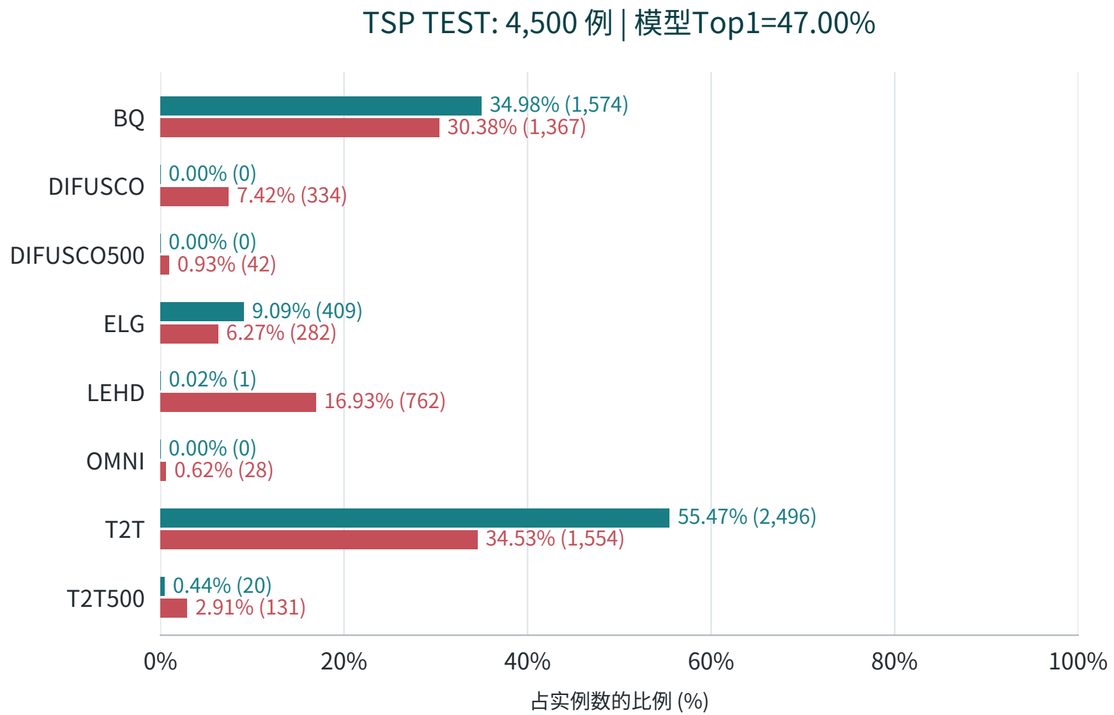
Normalization: new81000 TRAIN only; no TEST/LIB refit.

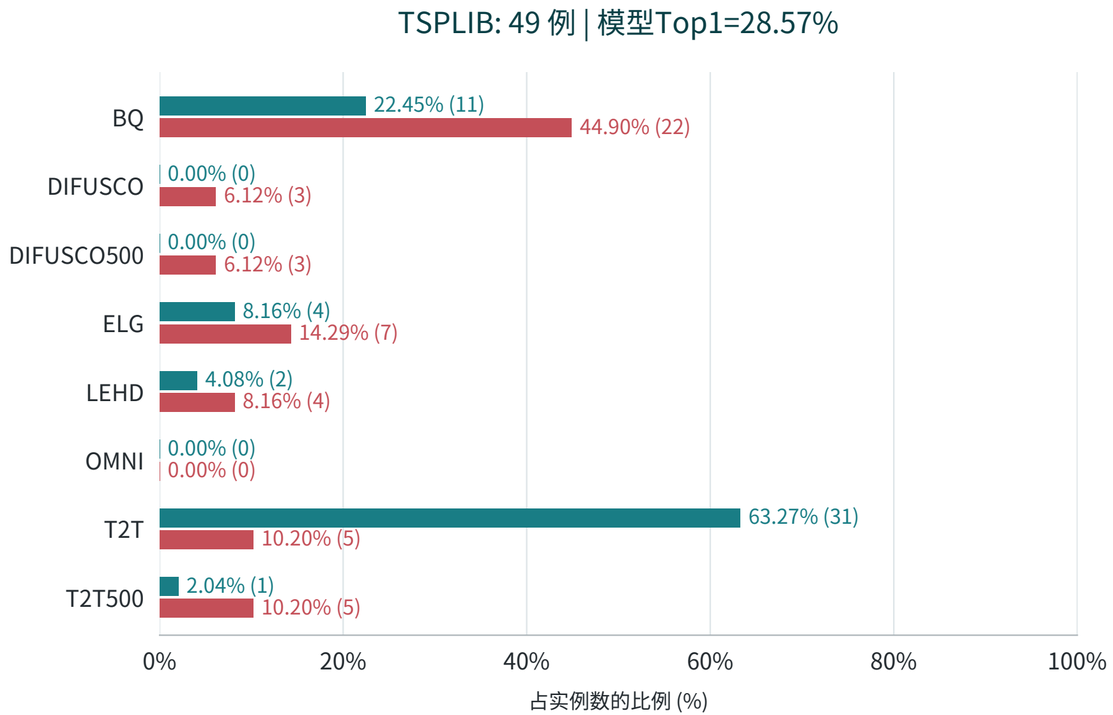
R101/A | TSP | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

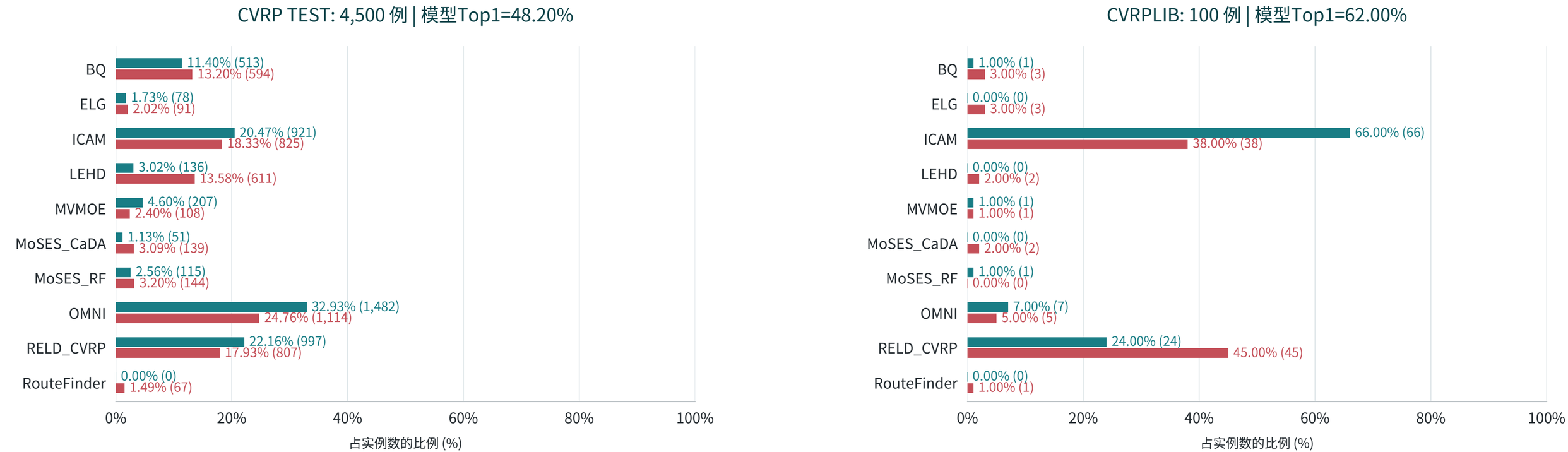
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | CVRP | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

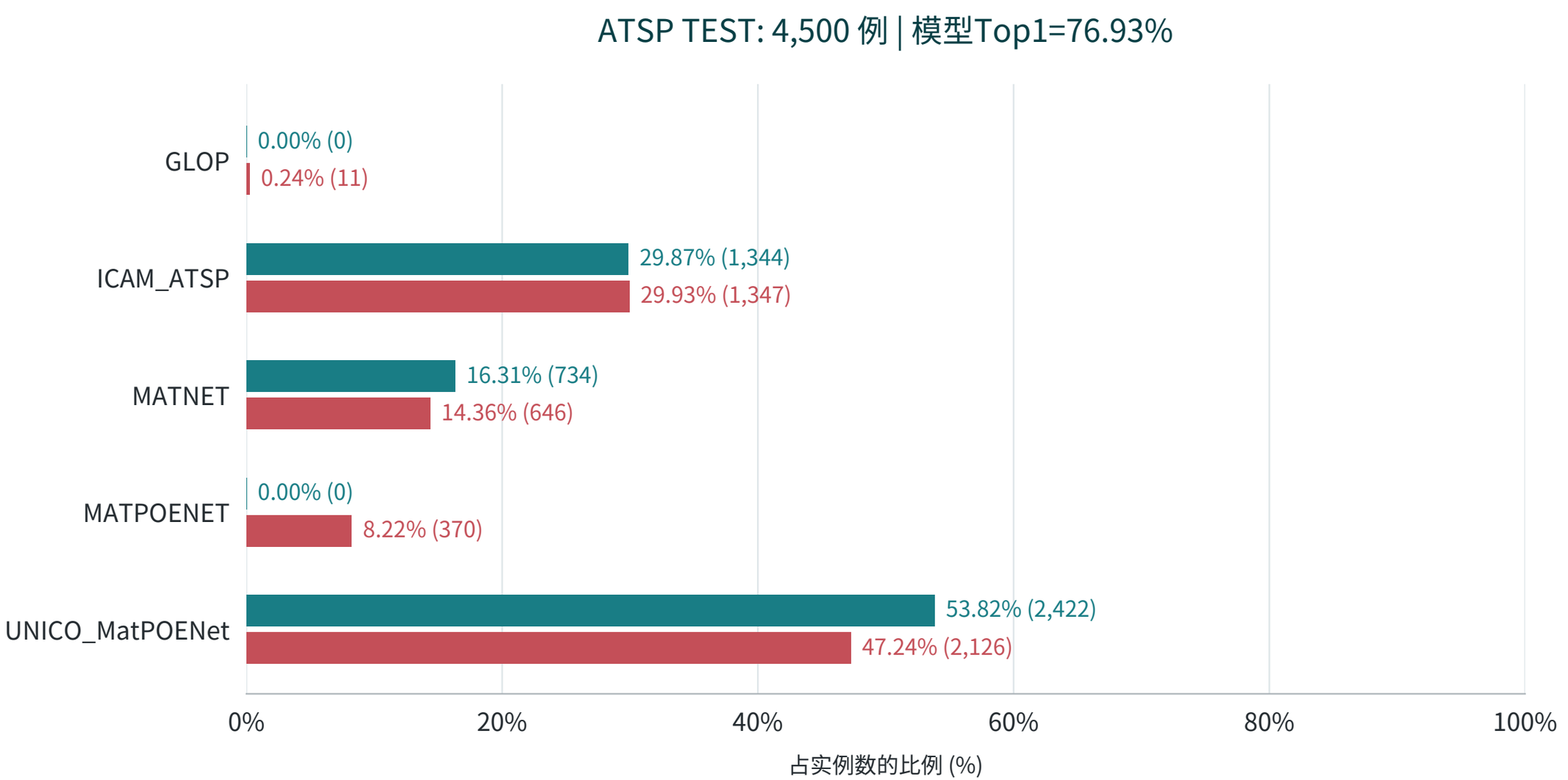
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | ATSP | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

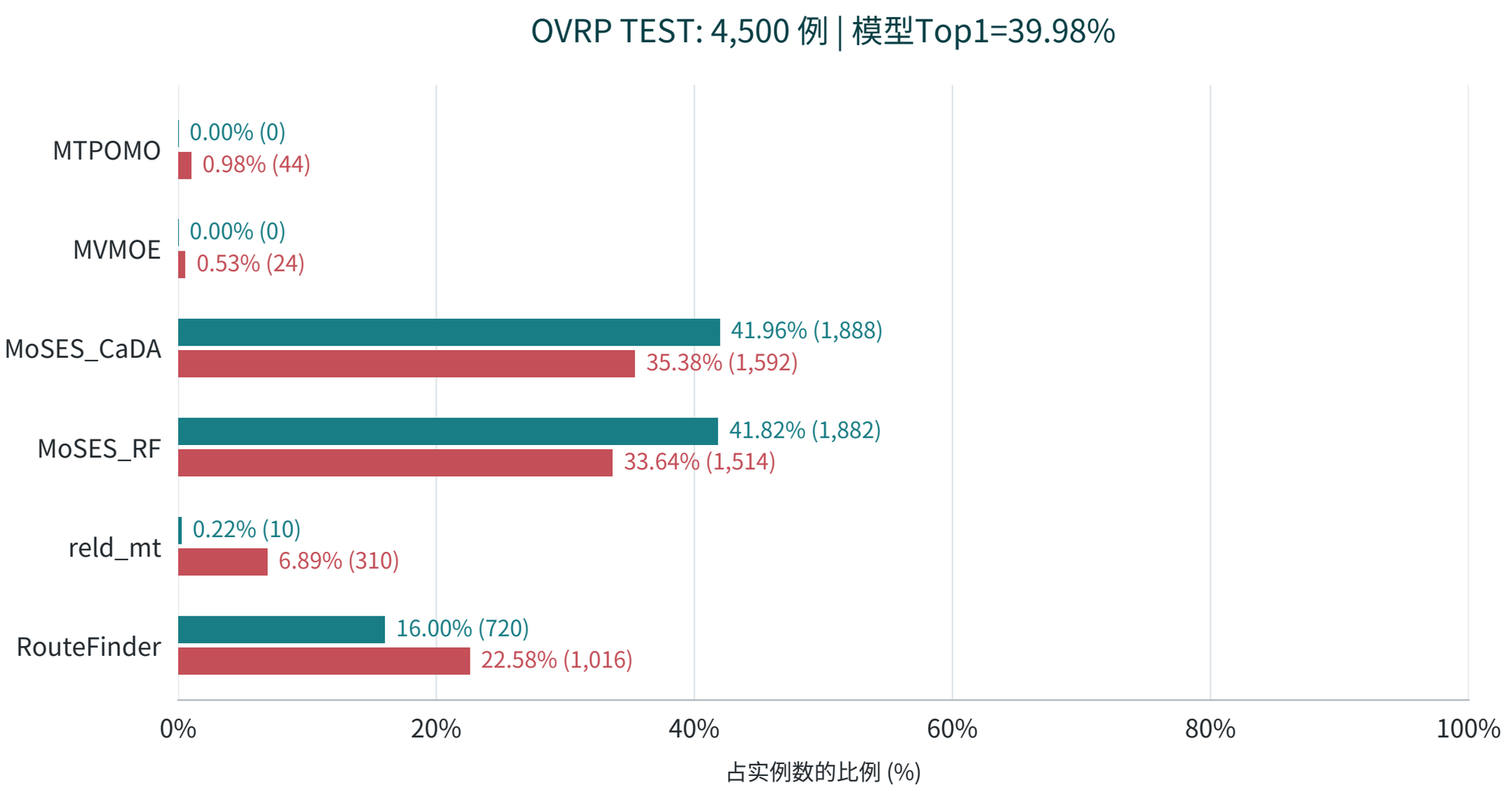
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRP | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

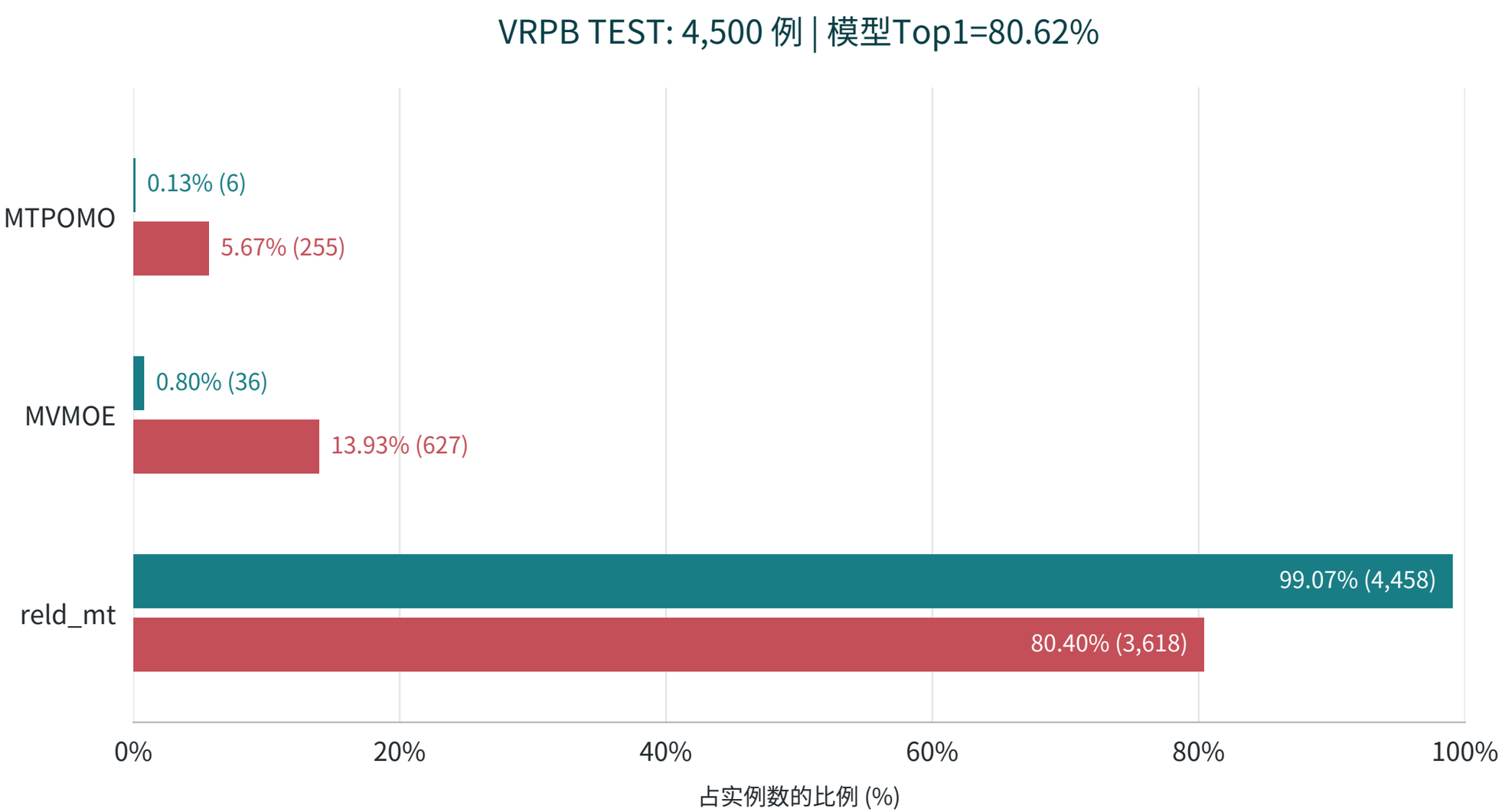
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPB | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

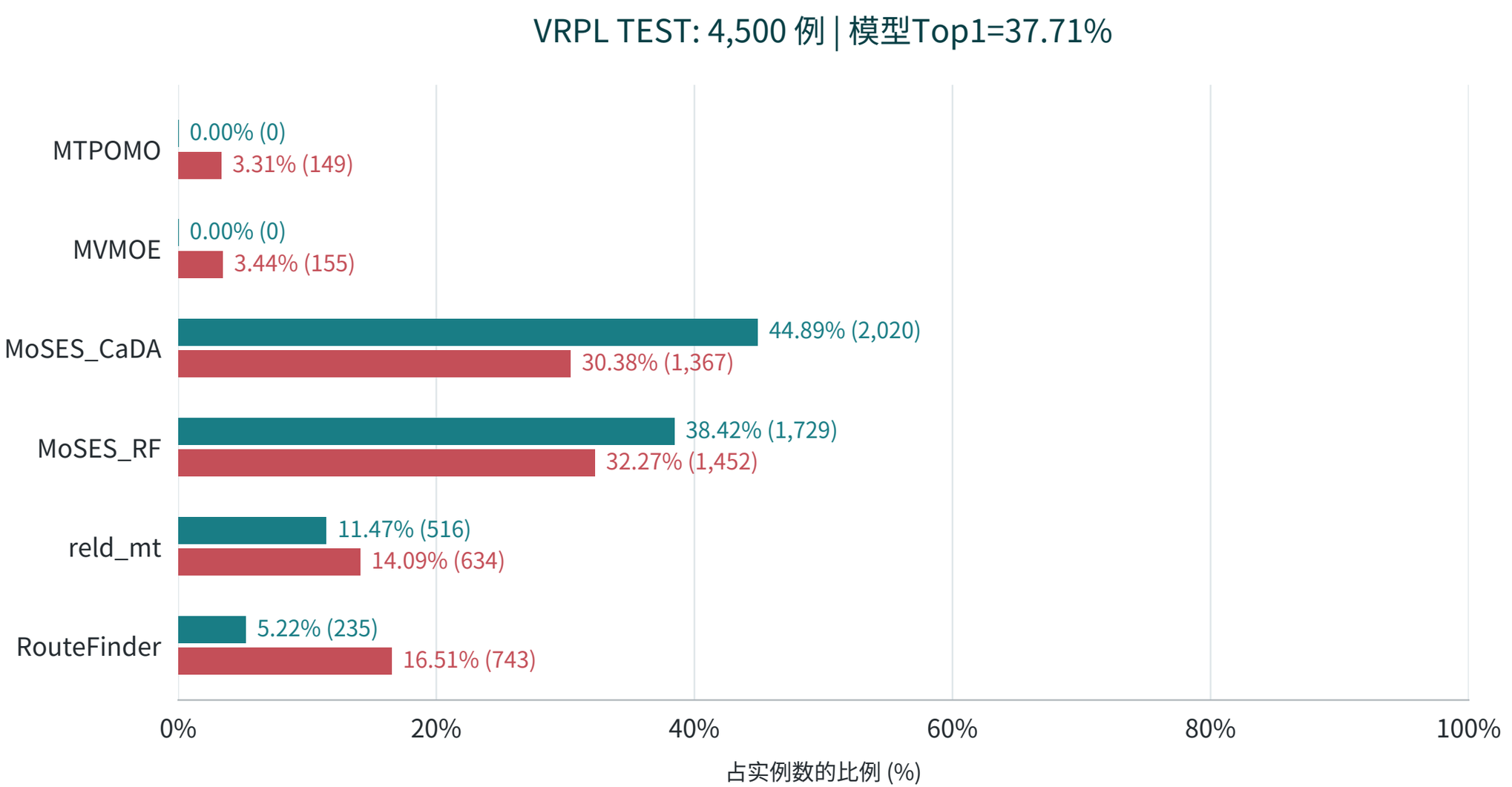
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPL | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

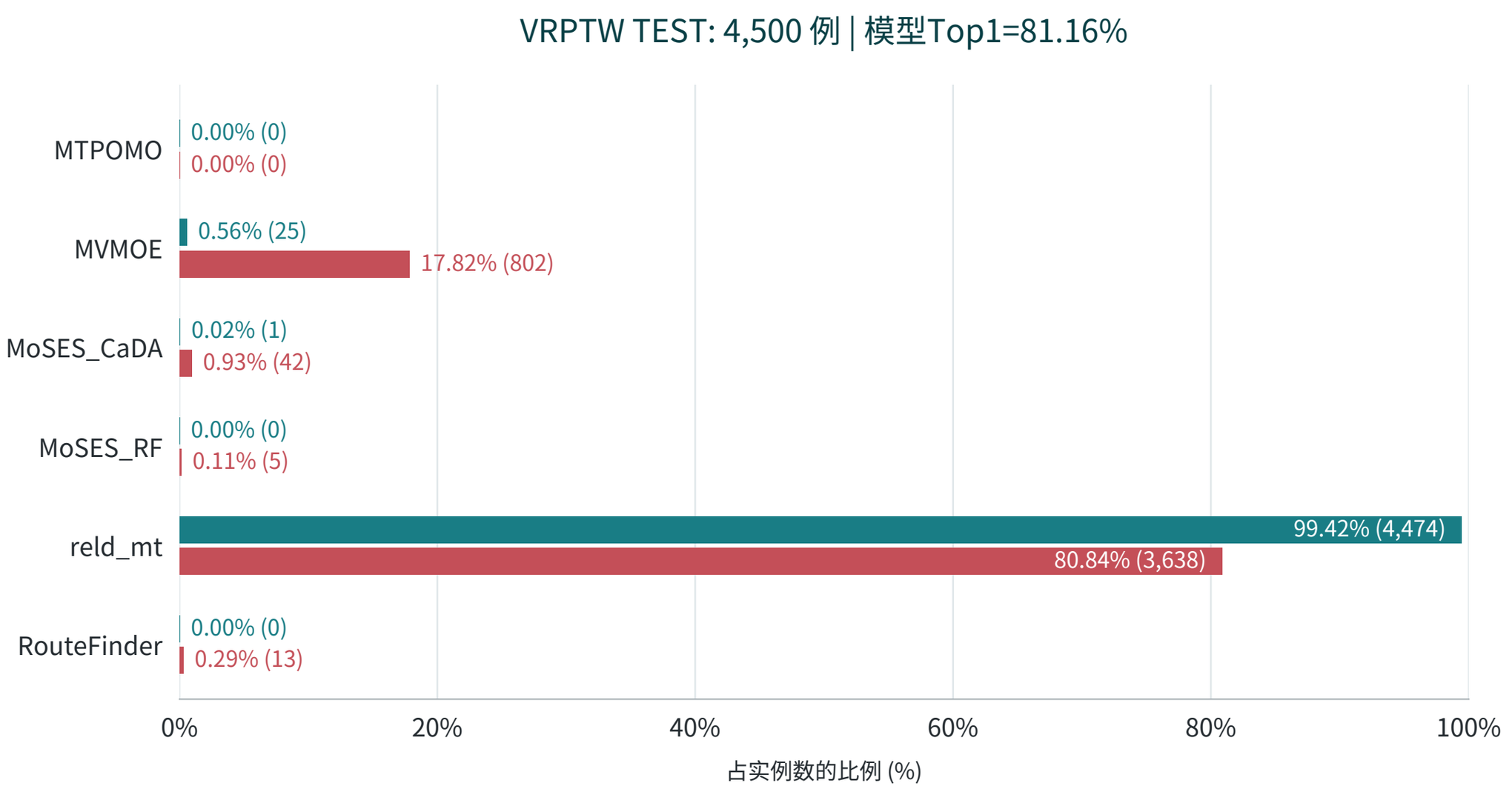
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

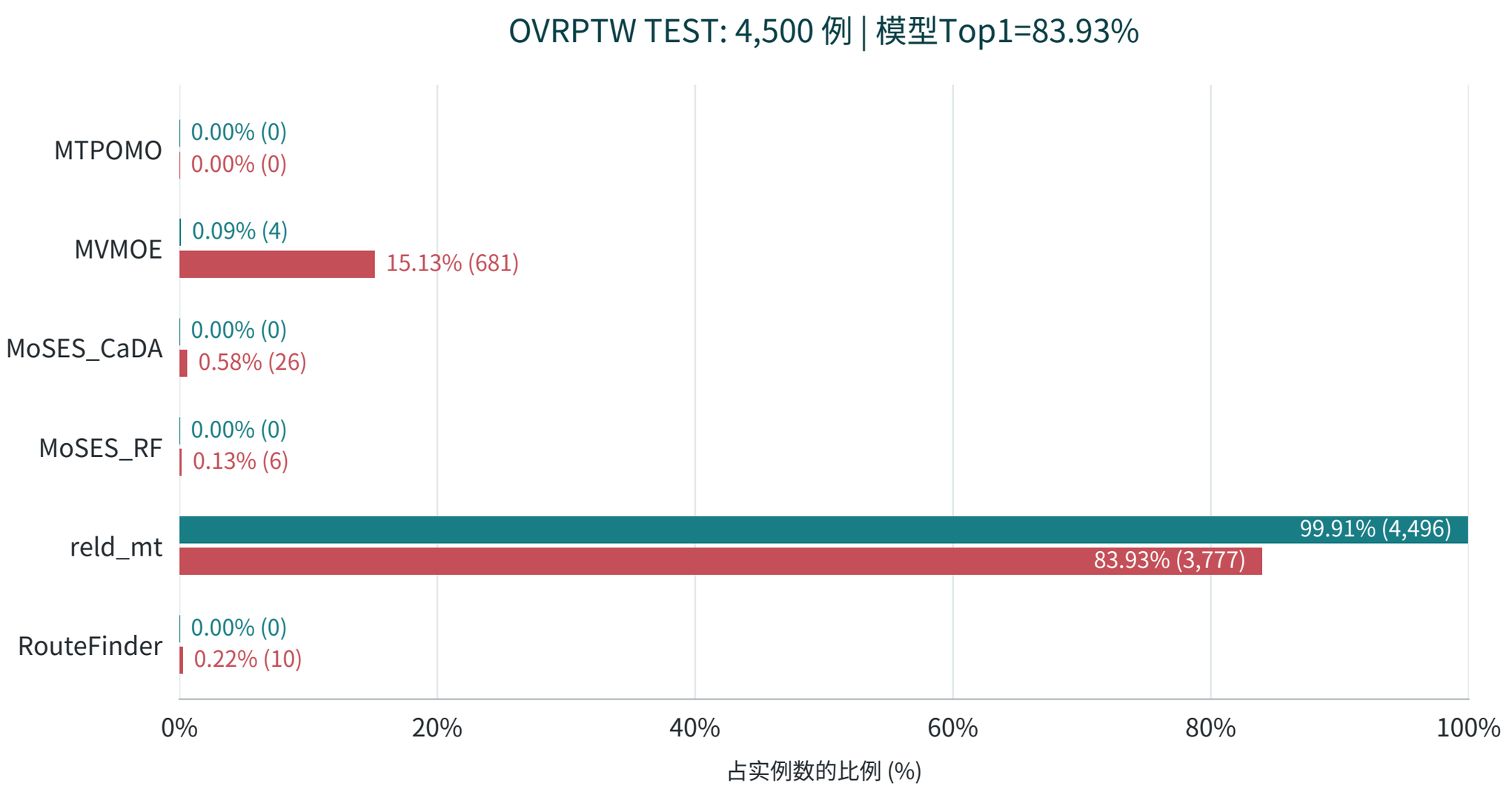
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

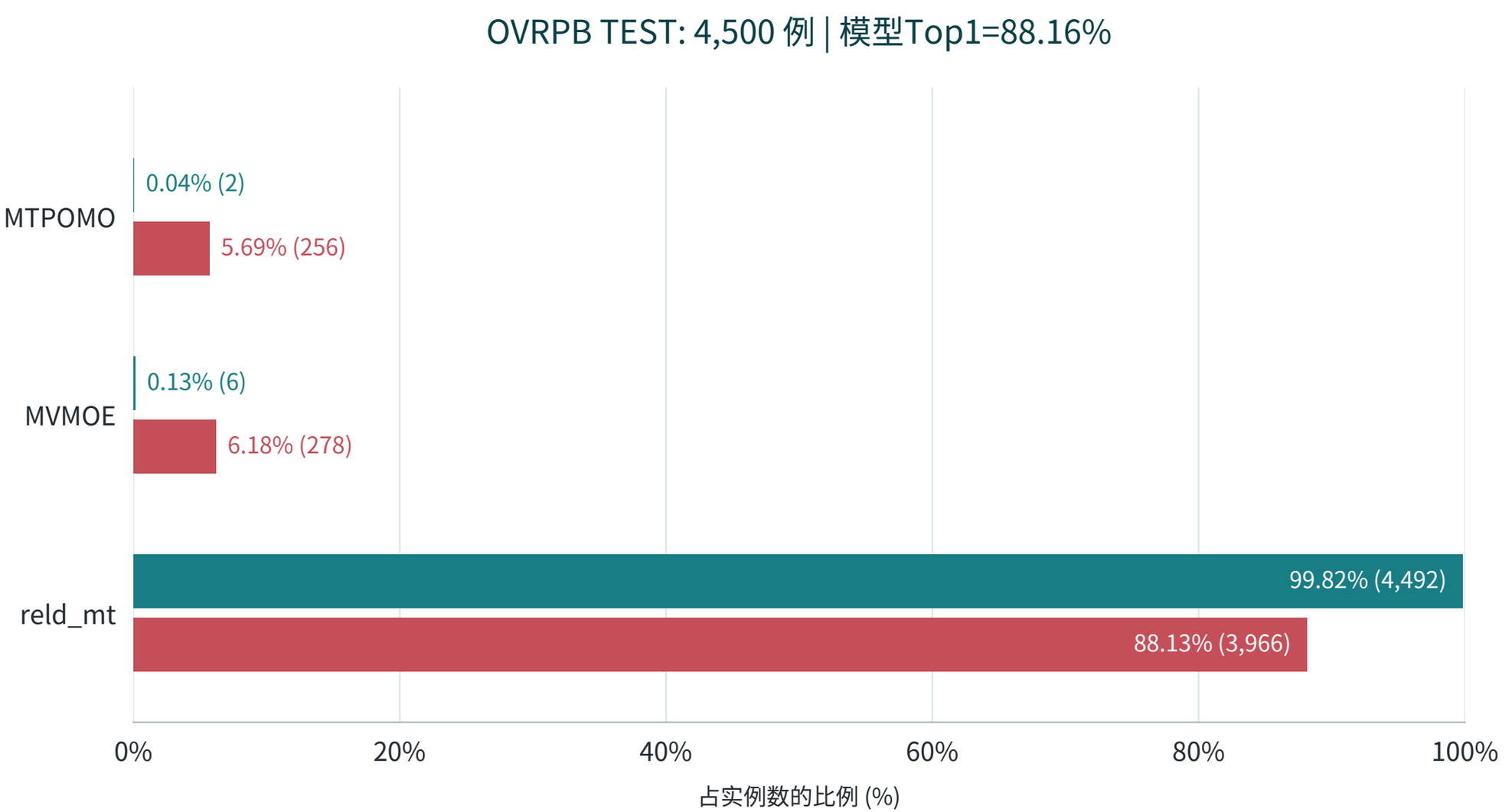
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPB | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

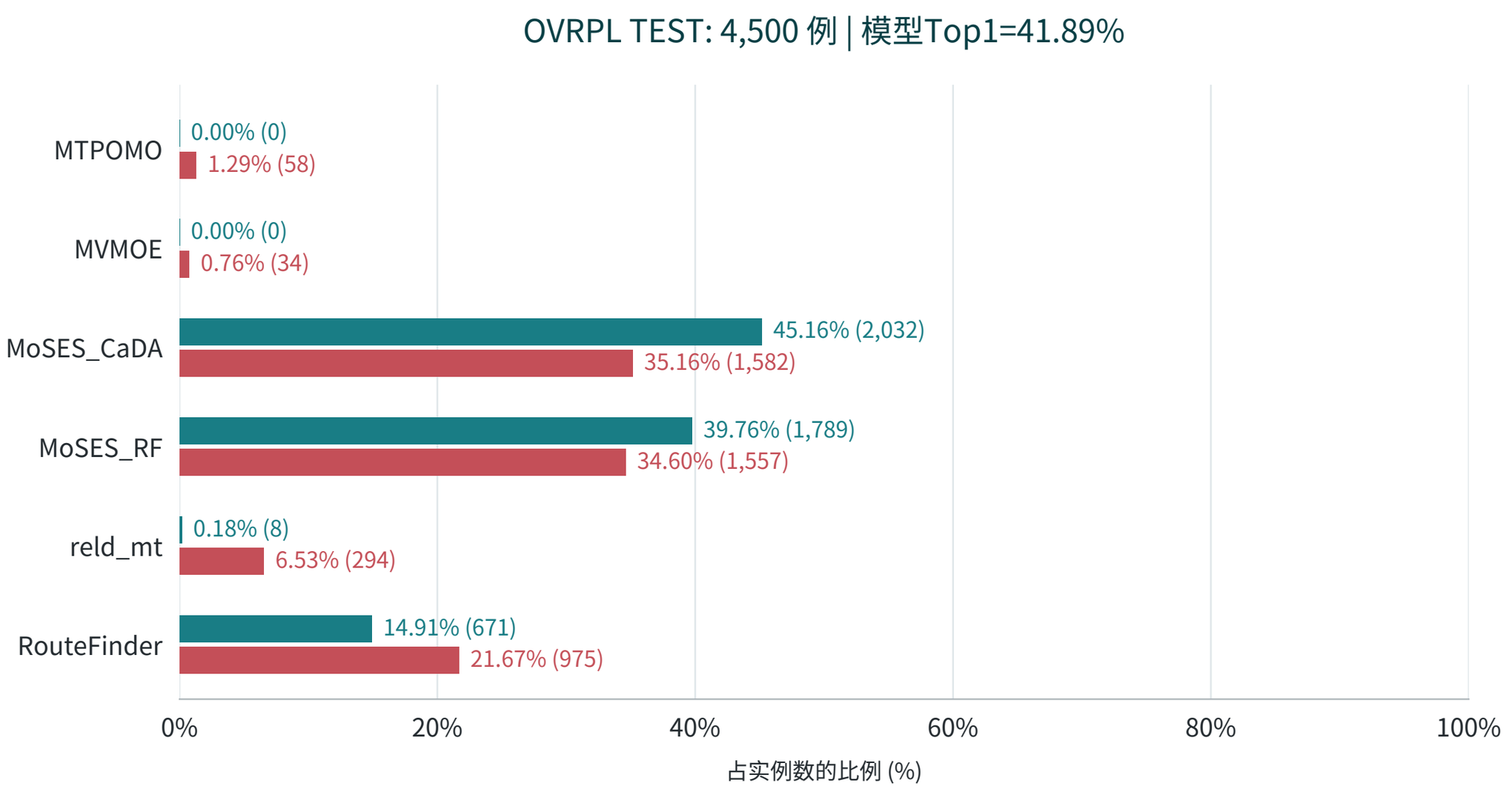
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPL | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

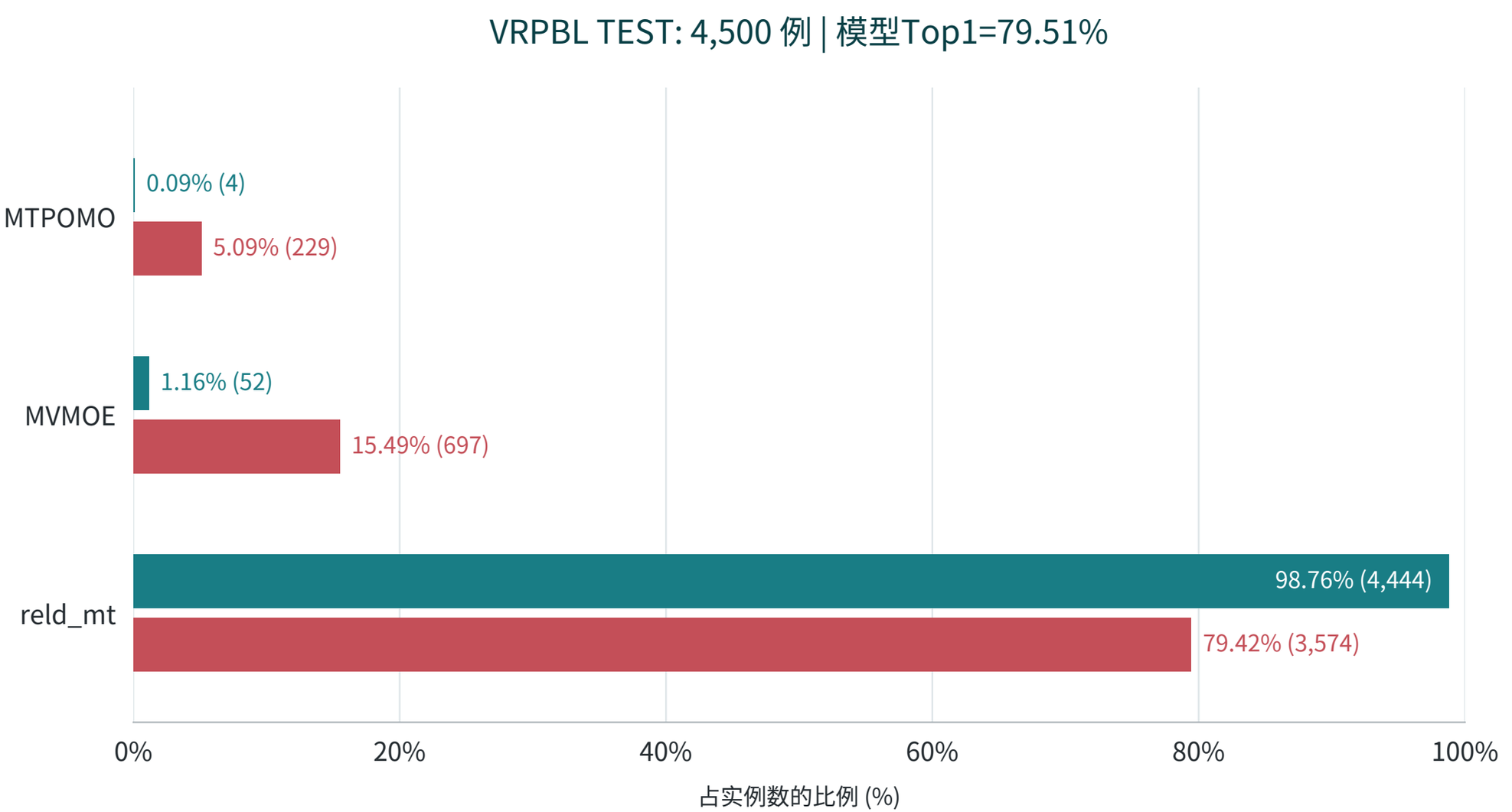
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPBL | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

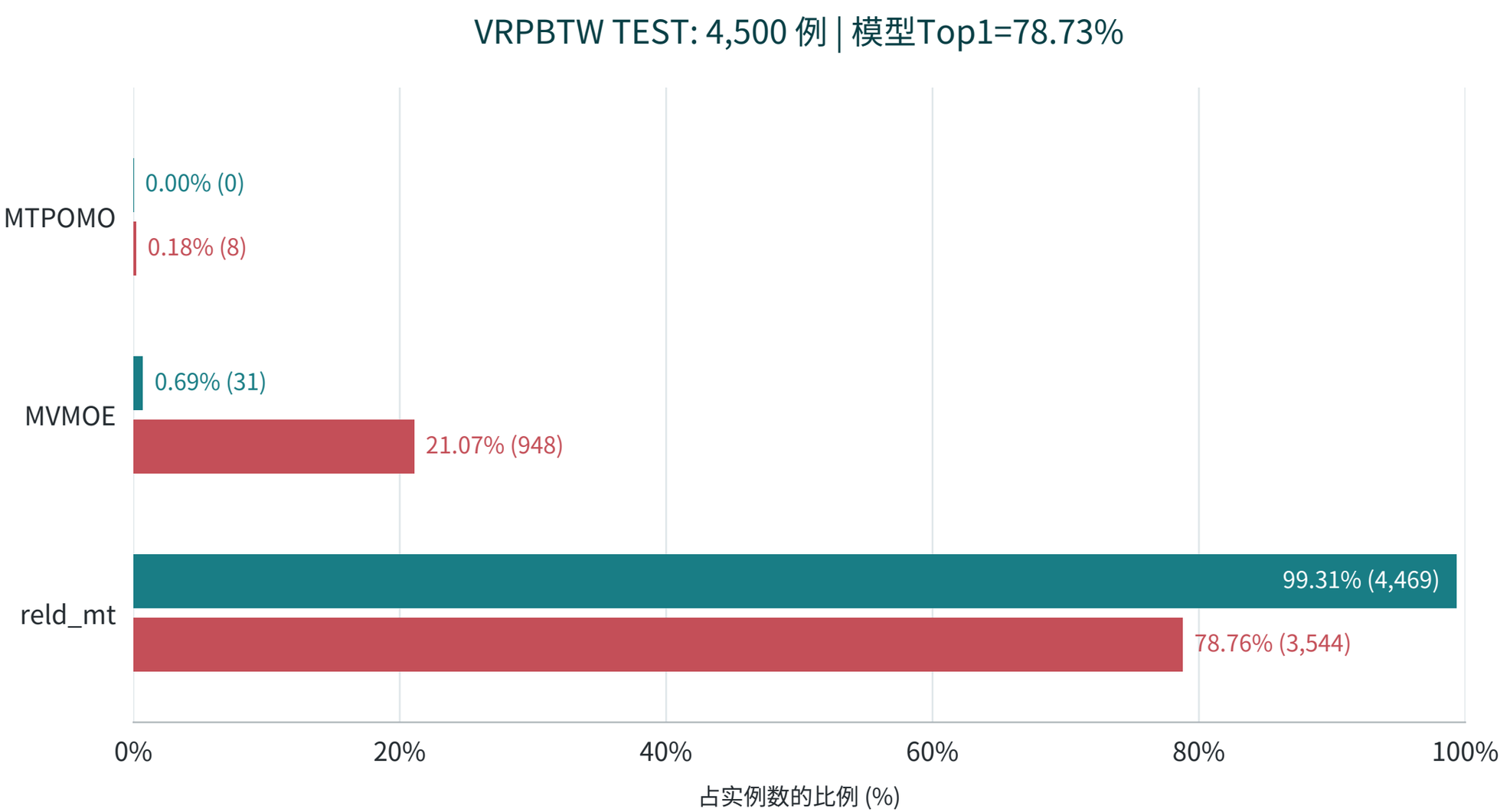
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPBTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

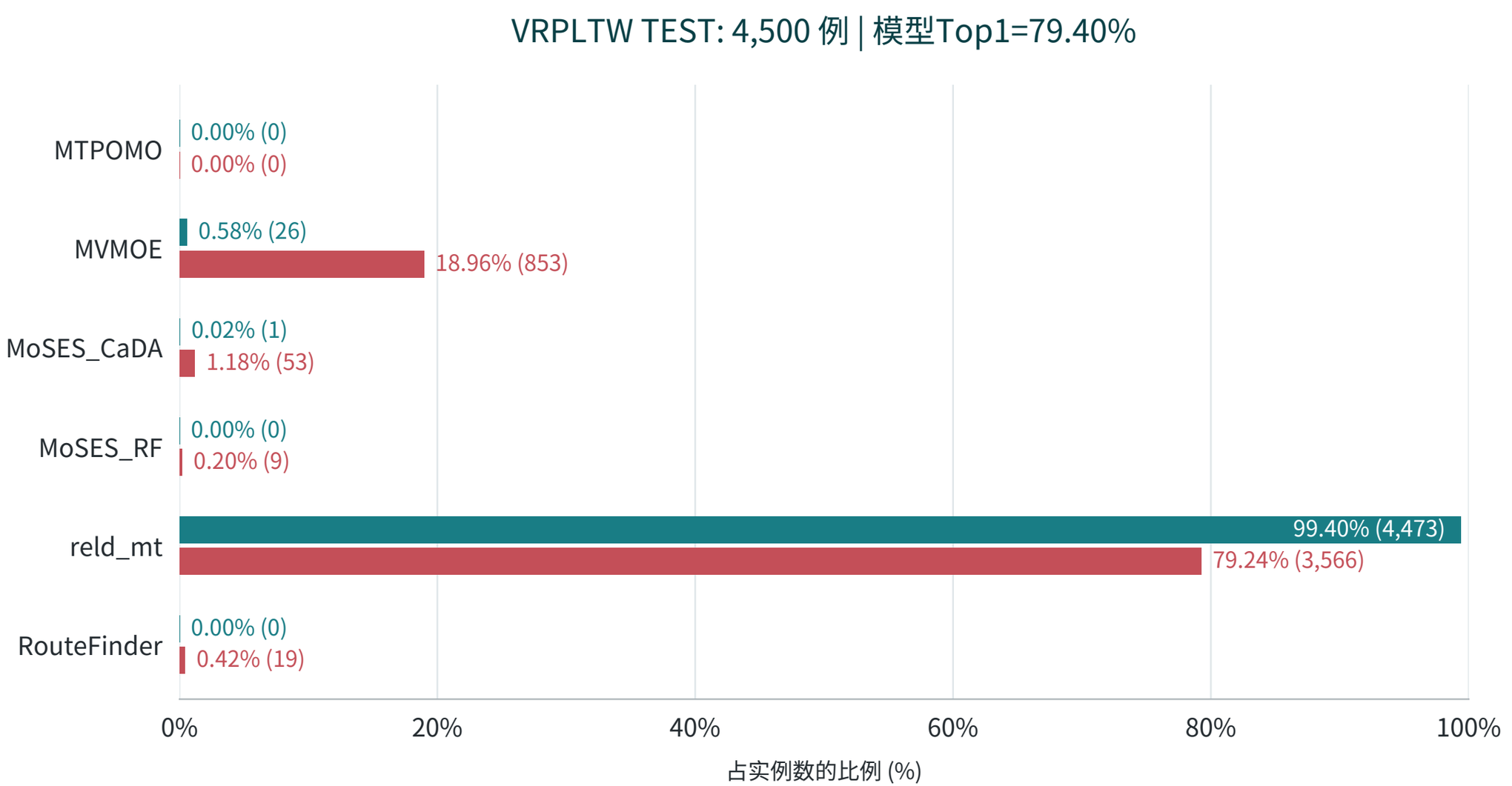
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPLTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

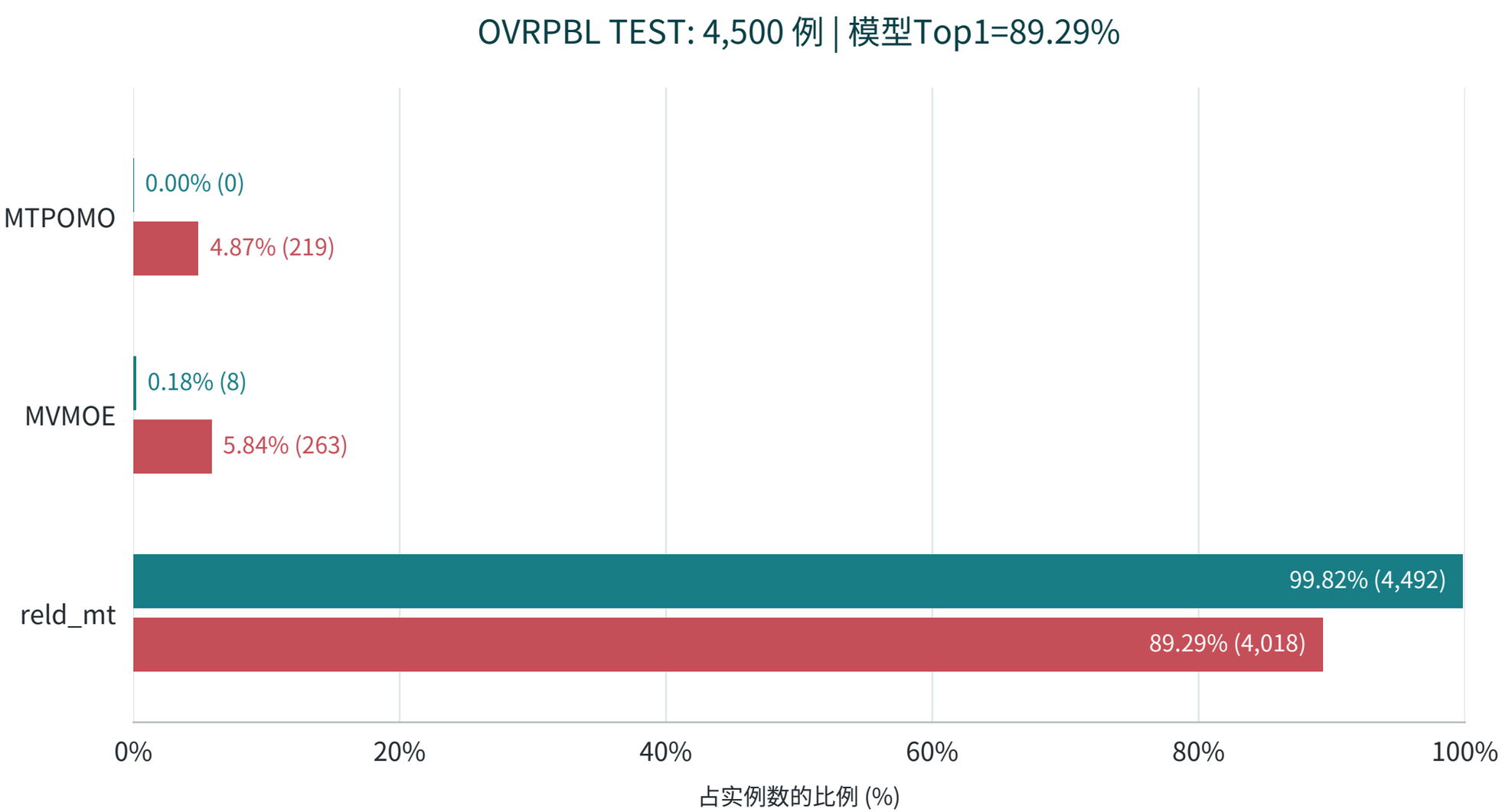
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPBL | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



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模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

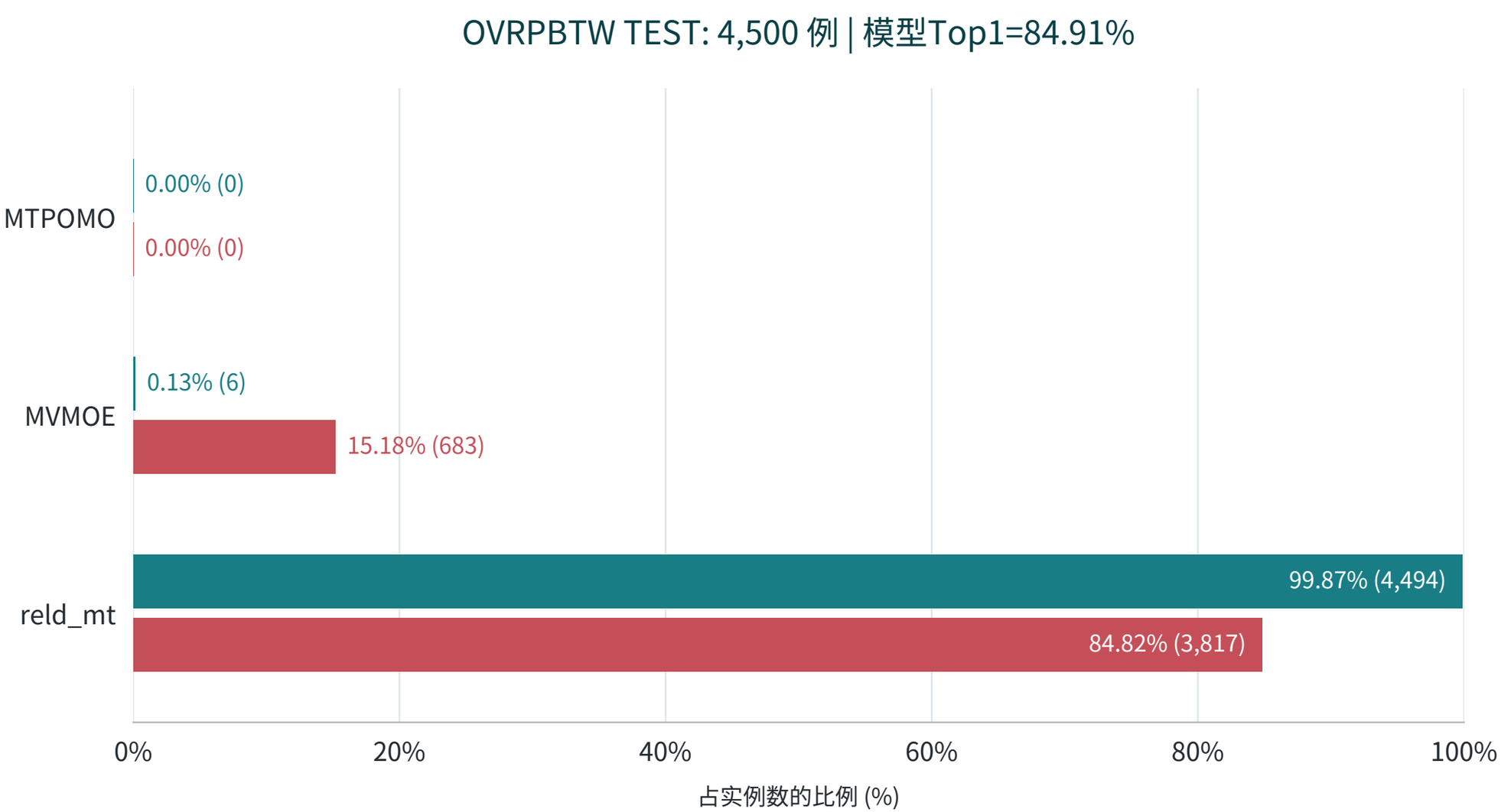
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPBTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

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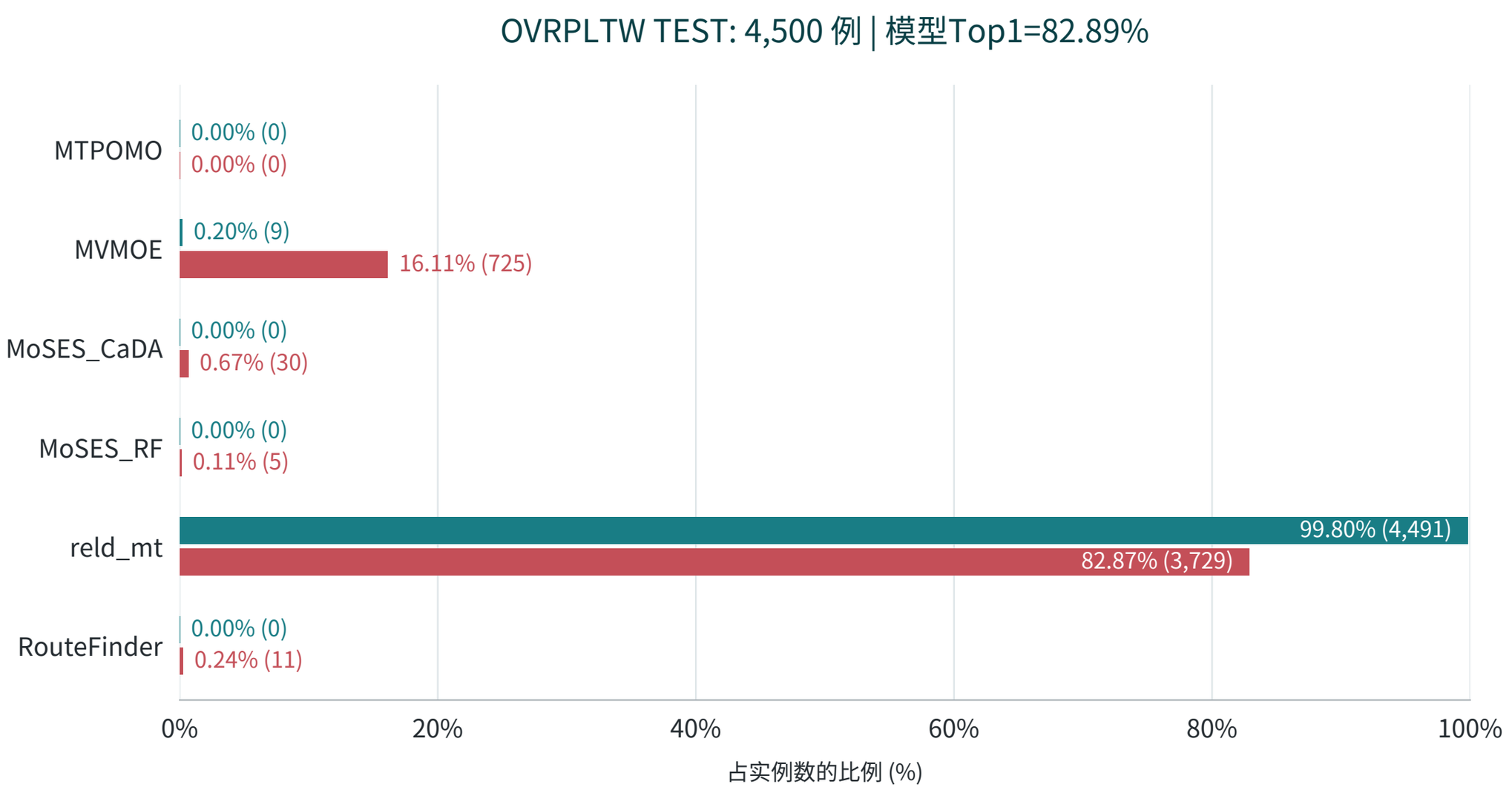
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPLTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

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模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

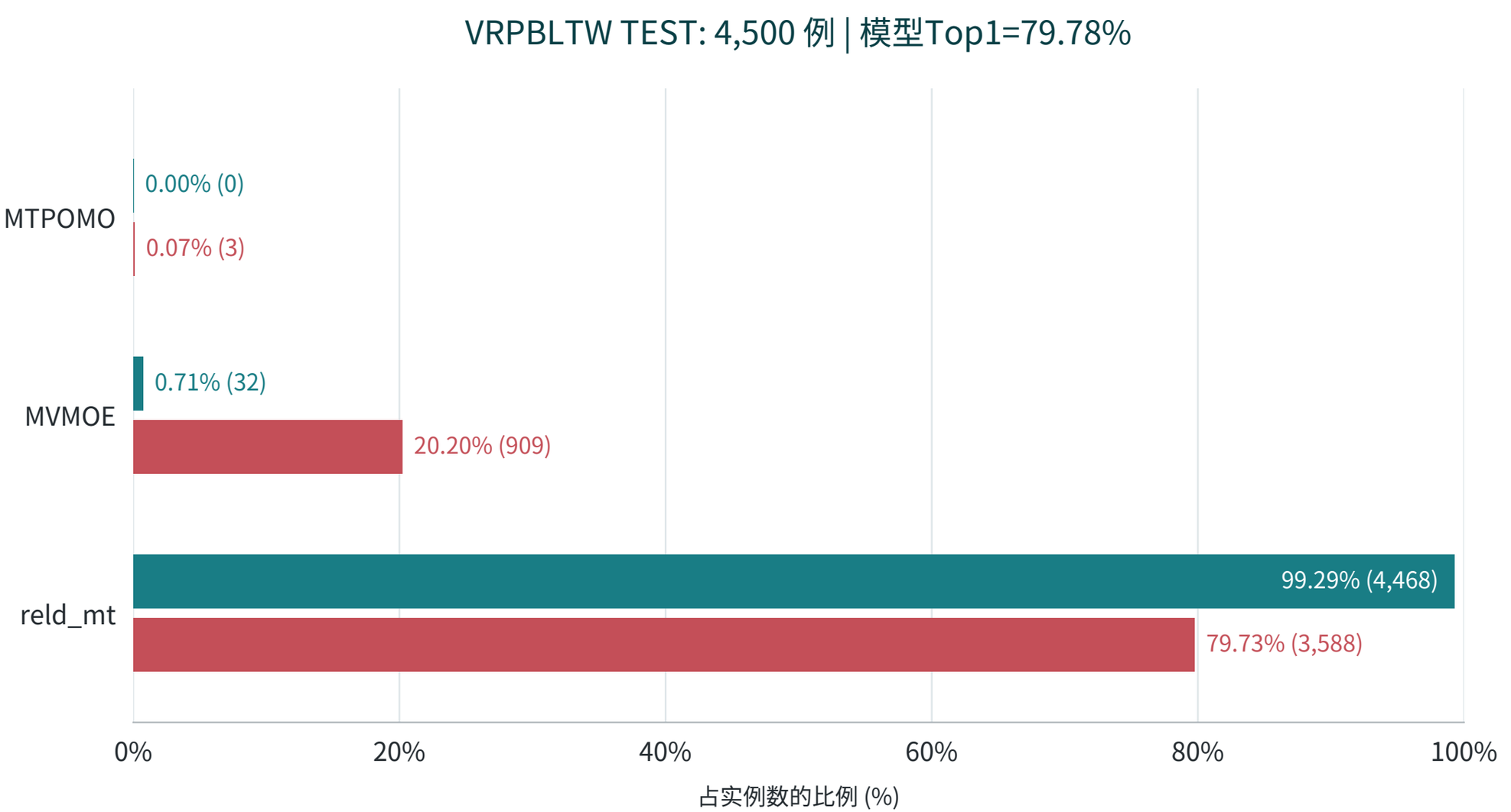
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPBLTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

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标签: 原生 native/FP64 ind; 仅读锁定预测



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模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

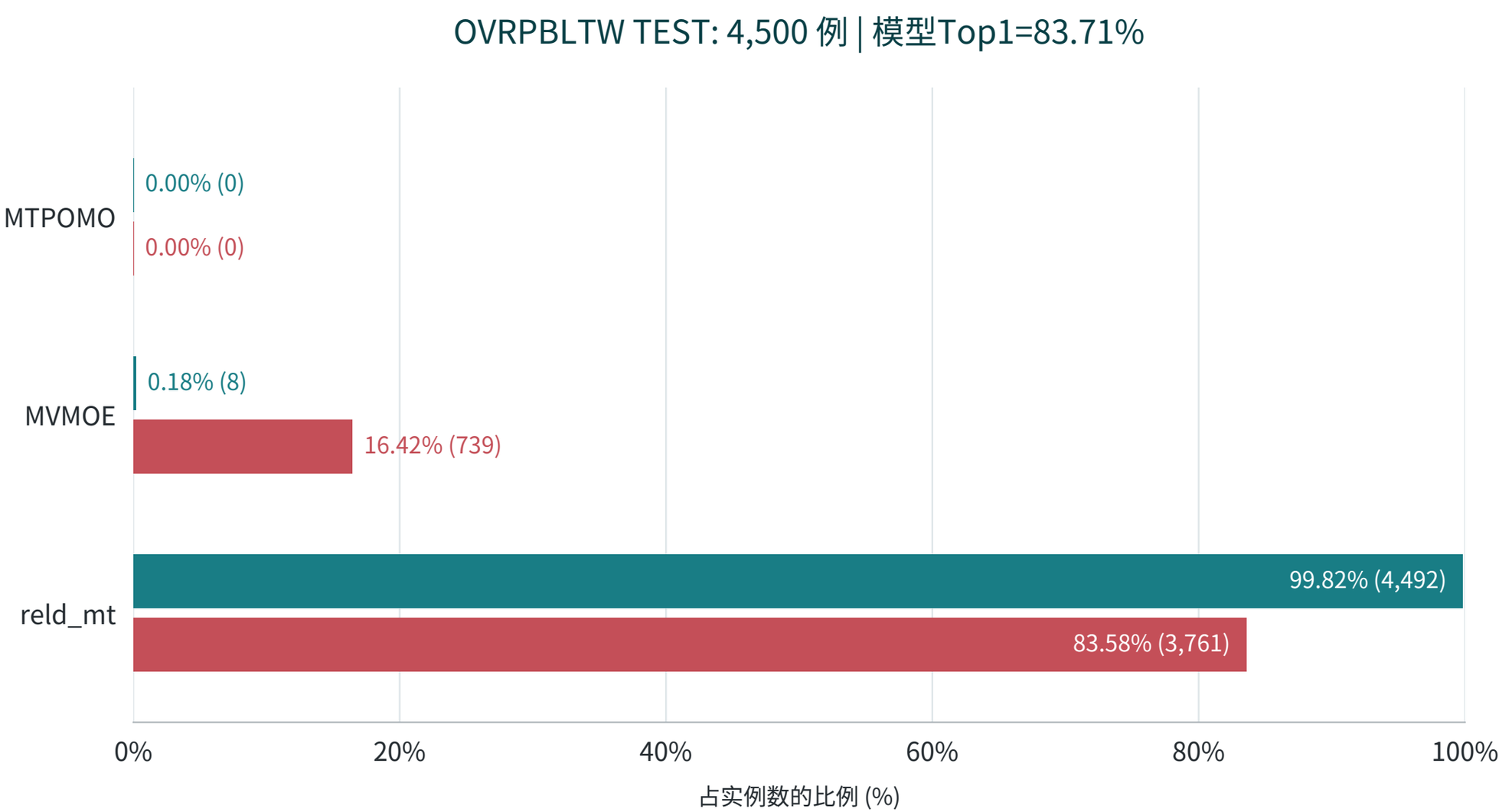
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPBLTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

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标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | Fixed target36 vs R89B

Native/FP64 winners; true rescues/losses and false-pick cost, not selection-frequency accuracy.

Split	True winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	407	82	699	140	0.4887
VAL	4625	15	2	39	8	0.3243
TEST	4598	19	4	30	9	0.4132

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.322359	22374	27074	0.000871852	0.008964
VAL	-0.096296	1290	1368	0.000277633	0.001933
TEST	-0.097531	1316	1395	0.000368085	0.003383

Split	New-false cost contribution	New-false cost delta	New-false regret pp
TRAIN	0.000733527	0.000026849	0.000463
VAL	0.000754442	0.000016672	0.000295
TEST	0.000963987	0.000032271	0.000726

Cost contributions: sum over newly false-picked instances / task n; ALL averages18 tasks.

R89B/R84A are historical, not12800+57600 budget-matched; zero-count movement is not success.

R101/A | Fixed target36 vs R84A

Native/FP64 winners; true rescues/losses and false-pick cost, not selection-frequency accuracy.

Split	True winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	407	204	94	199	0.4887
VAL	4625	15	8	4	9	0.3243
TEST	4598	19	11	5	13	0.4132

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.124897	22060	20239	-0.000212131	-0.002600
VAL	0.066667	1204	1150	-0.000039560	-0.000885
TEST	0.086420	1203	1133	-0.000004506	-0.001817

Split	New-false cost contribution	New-false cost delta	New-false regret pp
TRAIN	0.001015246	0.000038800	0.000693
VAL	0.000743371	0.000011176	0.000210
TEST	0.001231555	0.000028942	0.000849

Cost contributions: sum over newly false-picked instances / task n; ALL averages18 tasks.

R89B/R84A are historical, not12800+57600 budget-matched; zero-count movement is not success.

R101/A | Original student latency

First128 GPU-resident original TEST inputs/task; three synchronized repetitions.

Problem	A student ms/row
TSP	0.243893
CVRP	0.291117
ATSP	0.065964
OVRP	0.064915
VRPB	0.065113
VRPL	0.065013
VRPTW	0.064890
OVRPTW	0.065490
OVRPB	0.064941
OVRPL	0.064911
VRPBL	0.065333
VRPBTW	0.065099
VRPLTW	0.064982
OVRPBL	0.065081
OVRPBTW	0.065137
OVRPLTW	0.064811
VRPBLTW	0.065158
OVRPBLTW	0.065094

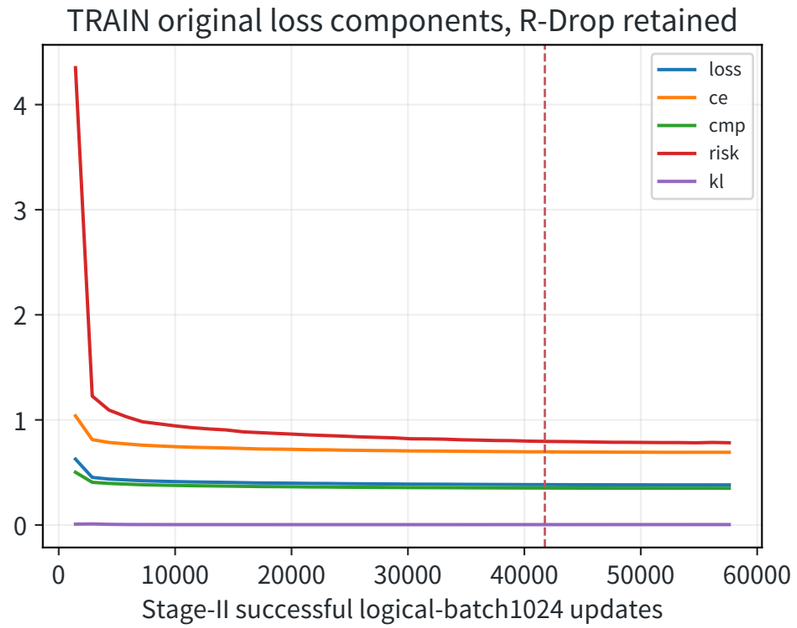
Original1842953-parameter student only; no online donor, teacher feature I/O or solver decoder.

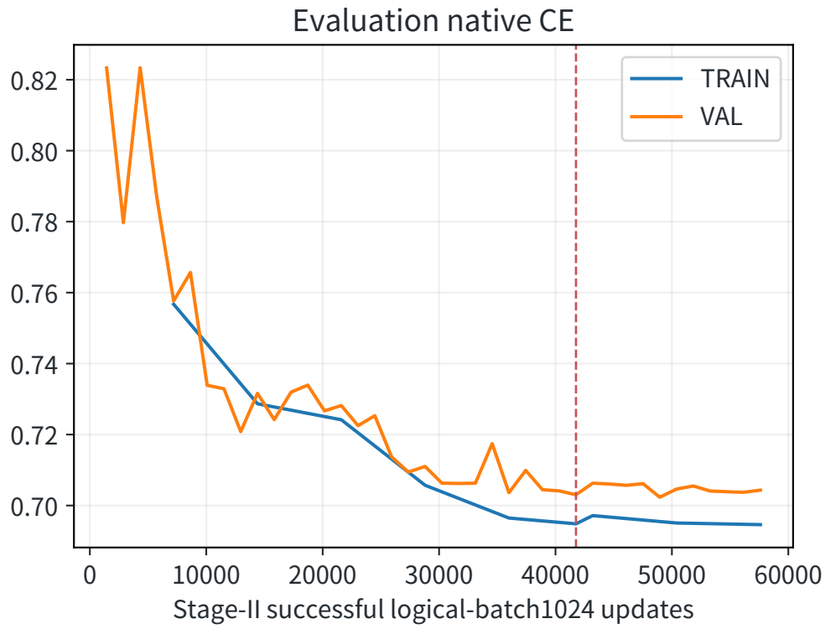
Excludes dataset I/O, host transfer and Stage-I pretraining; not single-instance/E2E latency.

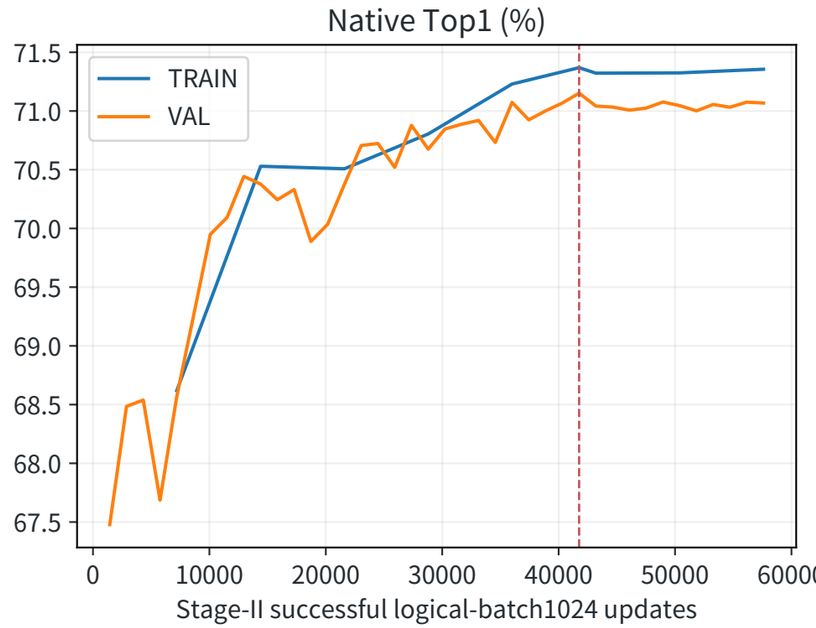
No latency or speedup claim is inferred from R94 cached-selector numbers.

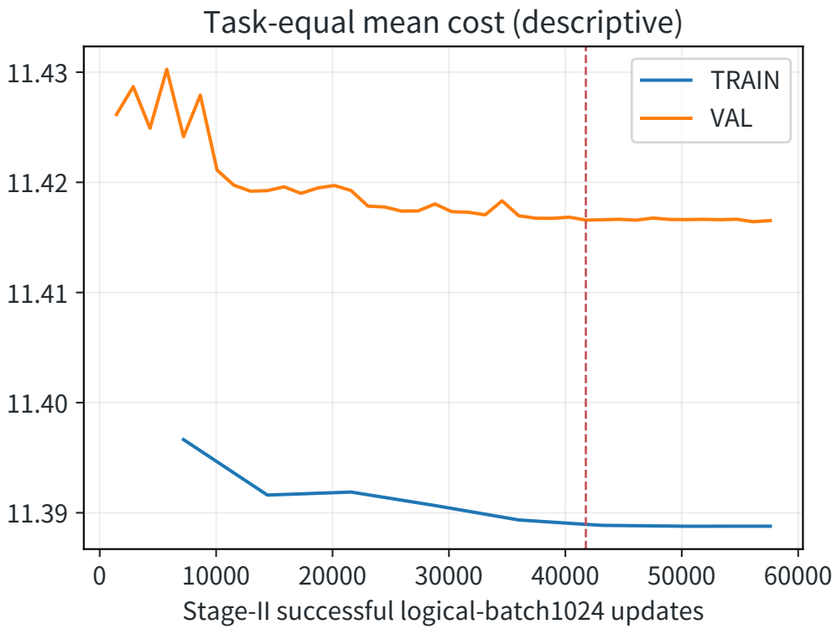
Hardware is disclosed, not bitwise matched; no model reselection based on timing.

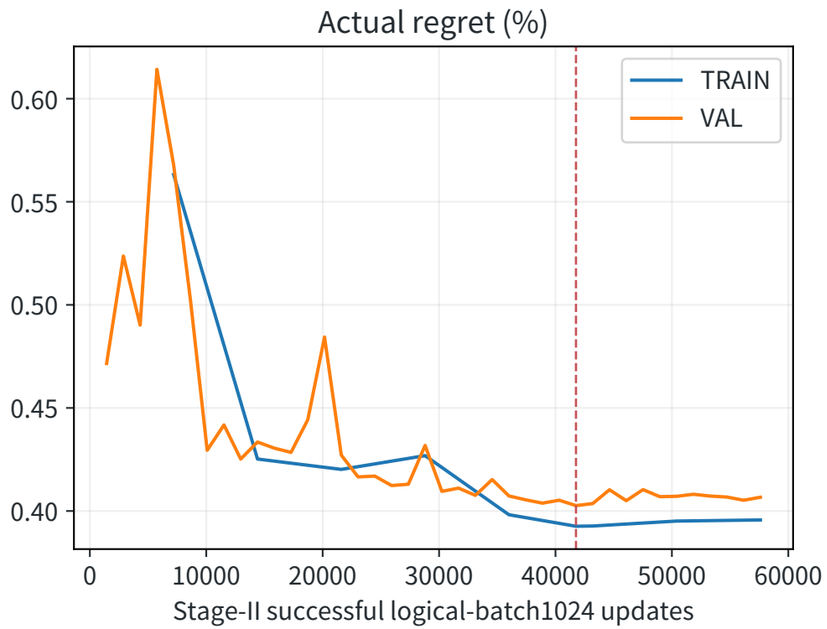
R101/A | Stage II: original R77, no hint loss or teacher inputs

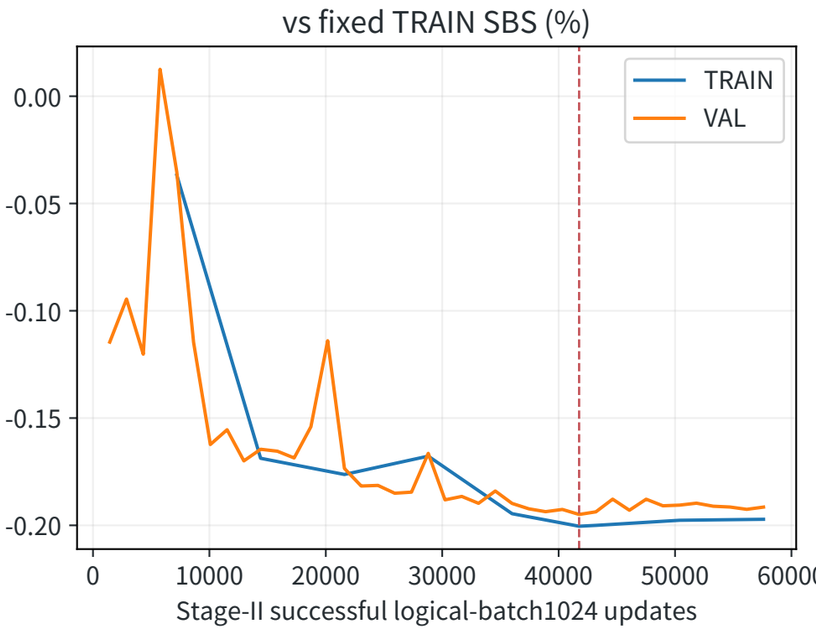






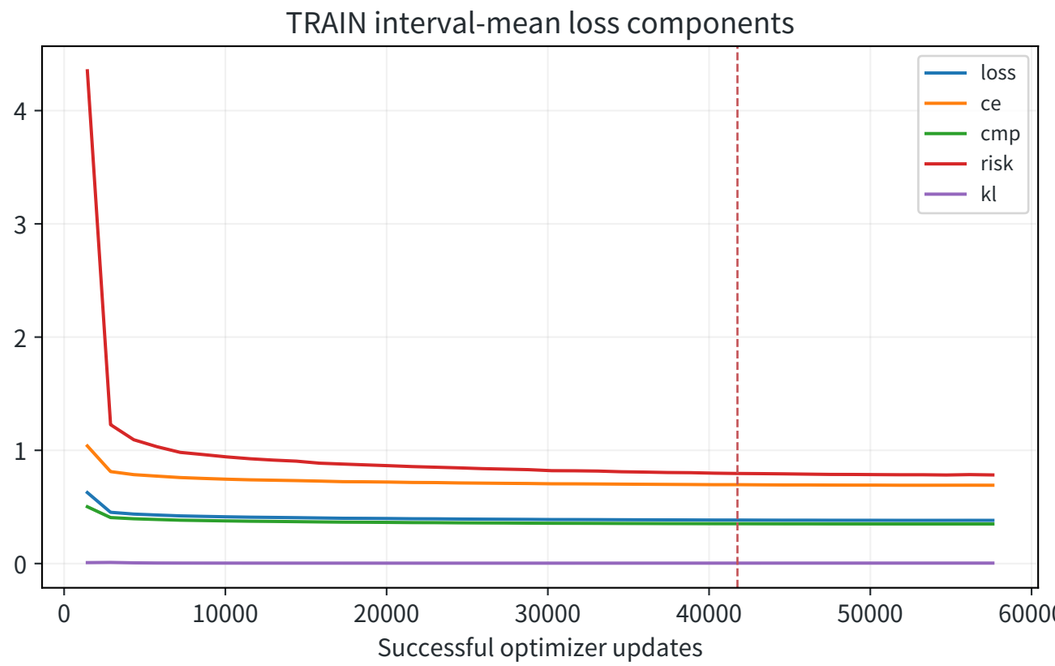


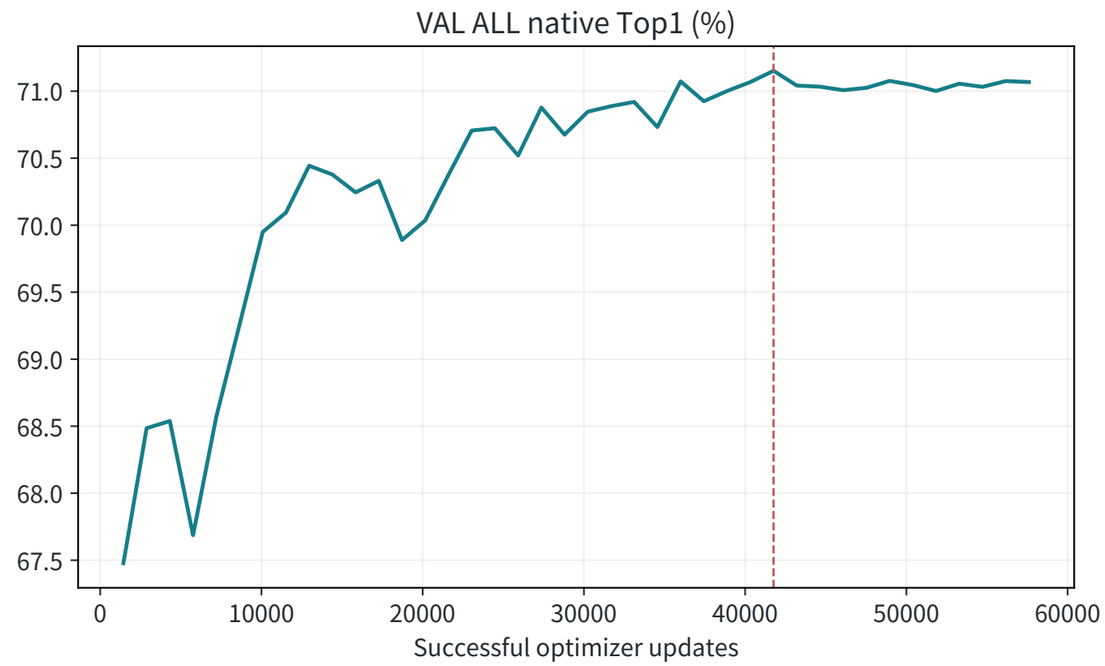


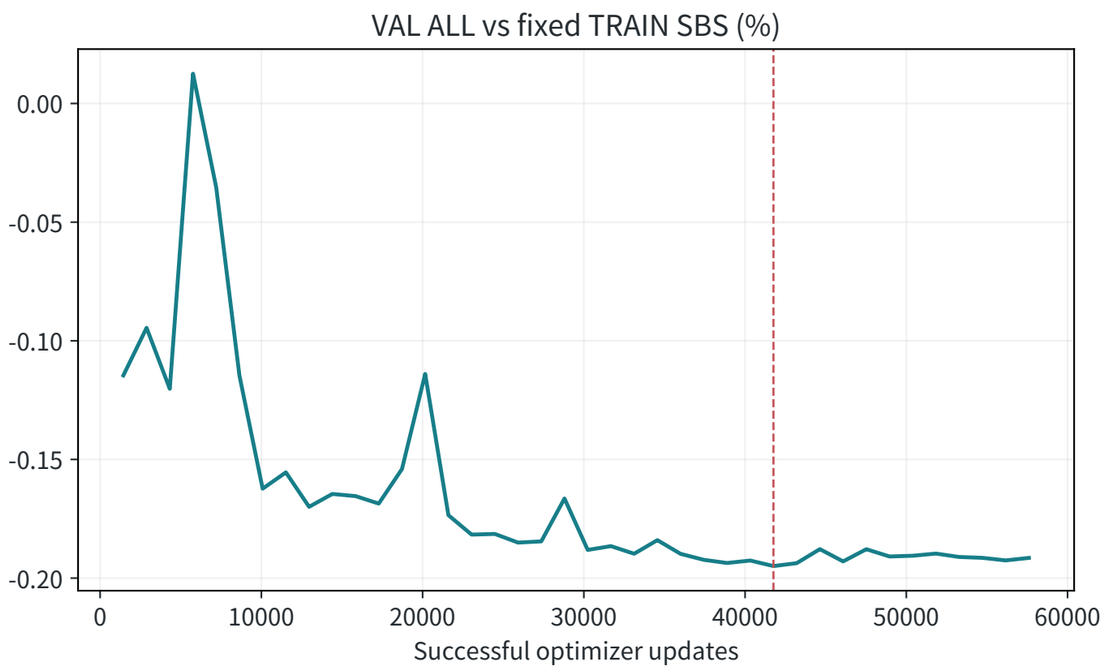


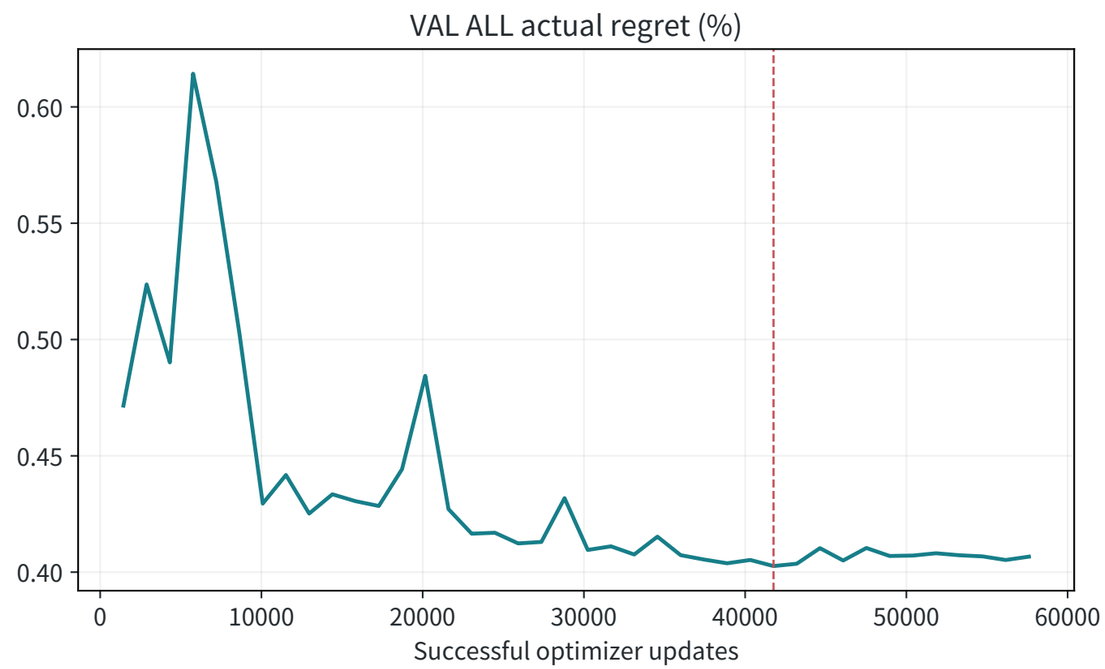
R101/A | Saved Training History

Successful-update axis. Training losses and full VAL traces are descriptive only; the checkpoint is already locked.



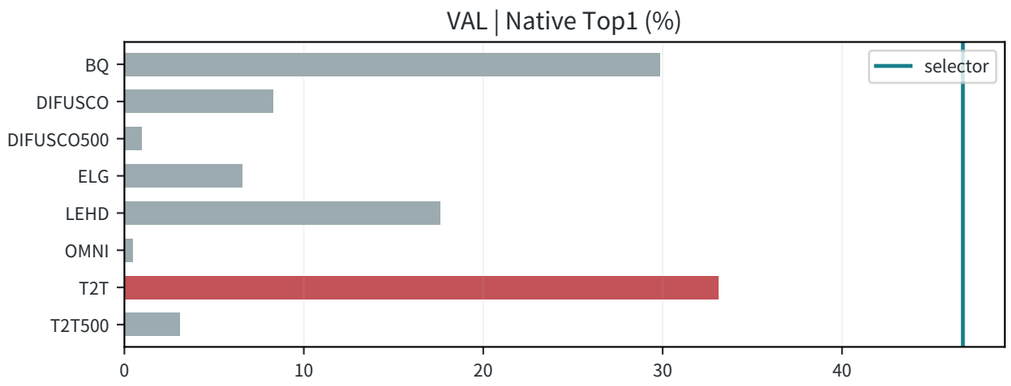


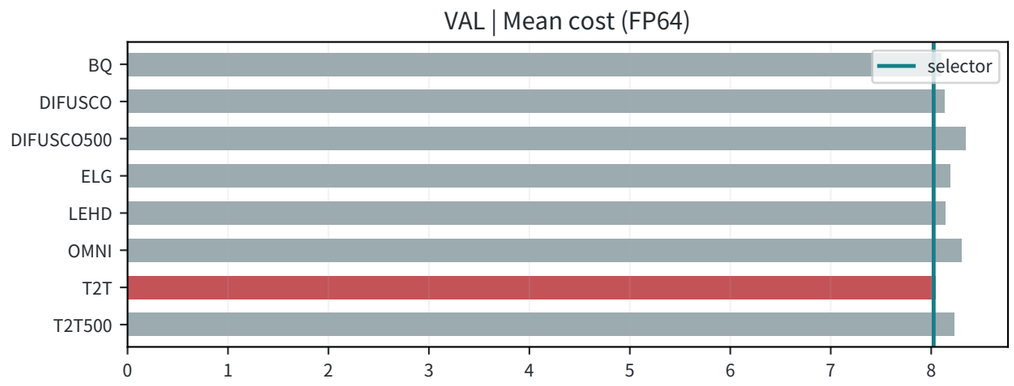


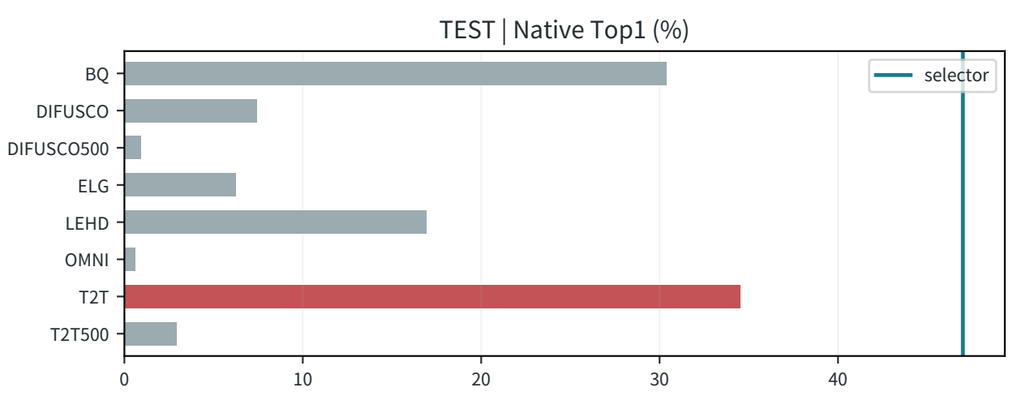


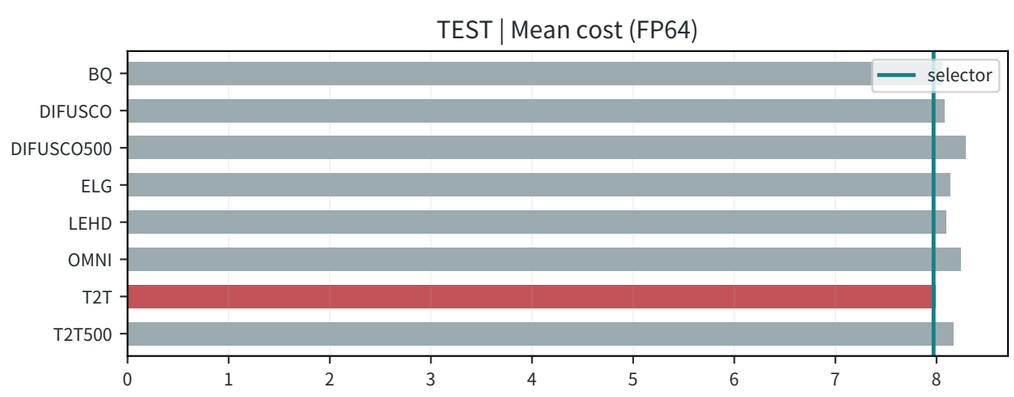
R101/A | TSP: Selector vs Single Methods

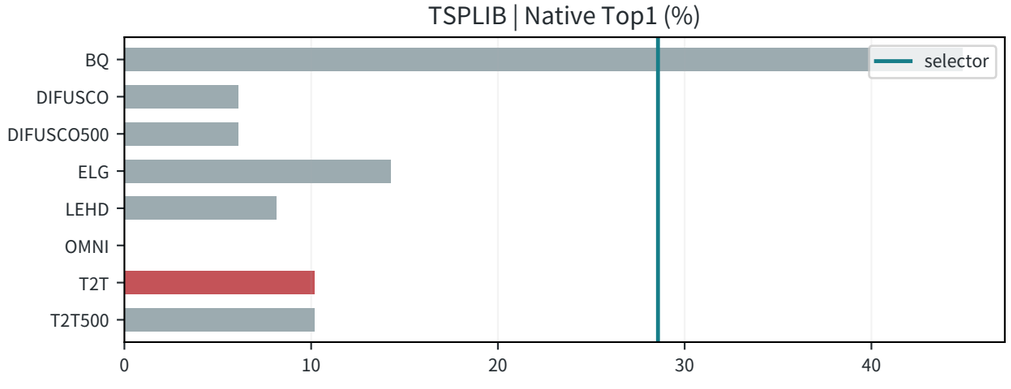
Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.

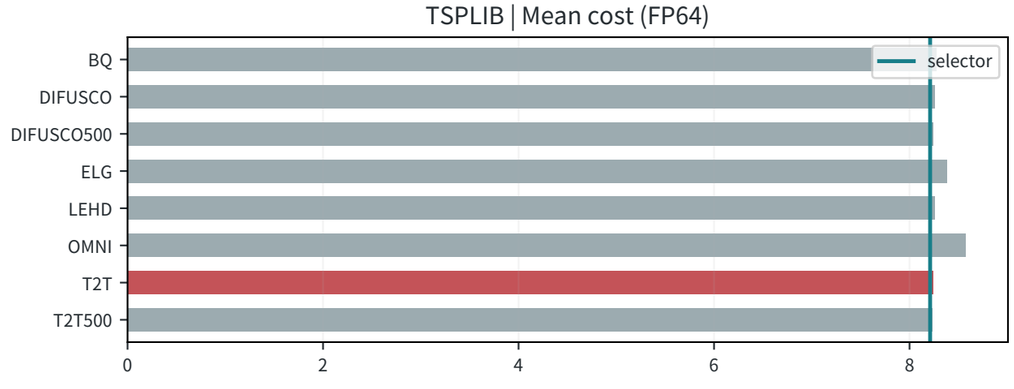






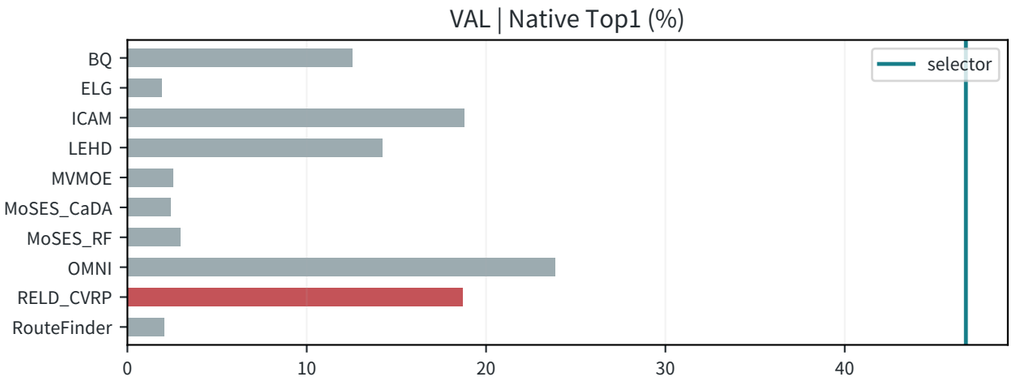


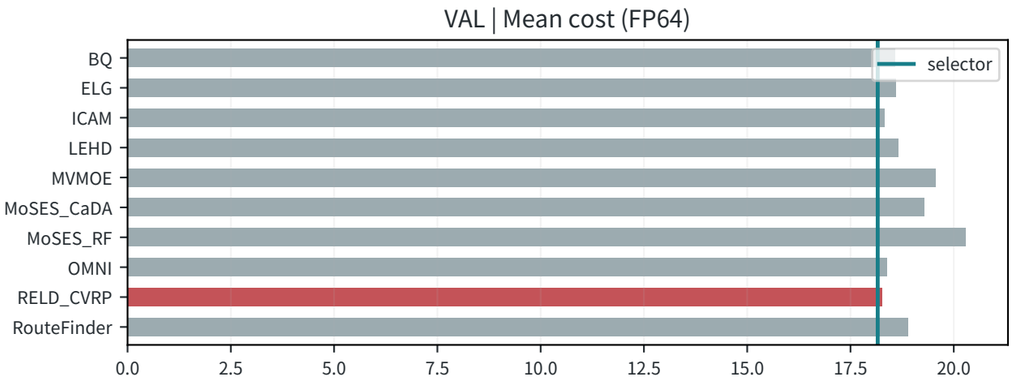


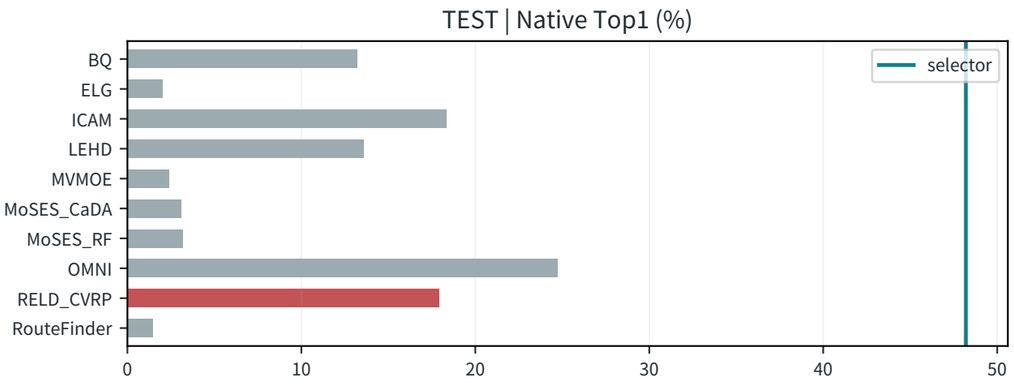


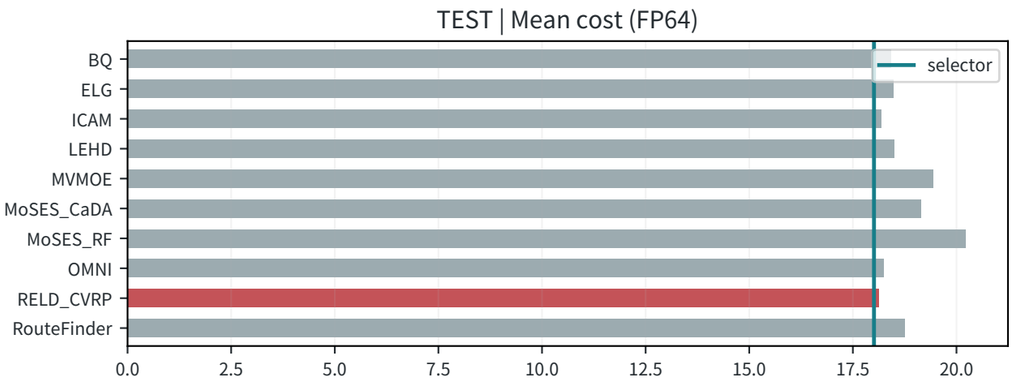
R101/A | CVRP: Selector vs Single Methods

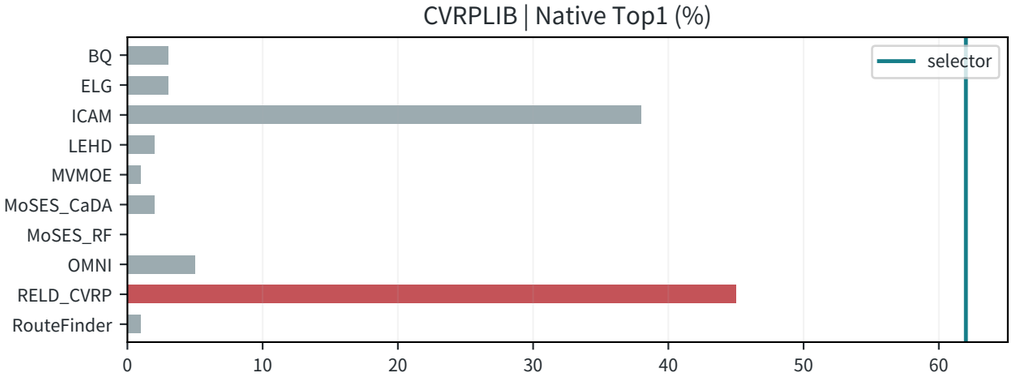
Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.

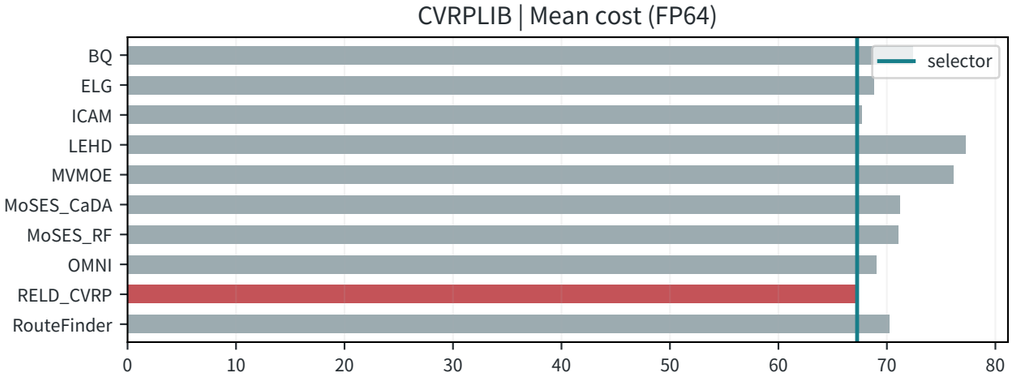






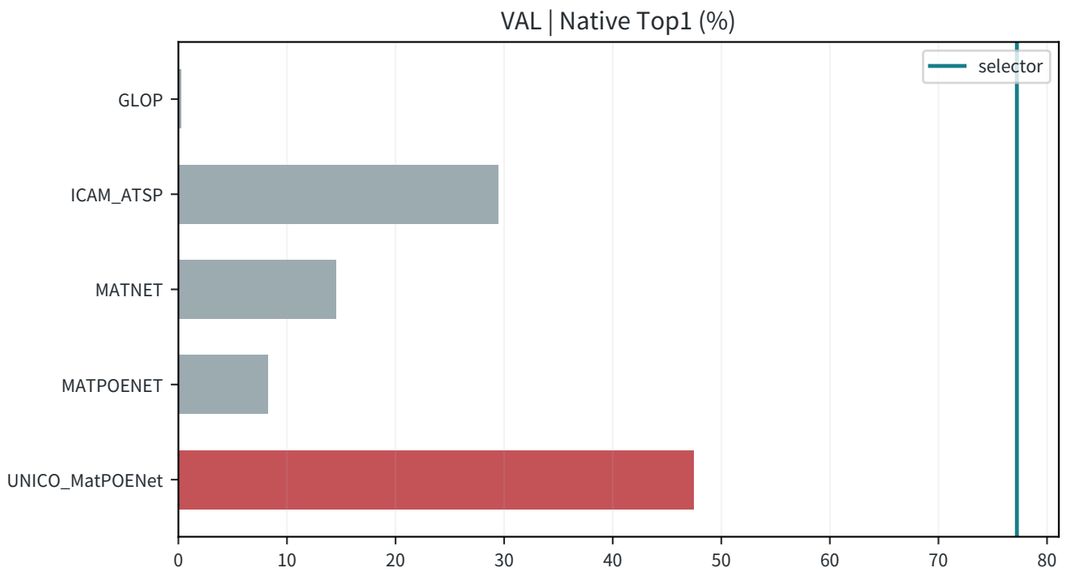


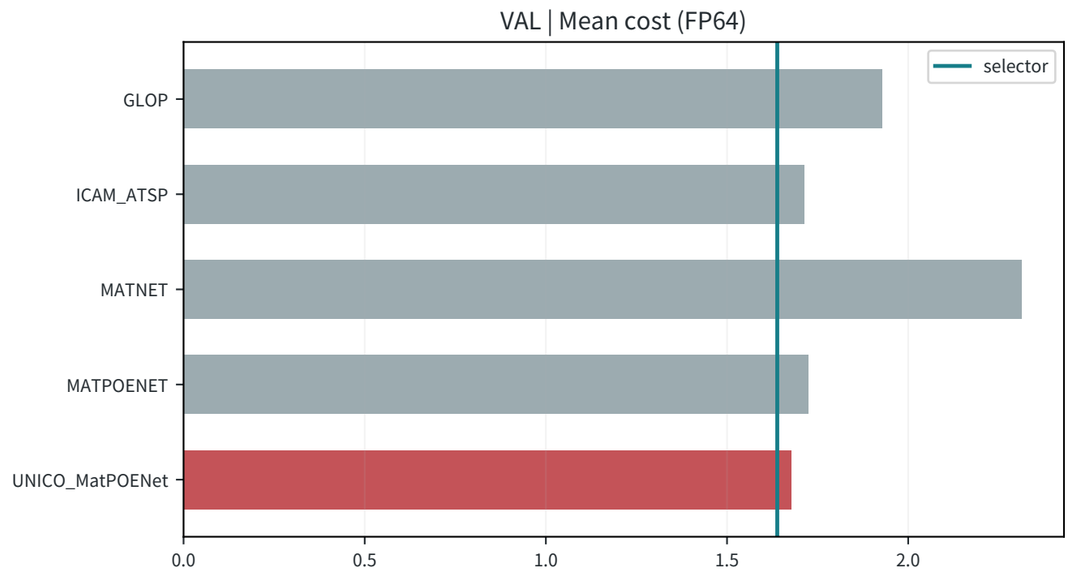


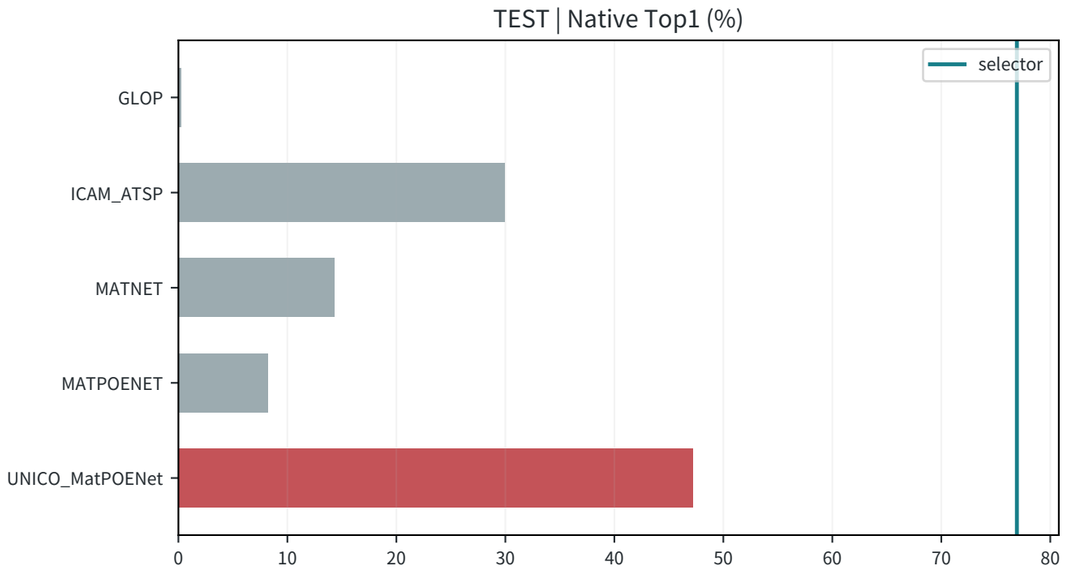


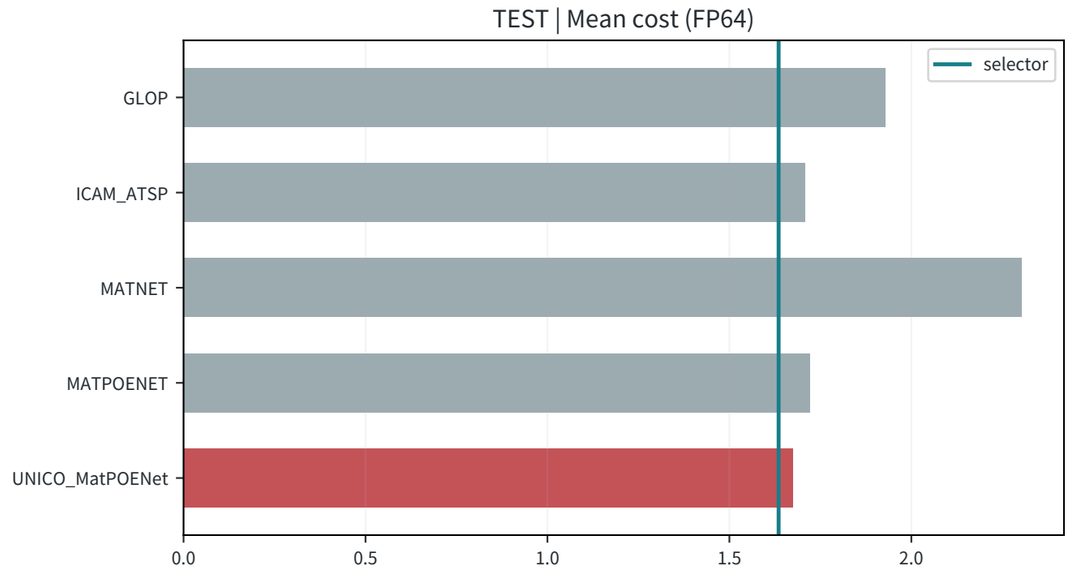
R101/A | ATSP: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



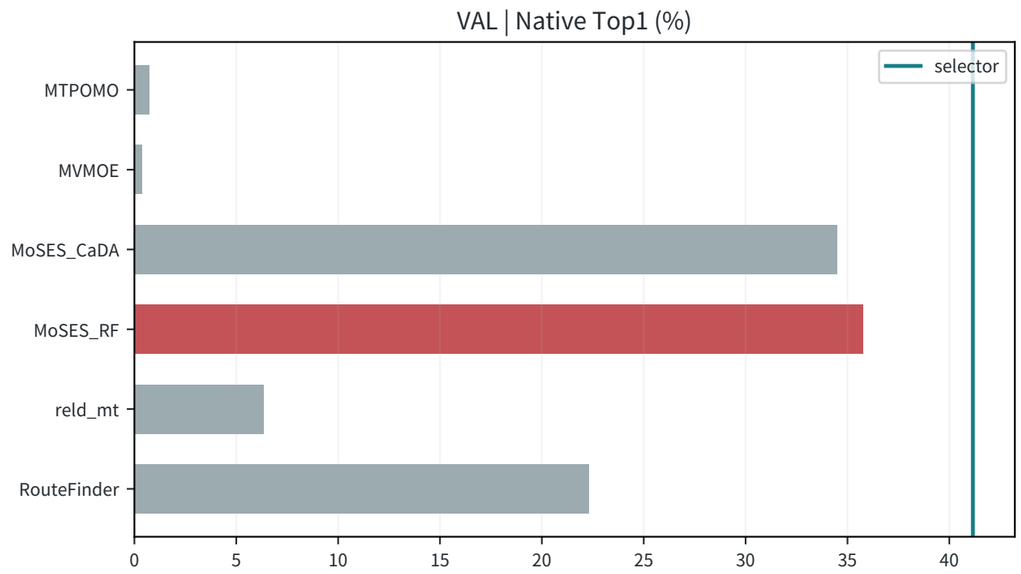


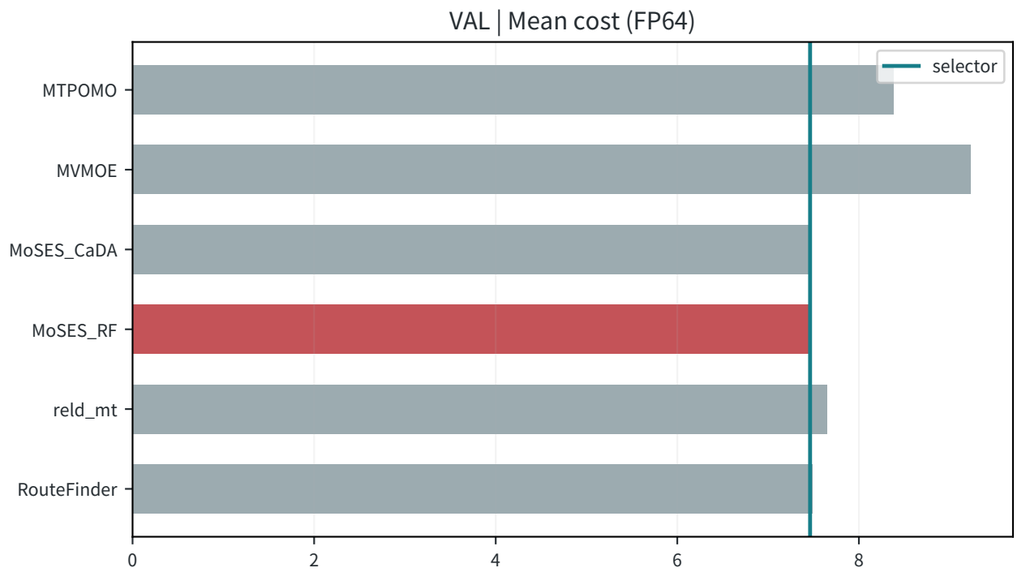


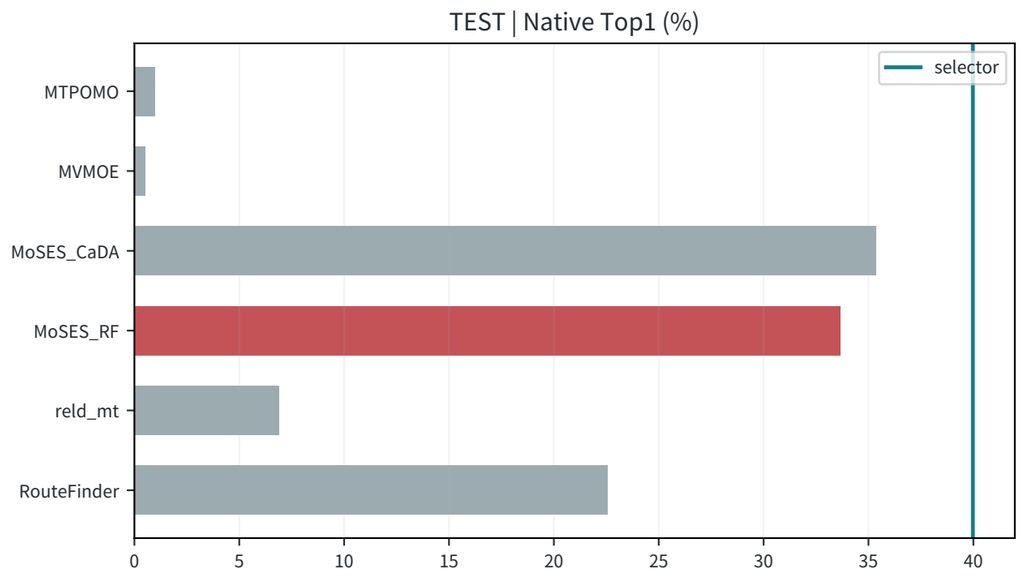


R101/A | OVRP: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



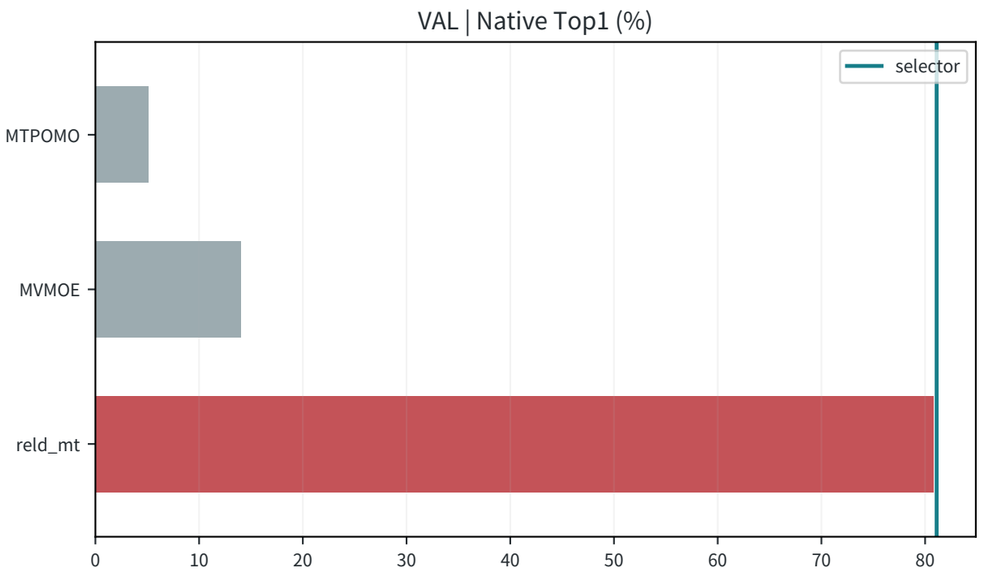


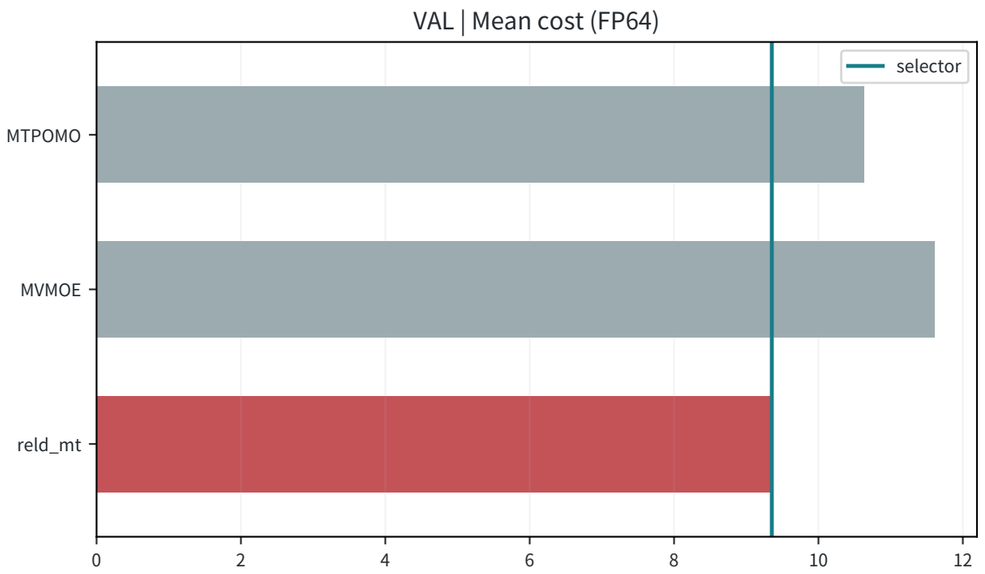


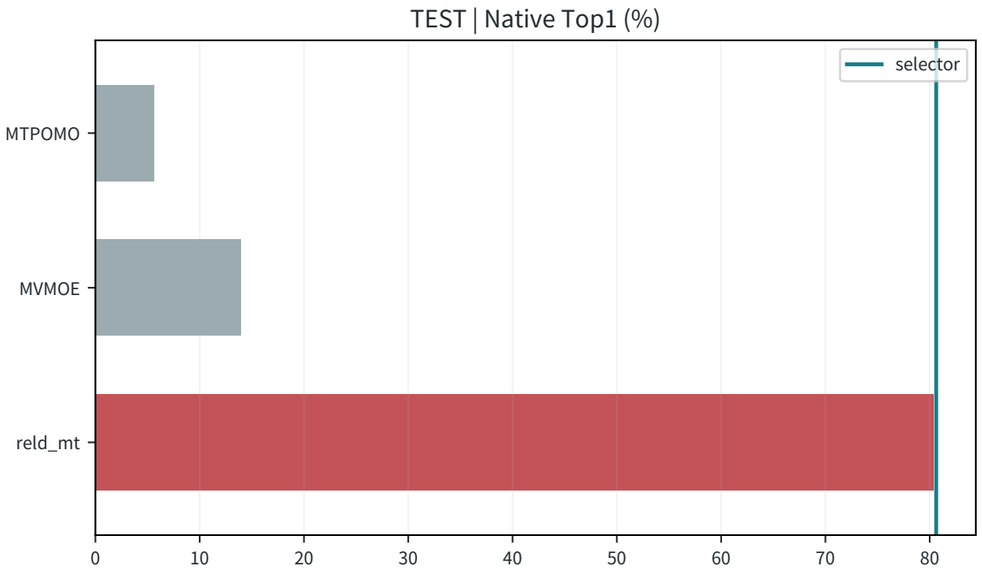


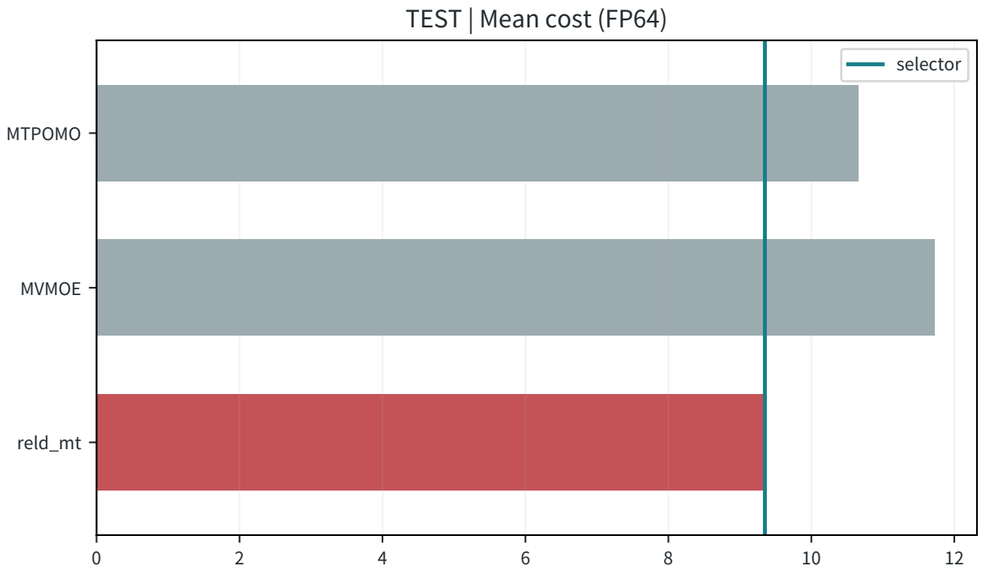
R101/A | VRPB: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



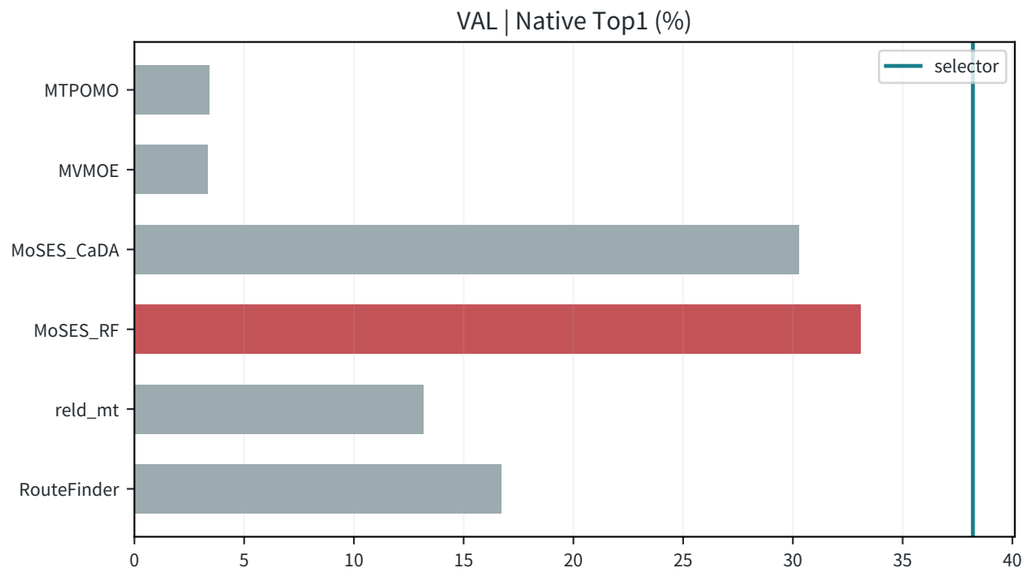


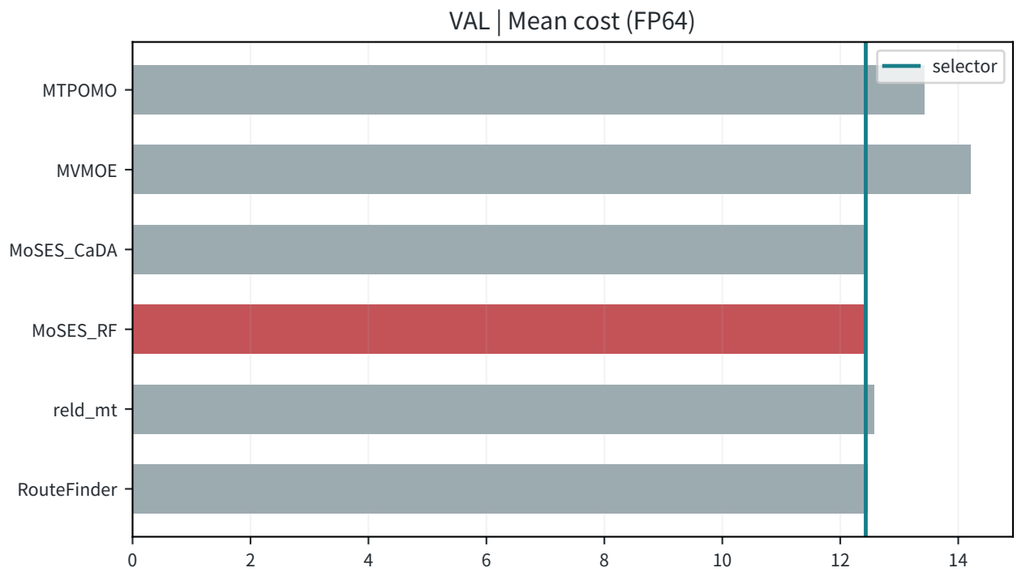


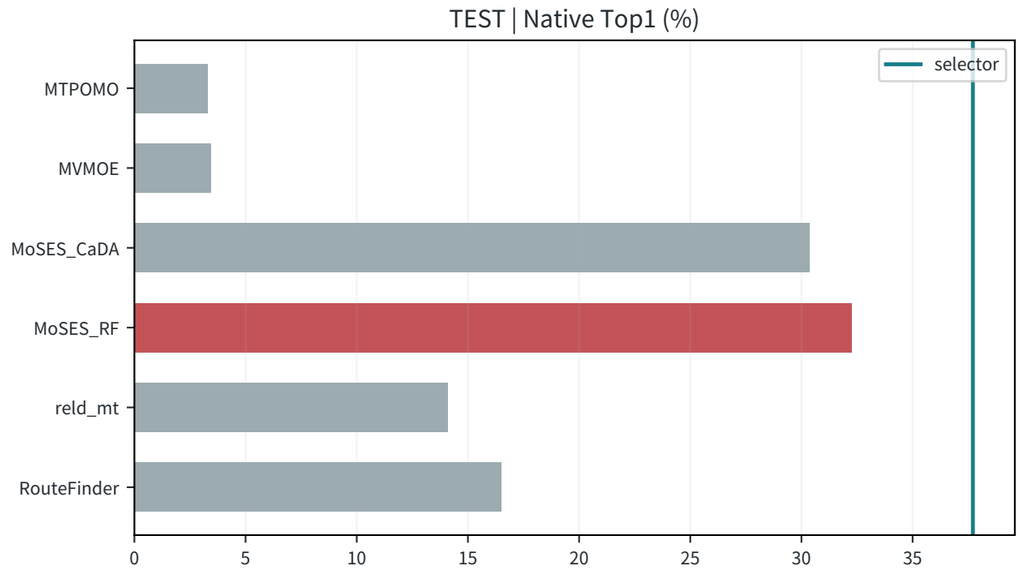


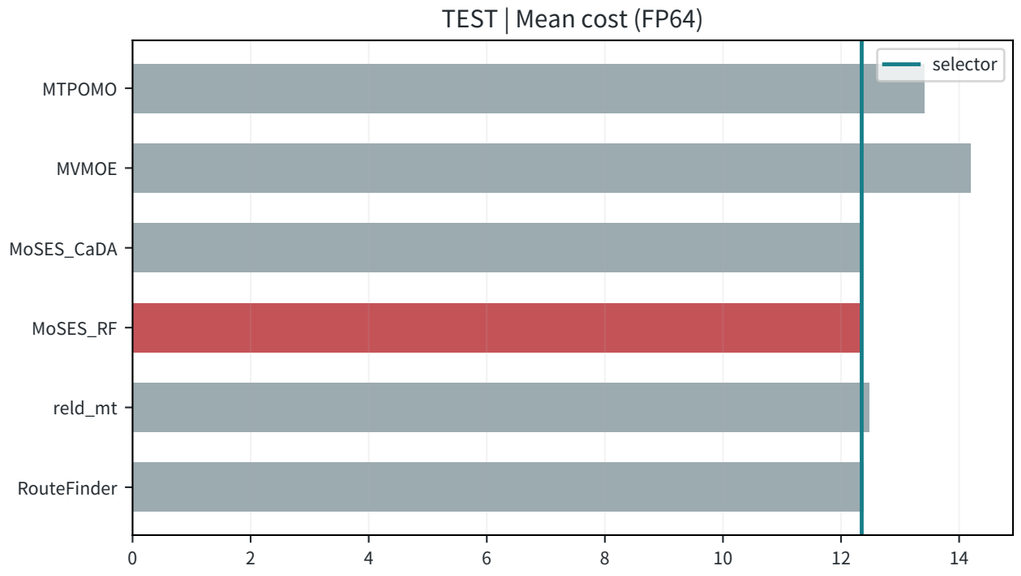
R101/A | VRPL: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



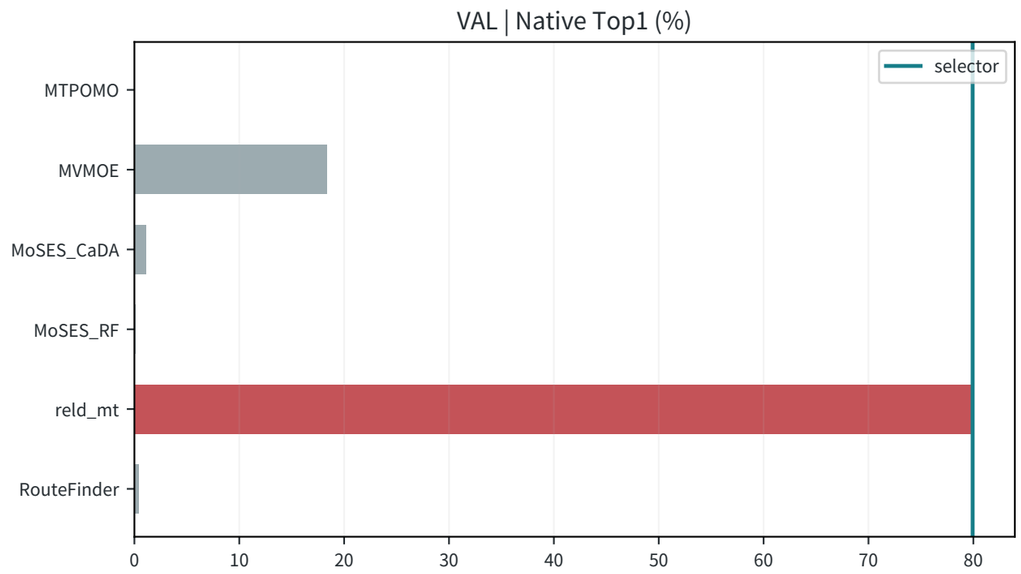


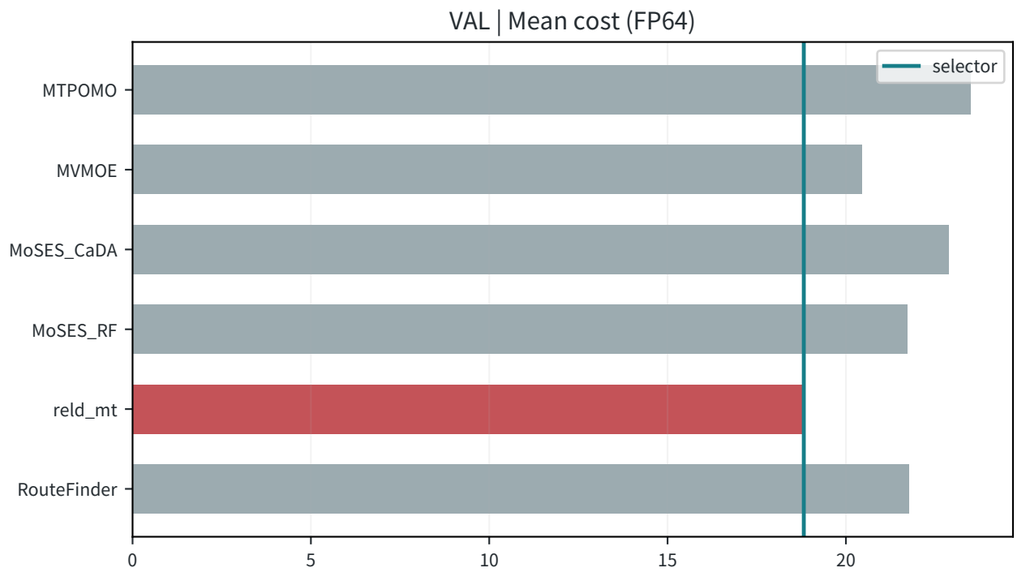


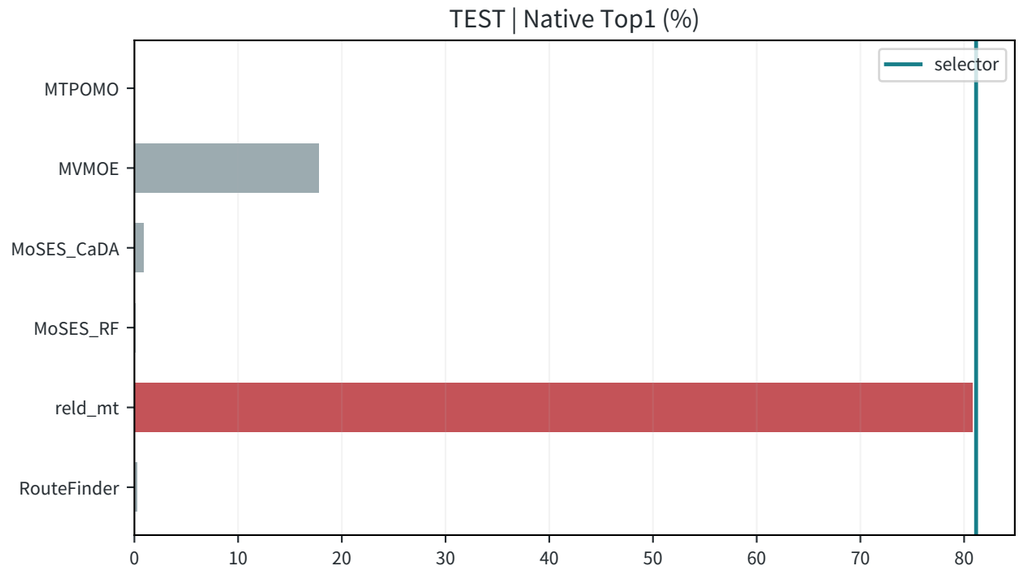


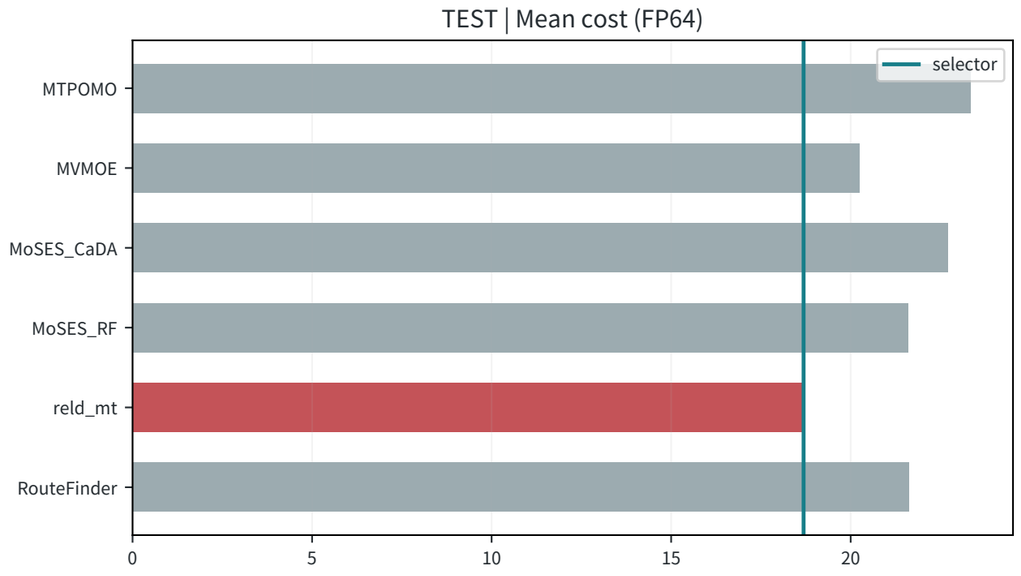
R101/A | VRPTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



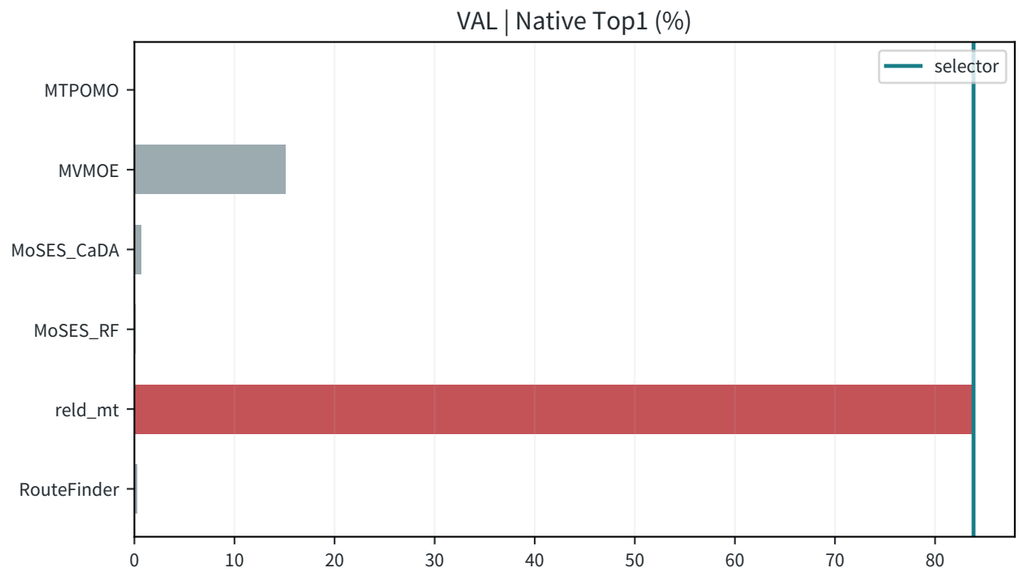


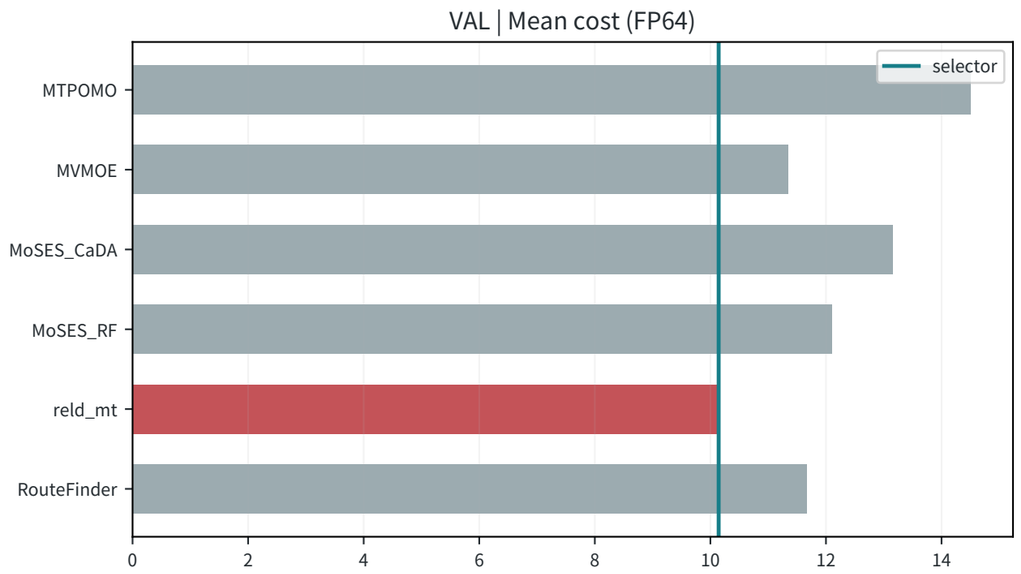


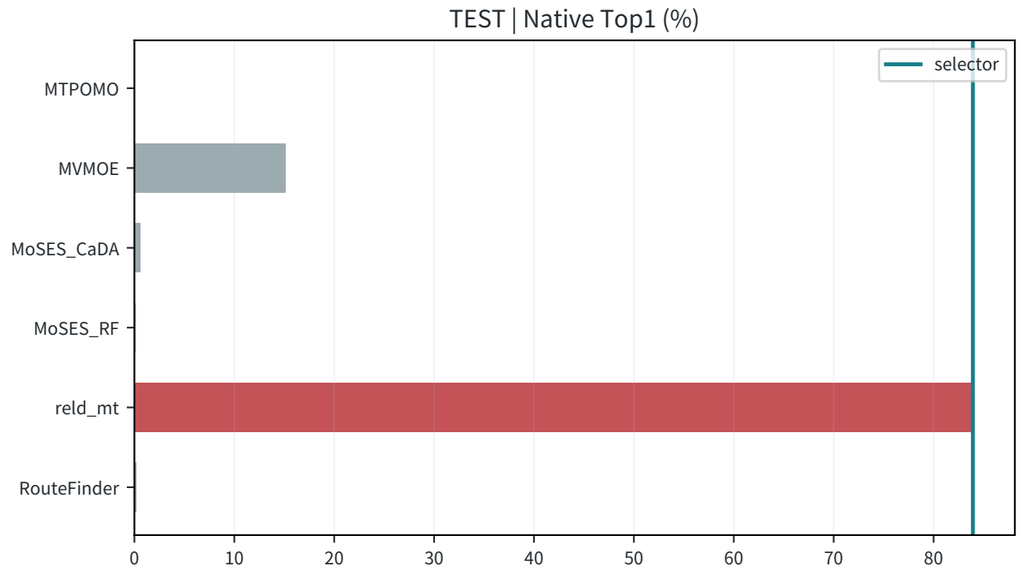


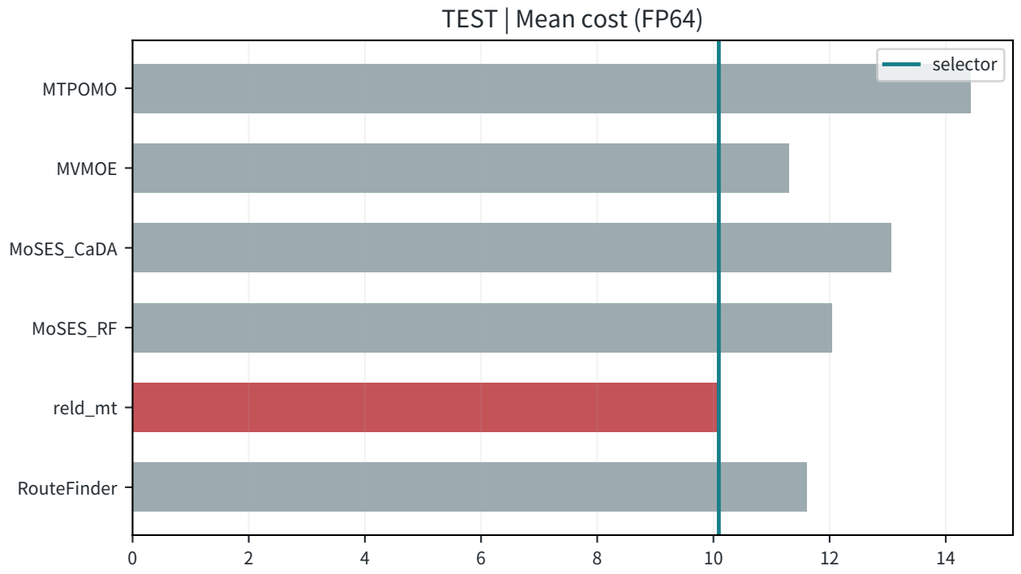
R101/A | OVRPTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



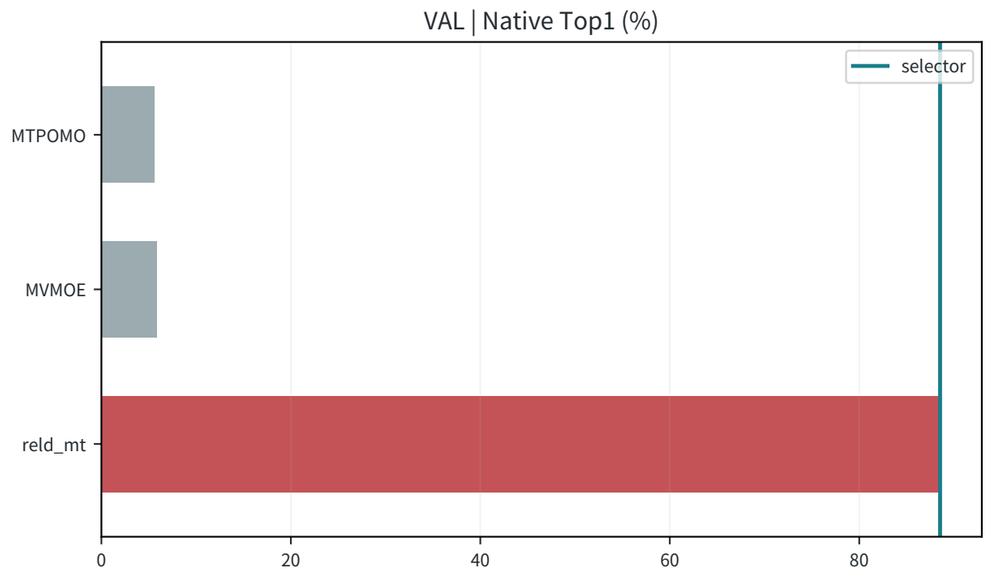


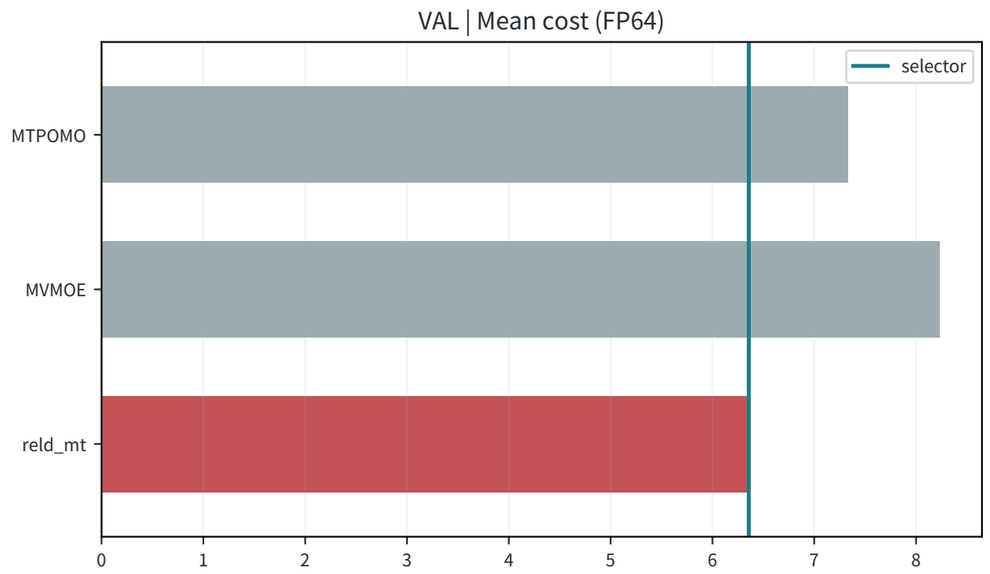


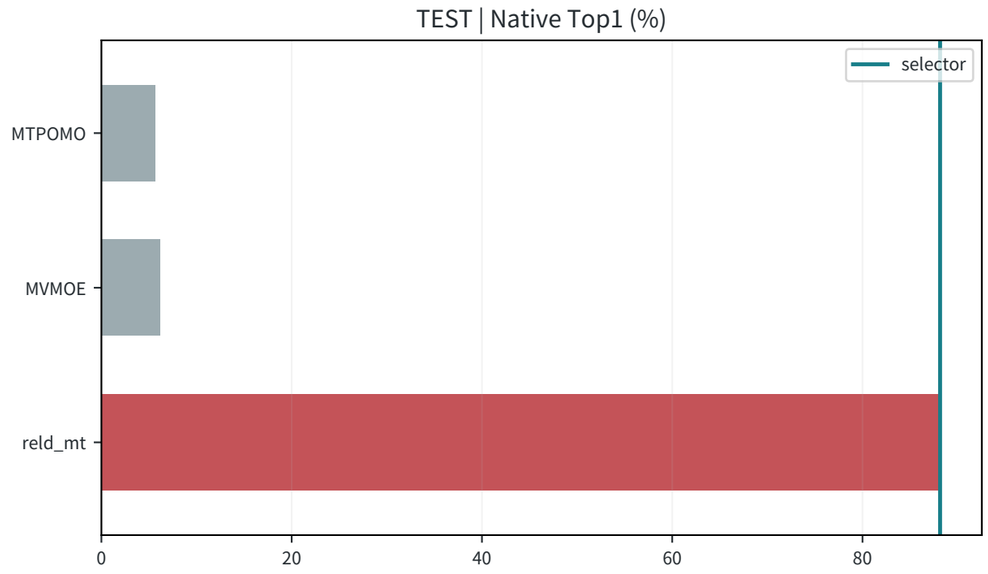


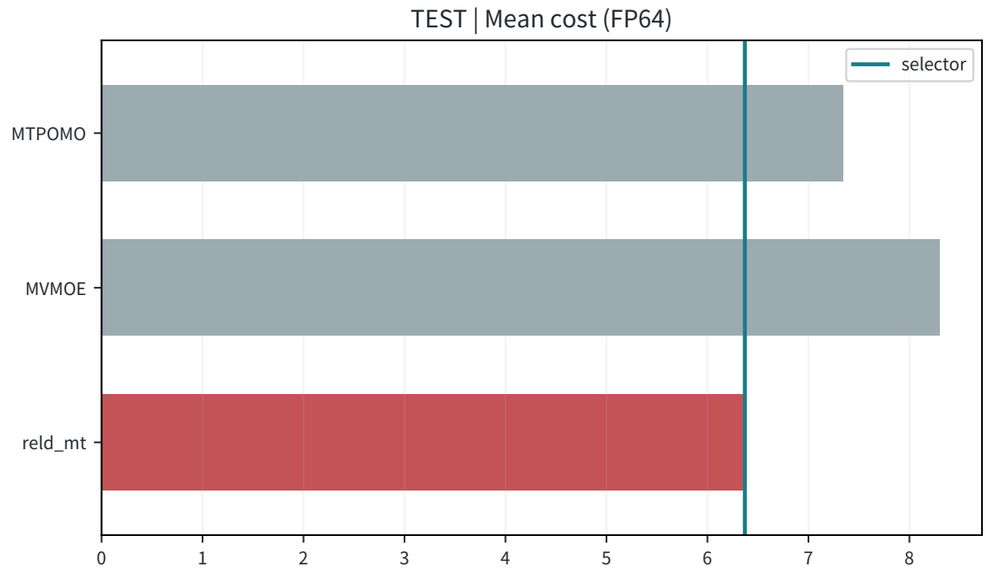
R101/A | OVRPB: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



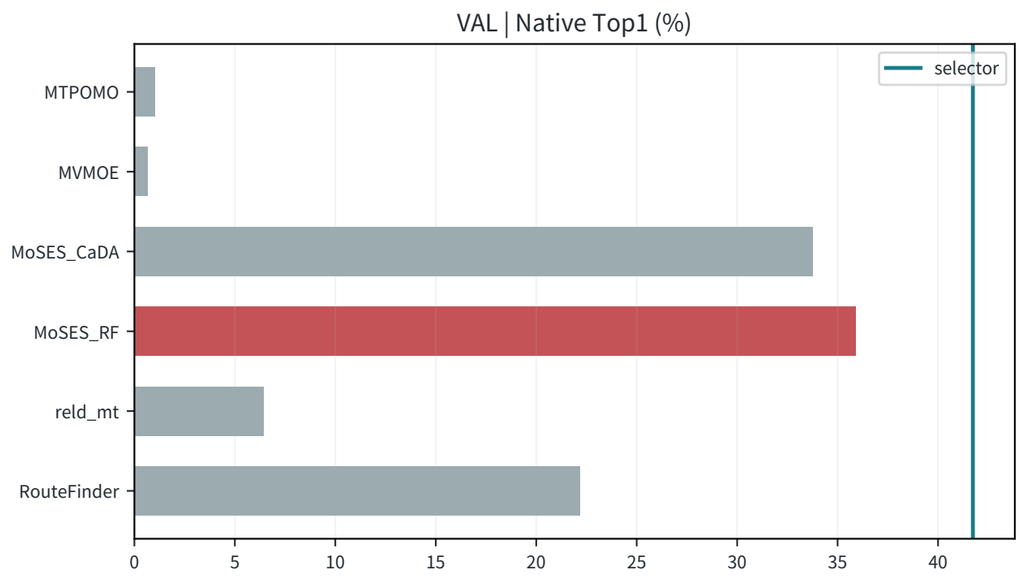


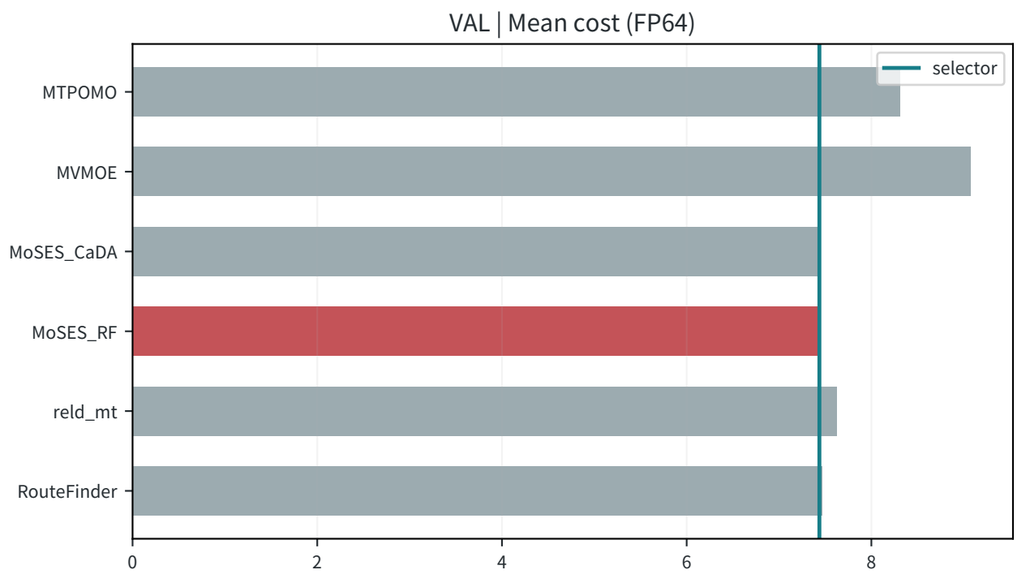


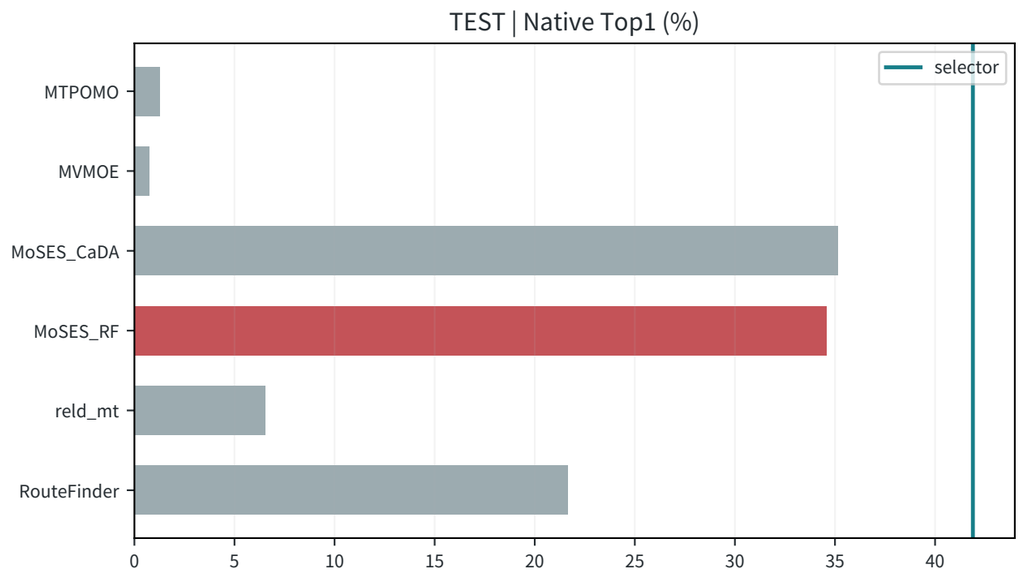


R101/A | OVRPL: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



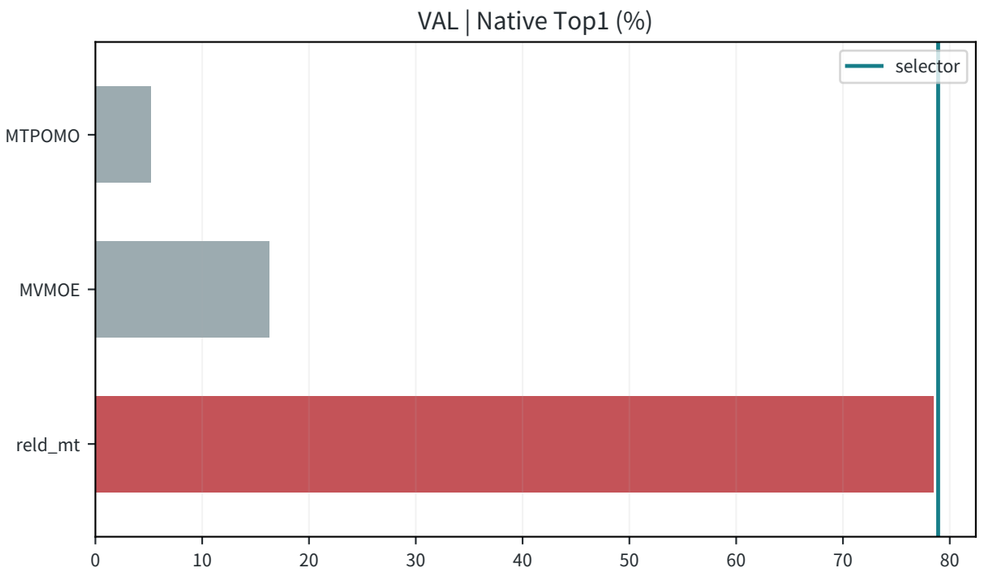


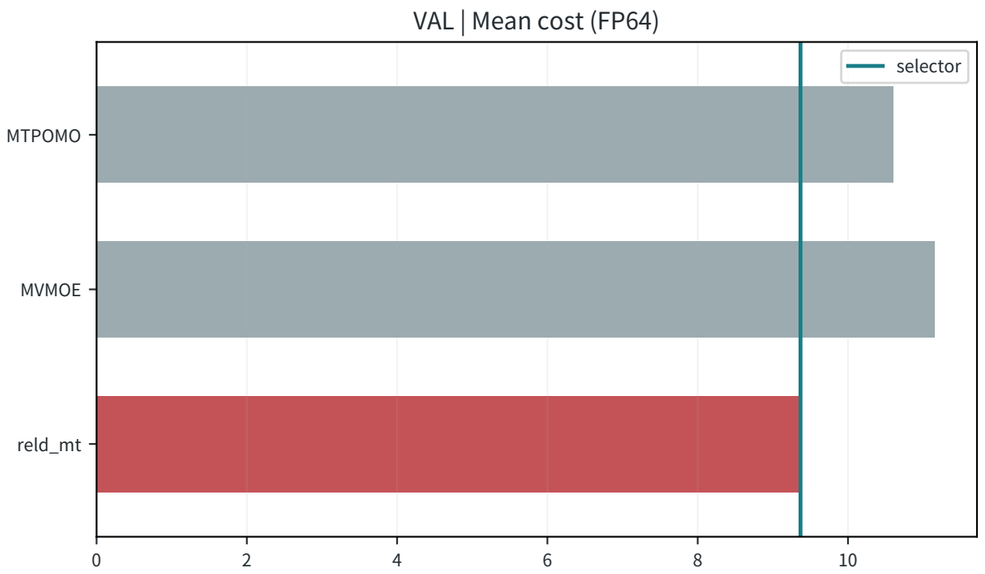


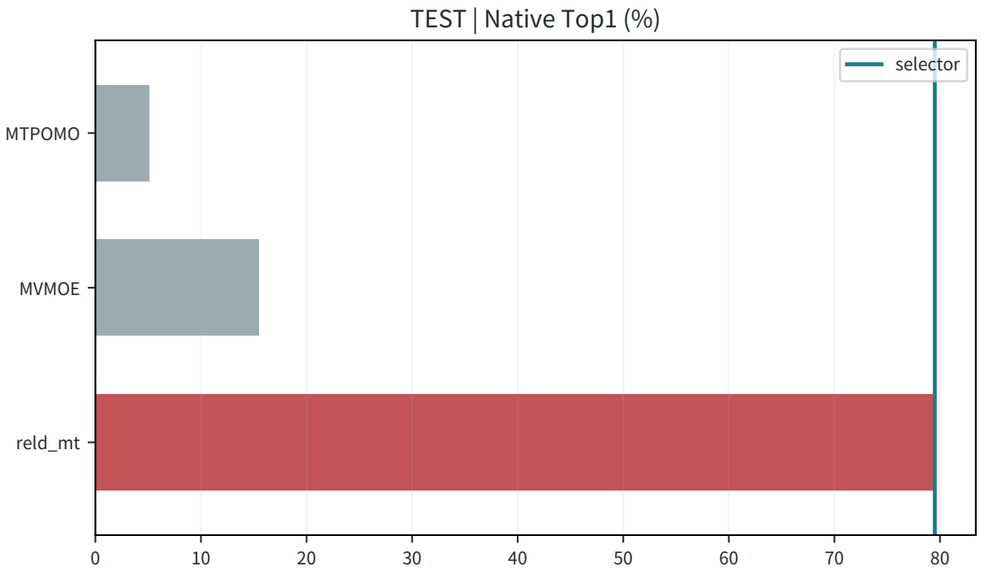


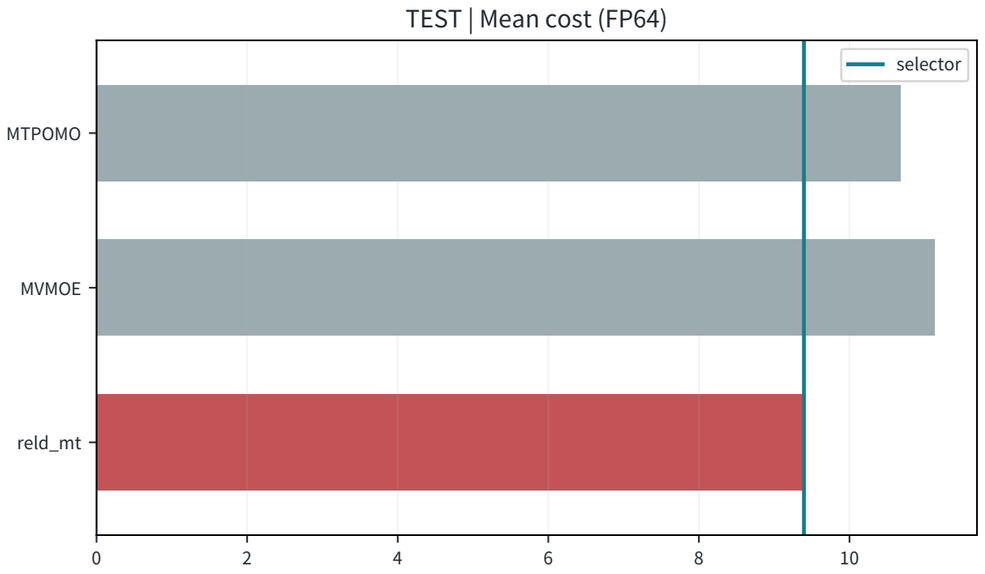
R101/A | VRPBL: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



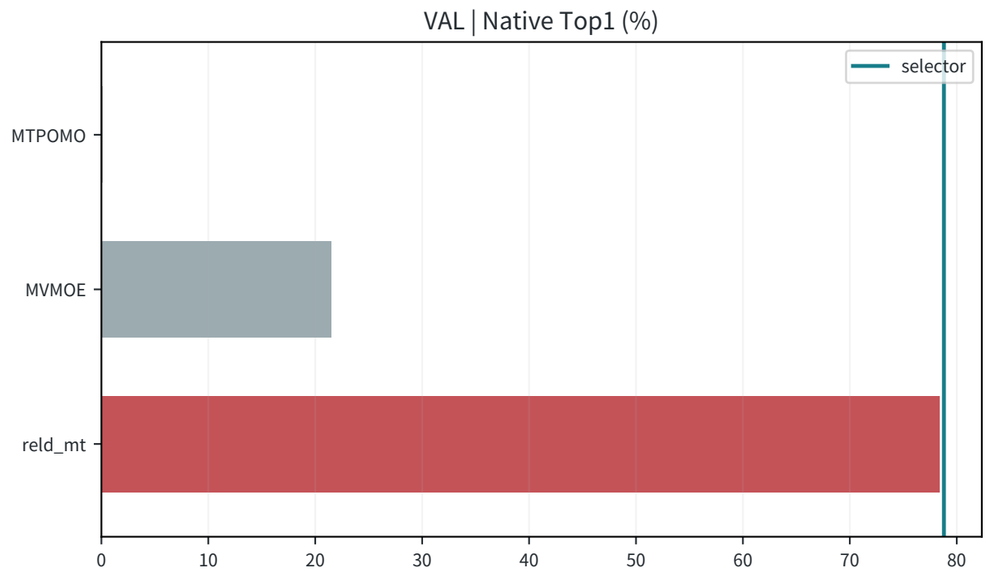


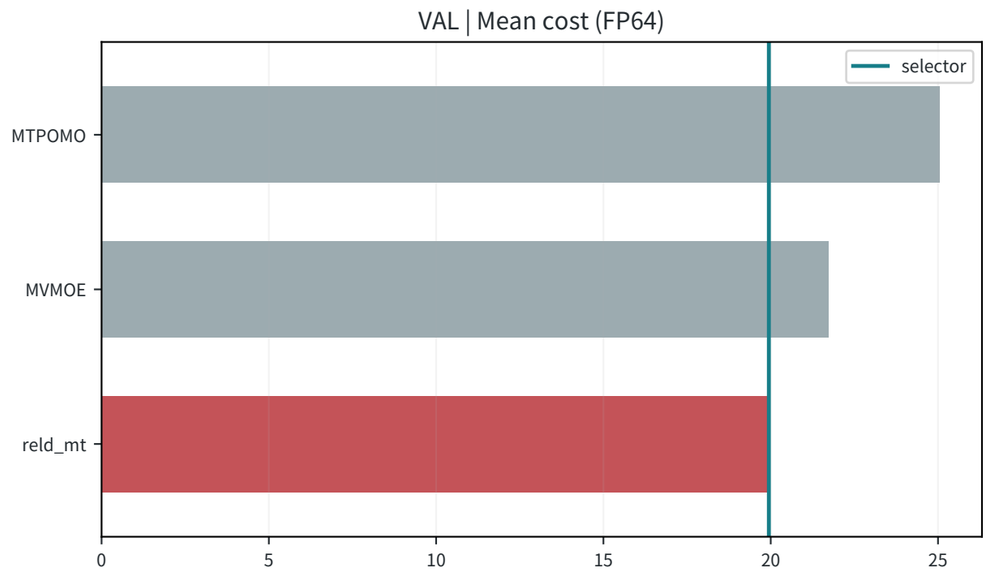


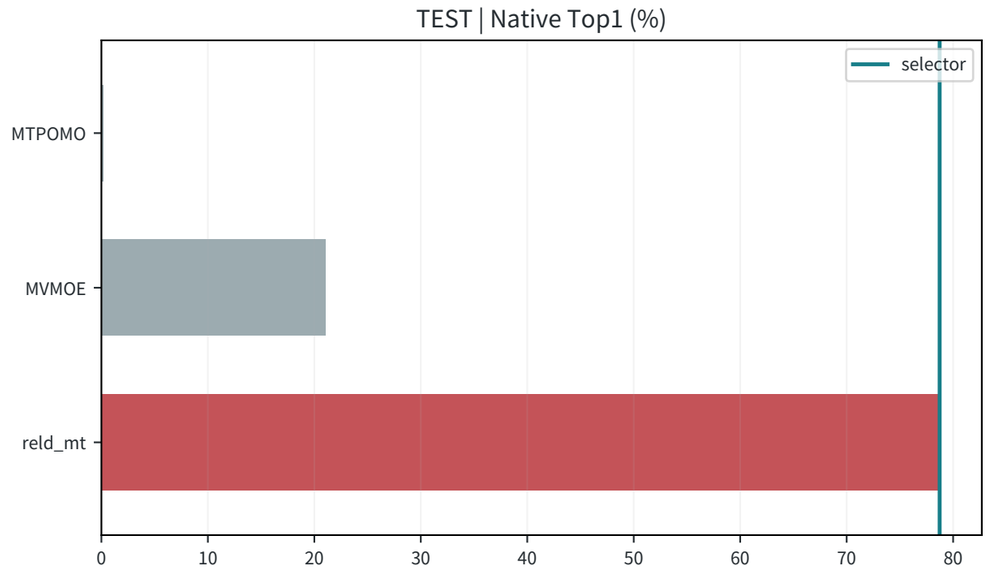


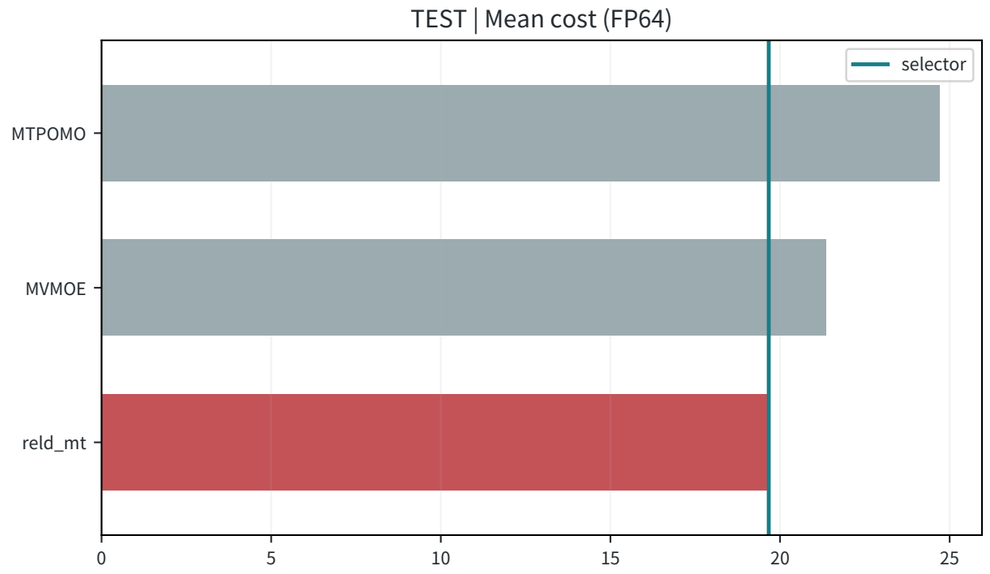
R101/A | VRPBTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



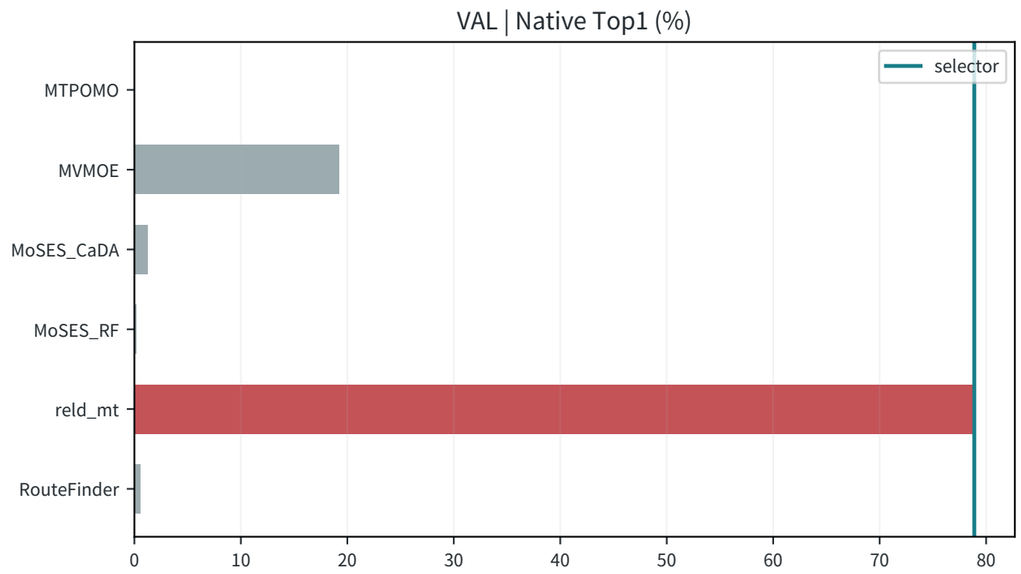


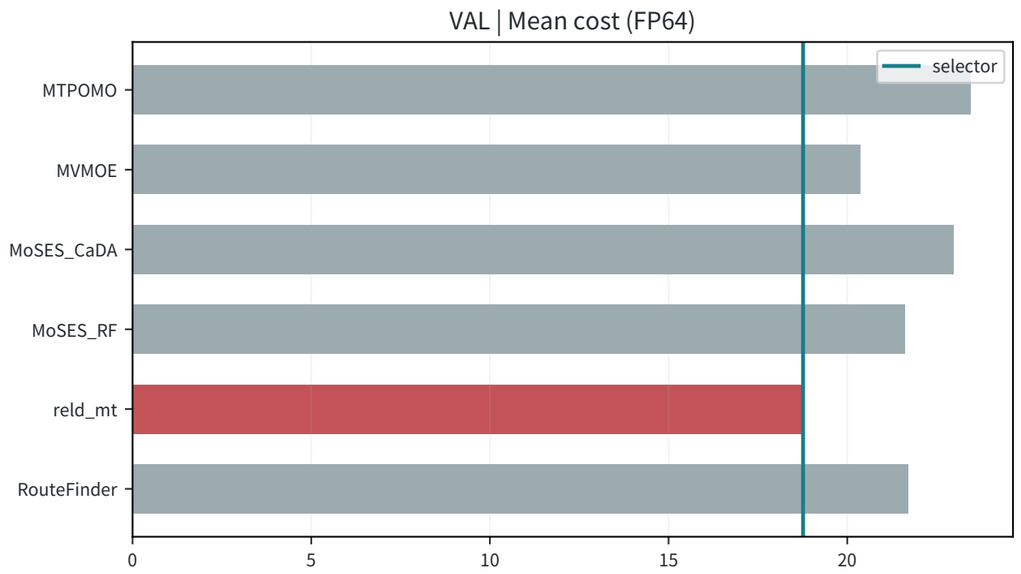


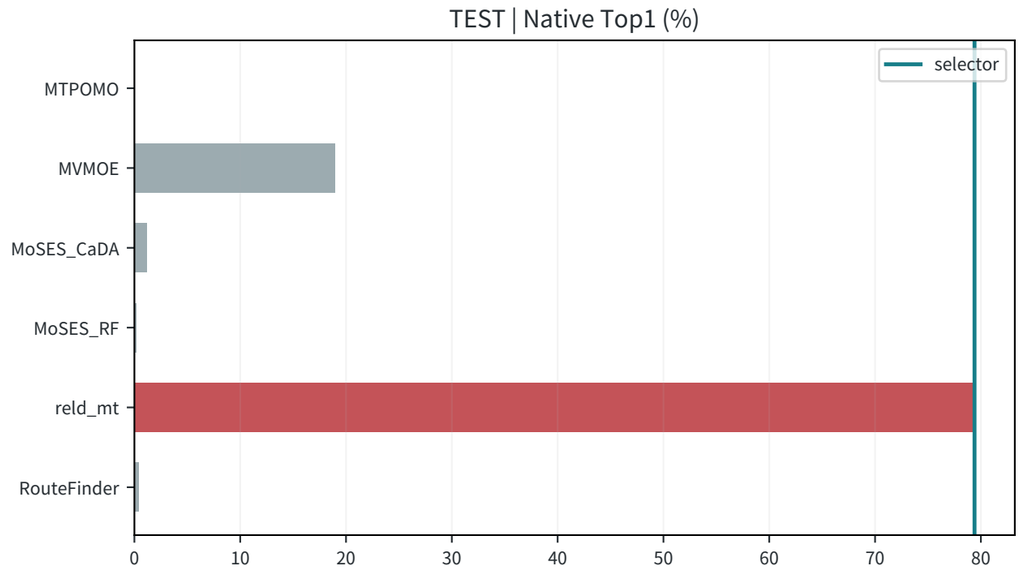


R101/A | VRPLTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



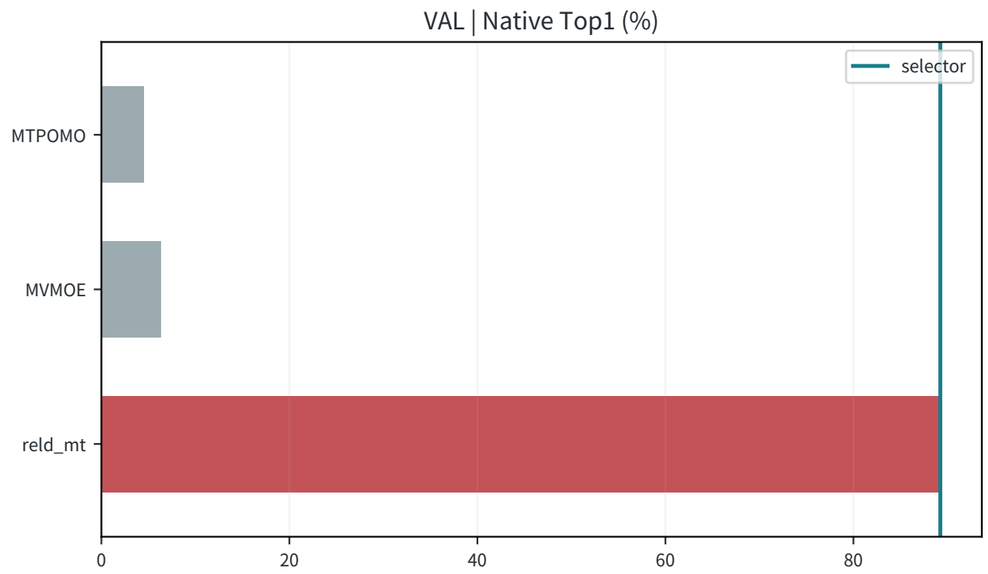


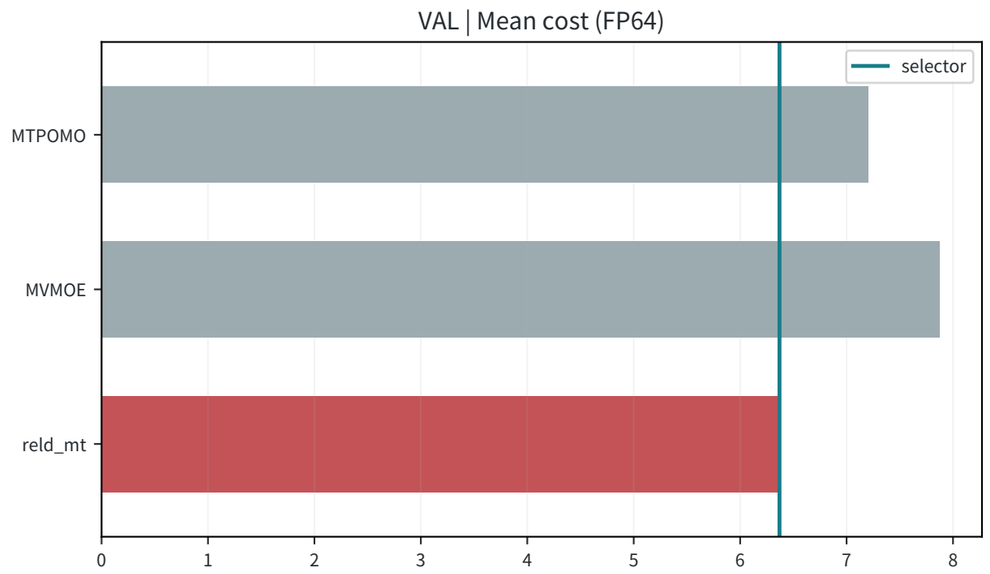


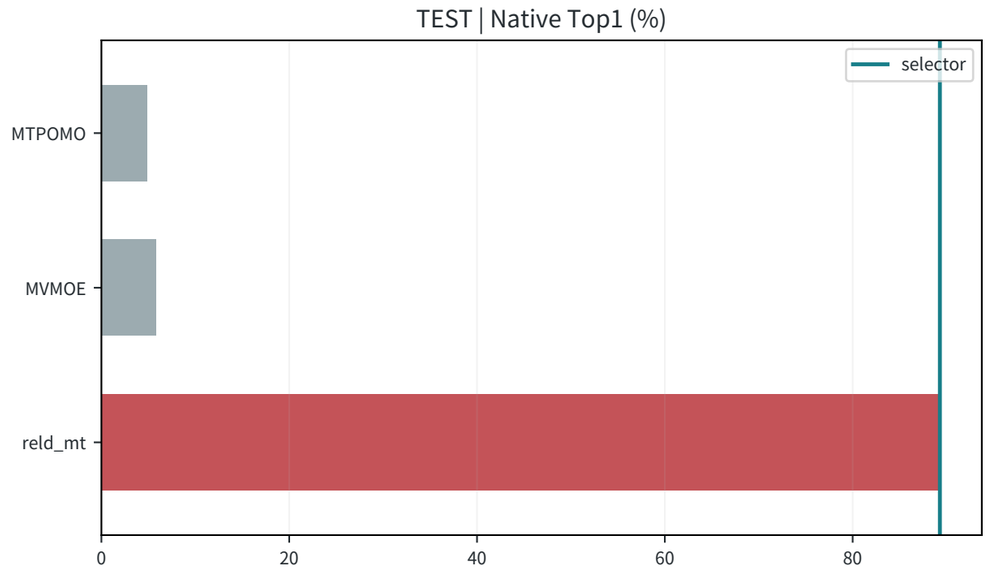


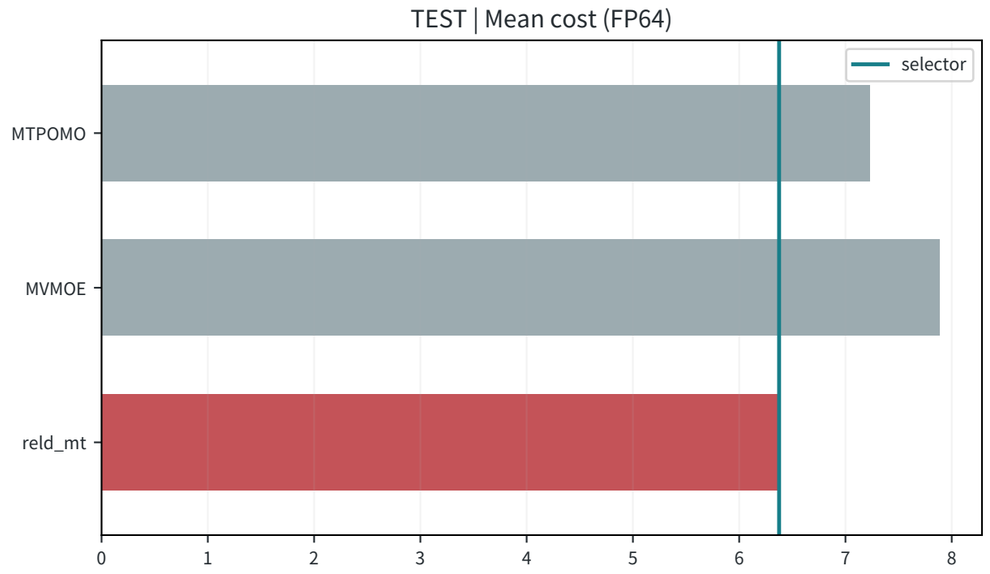
R101/A | OVRPBL: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



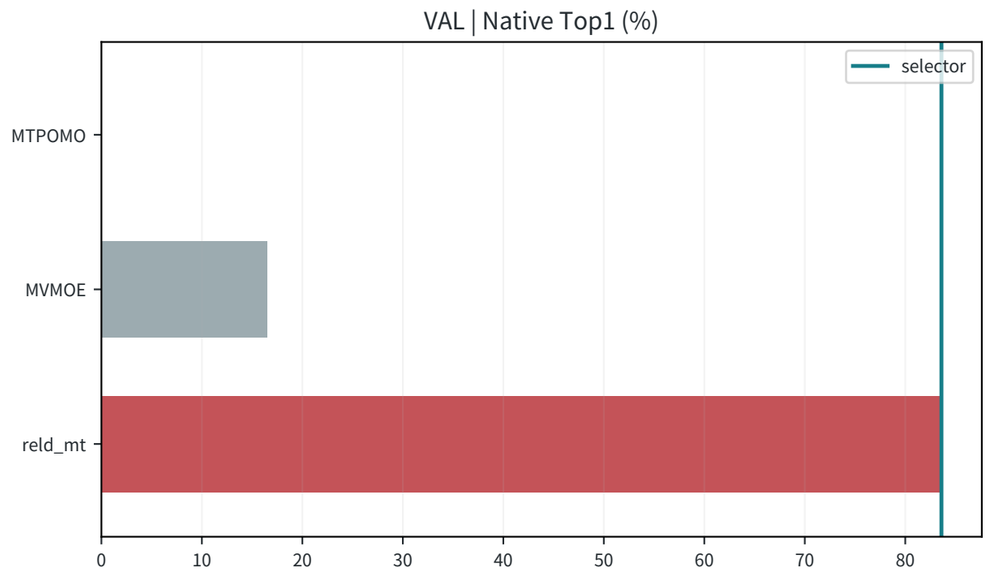


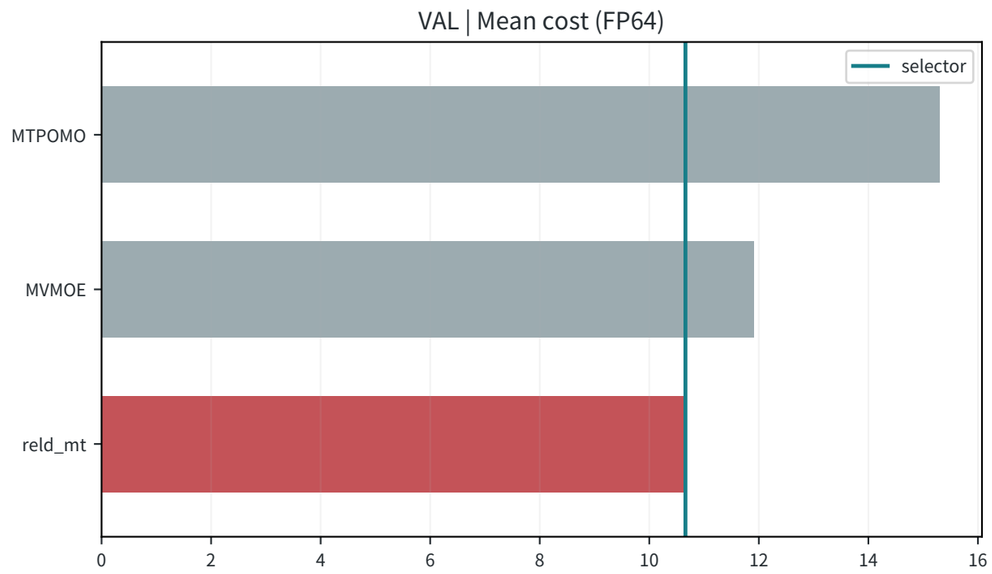


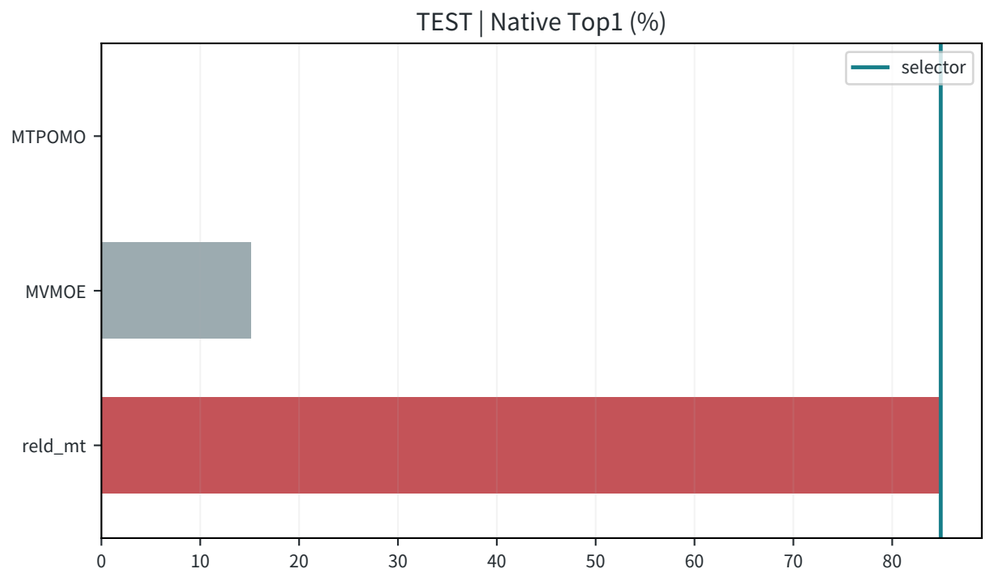


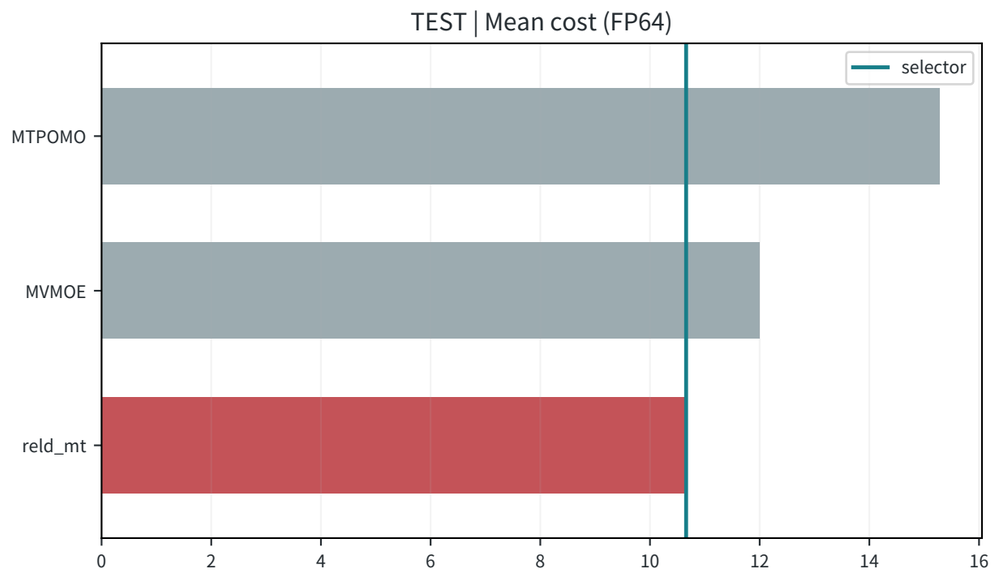
R101/A | OVRPBTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



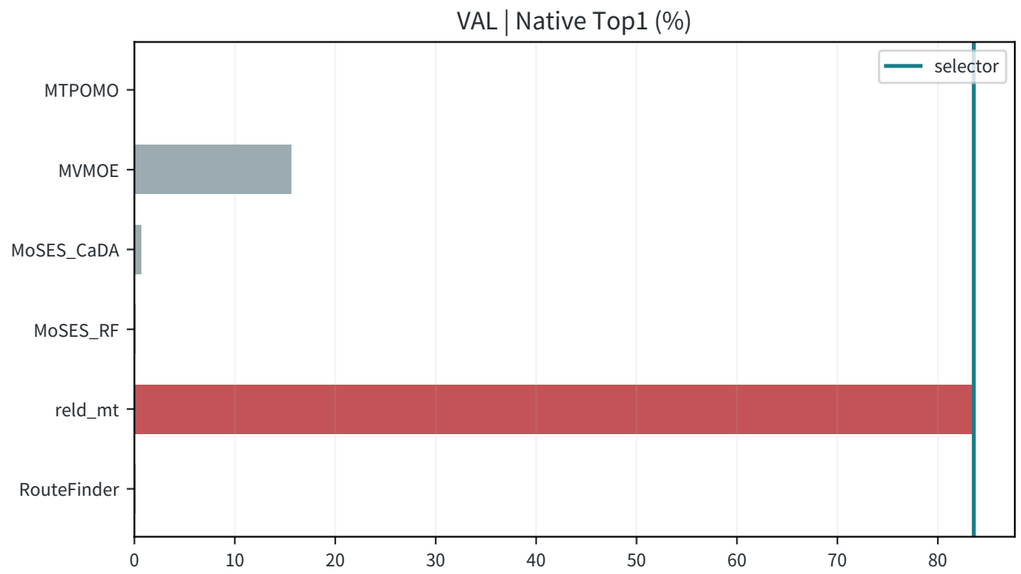


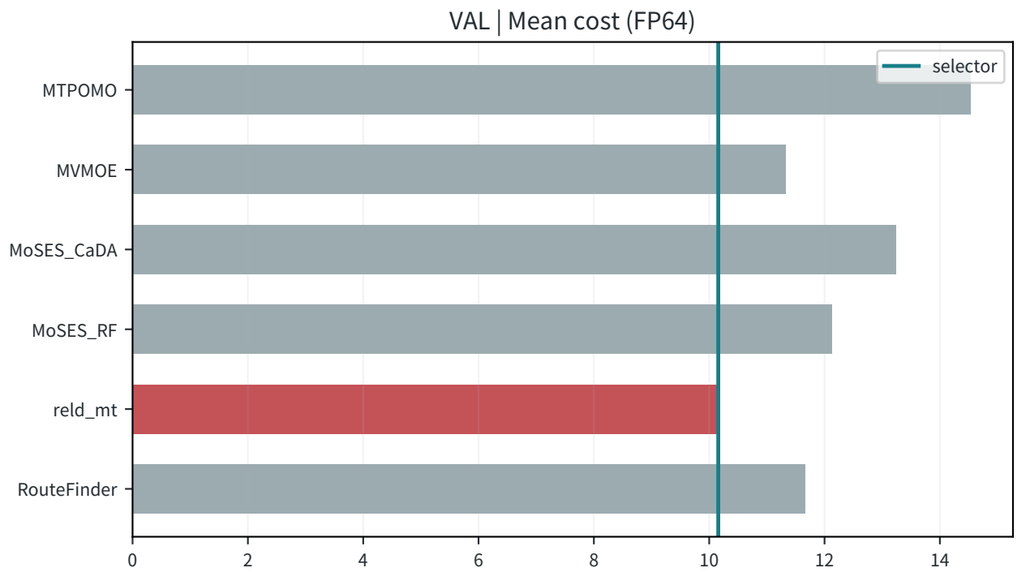


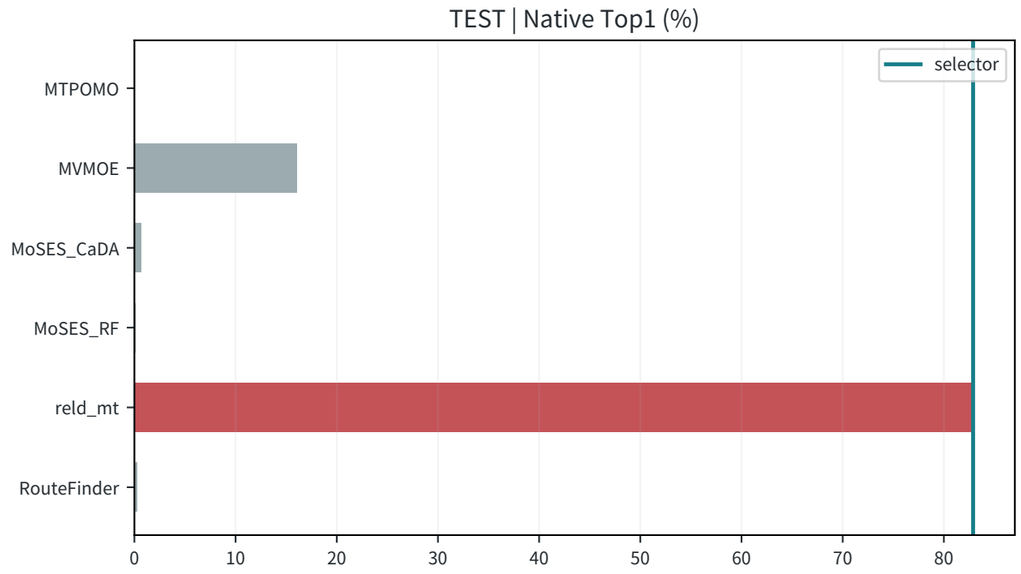


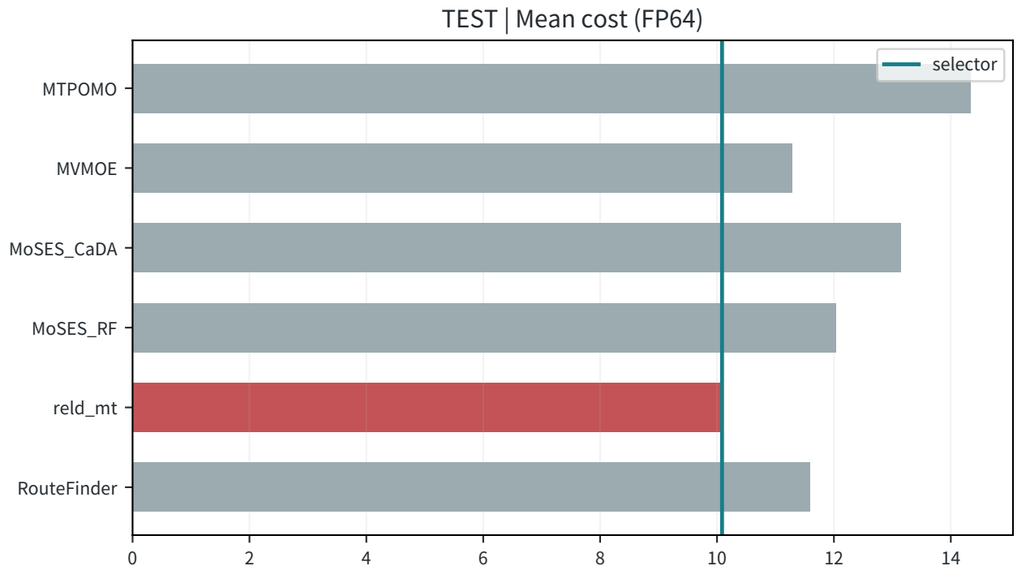
R101/A | OVRPLTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



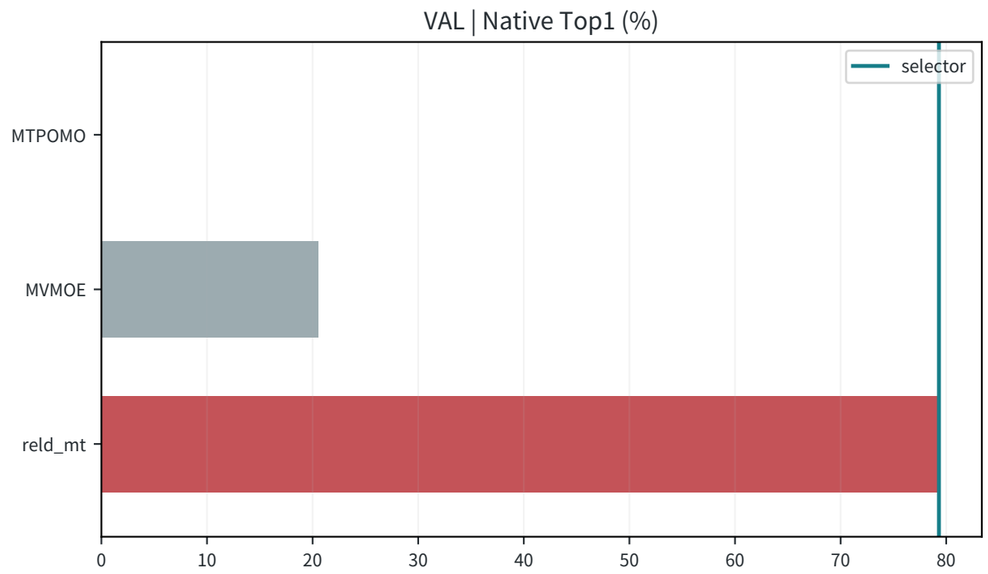


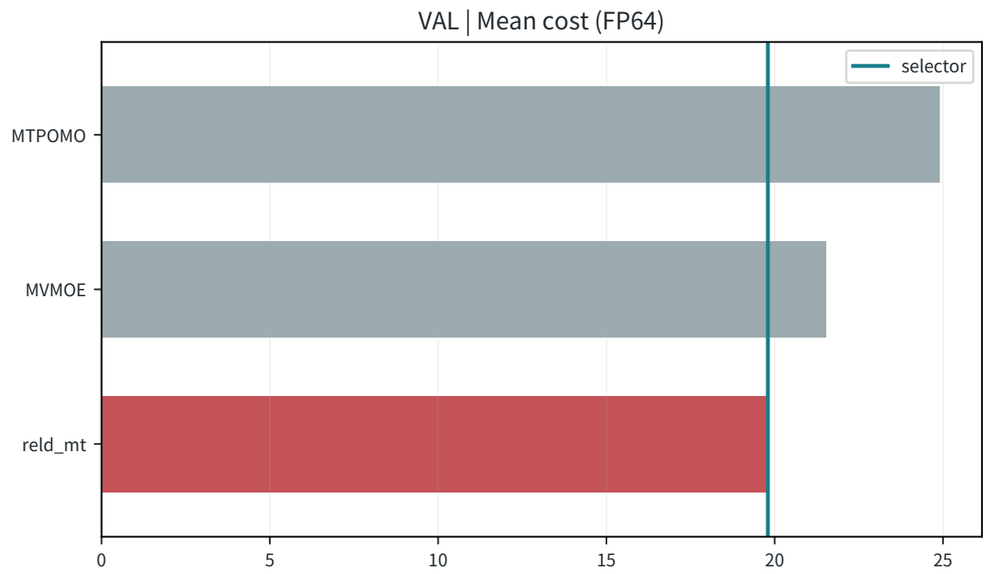


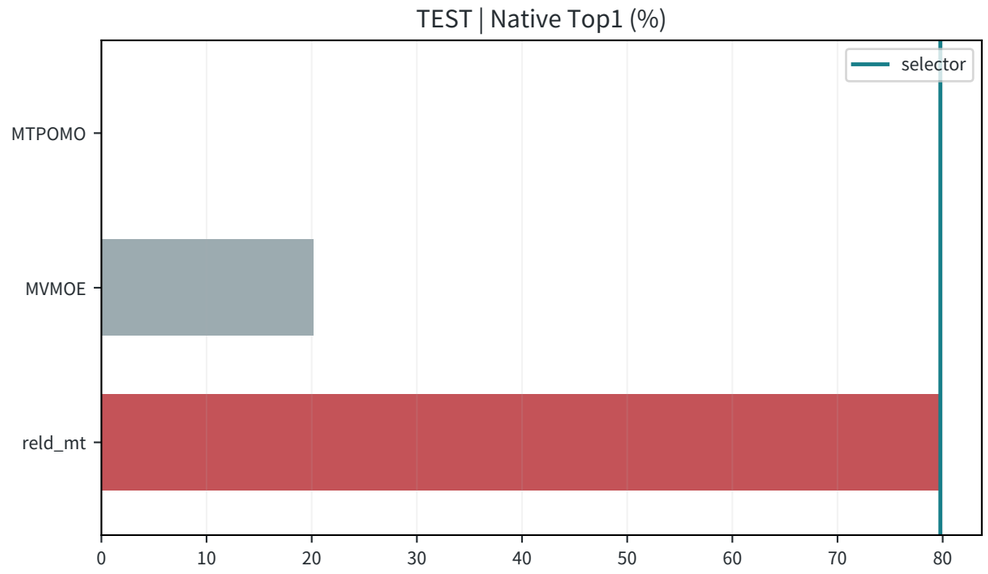


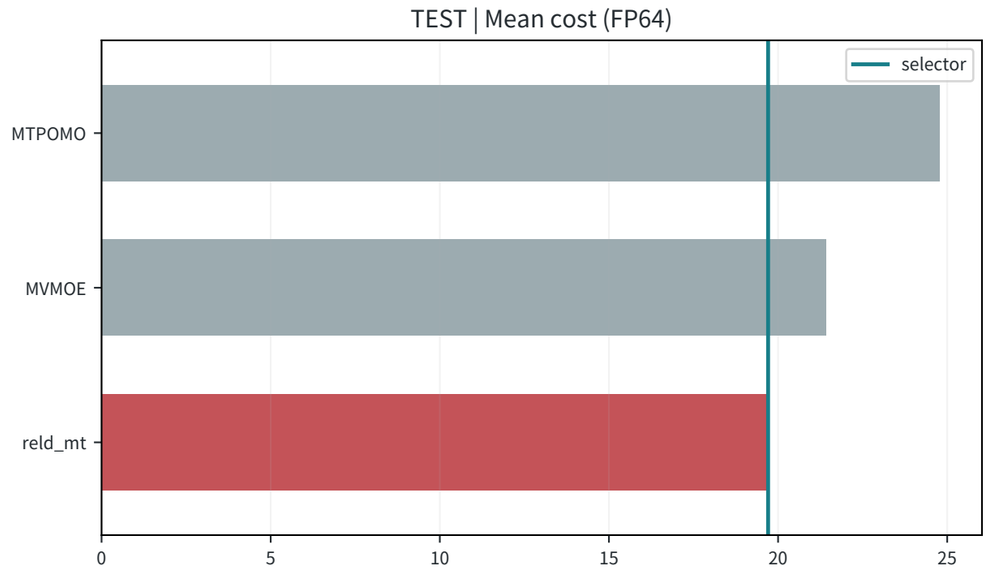
R101/A | VRPBLTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



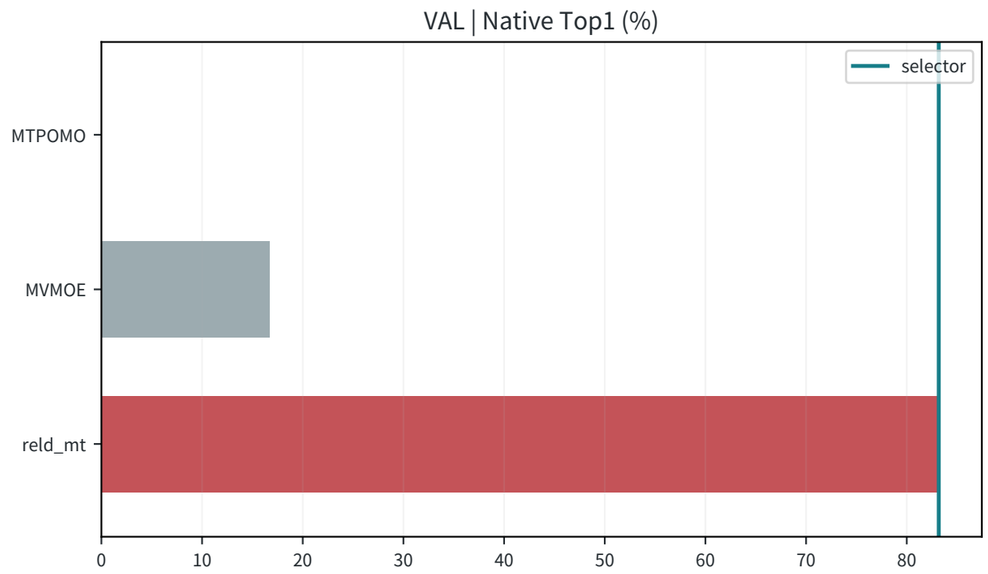


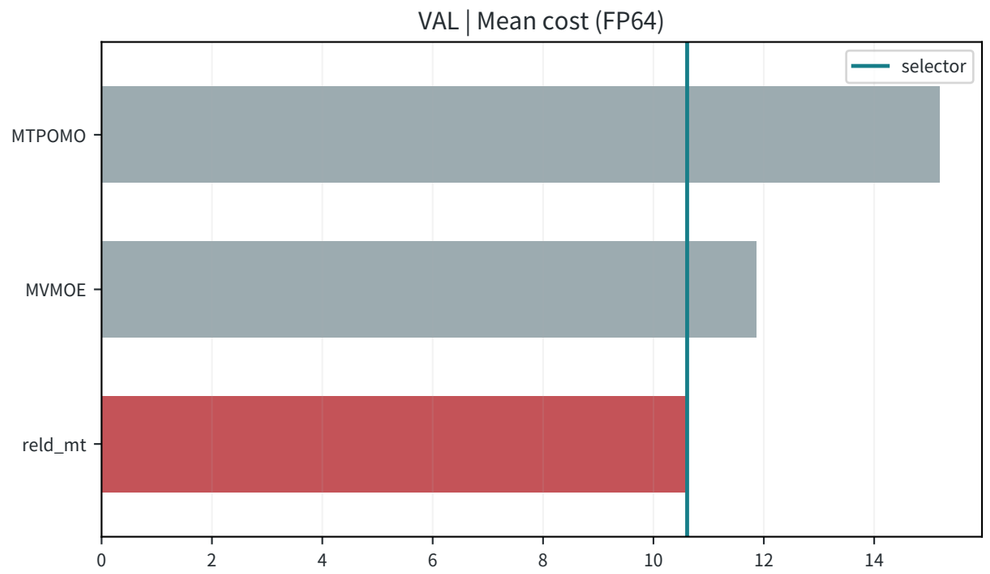


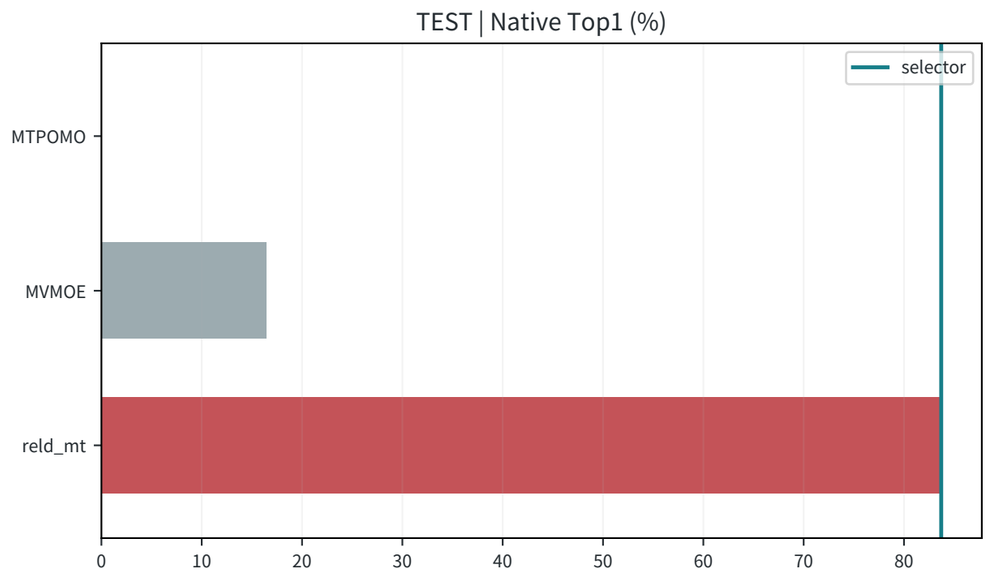


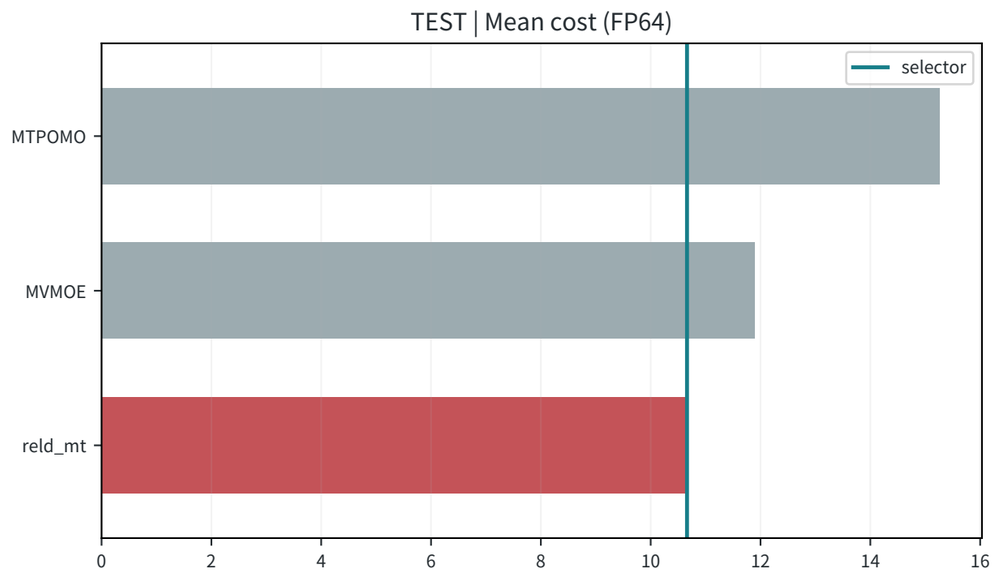
R101/A | OVRPBLTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.









R101/B | Two-stage provenance

Fixed TRAIN hint endpoint, then independently reset original full18 learning.

Teacher: pretrained. Stage I:12800 updates, logical batch1024, seed2; CVRP+15MVRP TRAIN81000/task only.

Separate non-affine per-node LayerNorm MSE; one dropout forward. No outcome labels, learned adapter or teacher VAL/TEST. Fixed endpoint, not validation selected.

Stage II: fresh original R84/R88B student; ONLY instance_encoder transferred. All non-encoder parameters remain identical fresh R84. Optimizer, scaler, RNG and task/sample streams reset.

Original1842953 parameters; no added heads or features. R77 CE .35 / comparison .35 / risk .02 / R-Drop .35. No hint loss, teacher arrays or online donor in Stage II or deployment.

Both A/B have12800+57600 updates; R89B/R84A are NOT total-budget-matched. B teacher has additional external solver pretraining history. The paper memo provides conditional motivation only.

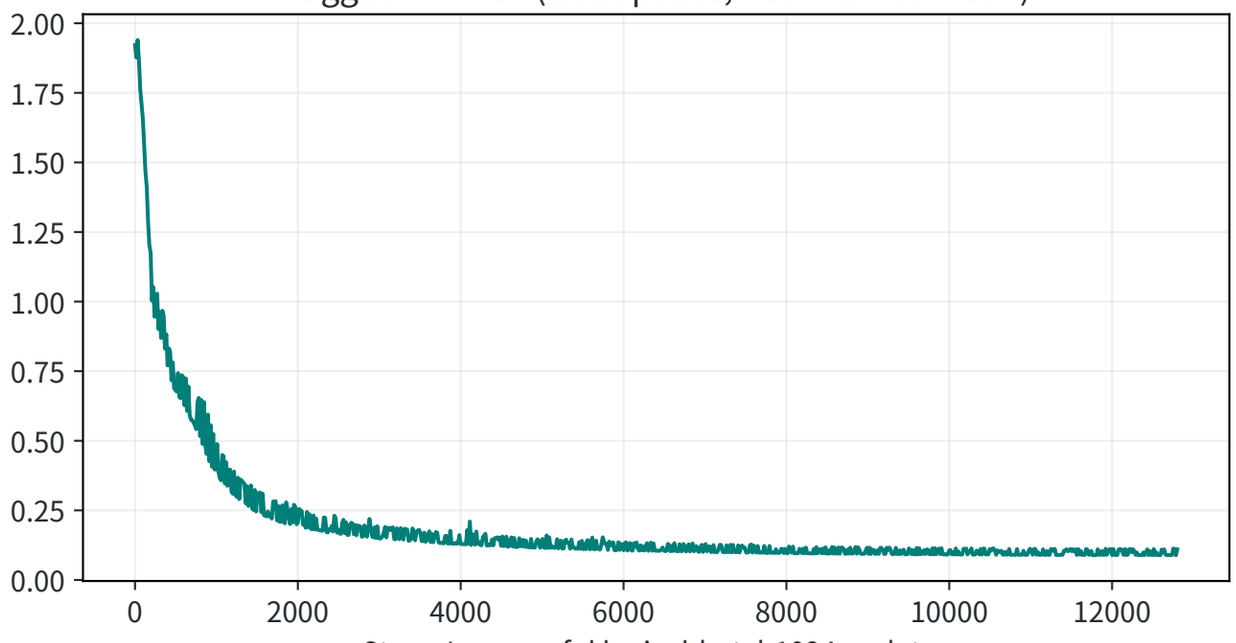
Native/FP64 ind winners; legal-pool Oracle and TRAIN-fitted SBS; ALL is task-equal. Development-exposed TEST; no TEST/LIB checkpoint selection.

Stage-I encoder SHA256: 6461d4d78289f6cbb92f534034602eea4eba389fc606f1fc0b386ed2662c657a

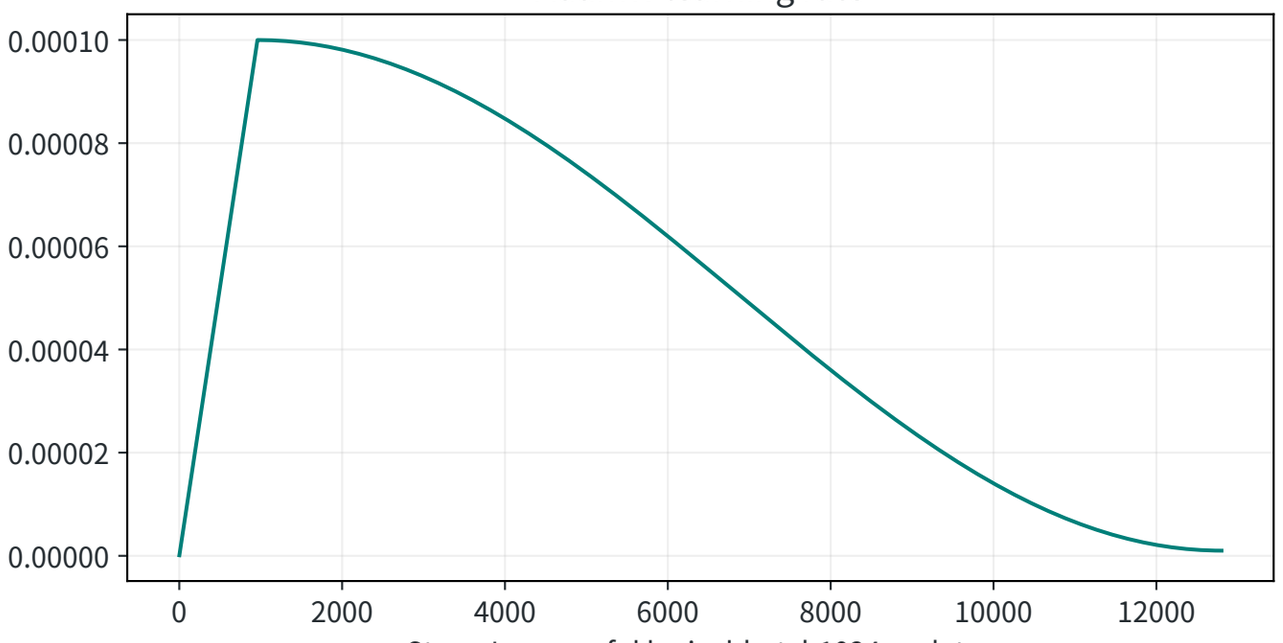
Unchanged non-encoder SHA256: 970a8aa2a548f9034f34b73d060662f71808e46359335185ce07bfc5b1bb2fa6

R101/B | TRAIN-only pretrained node hints; fixed12800 endpoint

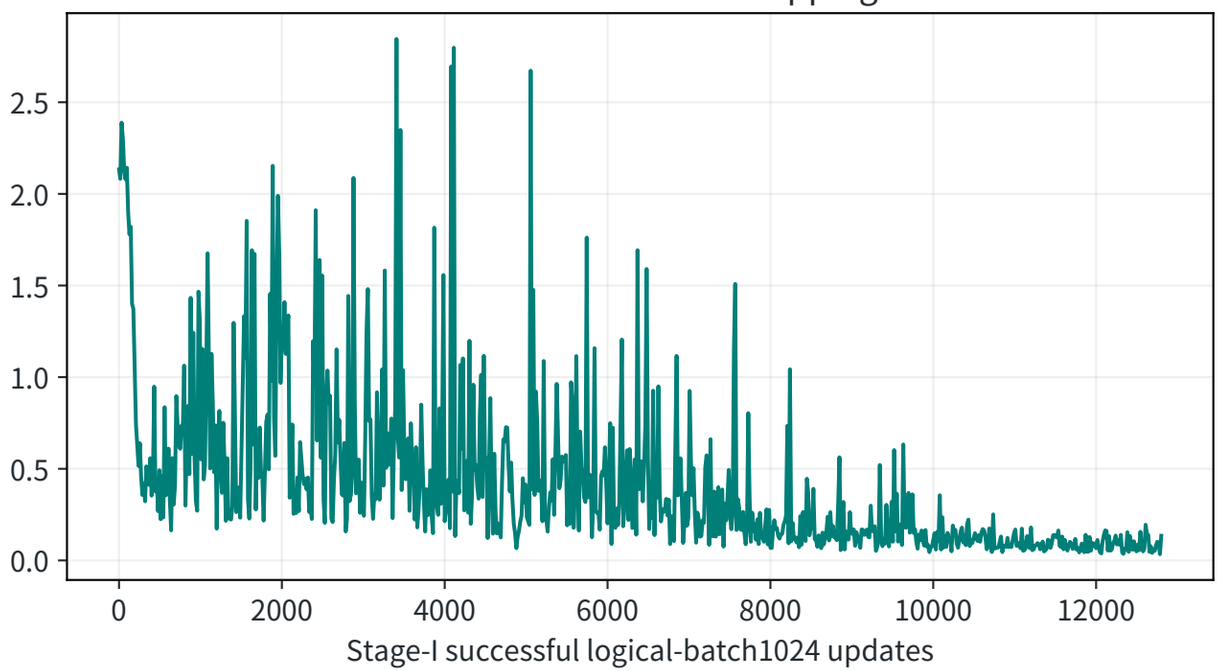
Logged LN-MSE (last update, not interval mean)



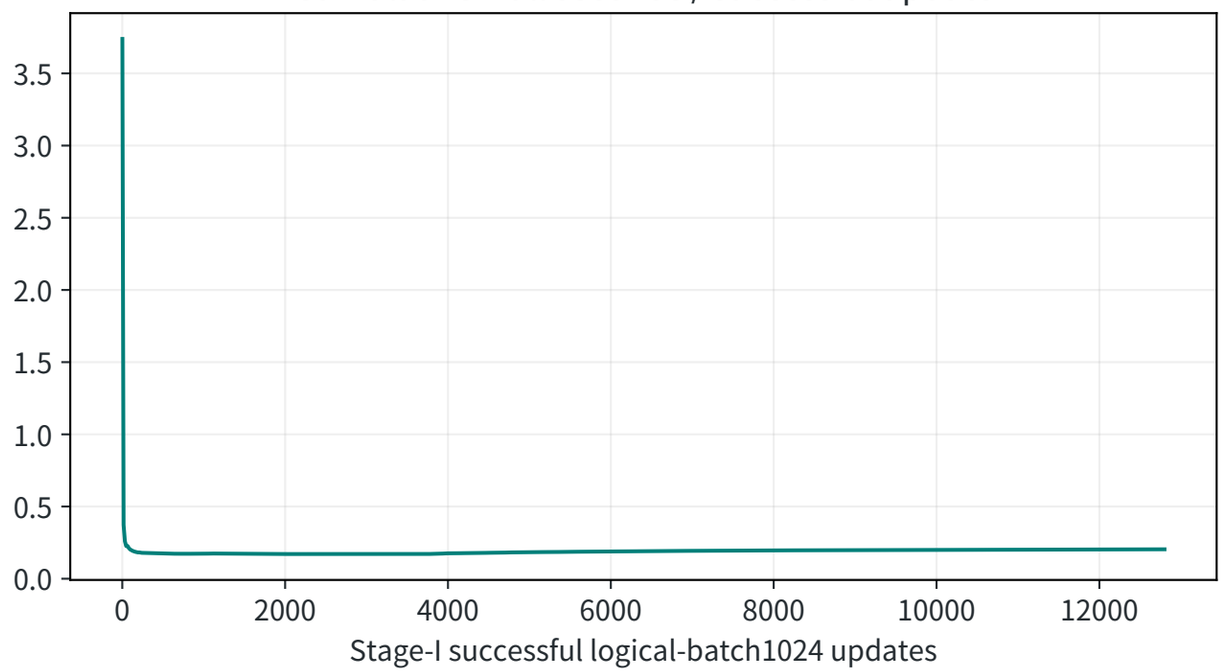
AdamW learning rate



Gradient norm before clipping



Cumulative wall seconds / successful update



R101/B | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 27 | update 38880 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.2012	66.2037	11.416300	11.427978	11.371692	-0.1965	+0.3911	0.4000
TEST ALL	71.4346	66.1802	11.373450	11.385299	11.329932	-0.2000	+0.3831	0.3910

Dataset	n	NSS Top1 (%)	R101 Top1 (%)	NSS cost	R101 cost
TSP TEST	4500	45.888889	47.400000	7.970400719	7.970903724
CVRP TEST	4500	45.977778	47.822222	18.004802368	18.008762029
TSPLIB	49	20.408163	30.612245	8.217907074	8.195489915
CVRPLIB	100	70.000000	63.000000	67.101111717	67.248113565

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original R77 native CE

R101/B | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.2012	66.2037	11.416300	11.427978	11.371692	-0.1965	+0.3911	0.4000	per-problem TRAIN SBS
TSP	47.0667	33.1111	8.024764	8.045354	7.990533	-0.2559	+0.4284	0.4168	T2T
CVRP	46.8444	18.7111	18.157905	18.269377	18.064087	-0.6102	+0.5194	0.6885	RELD_CVRP
ATSP	76.9778	47.4889	1.638750	1.676905	1.632471	-2.2753	+0.3847	0.3807	UNICO_MatPOENet
OVRP	41.1556	35.7556	7.461777	7.469579	7.415777	-0.1044	+0.6203	0.6360	MoSES_RF
VRPB	81.1778	80.8444	9.354736	9.354876	9.318750	-0.0015	+0.3862	0.3861	reld_mt
VRPL	38.0444	33.0889	12.431904	12.437959	12.383444	-0.0487	+0.3913	0.3992	MoSES_RF
VRPTW	80.1333	79.9556	18.818619	18.819431	18.766139	-0.0043	+0.2797	0.2950	reld_mt
OVRPTW	83.8889	83.7778	10.143708	10.143730	10.116859	-0.0002	+0.2654	0.2643	reld_mt
OVRPB	88.5556	88.5111	6.357796	6.357254	6.341078	+0.0085	+0.2636	0.2777	reld_mt
OVRPL	41.6667	35.9111	7.437495	7.445709	7.390416	-0.1103	+0.6370	0.6536	MoSES_RF
VRPBL	78.9333	78.5111	9.367756	9.371523	9.328366	-0.0402	+0.4223	0.4238	reld_mt
VRPBTW	78.7778	78.4222	19.945192	19.950430	19.851084	-0.0263	+0.4741	0.4523	reld_mt
VRPLTW	78.9333	78.7556	18.763489	18.765871	18.707843	-0.0127	+0.2974	0.3115	reld_mt
OVRPBL	89.4222	89.2000	6.368941	6.370093	6.352824	-0.0181	+0.2537	0.2633	reld_mt
OVRPBTW	83.6667	83.4444	10.659787	10.661098	10.623245	-0.0123	+0.3440	0.3123	reld_mt
OVRPLTW	83.6222	83.4889	10.156412	10.157423	10.129818	-0.0100	+0.2625	0.2638	reld_mt
VRPBLTW	79.4444	79.4000	19.794736	19.797044	19.706420	-0.0117	+0.4482	0.4360	reld_mt
OVRPBLTW	83.3111	83.2889	10.609625	10.609955	10.571302	-0.0031	+0.3625	0.3389	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R101/B | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.4346	66.1802	11.373450	11.385299	11.329932	-0.2000	+0.3831	0.3910	per-problem TRAIN SBS
TSP	47.4000	34.5333	7.970904	7.989621	7.937593	-0.2343	+0.4197	0.4137	T2T
CVRP	47.8222	17.9333	18.008762	18.124983	17.918509	-0.6412	+0.5037	0.6601	RELD_CVRP
ATSP	76.8889	47.2444	1.635283	1.674207	1.629441	-2.3249	+0.3585	0.3578	UNICO_MatPOENet
OVRP	41.4889	33.6444	7.445720	7.455750	7.400487	-0.1345	+0.6112	0.6191	MoSES_RF
VRPB	80.7333	80.4000	9.349268	9.351810	9.315298	-0.0272	+0.3647	0.3696	reld_mt
VRPL	37.9556	32.2667	12.349619	12.356138	12.300958	-0.0528	+0.3956	0.4030	MoSES_RF
VRPTW	81.1333	80.8444	18.688695	18.690545	18.637408	-0.0099	+0.2752	0.2839	reld_mt
OVRPTW	83.9778	83.9333	10.092510	10.093063	10.068135	-0.0055	+0.2421	0.2329	reld_mt
OVRPB	88.0667	88.1333	6.371061	6.370826	6.352225	+0.0037	+0.2965	0.3069	reld_mt
OVRPL	41.4667	34.6000	7.464103	7.472542	7.418033	-0.1129	+0.6211	0.6301	MoSES_RF
VRPBL	79.6667	79.4222	9.395508	9.396379	9.355967	-0.0093	+0.4226	0.4182	reld_mt
VRPBTW	79.0222	78.7556	19.662672	19.665725	19.577666	-0.0155	+0.4342	0.4307	reld_mt
VRPLTW	79.4222	79.2444	18.801794	18.803136	18.745143	-0.0071	+0.3022	0.3203	reld_mt
OVRPBL	89.2444	89.2889	6.375857	6.375741	6.359310	+0.0018	+0.2602	0.2736	reld_mt
OVRPBTW	84.9556	84.8222	10.658871	10.659591	10.624748	-0.0068	+0.3212	0.2930	reld_mt
OVRPLTW	82.9778	82.8667	10.085927	10.086112	10.059001	-0.0018	+0.2677	0.2681	reld_mt
VRPBLTW	79.8667	79.7333	19.704221	19.707068	19.613284	-0.0144	+0.4637	0.4440	reld_mt
OVRPBLTW	83.7333	83.5778	10.661334	10.662149	10.625574	-0.0076	+0.3366	0.3134	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R101/B | LIB and locked provenance

Current R101 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	30.6122	10.2041	8.195490	8.240435	8.142167	-0.5454	+0.6549	0.6673
CVRPLIB	63.0000	45.0000	67.248114	67.245364	66.945093	+0.0041	+0.4526	0.5681

SHA256: cd1f5e55d852c2460d3949825f1a7f923b685295146e93a542c753611ef0683b

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, original R88B physical TSP node0 slot3; original R77 hand34/methodID, joint2, score MLP/pair differences/maximin; stable ascending global-solver-ID ties; original inputs only; no teacher, hint, adapter or calibration

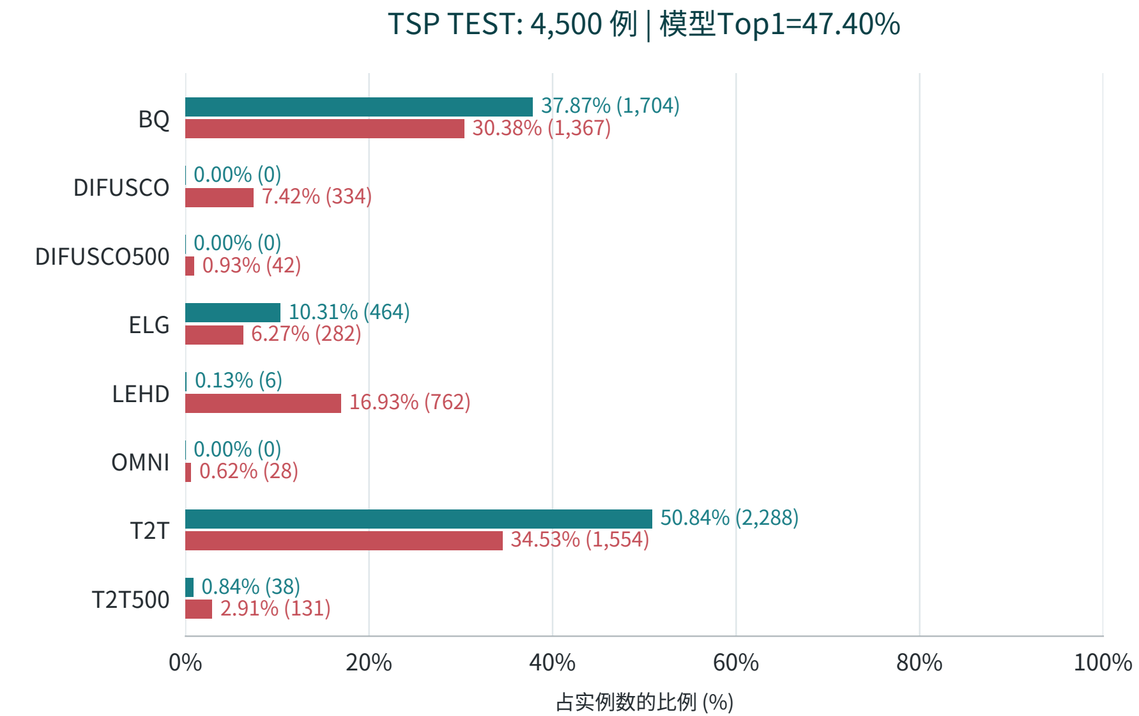
Normalization: new81000 TRAIN only; no TEST/LIB refit.

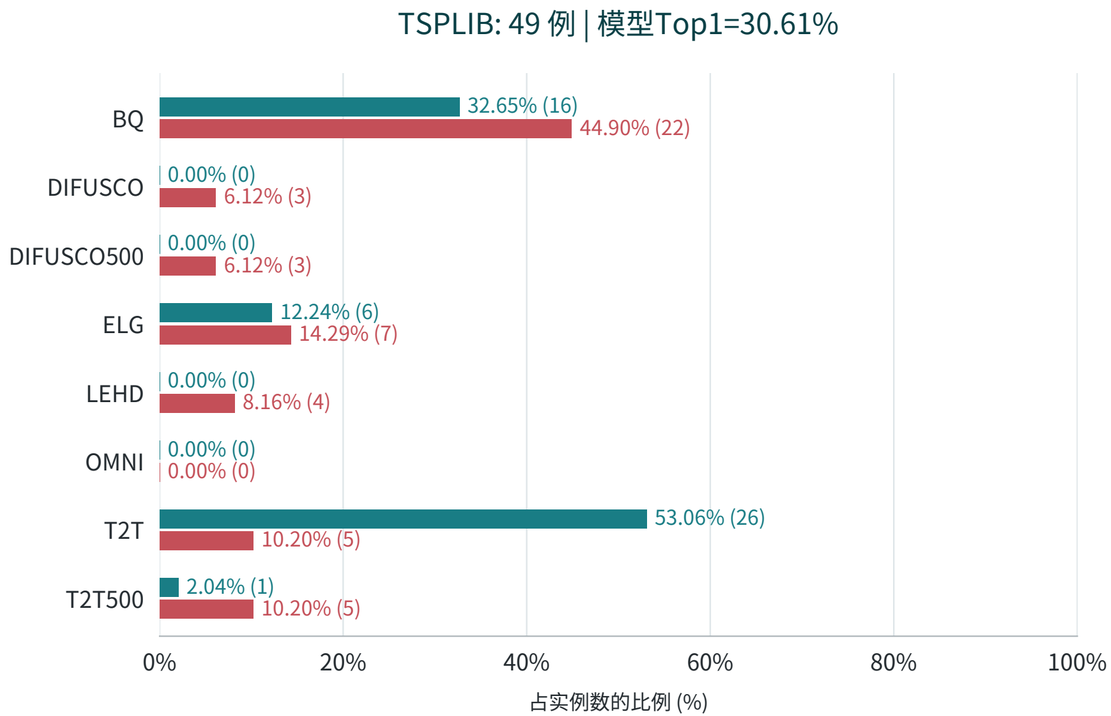
R101/B | TSP | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

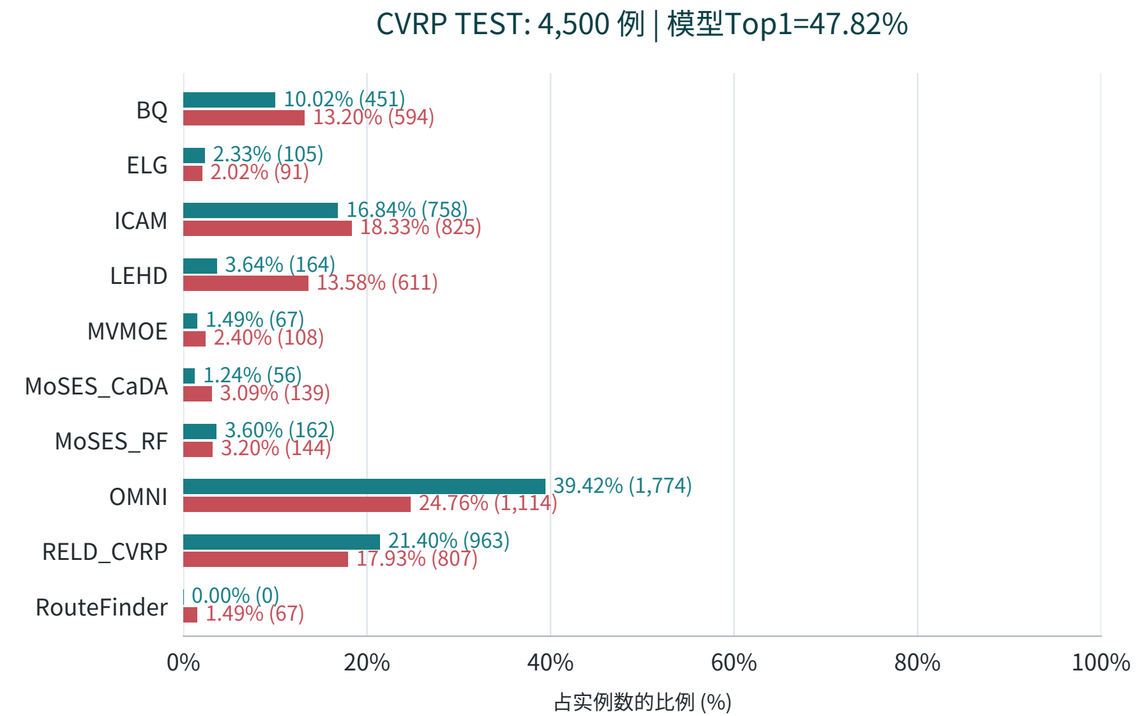
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

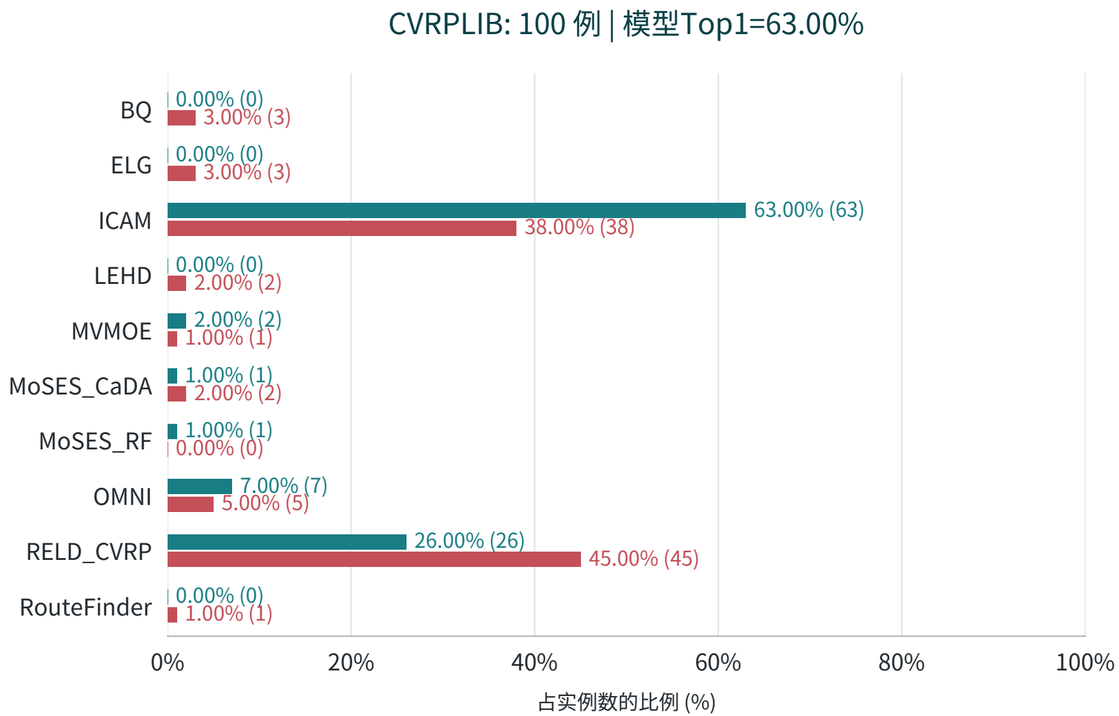
R101/B | CVRP | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

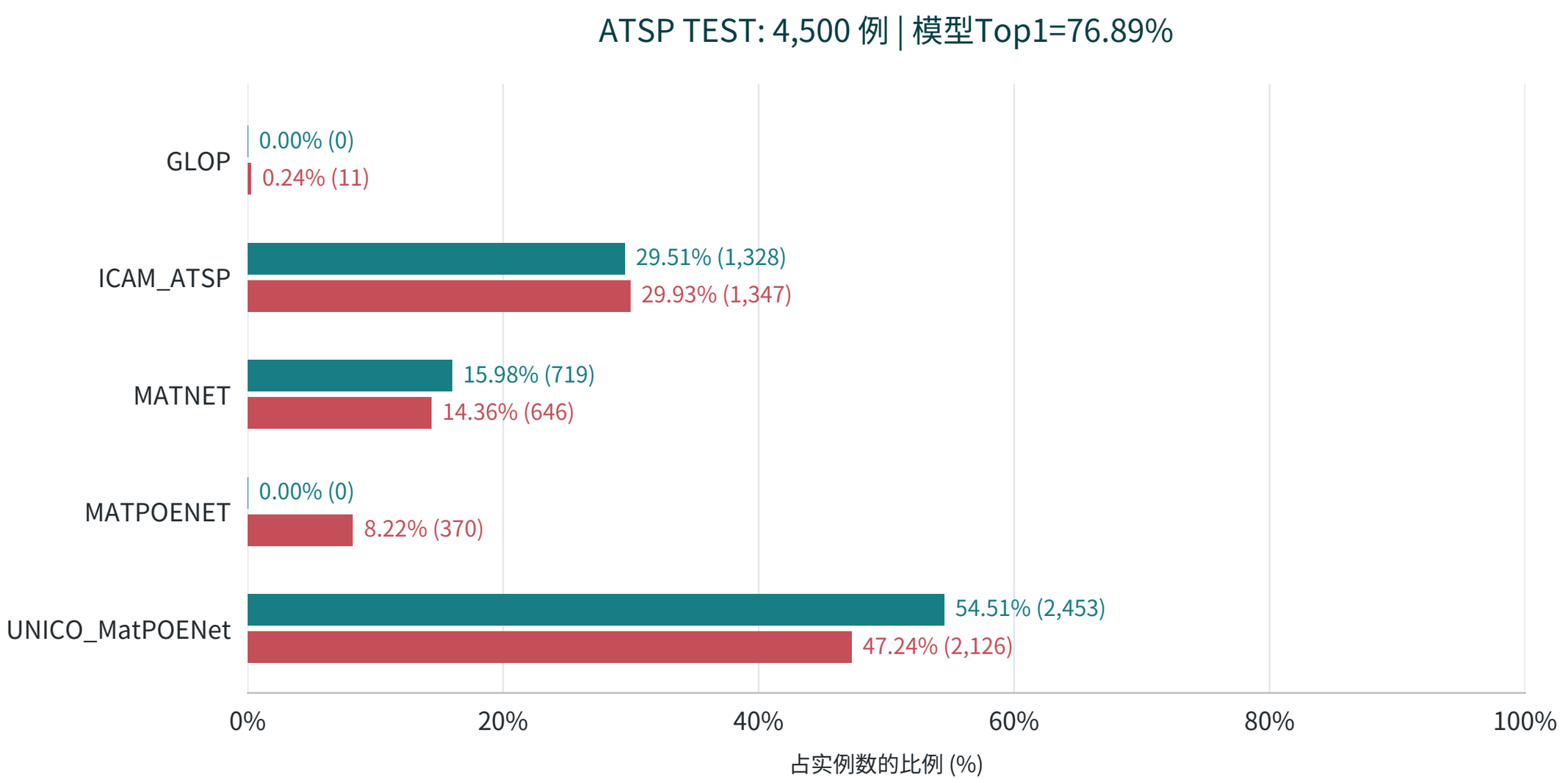
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | ATSP | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

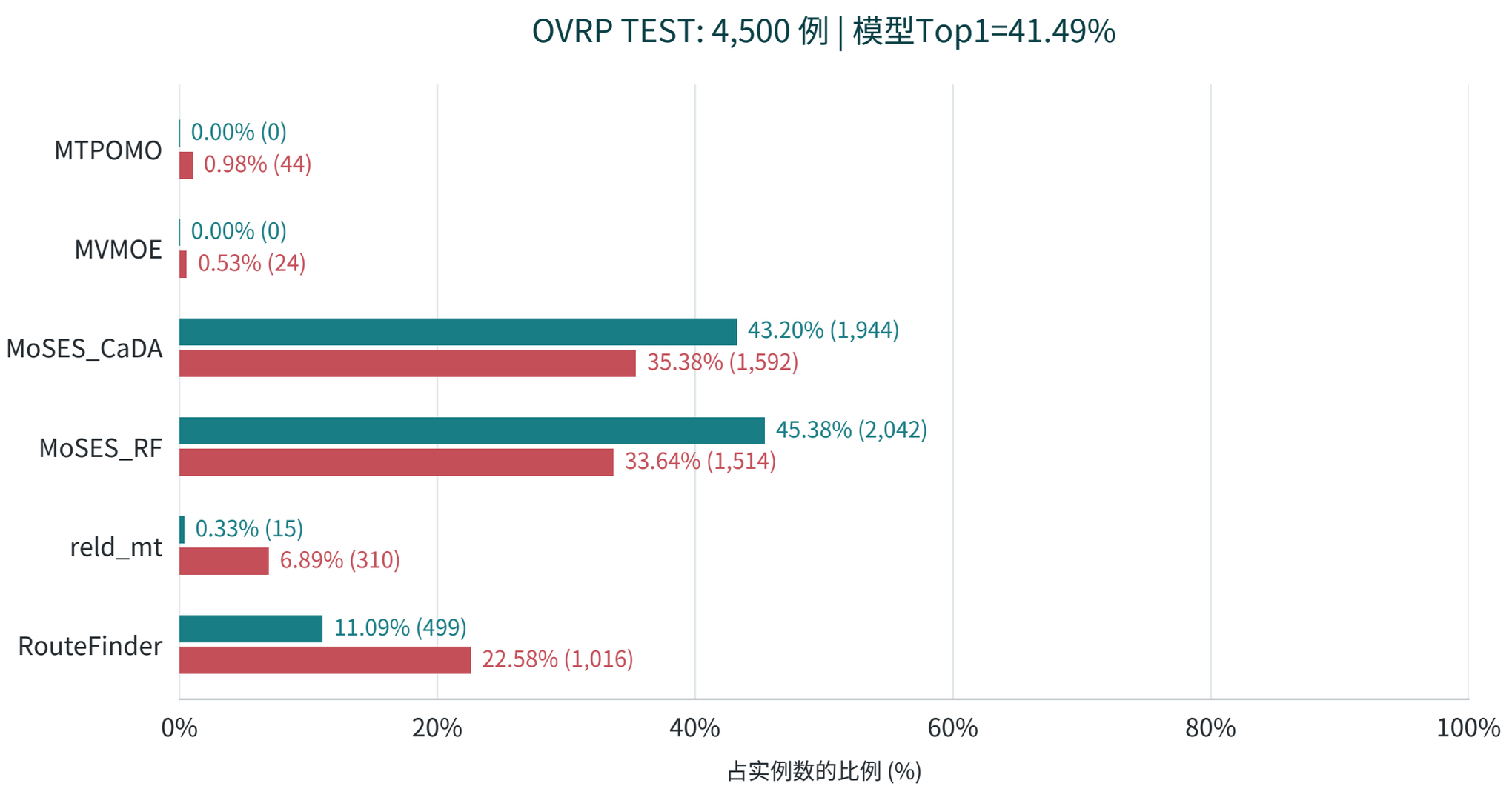
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRP | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

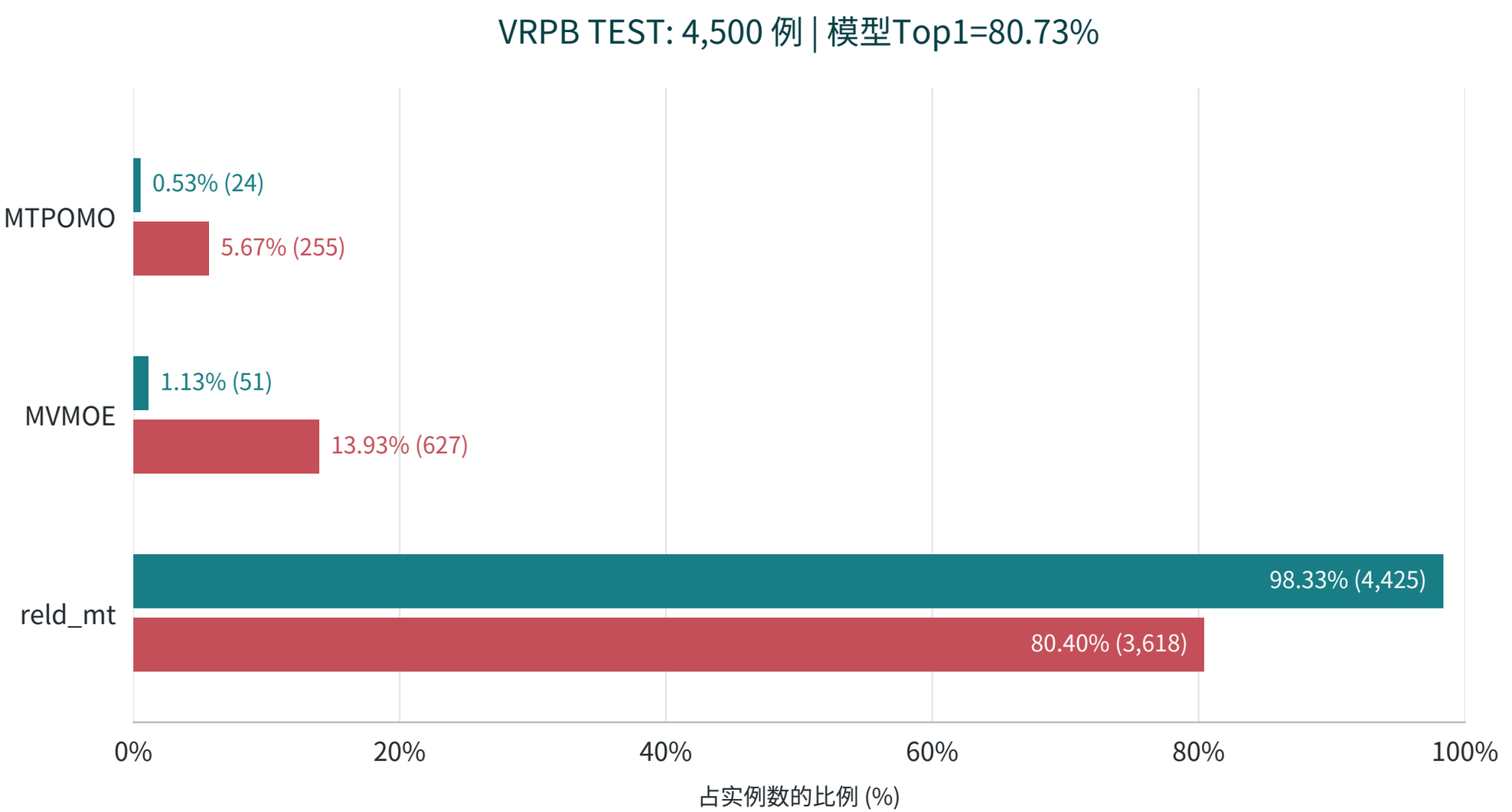
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPB | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

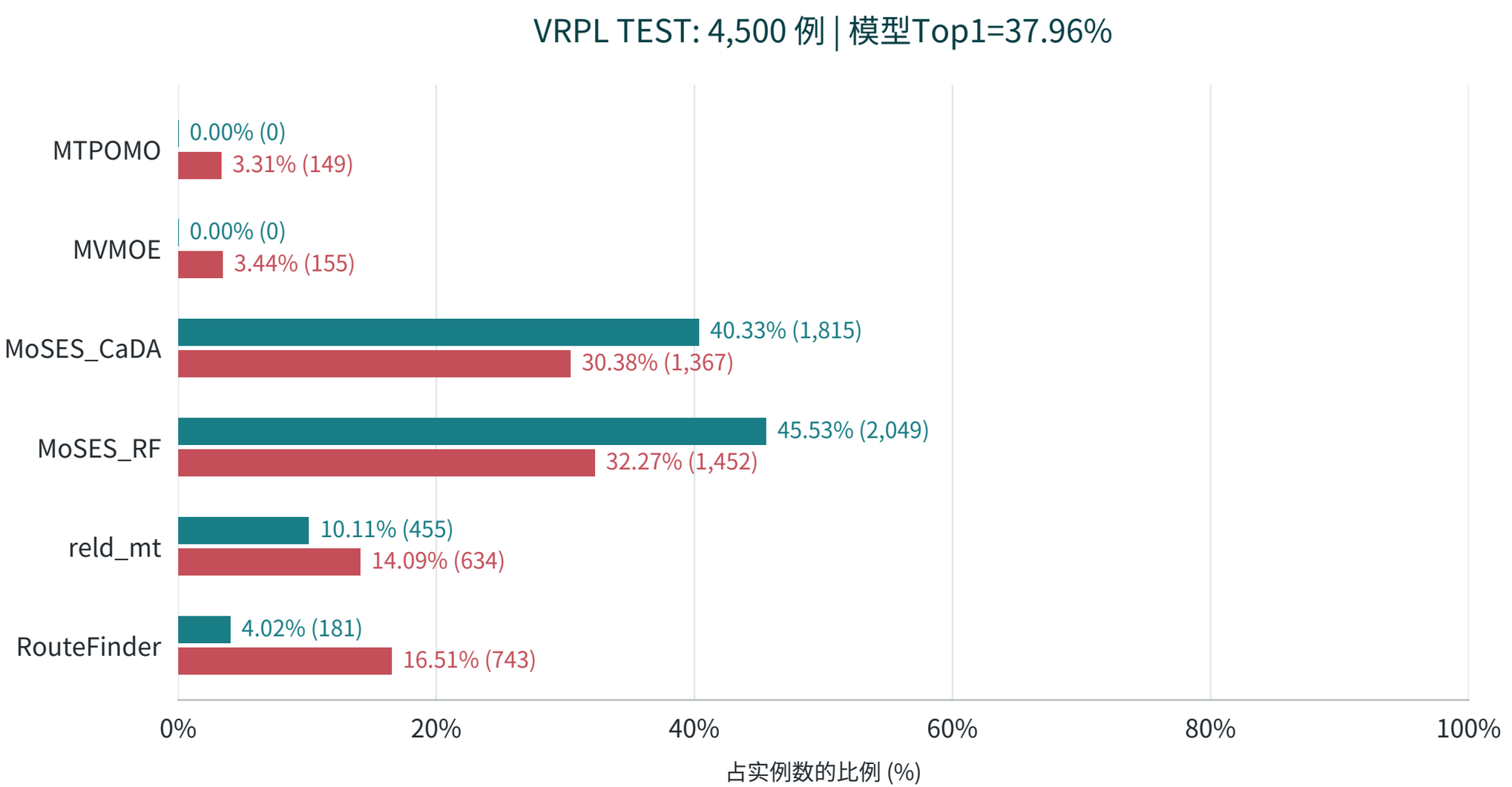
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPL | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

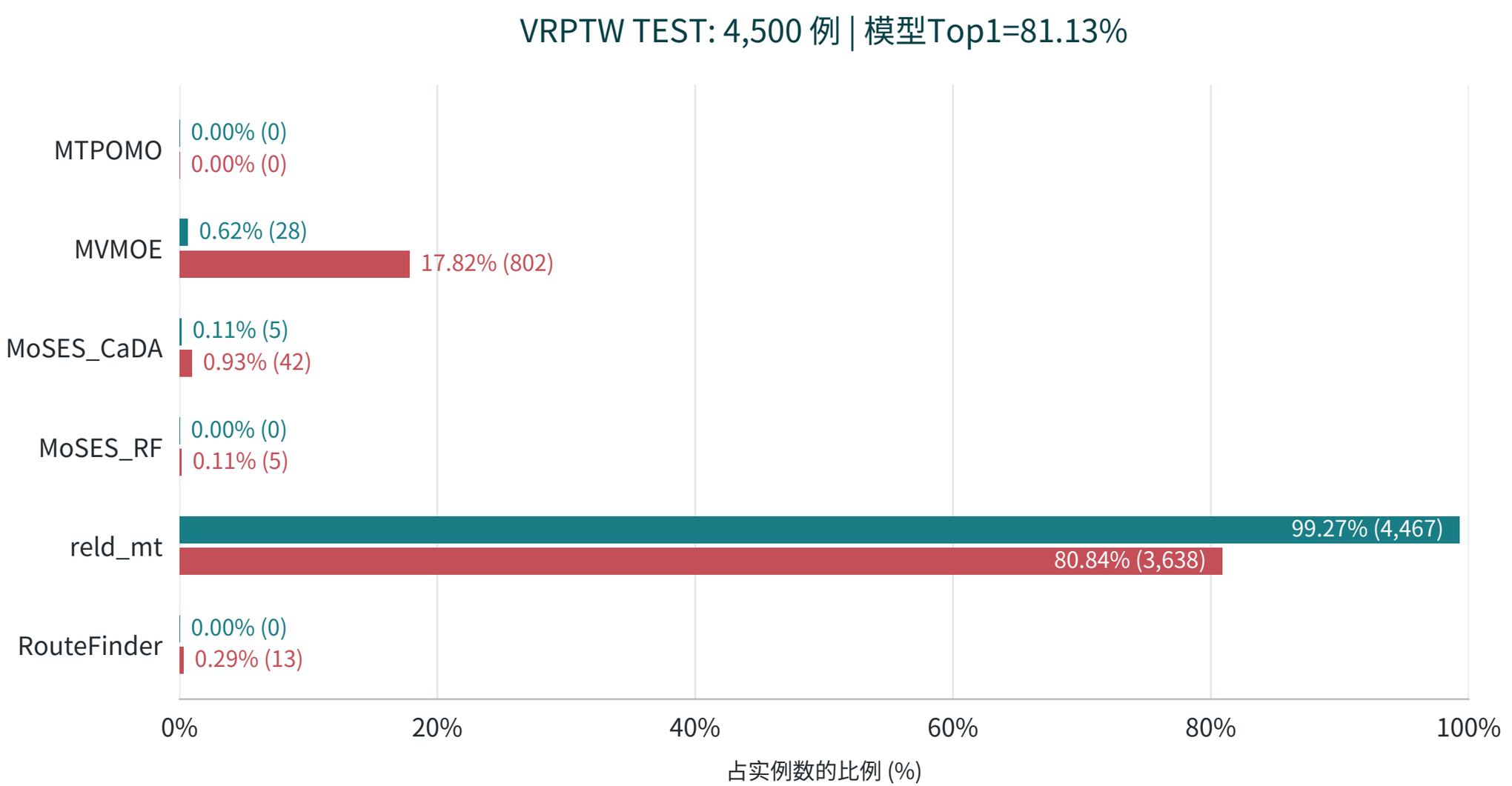
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

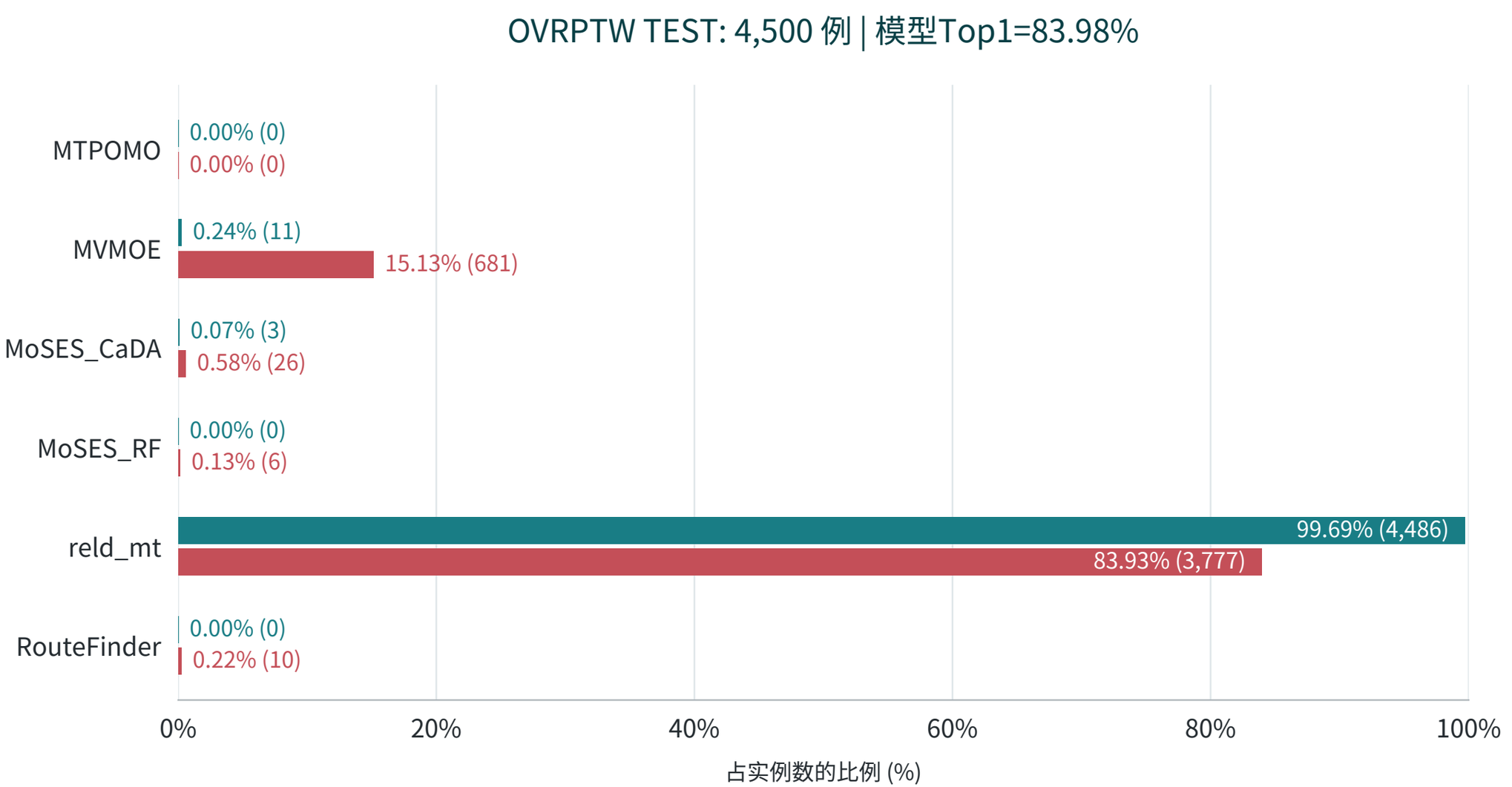
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

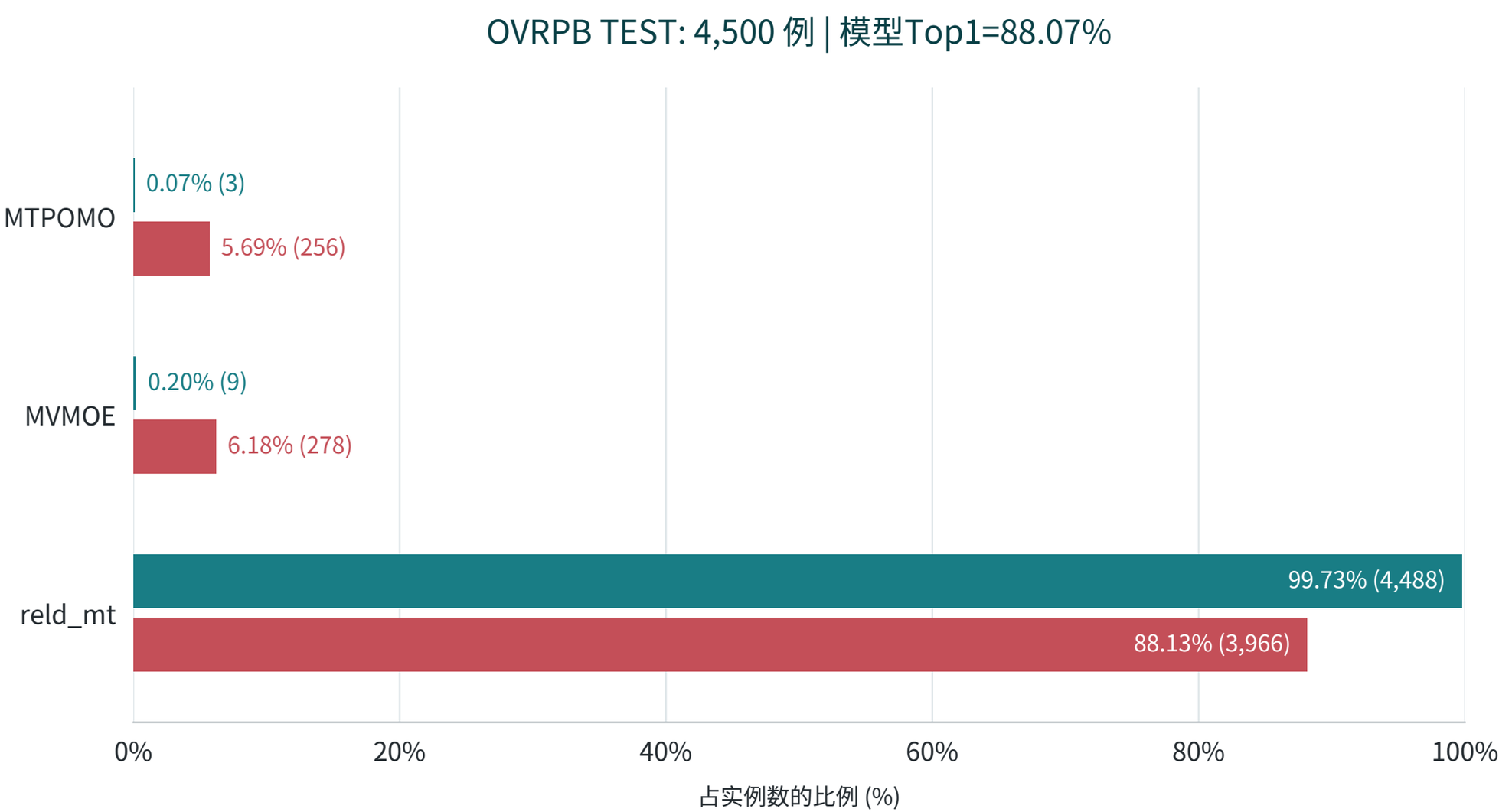
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPB | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

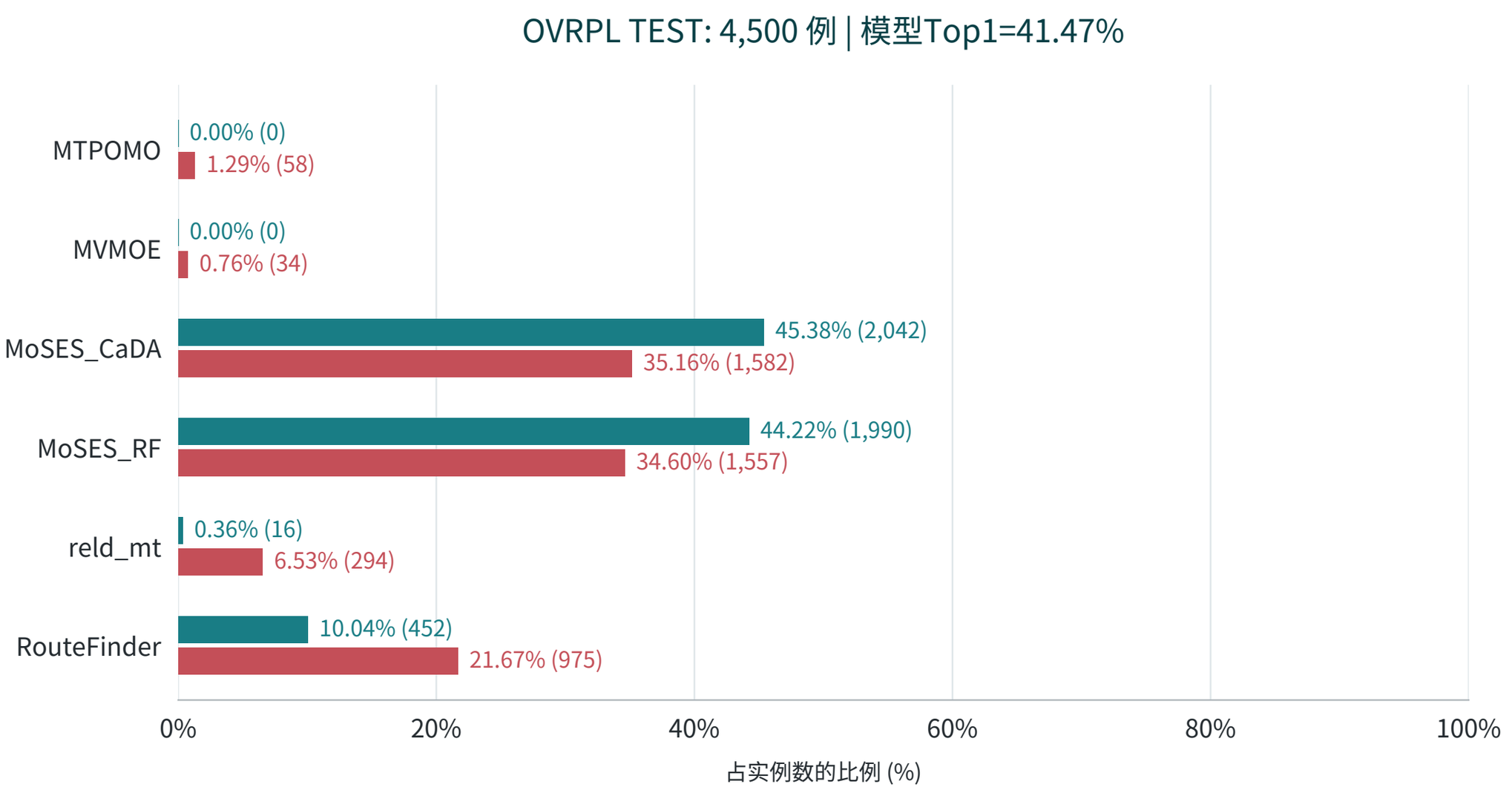
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPL | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

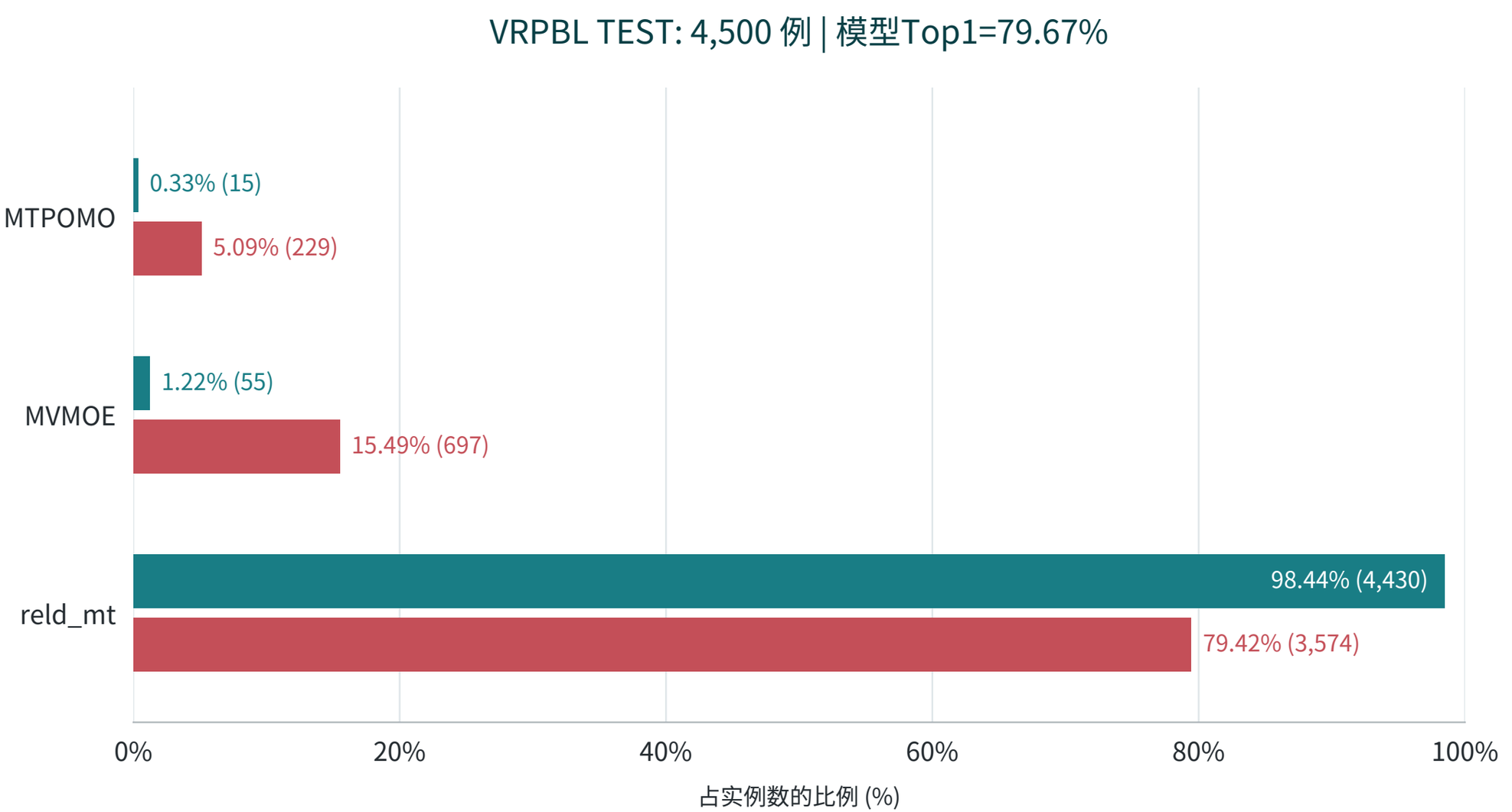
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPBL | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

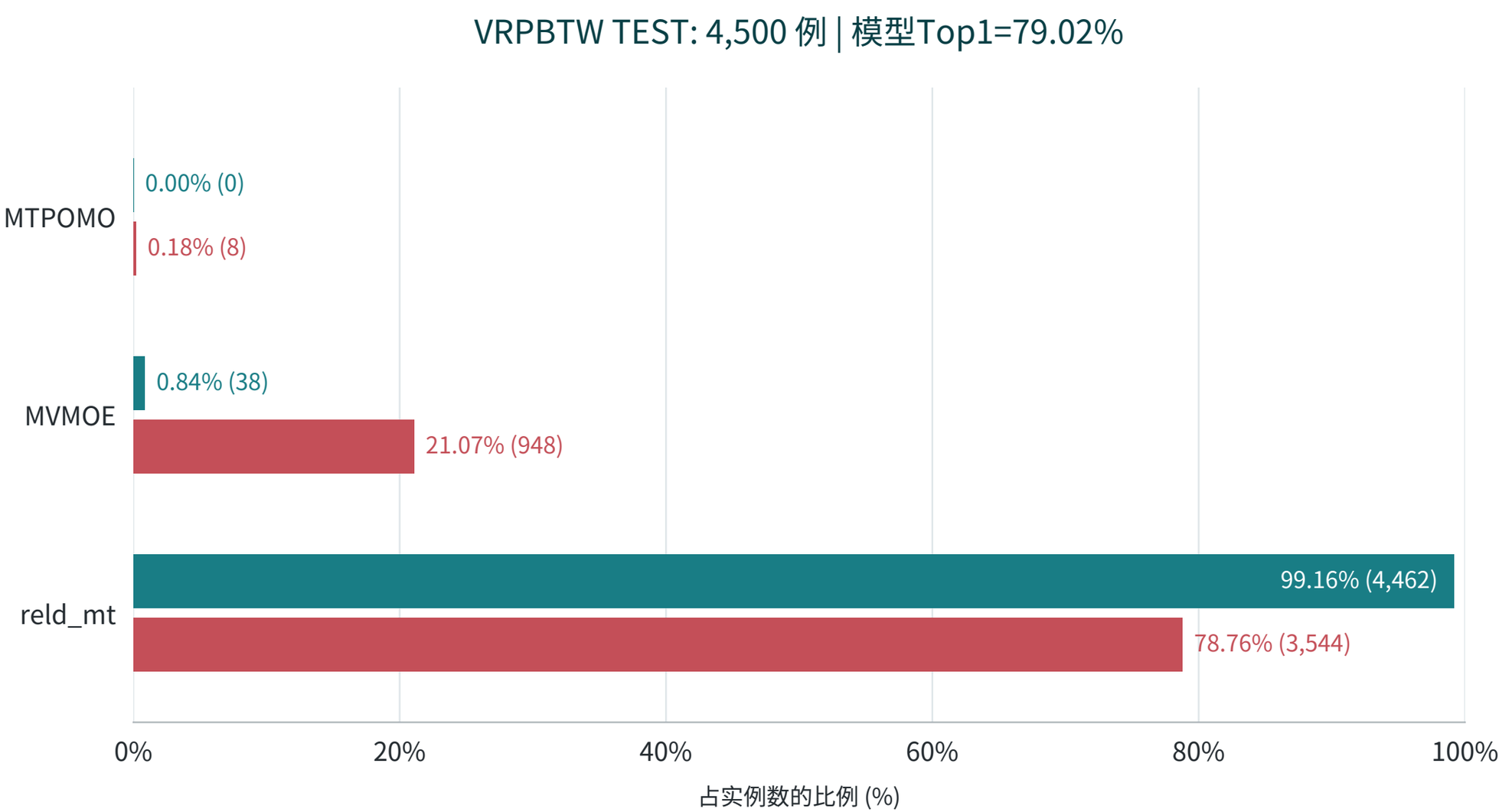
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPBTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

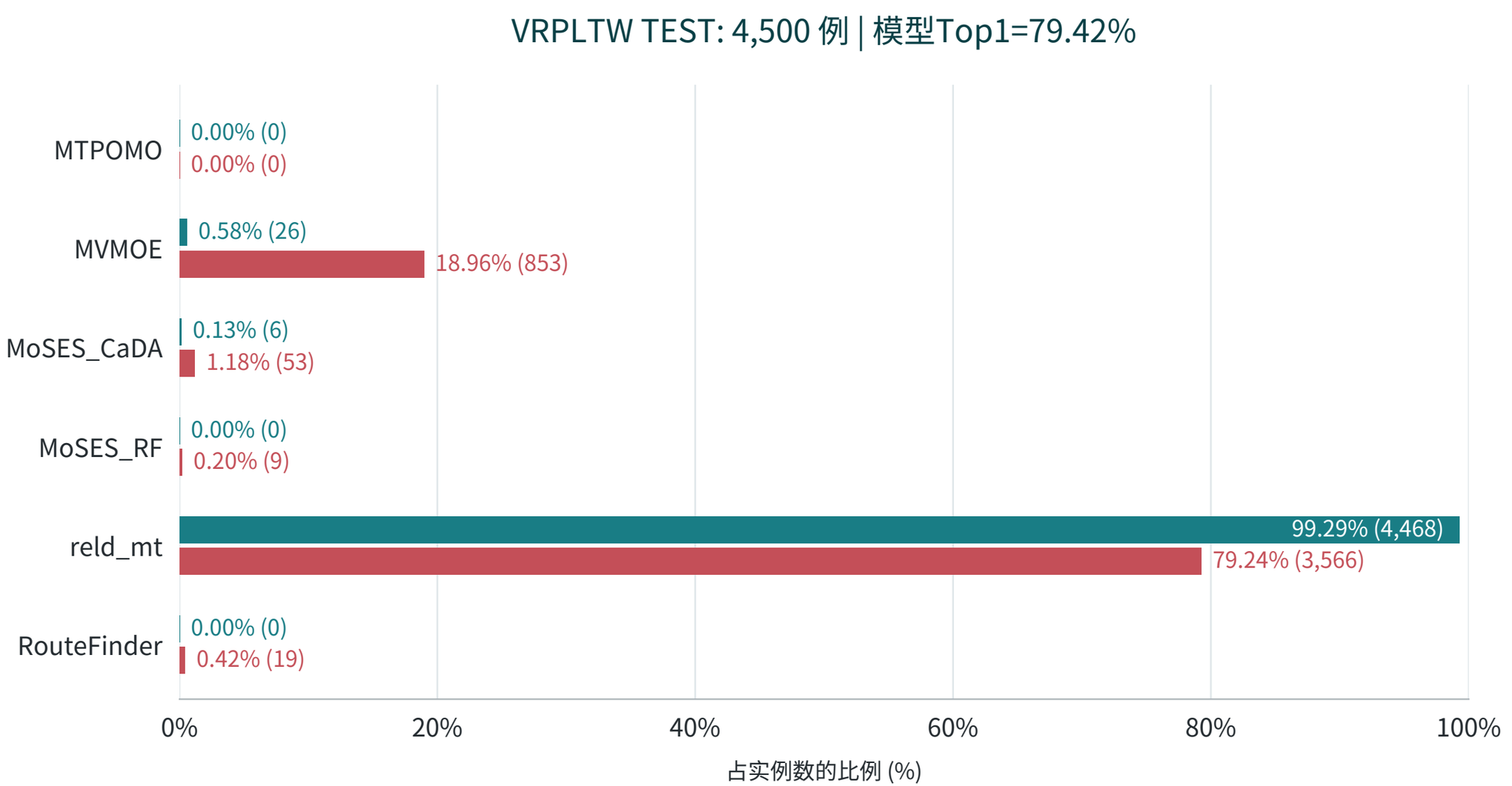
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPLTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

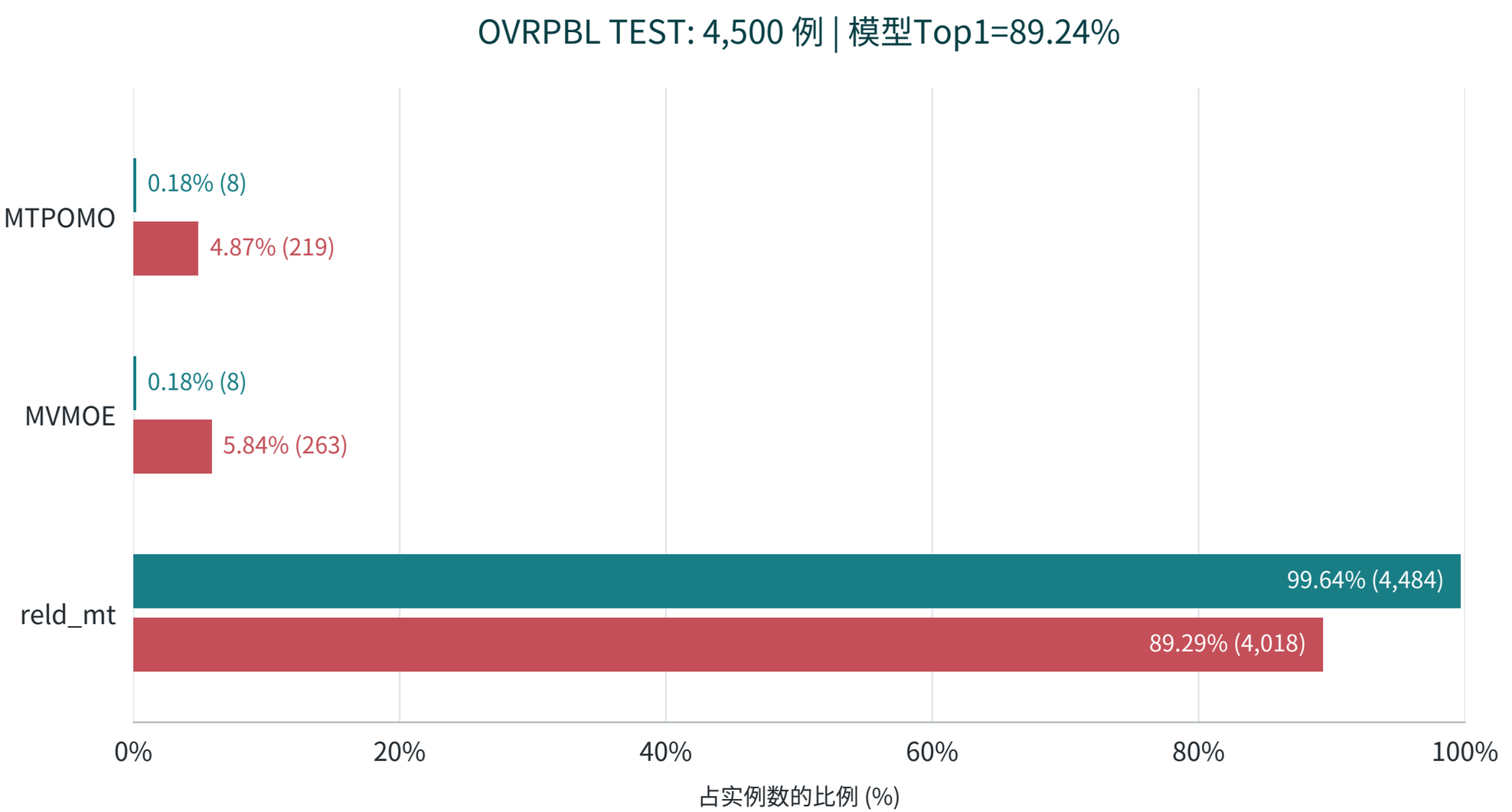
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPBL | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

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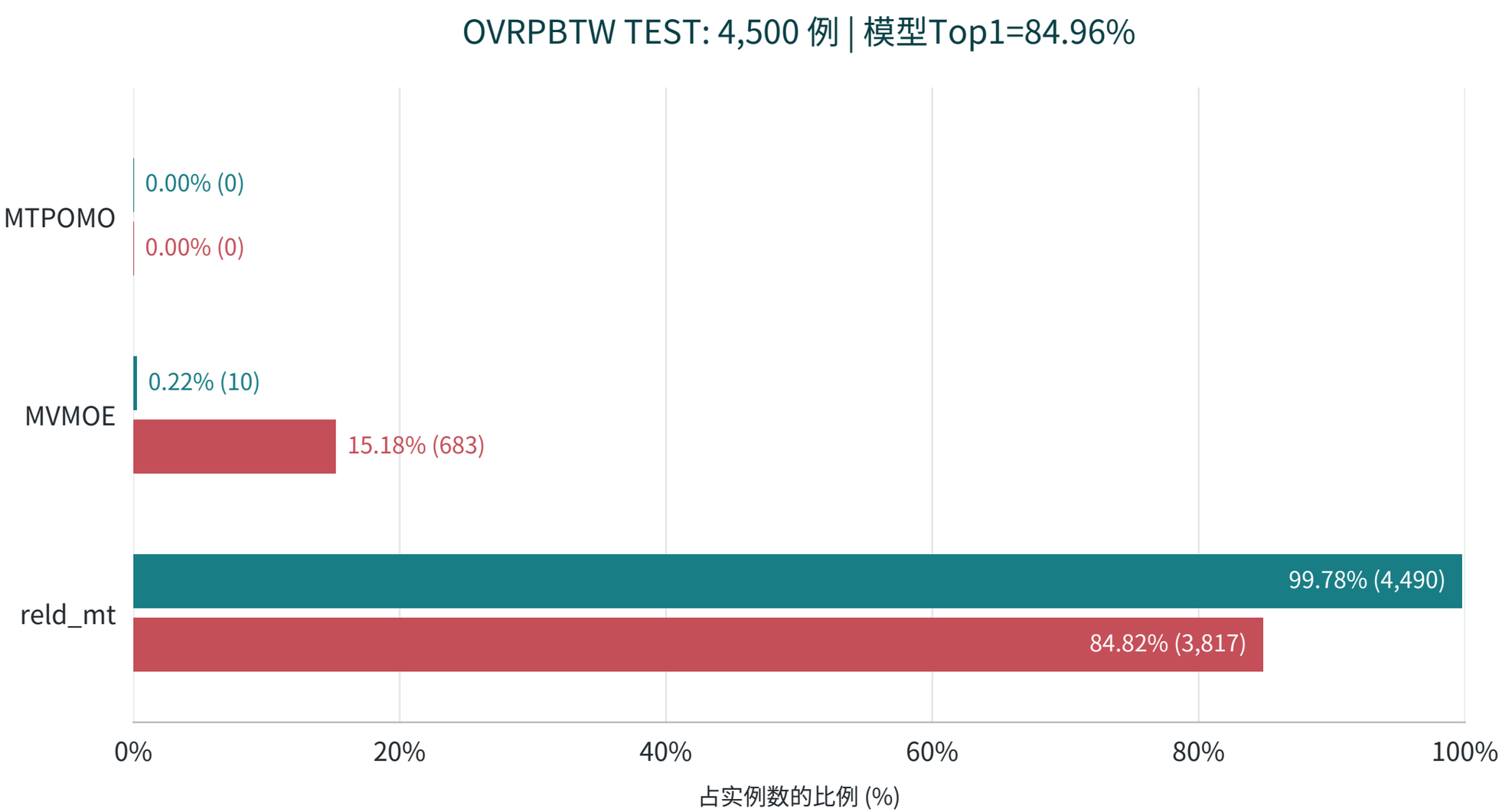
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPBTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

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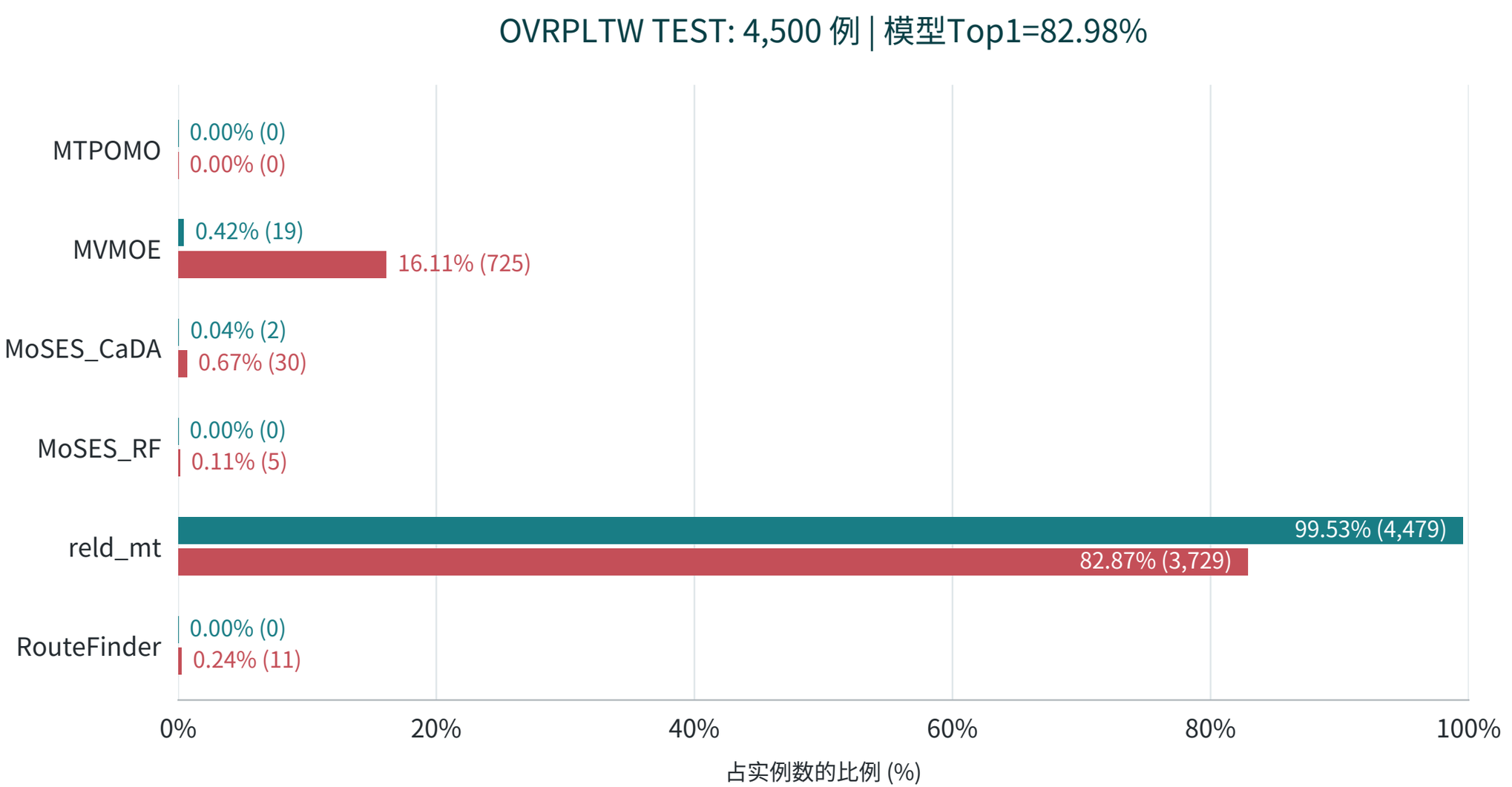
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPLTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

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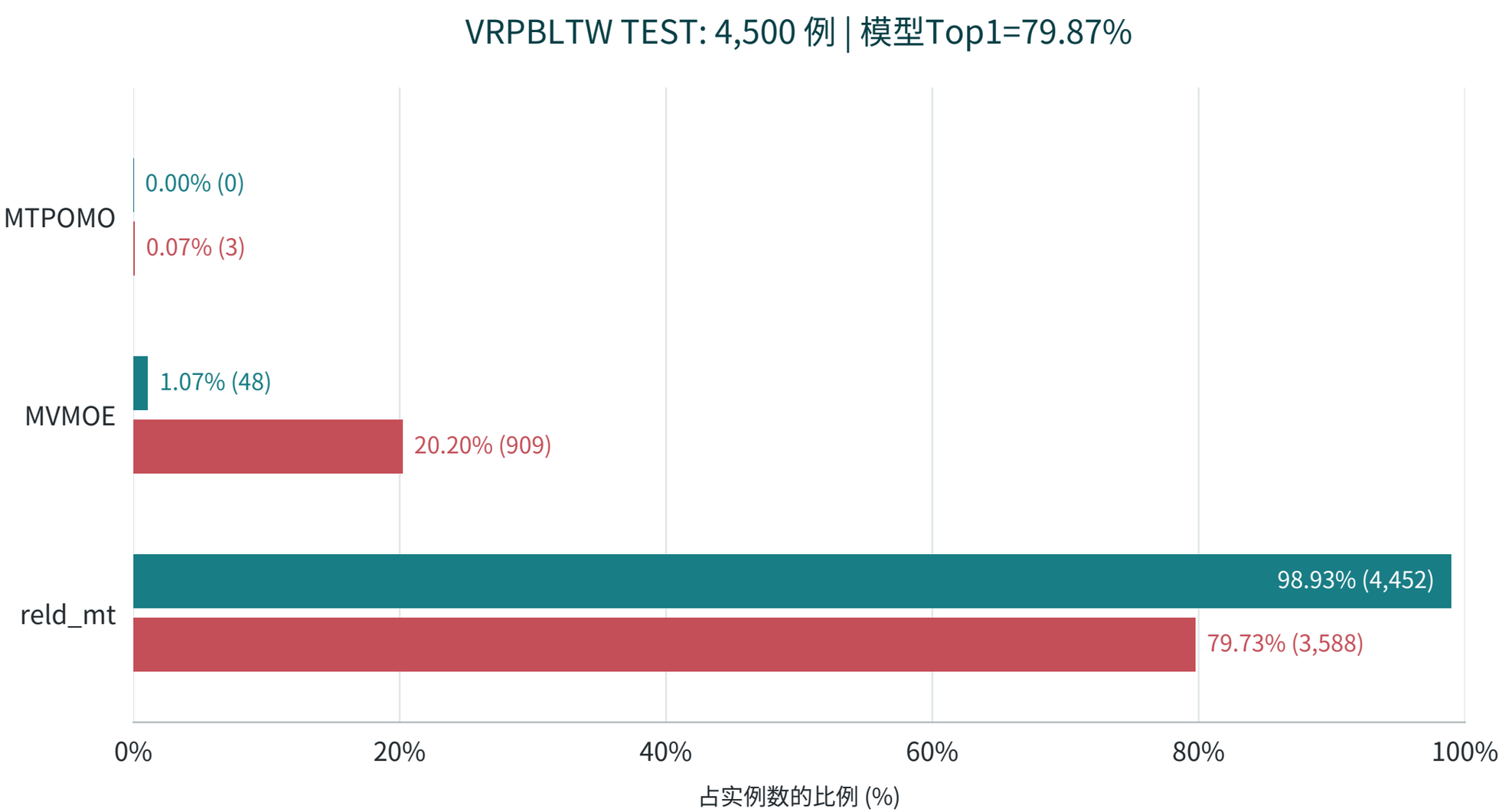
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | VRPBLTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



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模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

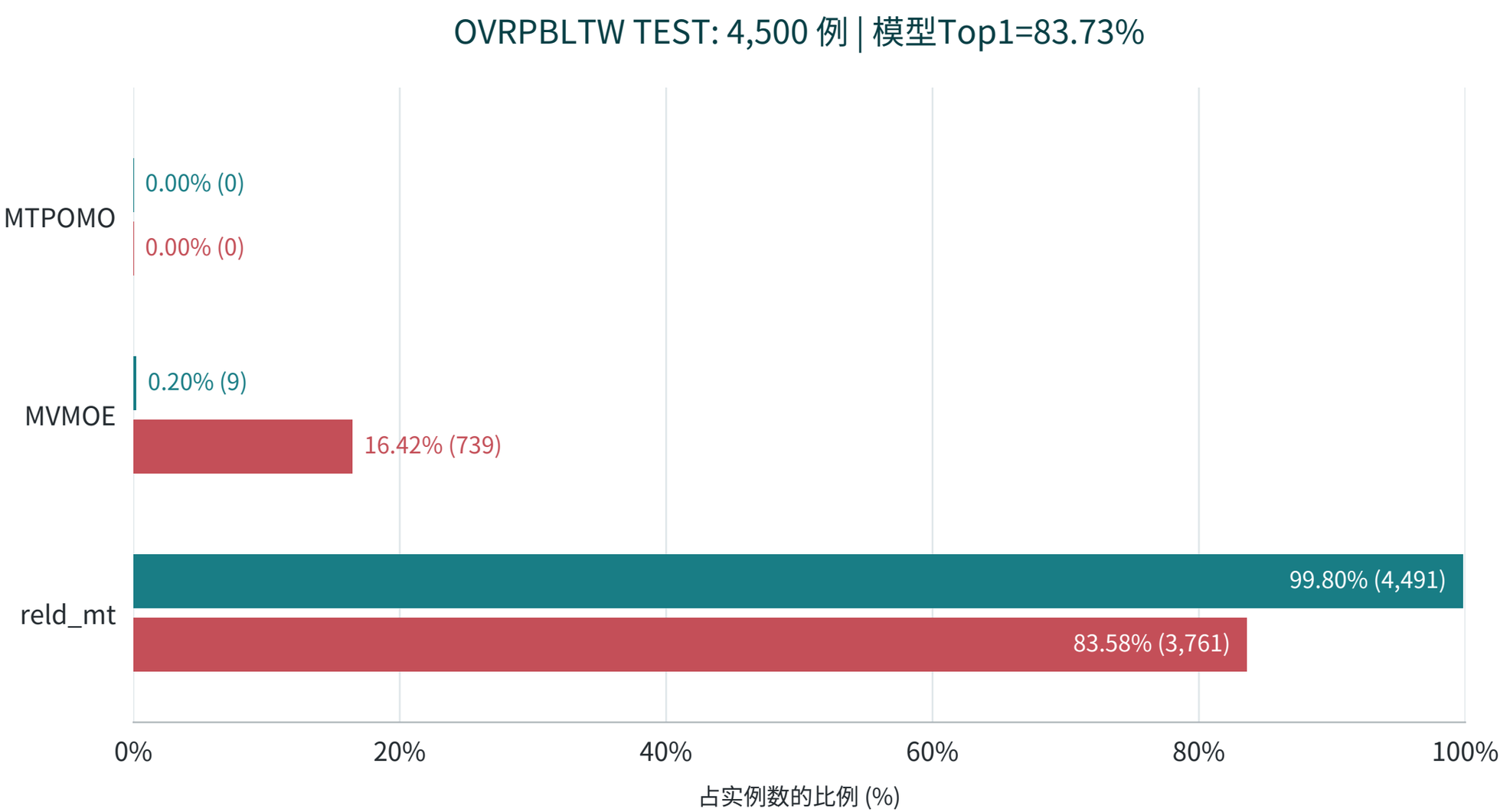
R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | OVRPBLTW | 方法分布

同一锁定模型: epoch 27 / update 38880 / SHA256 cd1f5e55d852c246

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

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标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

R101/B: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/B | Fixed target36 vs R89B

Native/FP64 winners; true rescues/losses and false-pick cost, not selection-frequency accuracy.

Split	True winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	919	384	489	408	1.1035
VAL	4625	52	24	24	23	1.1243
TEST	4598	50	25	20	29	1.0874

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.206036	20843	23847	0.000146933	0.002871
VAL	-0.046914	1213	1251	0.000005325	-0.000675
TEST	0.014815	1265	1253	-0.000019990	0.001375

Split	New-false cost contribution	New-false cost delta	New-false regret pp
TRAIN	0.002346857	0.000065298	0.001318
VAL	0.002741576	0.000082391	0.001062
TEST	0.002862585	0.000151793	0.002304

Cost contributions: sum over newly false-picked instances / task n; ALL averages18 tasks.

R89B/R84A are historical, not12800+57600 budget-matched; zero-count movement is not success.

R101/B | Fixed target36 vs R84A

Native/FP64 winners; true rescues/losses and false-pick cost, not selection-frequency accuracy.

Split	True winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	919	694	72	580	1.1035
VAL	4625	52	45	4	33	1.1243
TEST	4598	50	42	5	44	1.0874

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.241221	25963	22446	-0.000937050	-0.008693
VAL	0.116049	1421	1327	-0.000311869	-0.003493
TEST	0.198765	1417	1256	-0.000392582	-0.003826

Split	New-false cost contribution	New-false cost delta	New-false regret pp
TRAIN	0.003248679	0.000081773	0.001785
VAL	0.003489623	0.000131106	0.002195
TEST	0.004247930	0.000189303	0.003050

Cost contributions: sum over newly false-picked instances / task n; ALL averages18 tasks.

R89B/R84A are historical, not12800+57600 budget-matched; zero-count movement is not success.

R101/B | Original student latency

First128 GPU-resident original TEST inputs/task; three synchronized repetitions.

Problem	B student ms/row
TSP	0.242471
CVRP	0.291154
ATSP	0.061758
OVRP	0.061512
VRPB	0.061487
VRPL	0.061102
VRPTW	0.061006
OVRPTW	0.060956
OVRPB	0.061135
OVRPL	0.060924
VRPBL	0.060681
VRPBTW	0.061029
VRPLTW	0.060763
OVRPBL	0.061138
OVRPBTW	0.061167
OVRPLTW	0.061032
VRPBLTW	0.060978
OVRPBLTW	0.060994

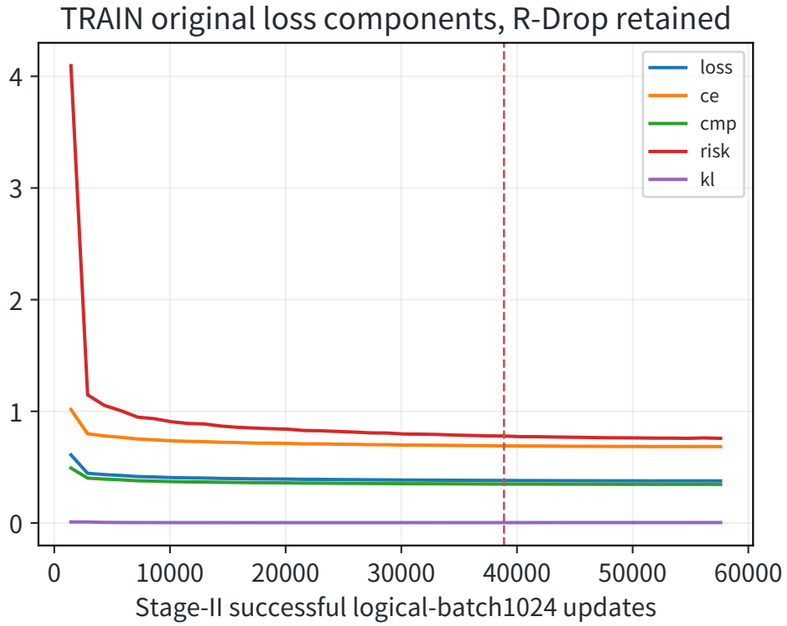
Original1842953-parameter student only; no online donor, teacher feature I/O or solver decoder.

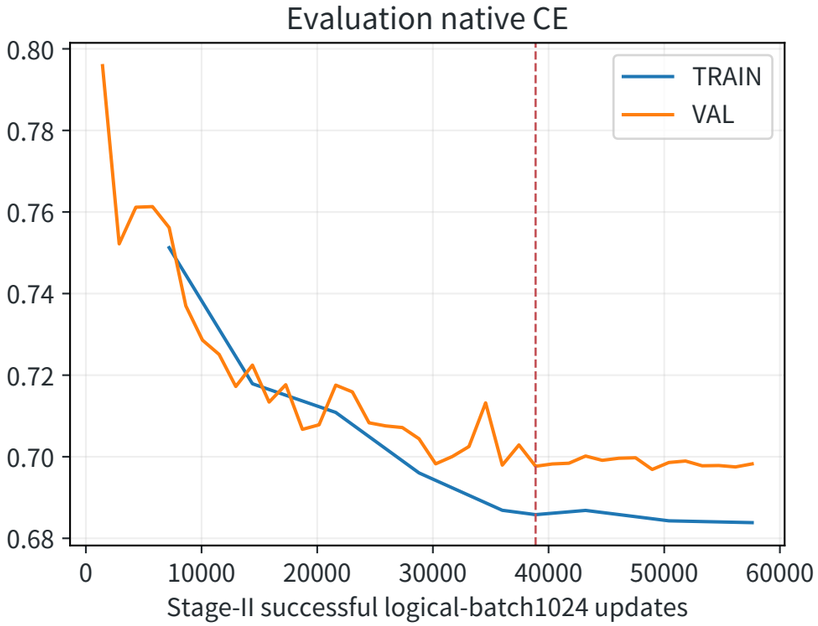
Excludes dataset I/O, host transfer and Stage-I pretraining; not single-instance/E2E latency.

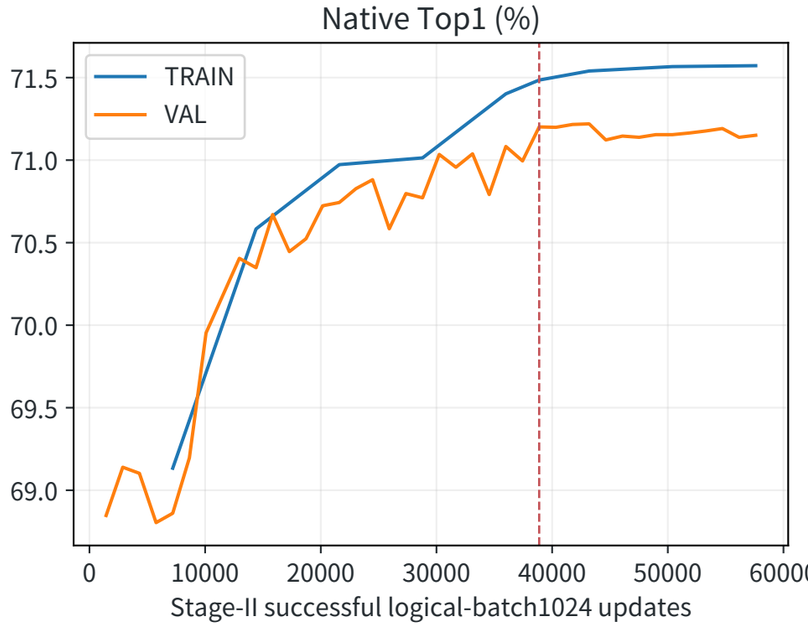
No latency or speedup claim is inferred from R94 cached-selector numbers.

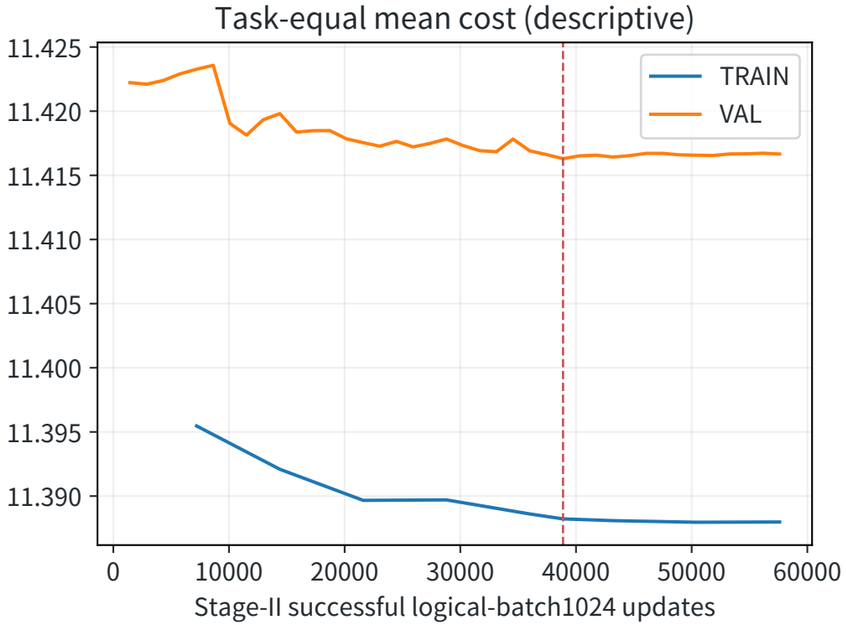
Hardware is disclosed, not bitwise matched; no model reselection based on timing.

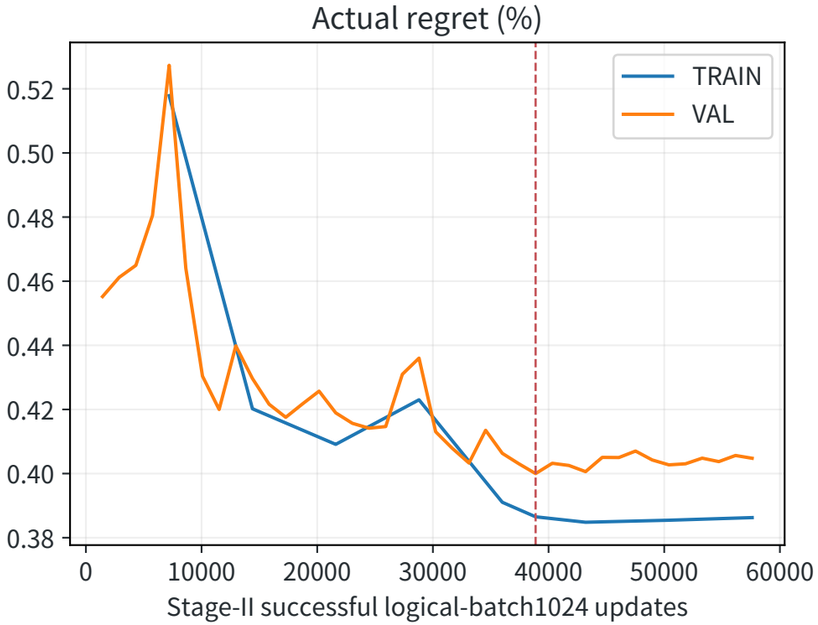
R101/B | Stage II: original R77, no hint loss or teacher inputs

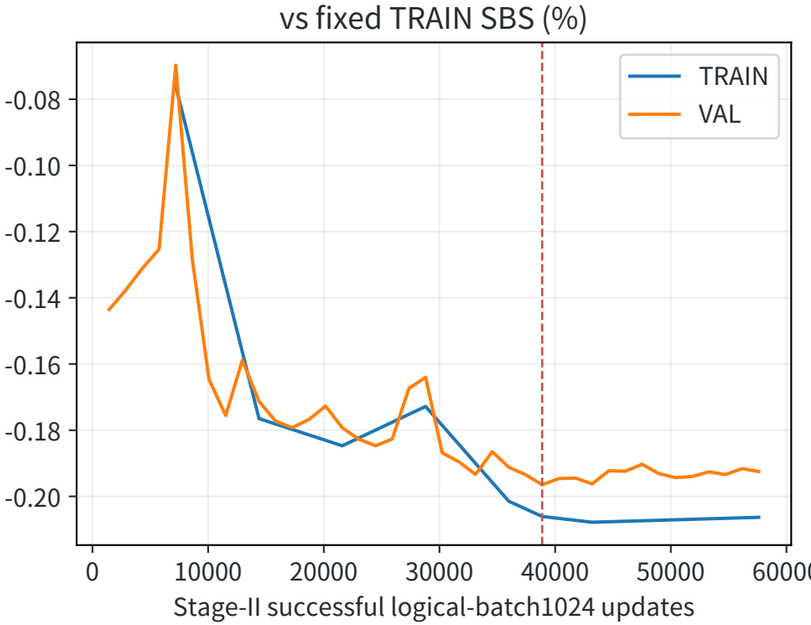






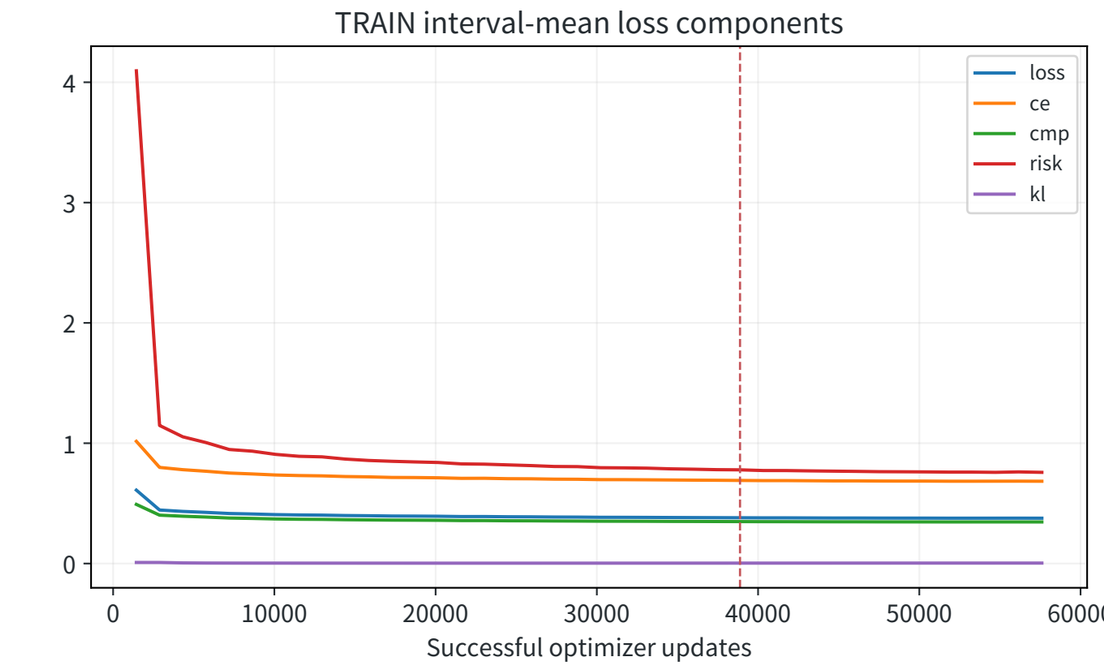


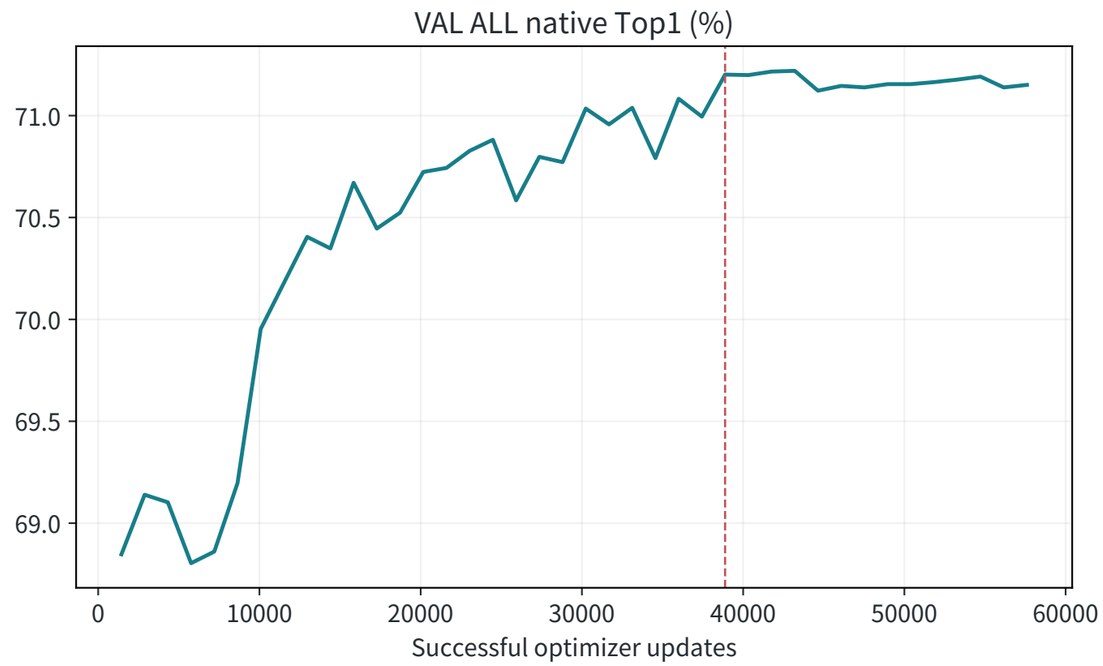


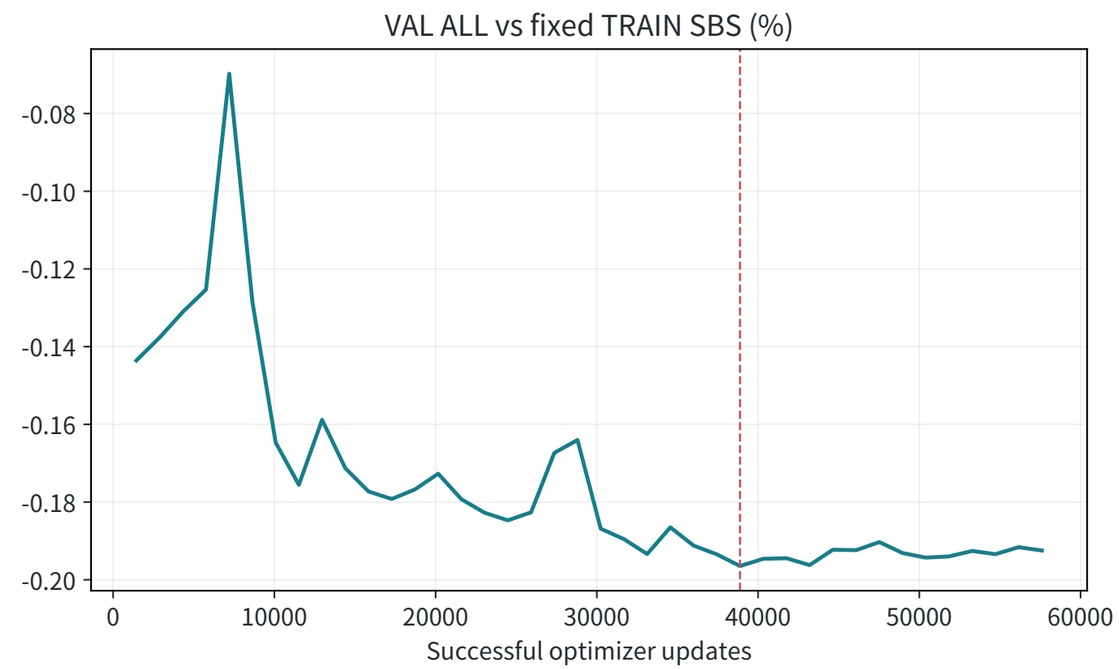


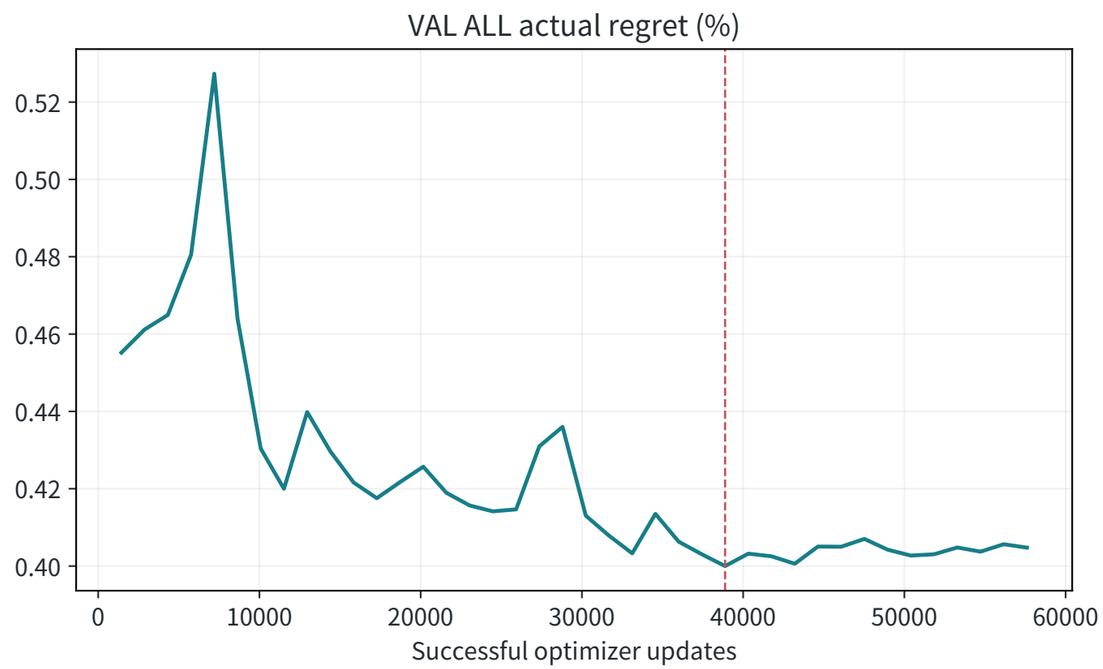
R101/B | Saved Training History

Successful-update axis. Training losses and full VAL traces are descriptive only; the checkpoint is already locked.



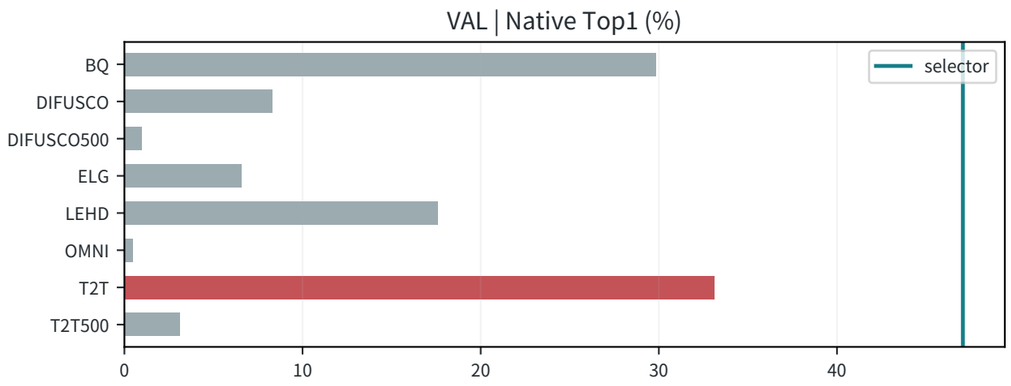


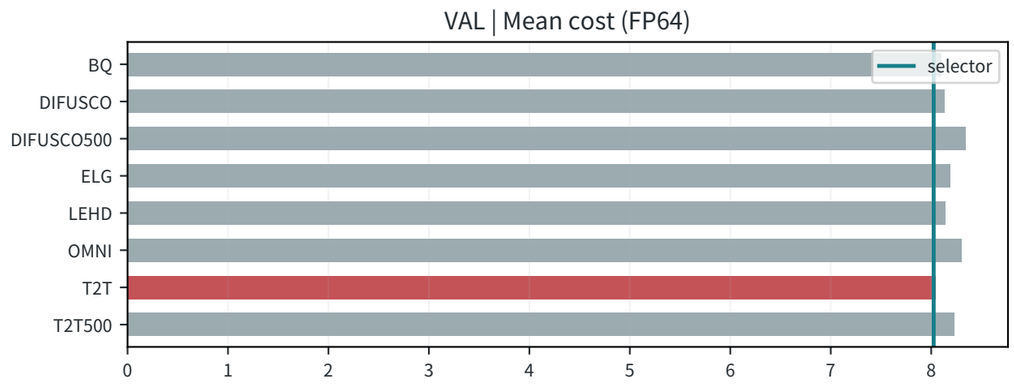


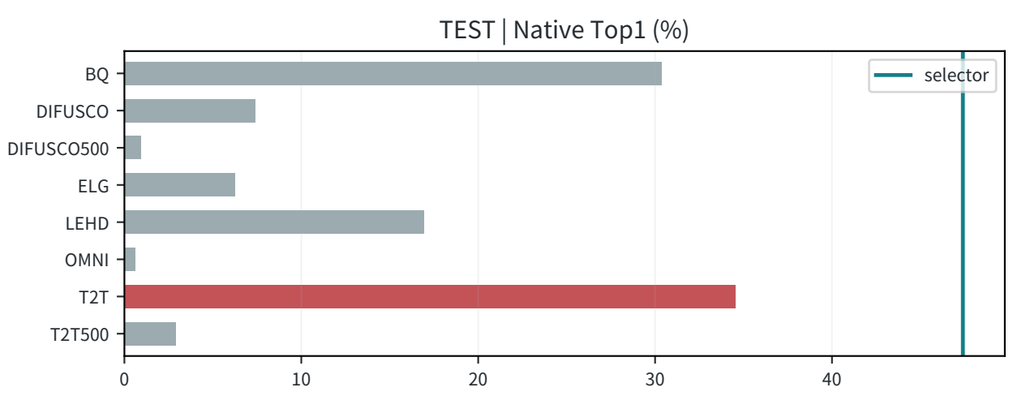


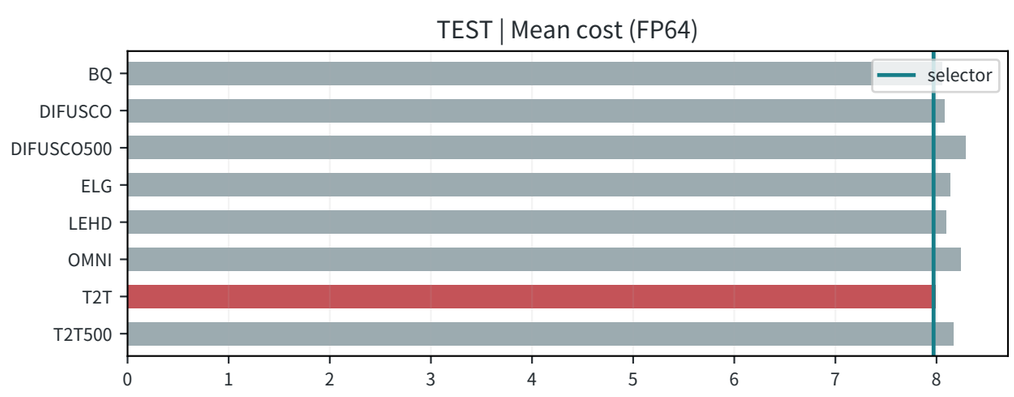
R101/B | TSP: Selector vs Single Methods

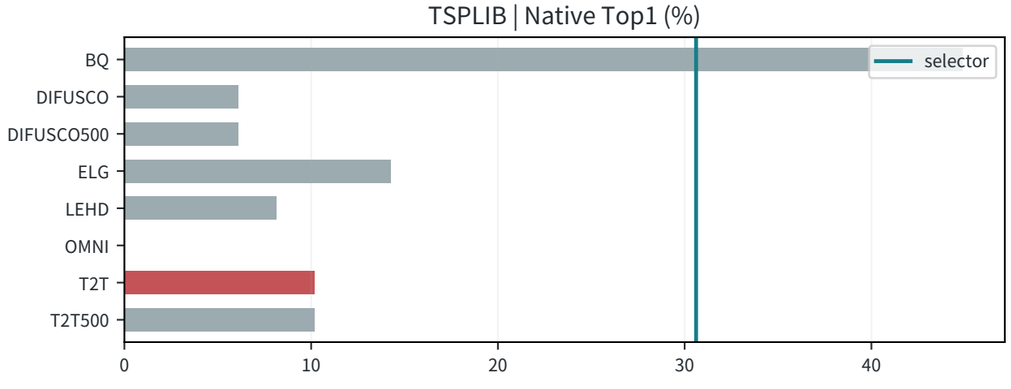
Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.

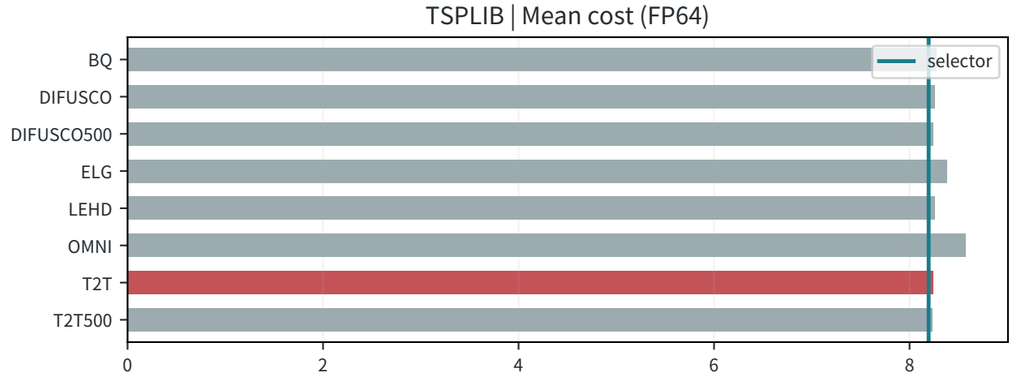






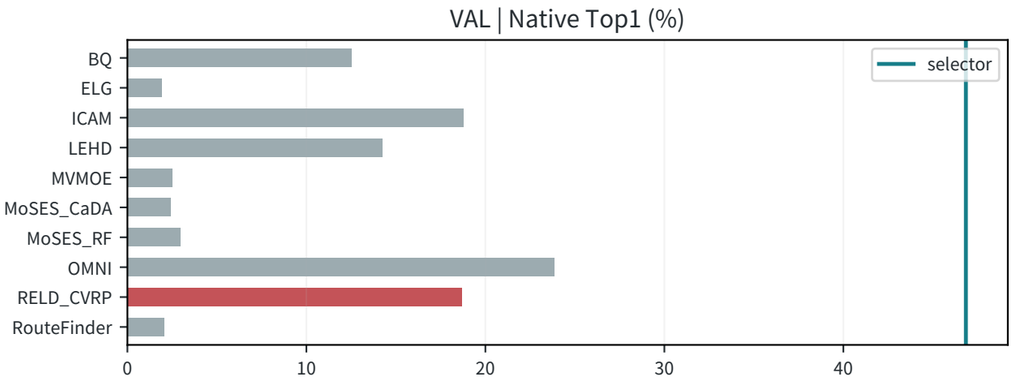


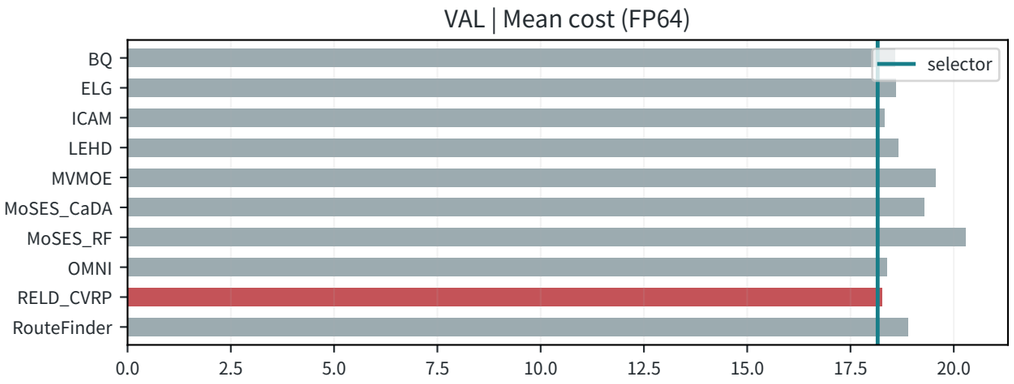


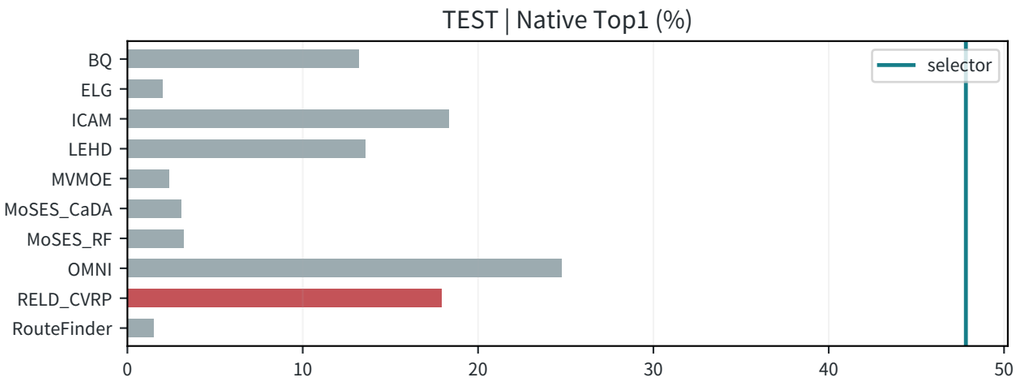


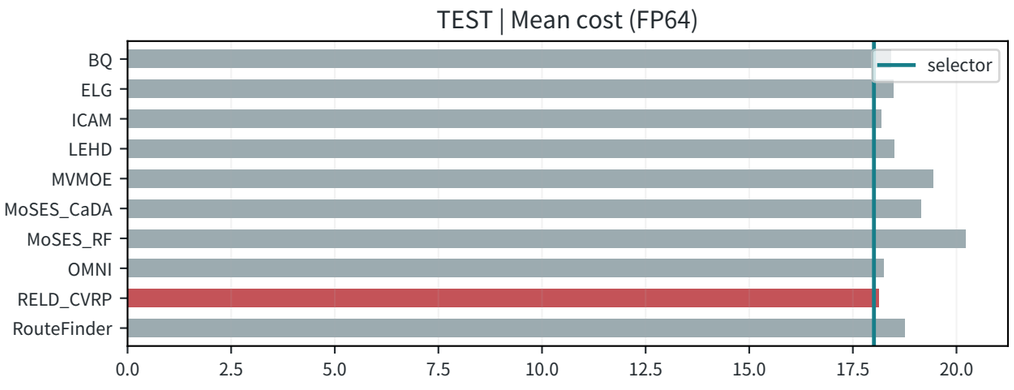
R101/B | CVRP: Selector vs Single Methods

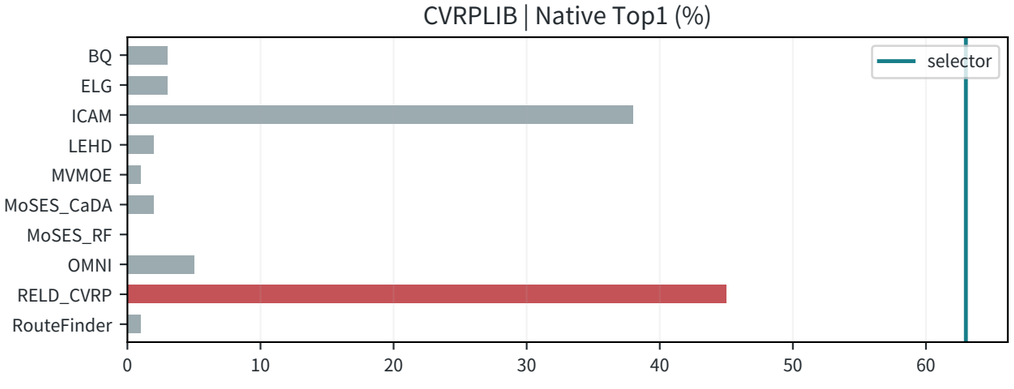
Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.

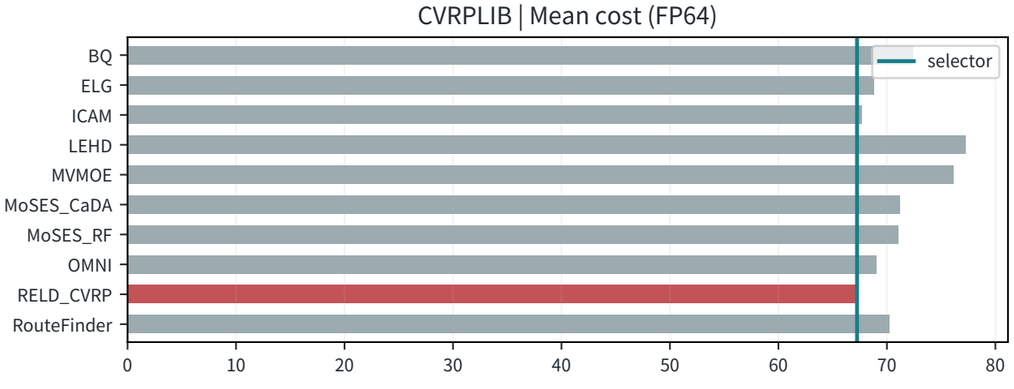






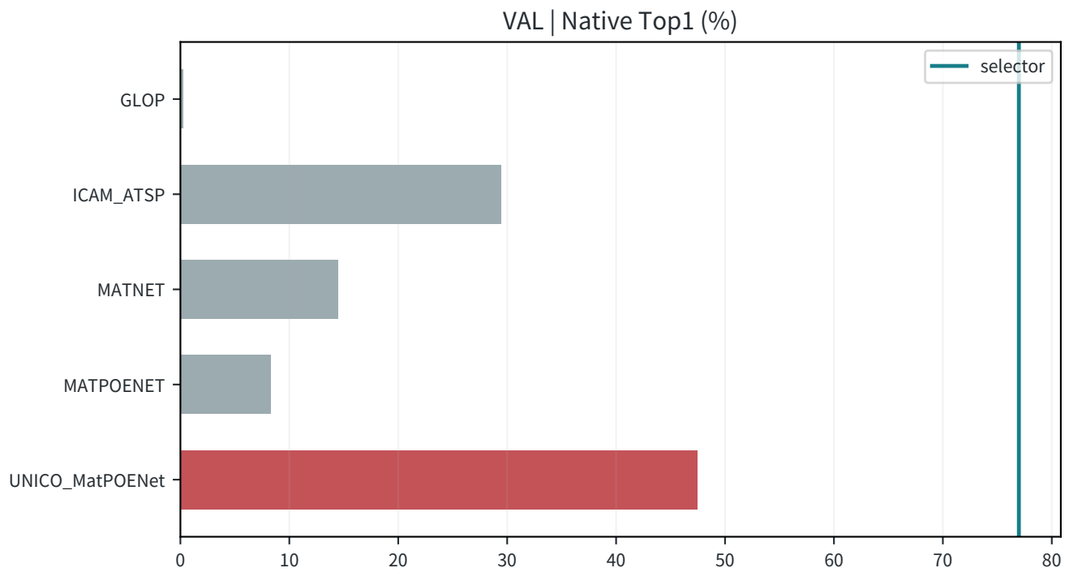


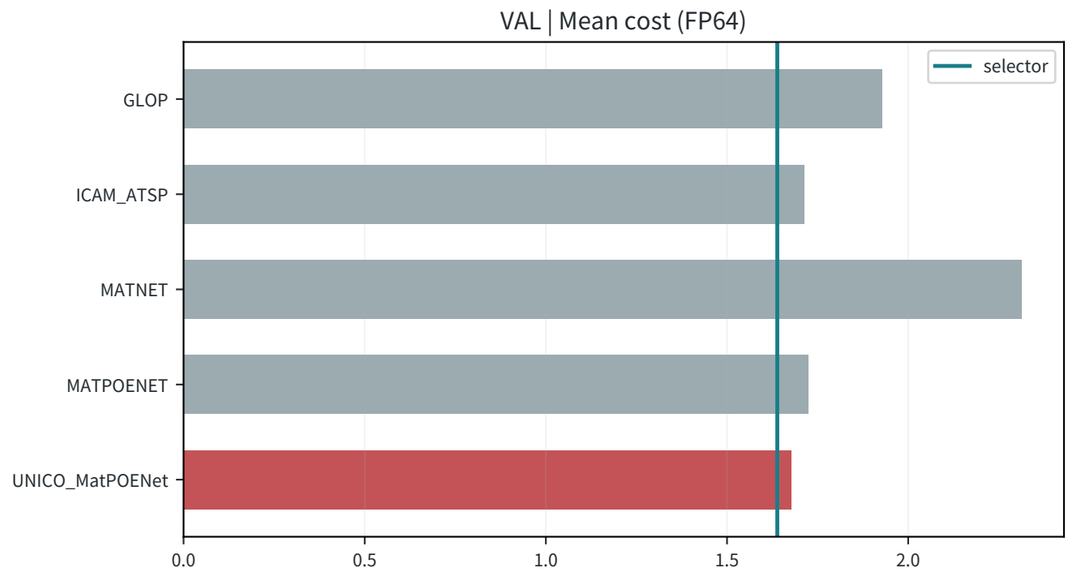


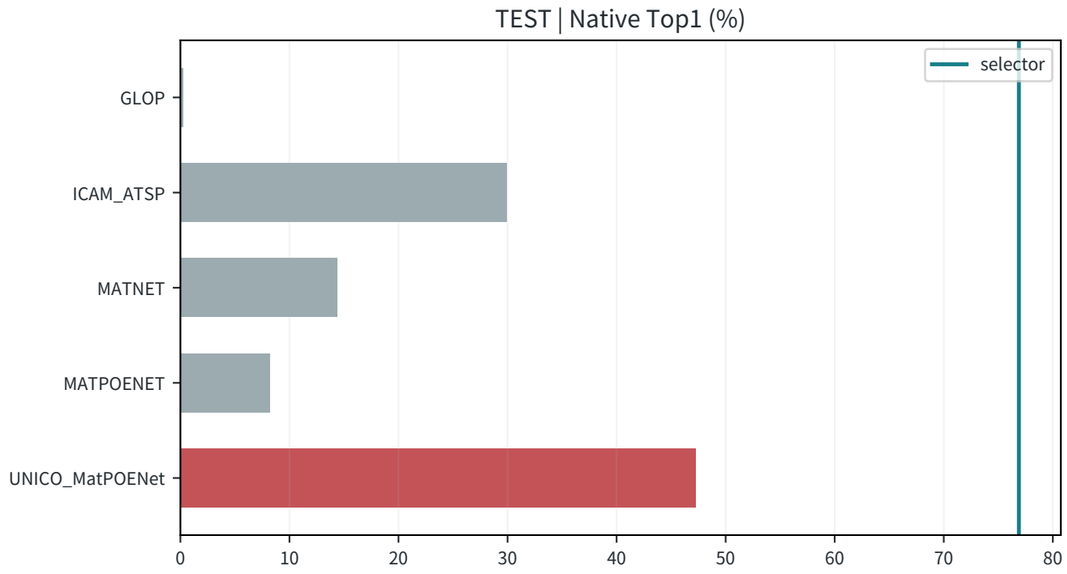


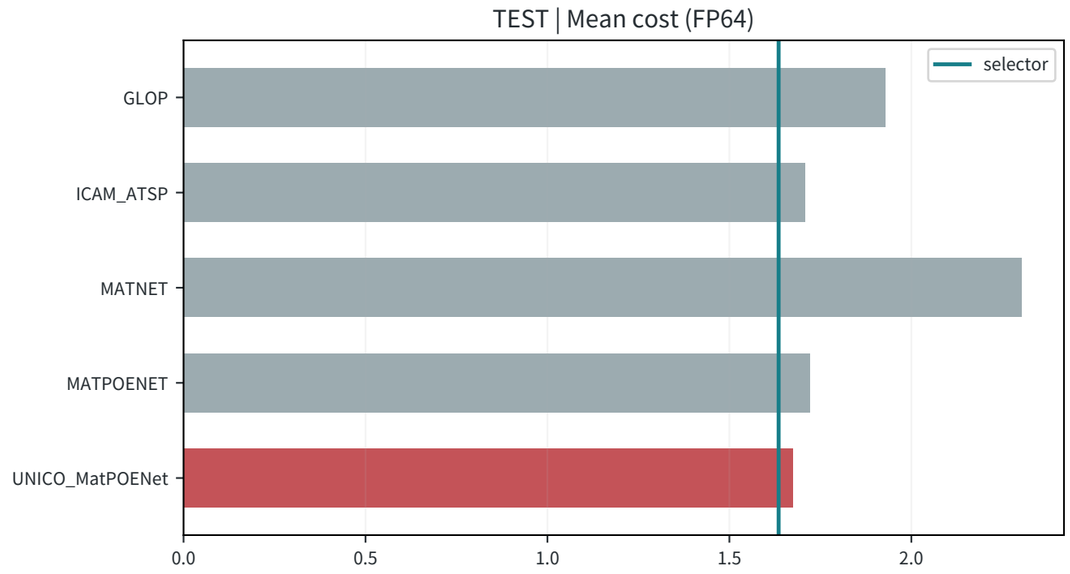
R101/B | ATSP: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



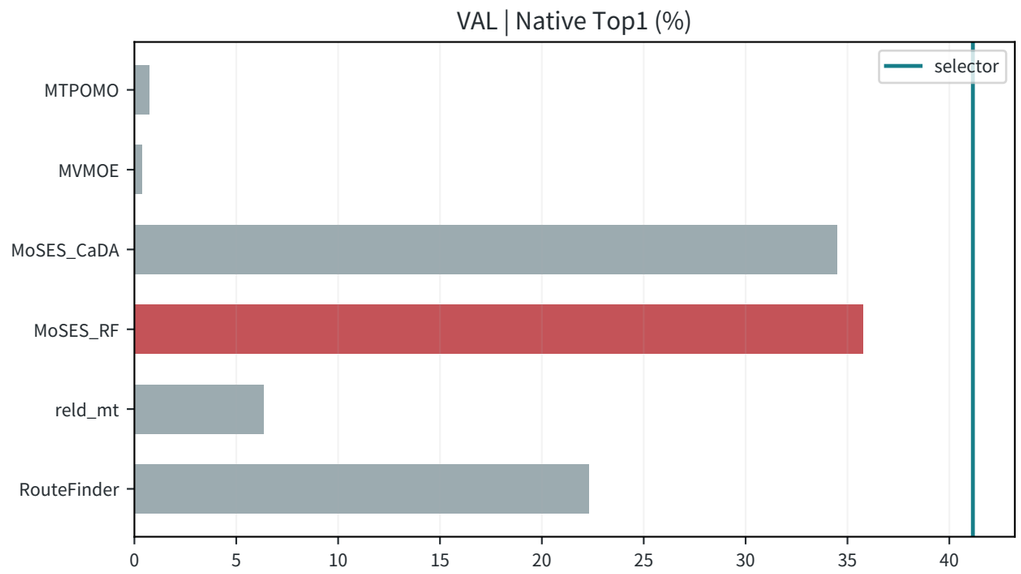


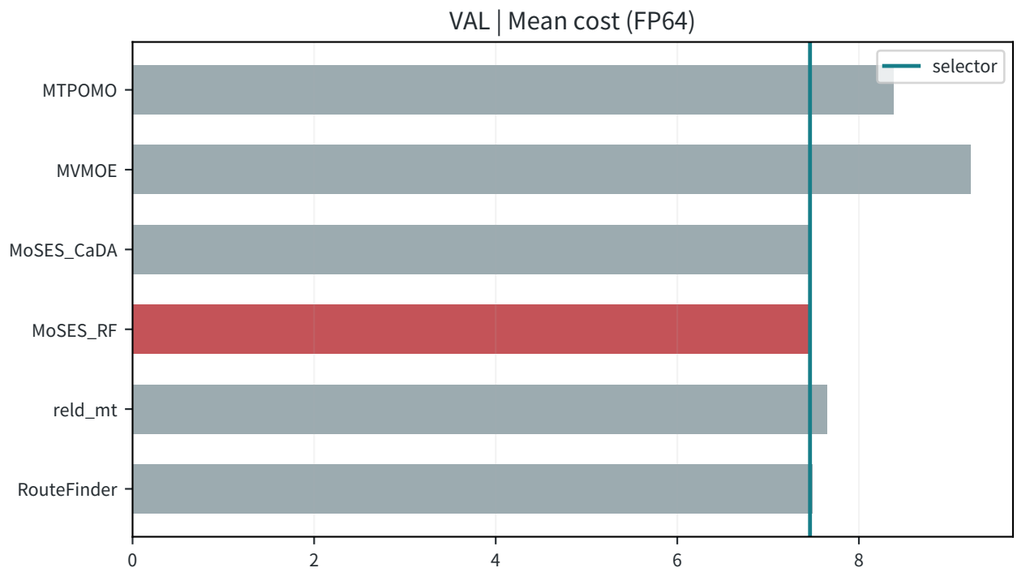


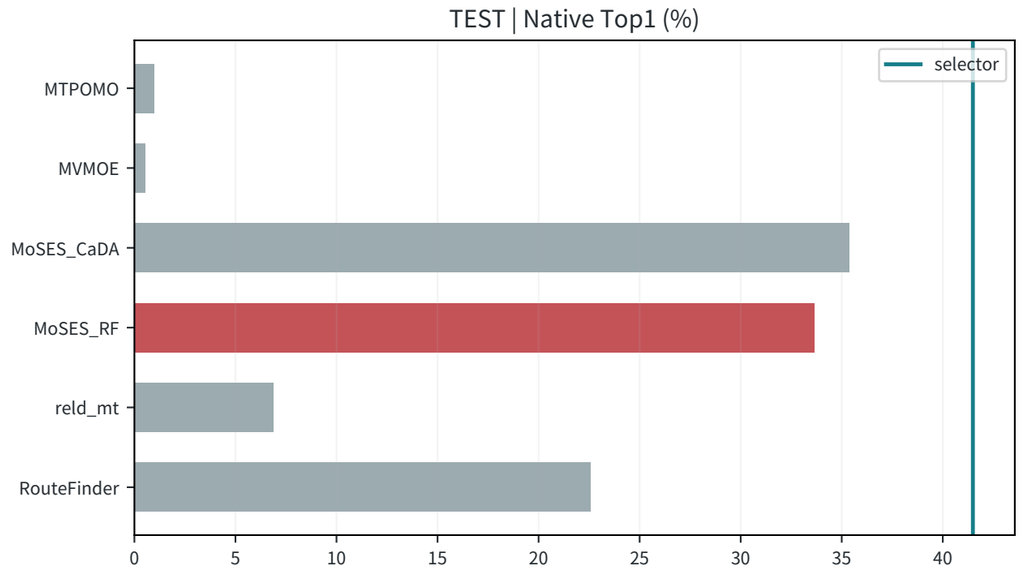


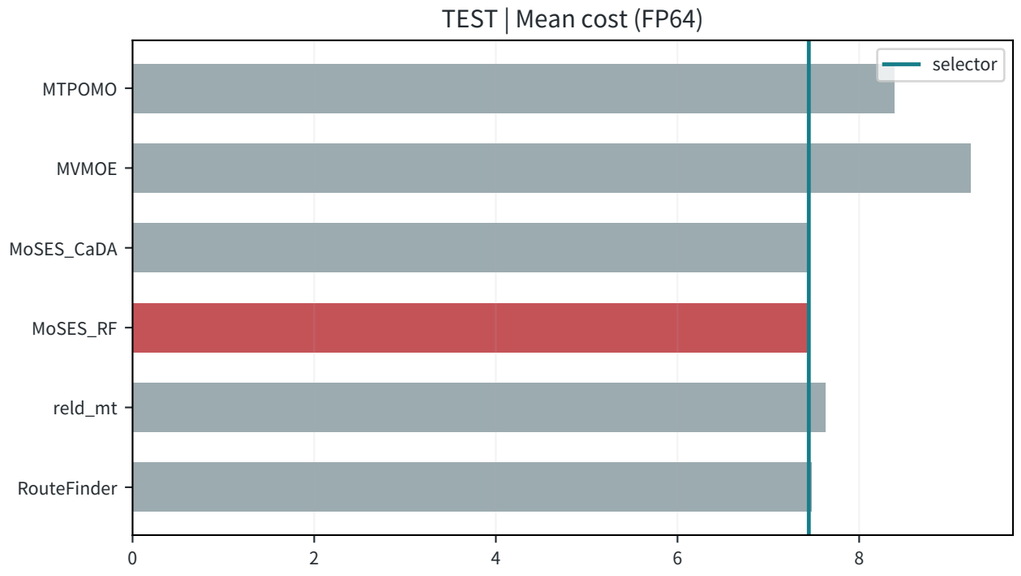
R101/B | OVRP: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



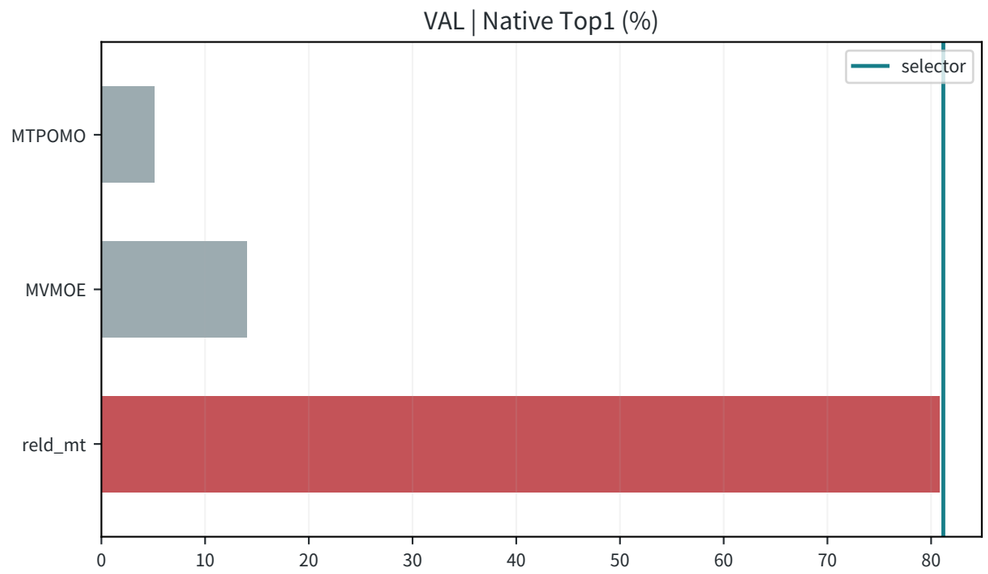


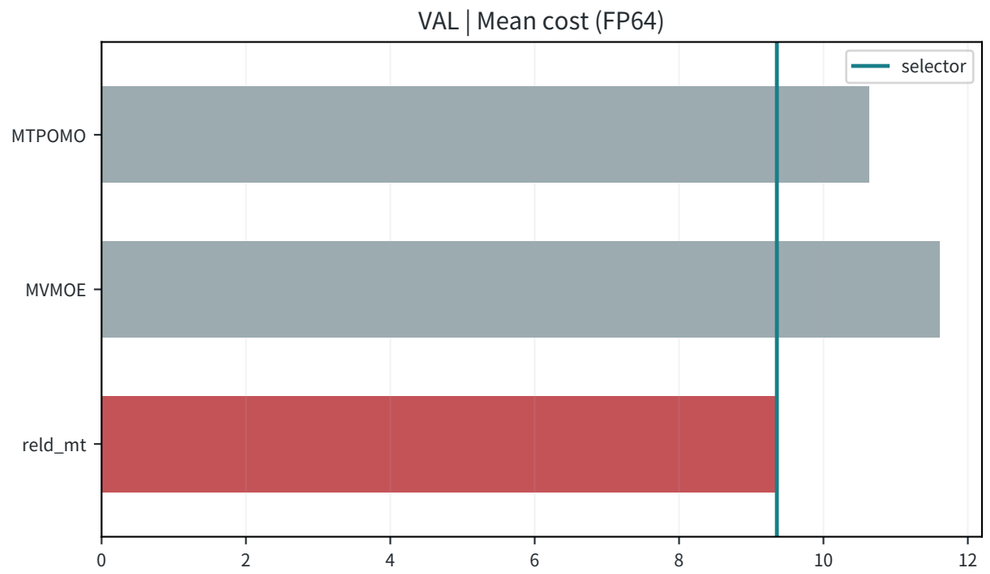


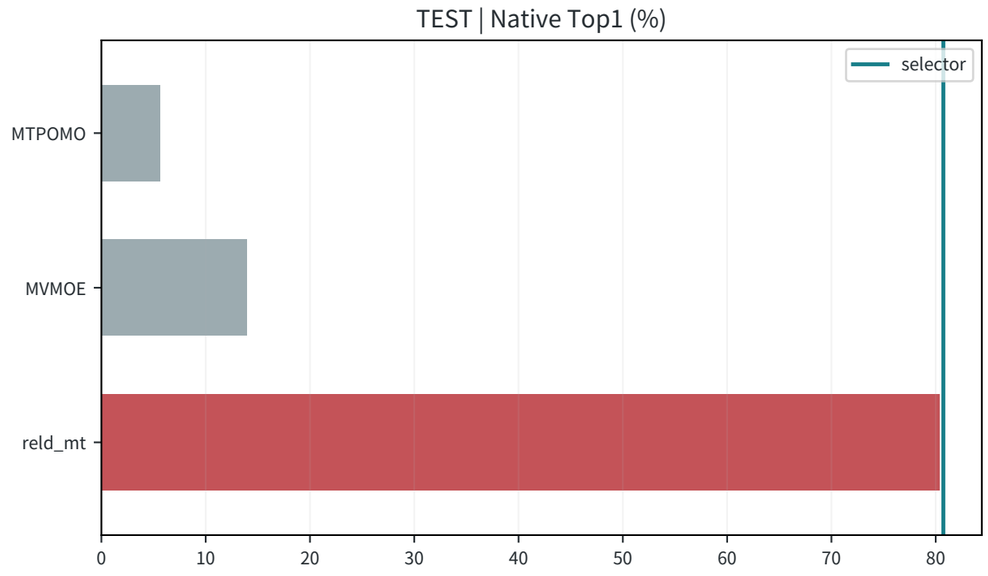


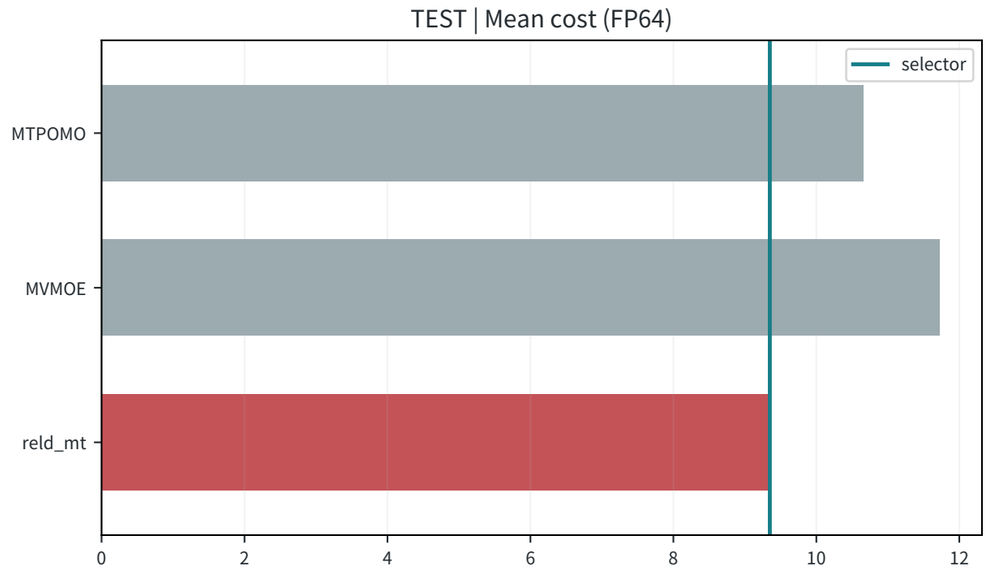
R101/B | VRPB: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



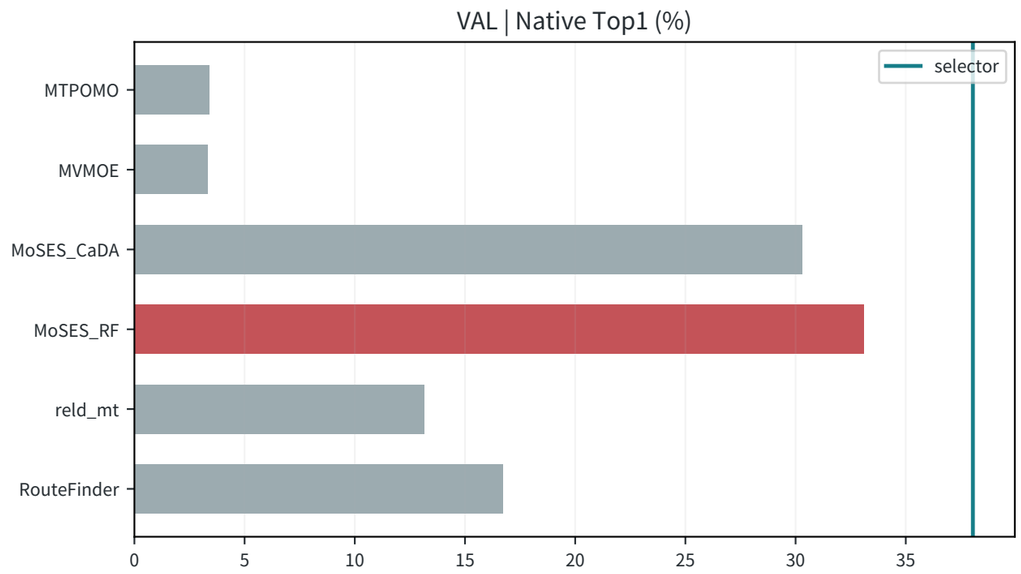


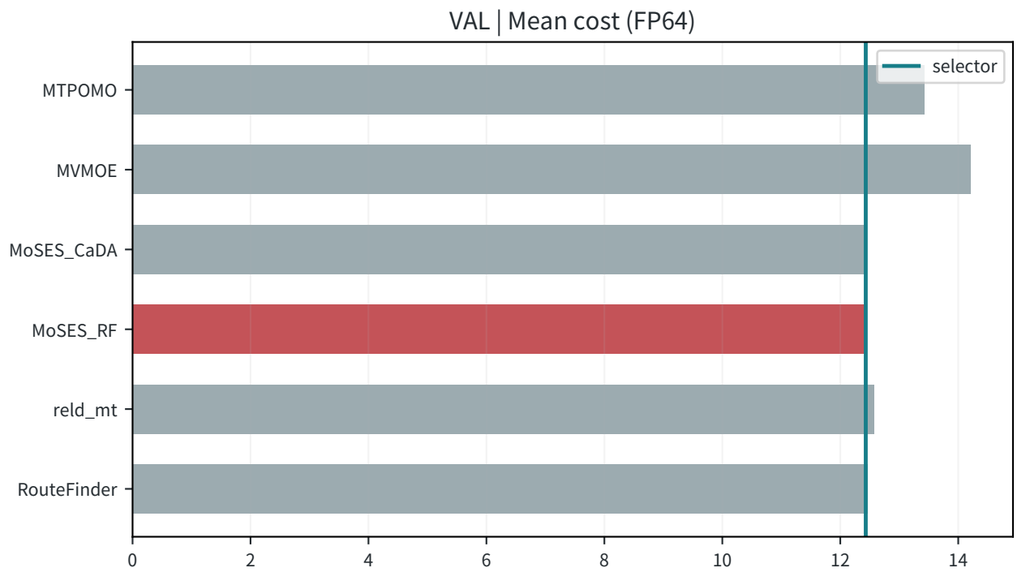


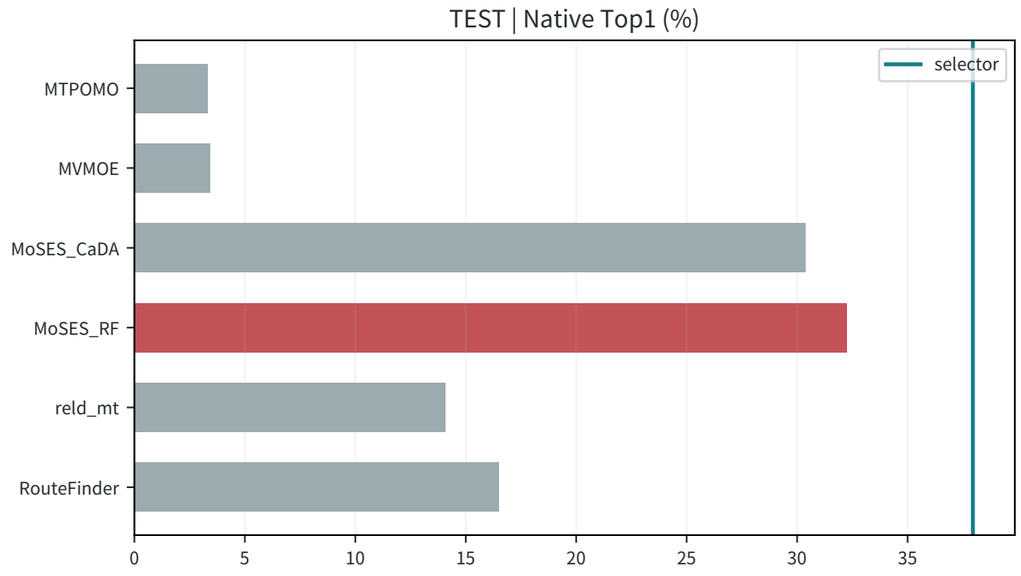


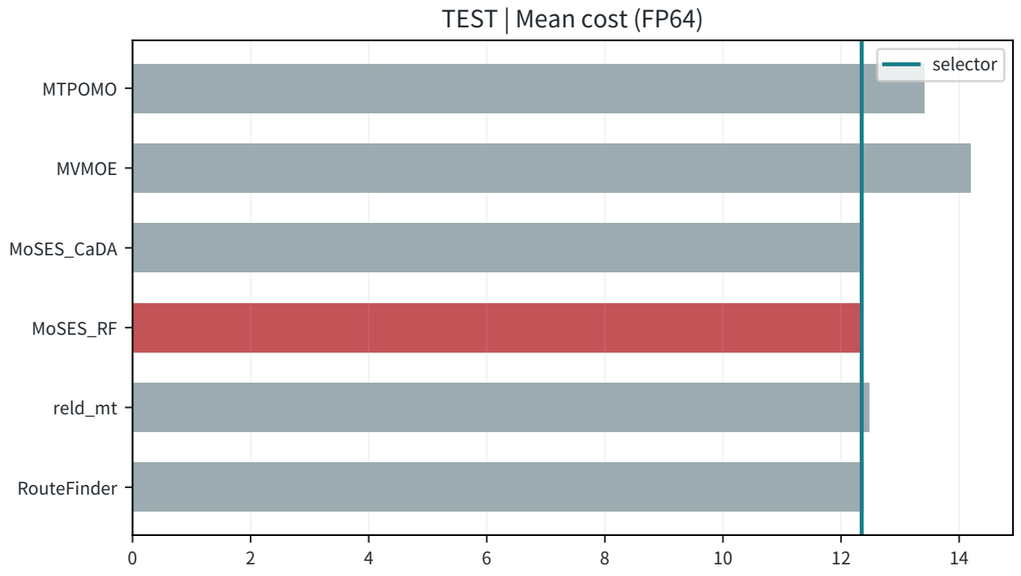
R101/B | VRPL: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



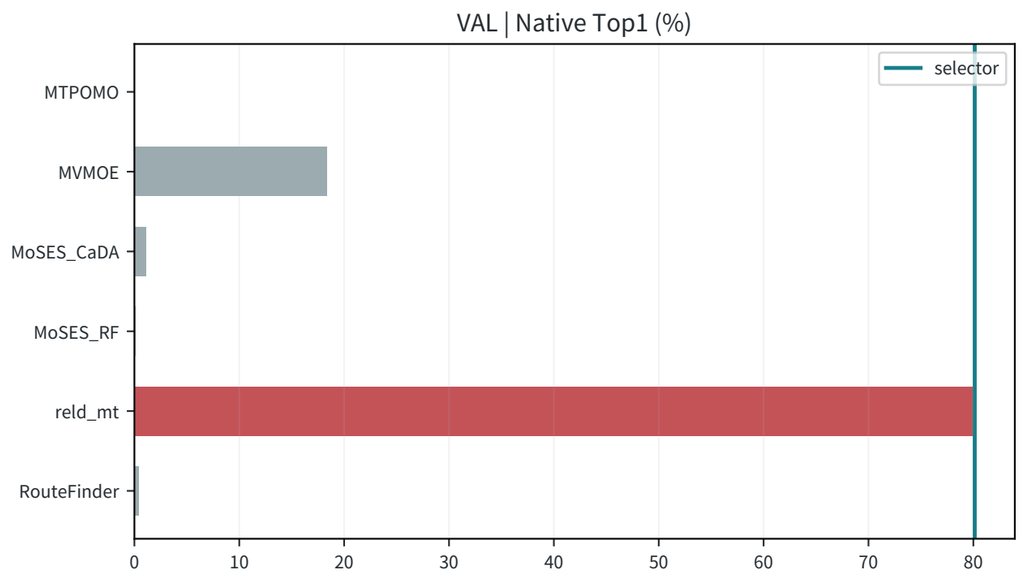


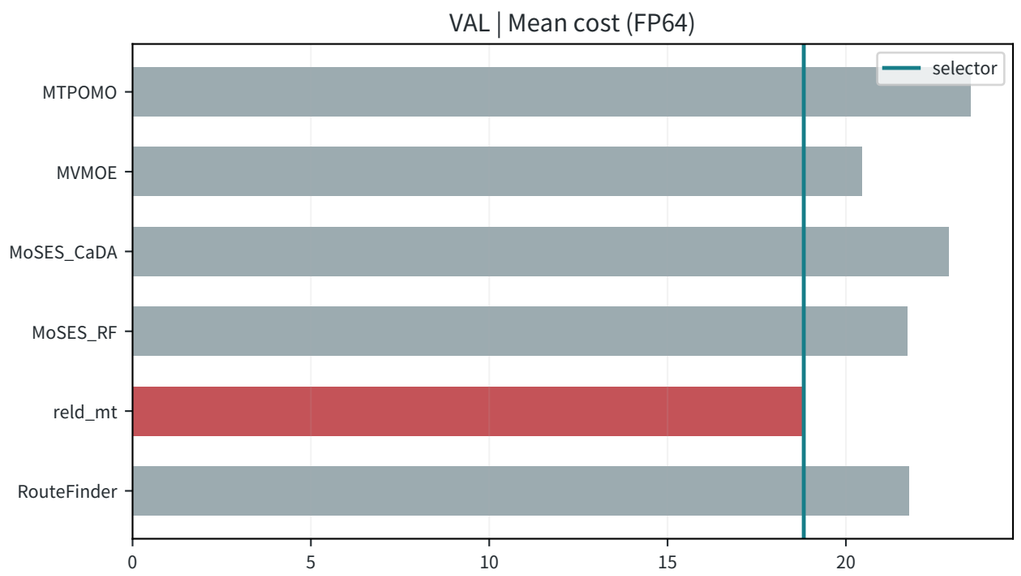


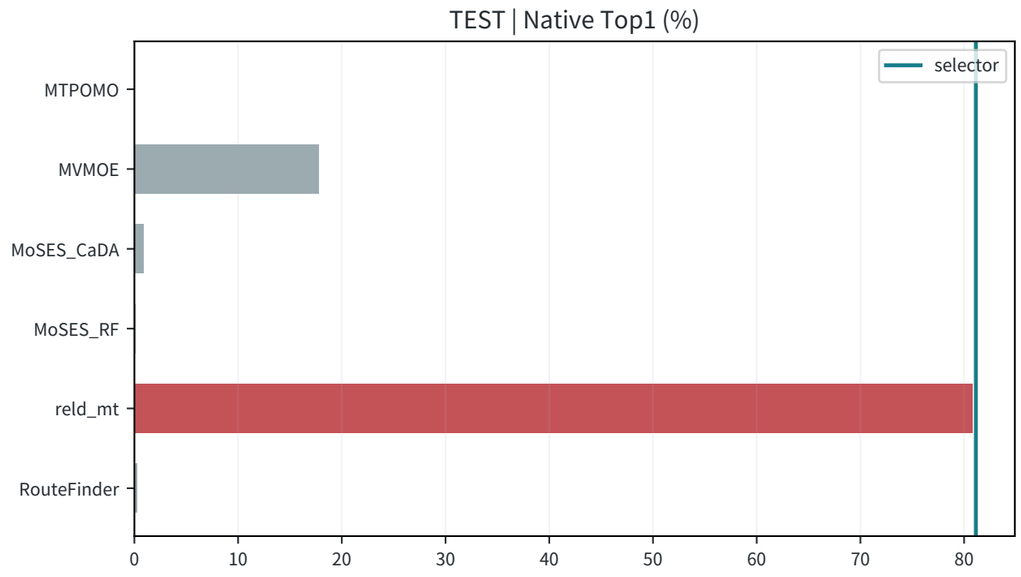


R101/B | VRPTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



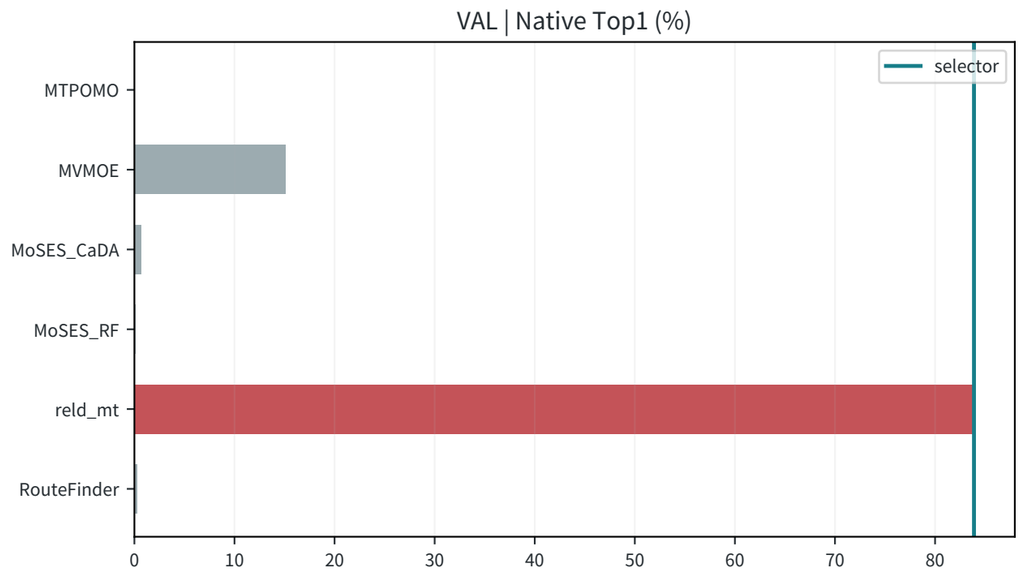


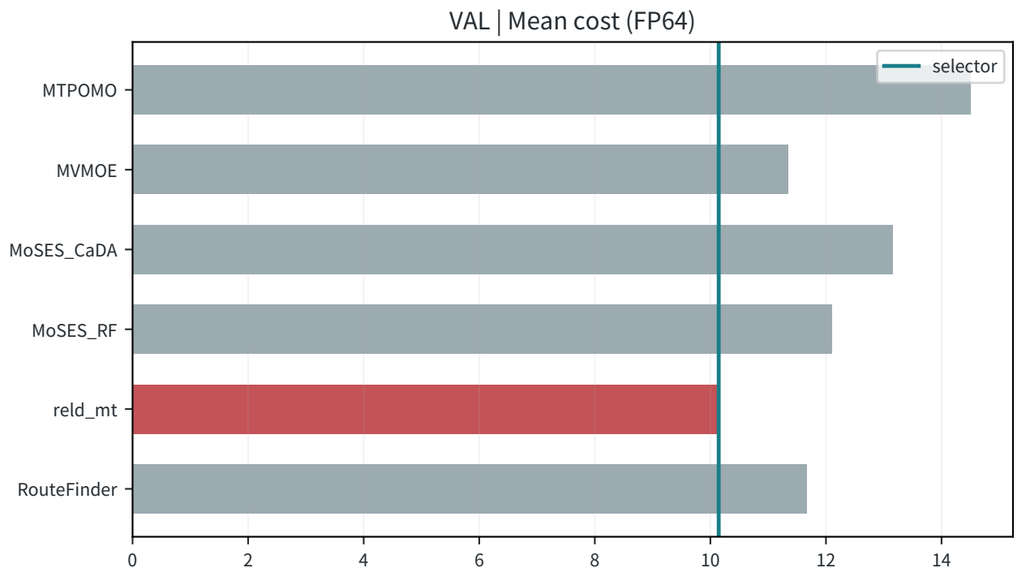


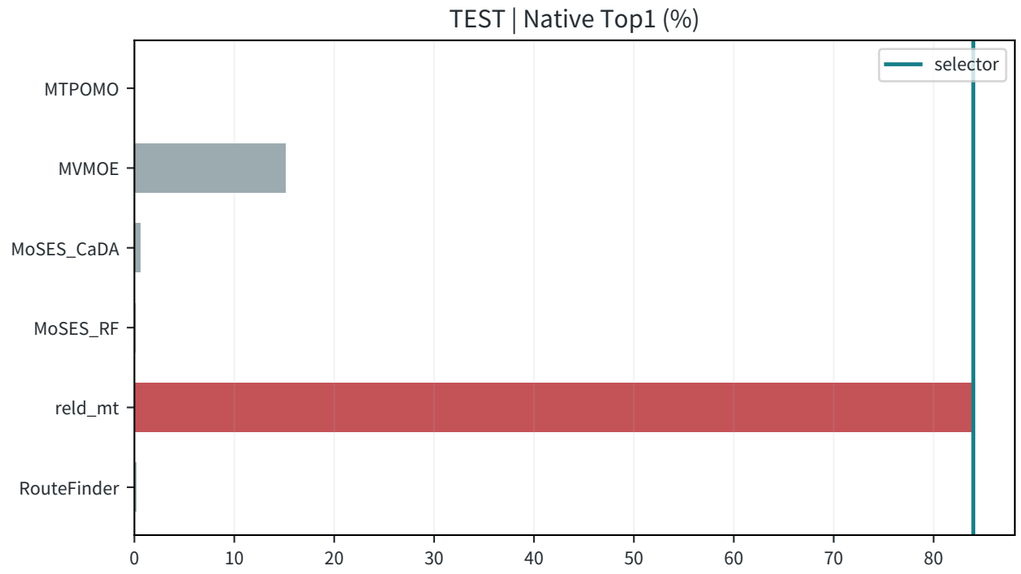


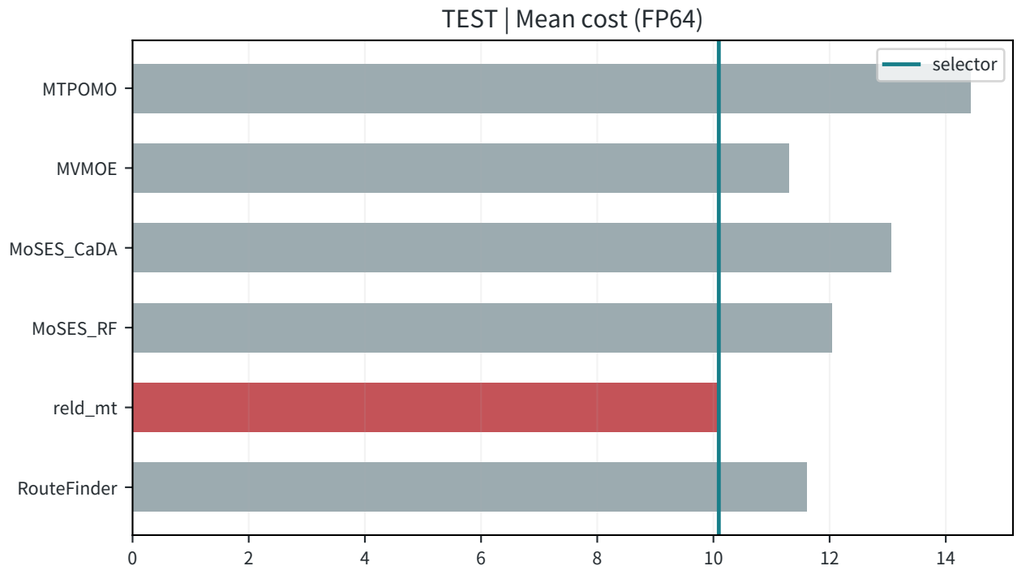
R101/B | OVRPTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



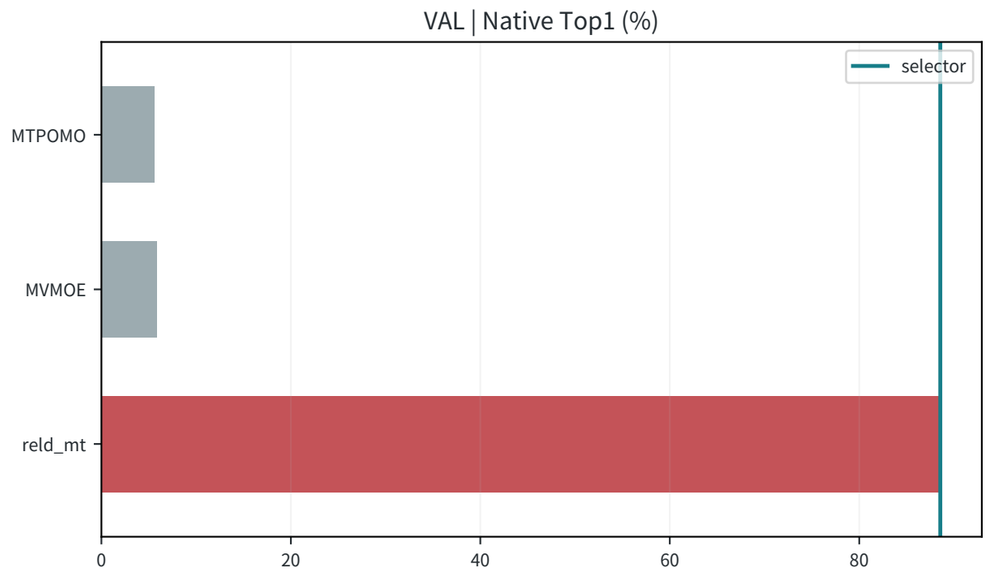


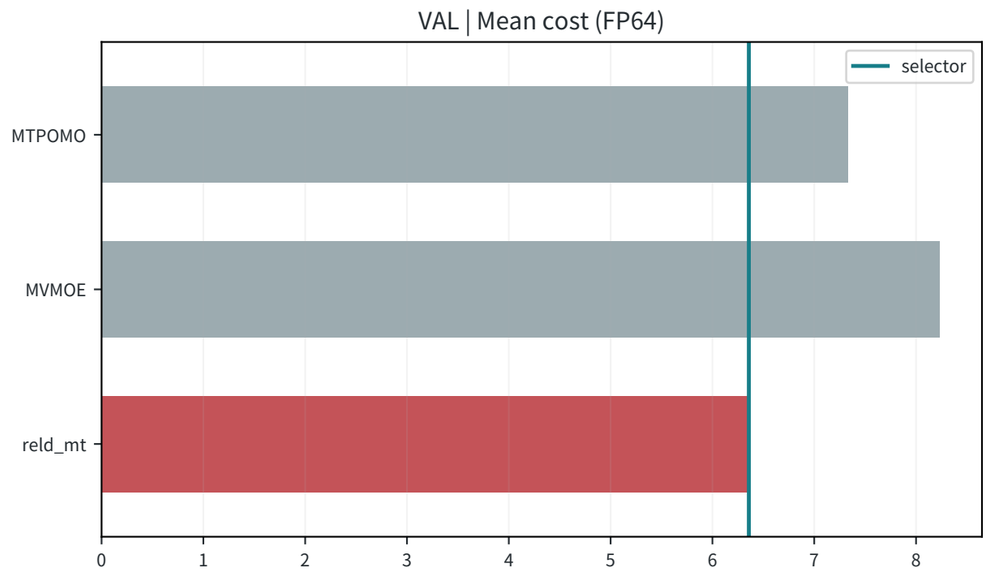


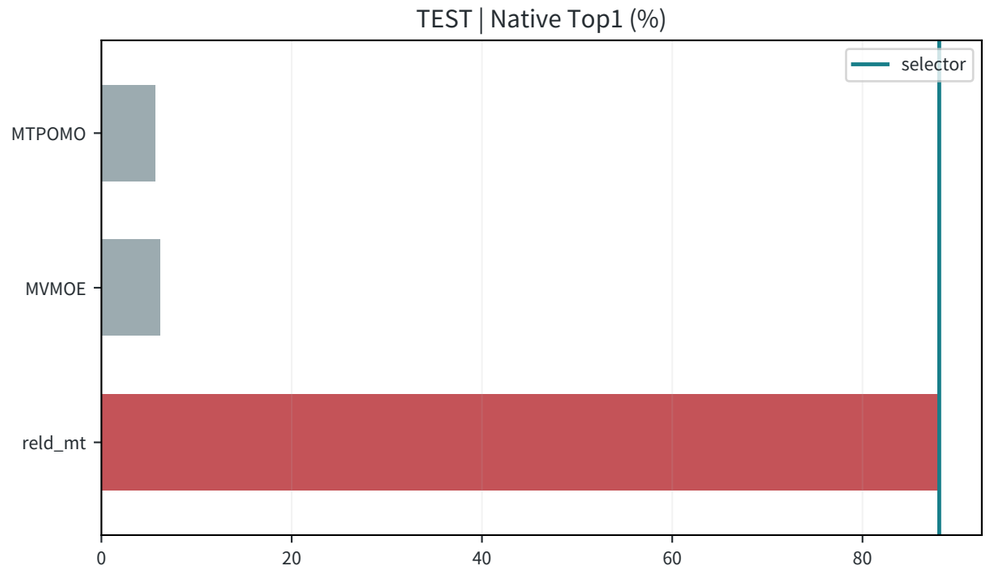


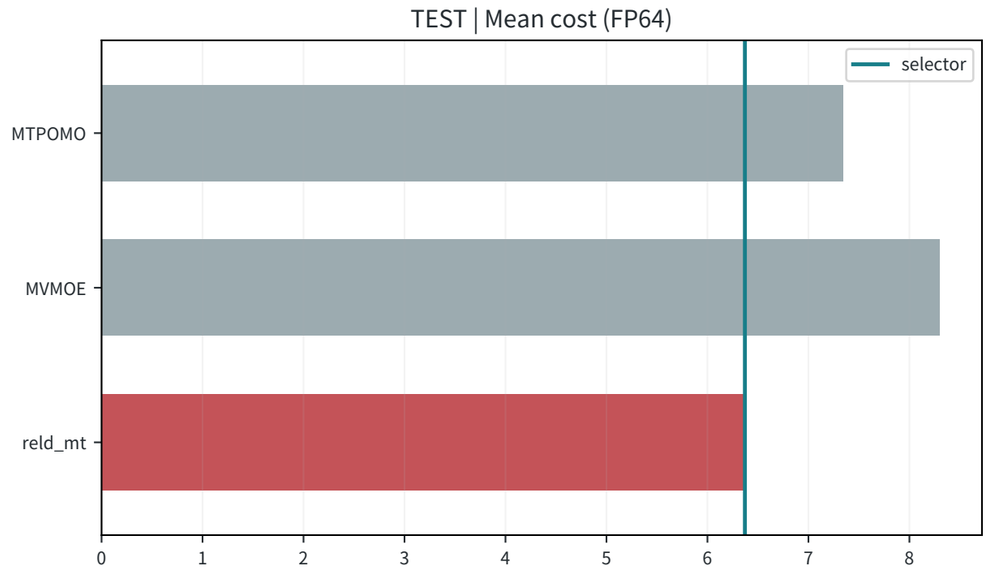
R101/B | OVRPB: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



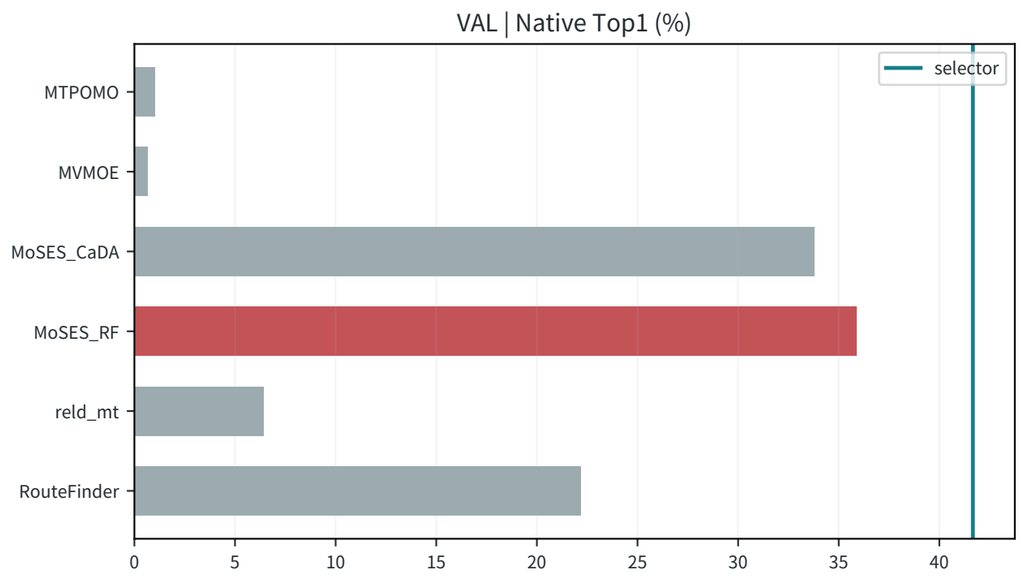


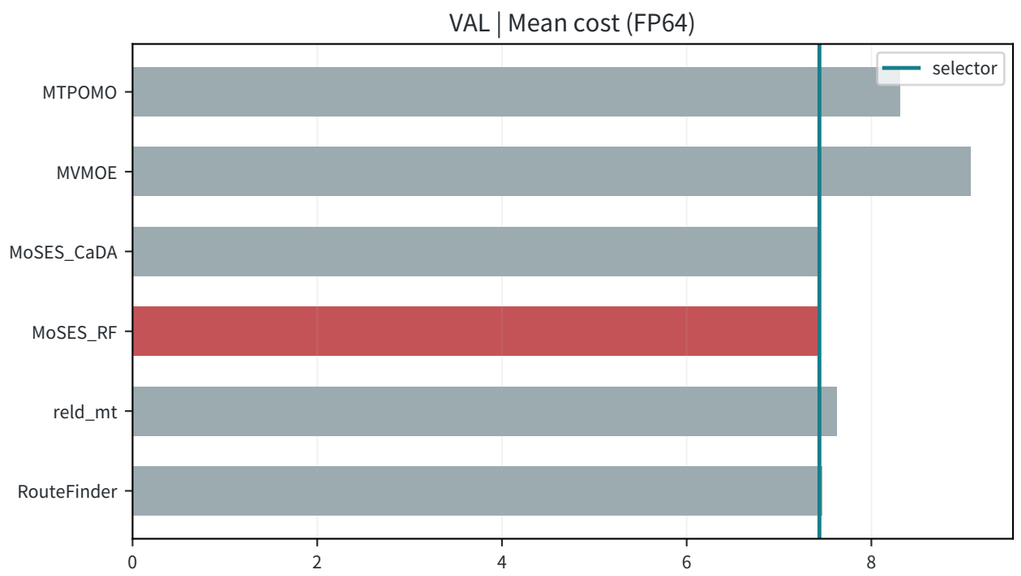


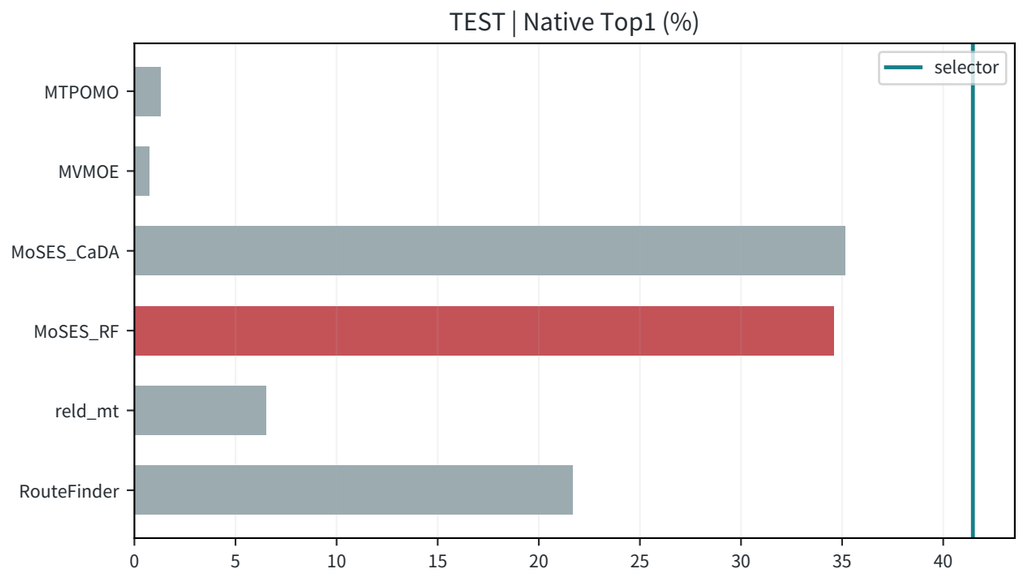


R101/B | OVRPL: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



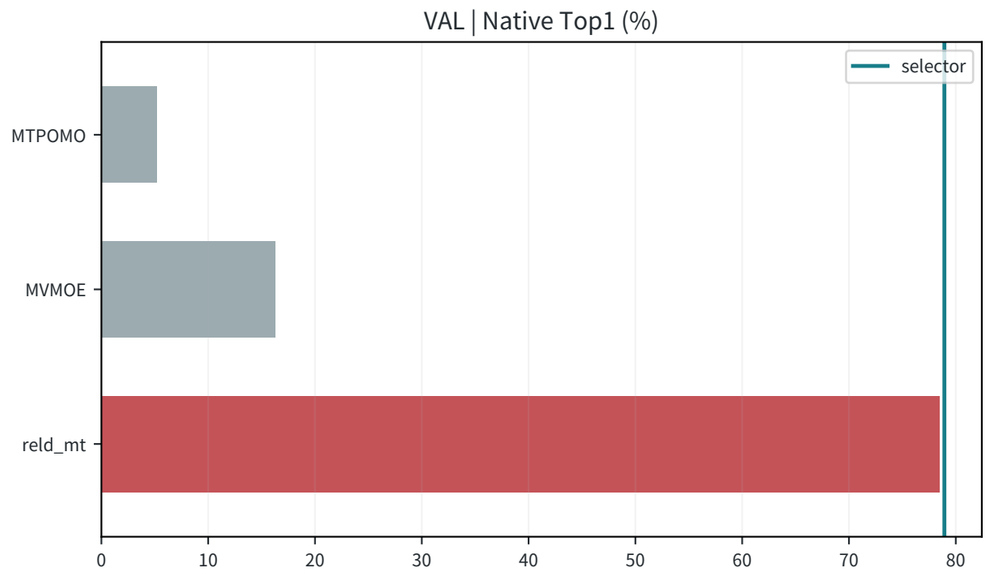


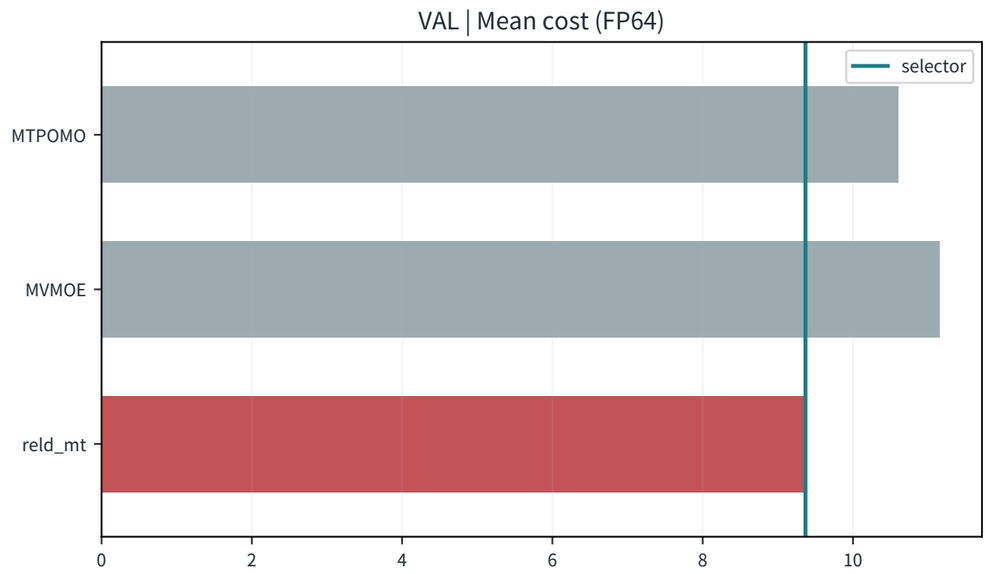


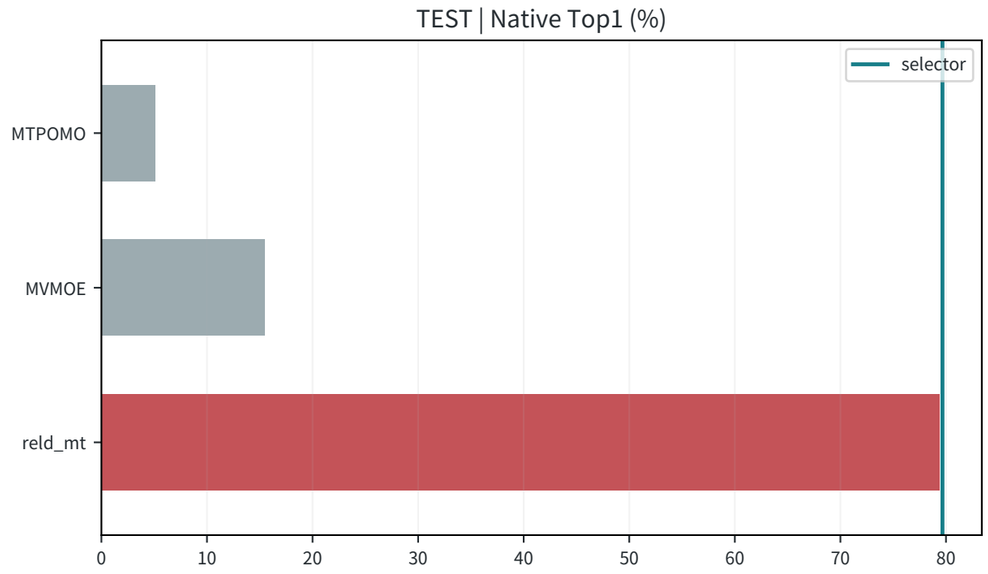


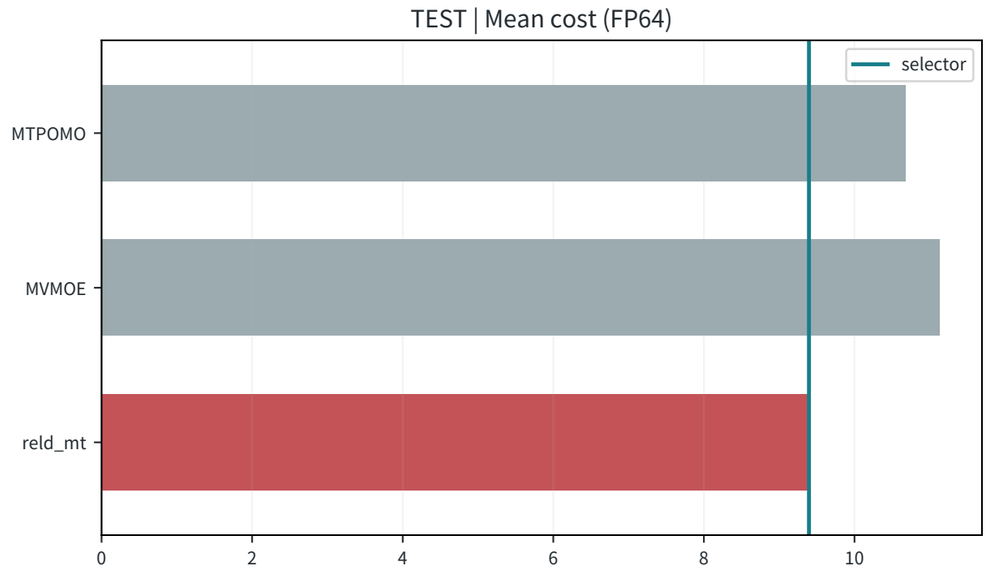
R101/B | VRPBL: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



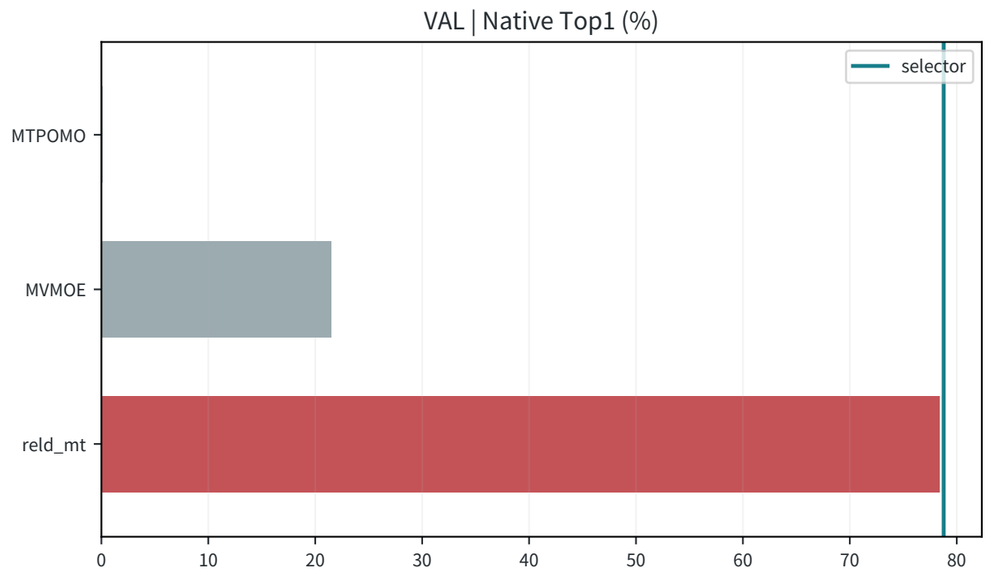


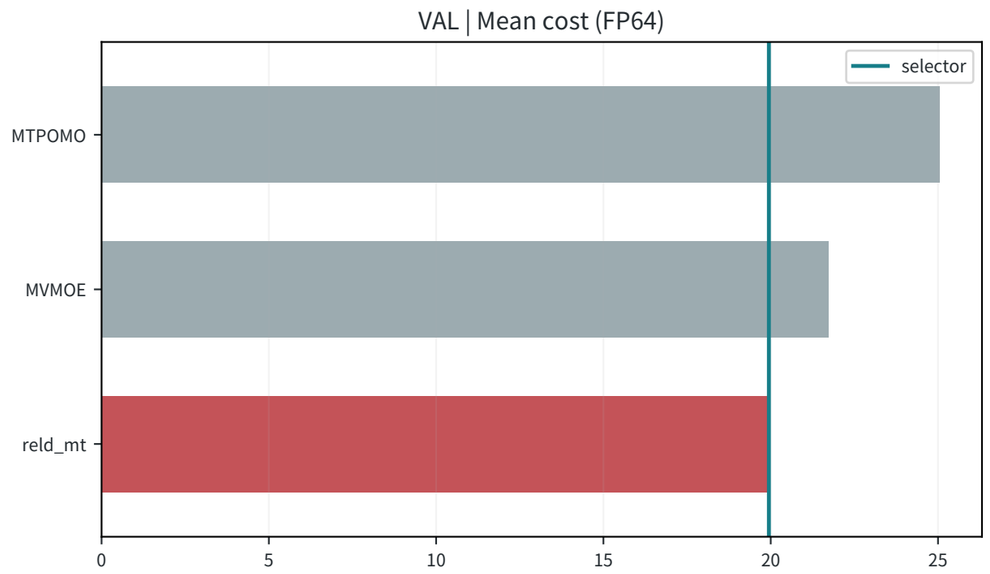


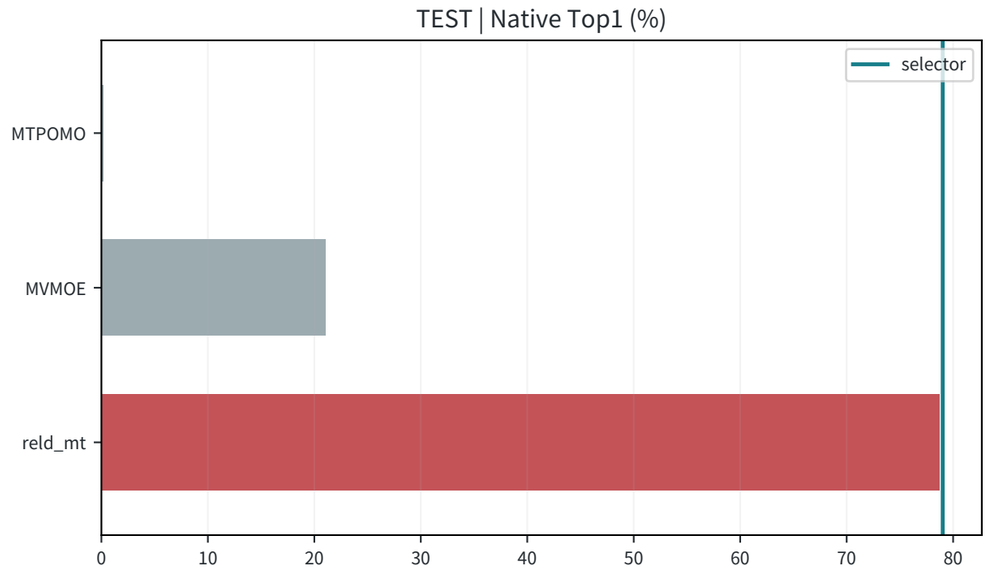


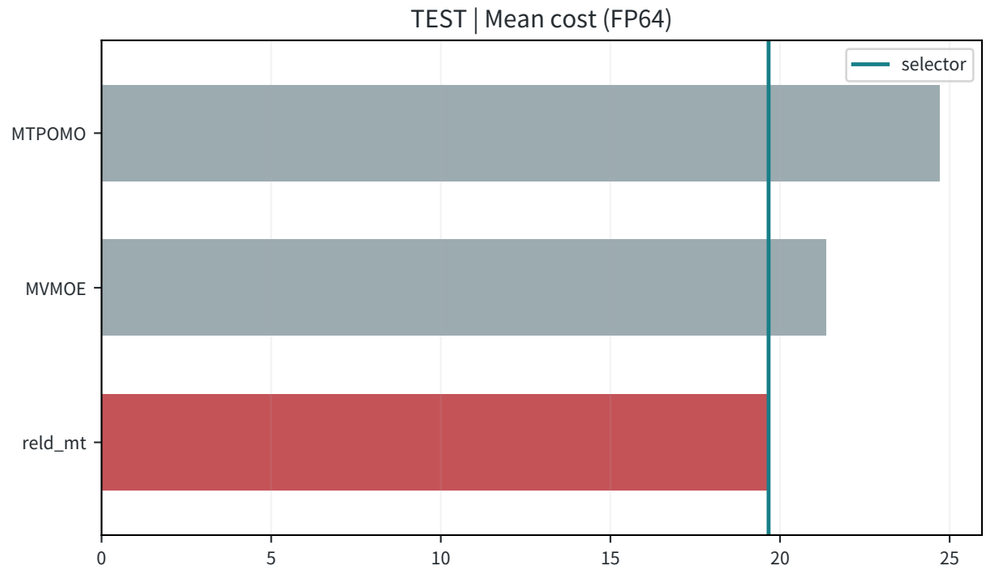
R101/B | VRPBTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



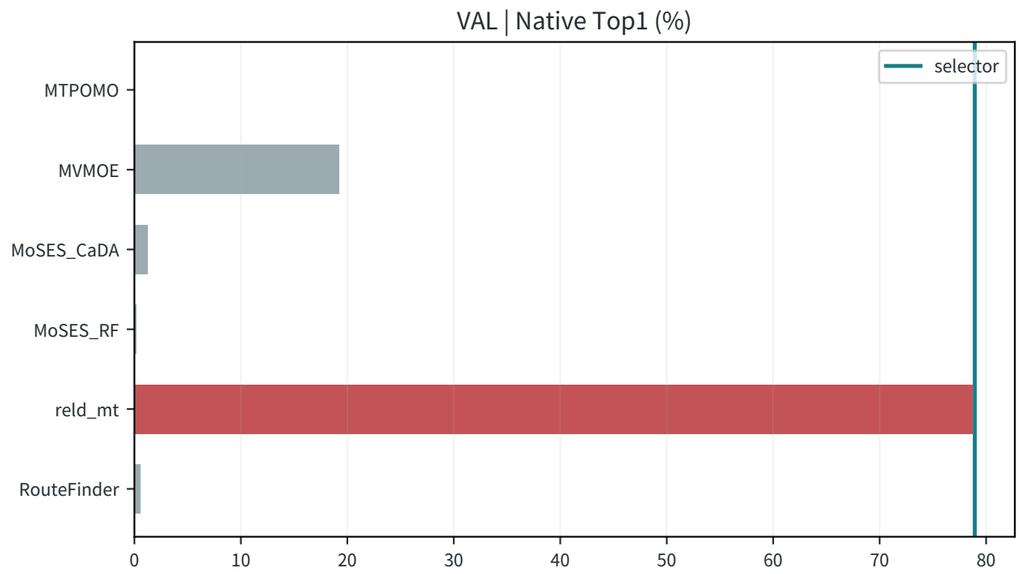


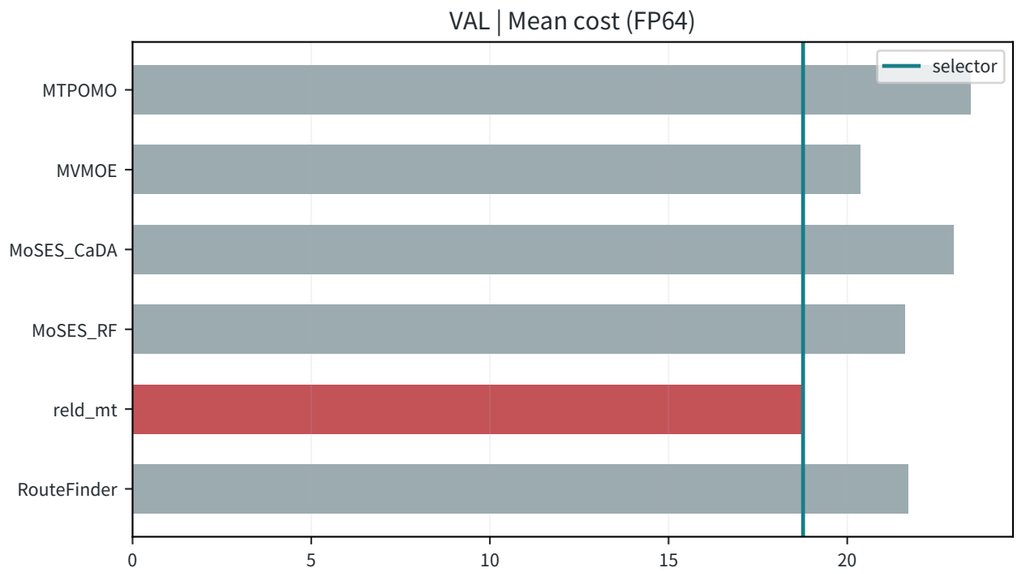


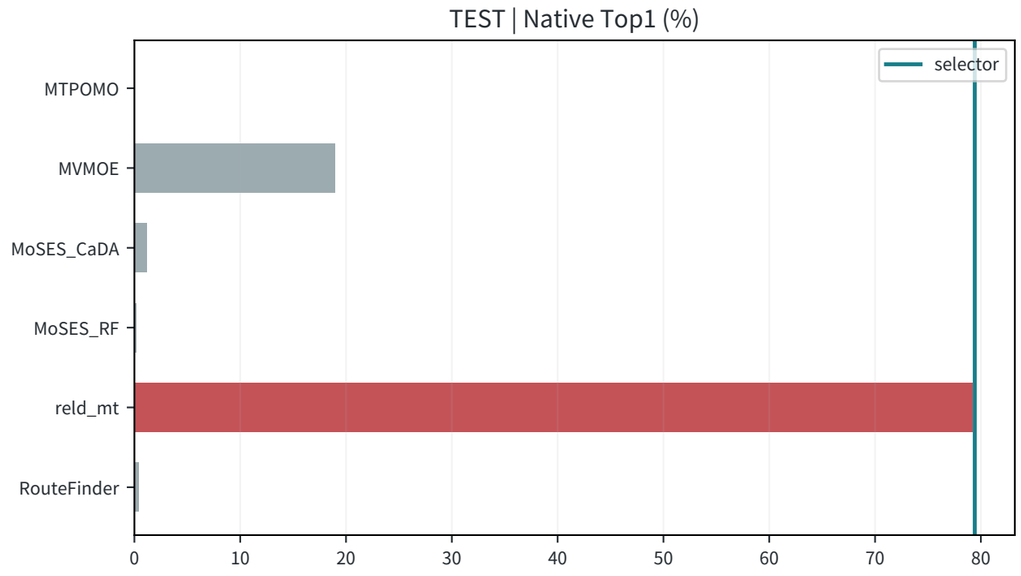


R101/B | VRPLTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



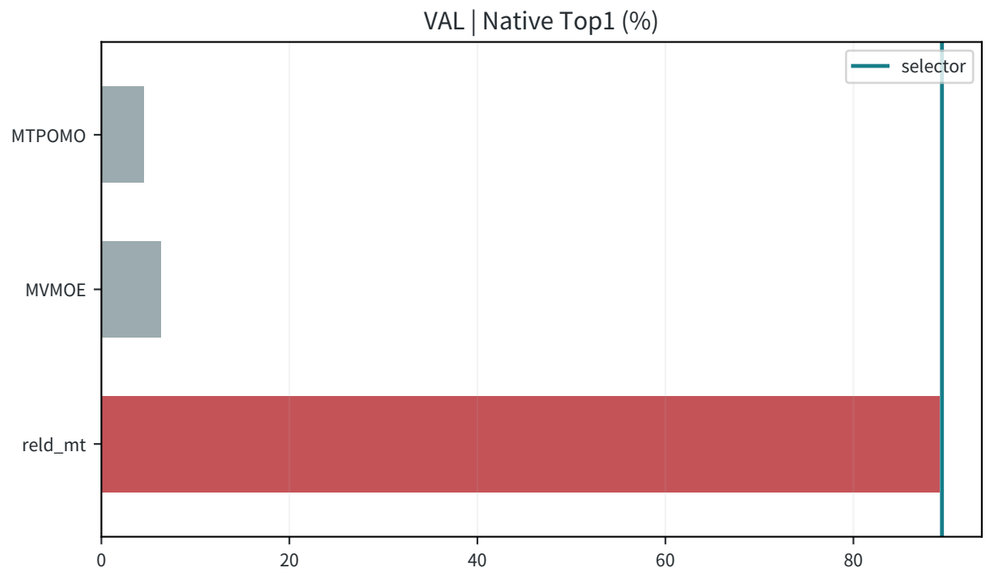


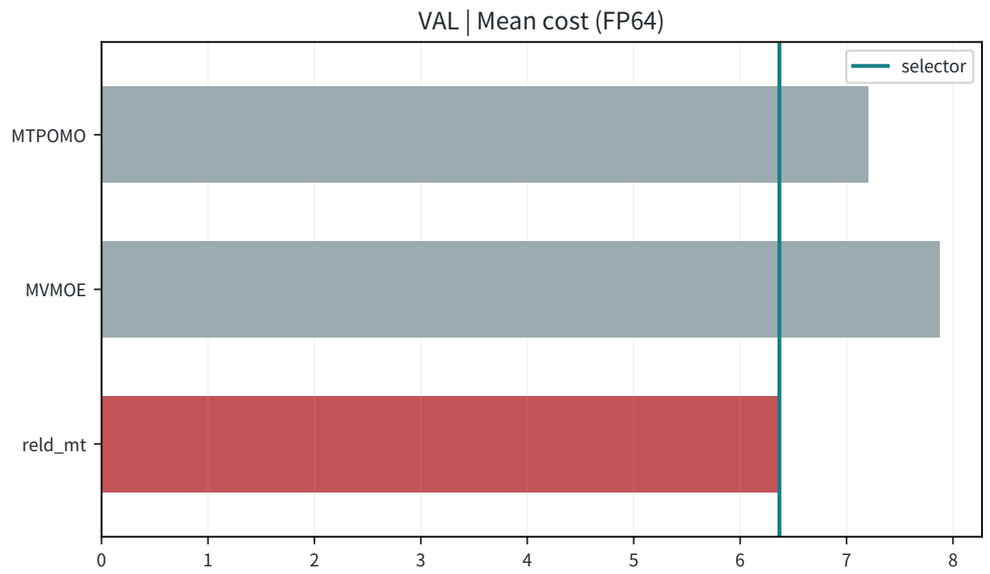


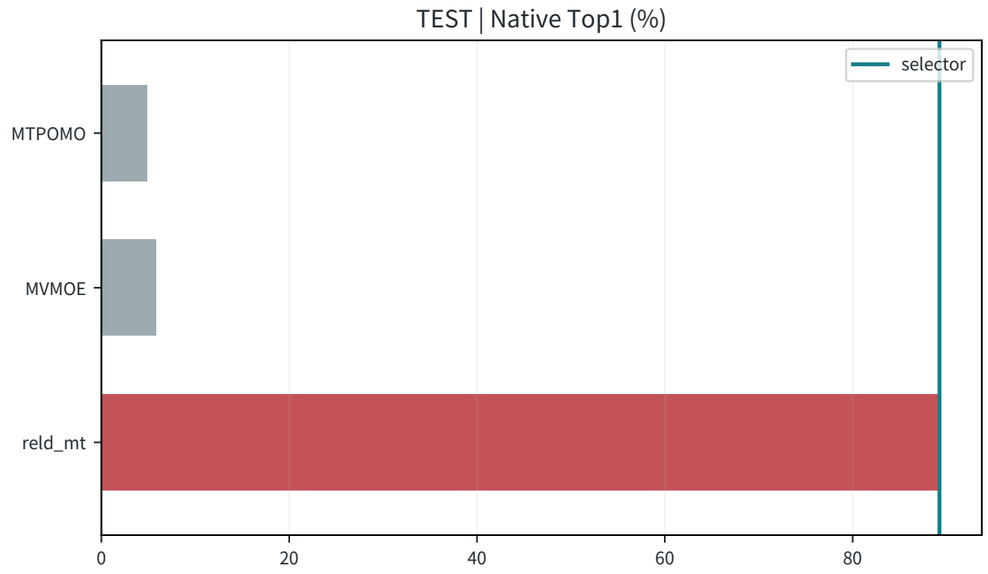


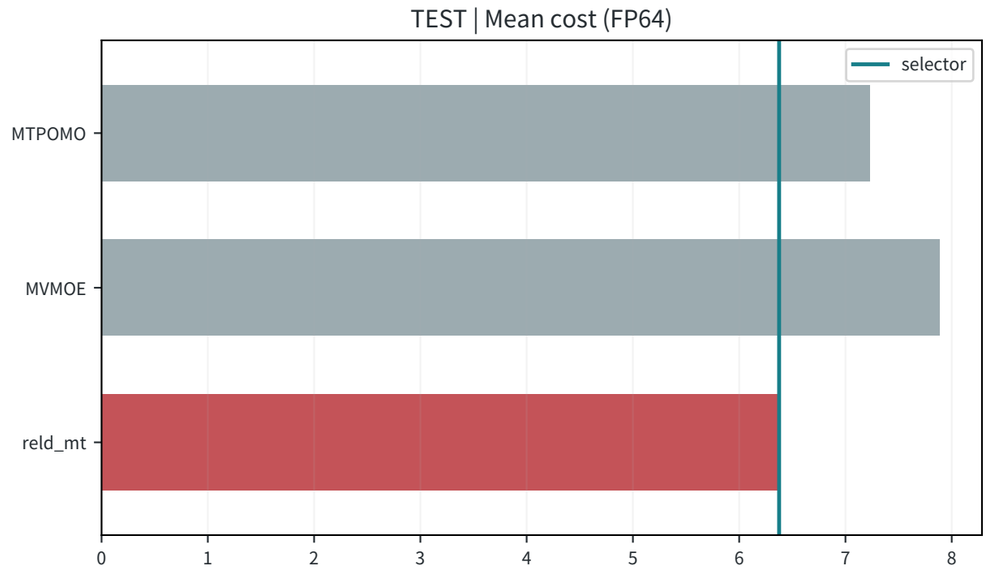
R101/B | OVRPBL: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



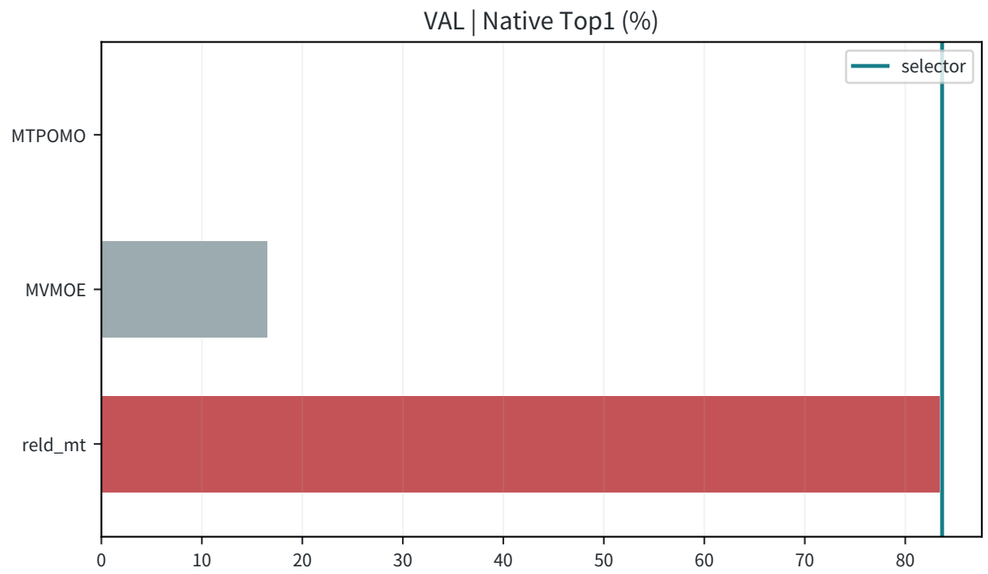


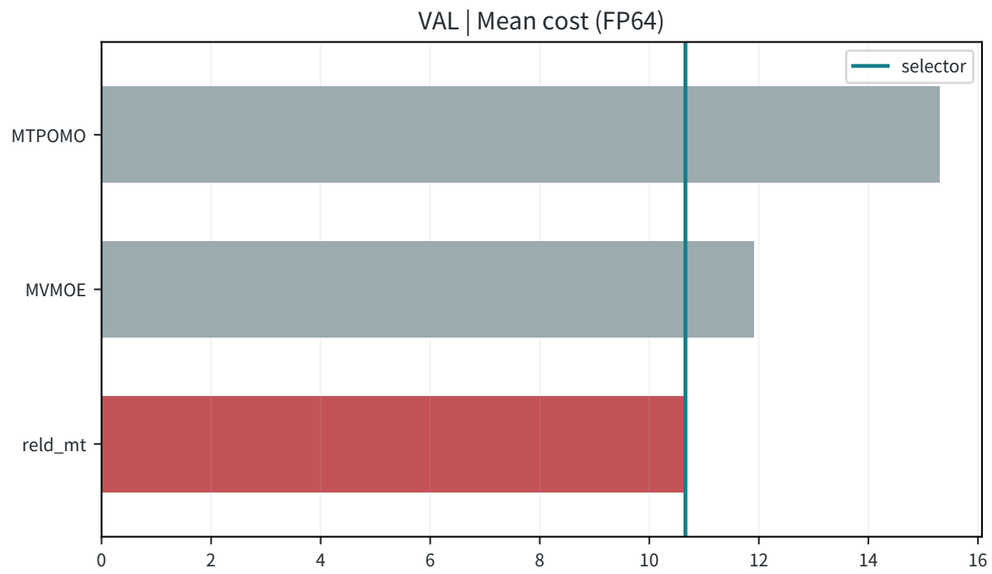


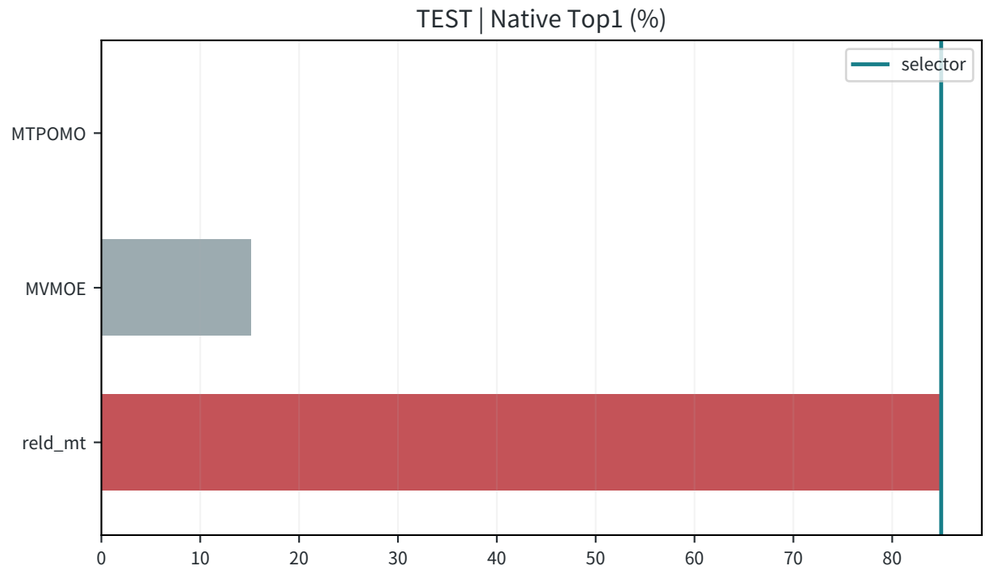


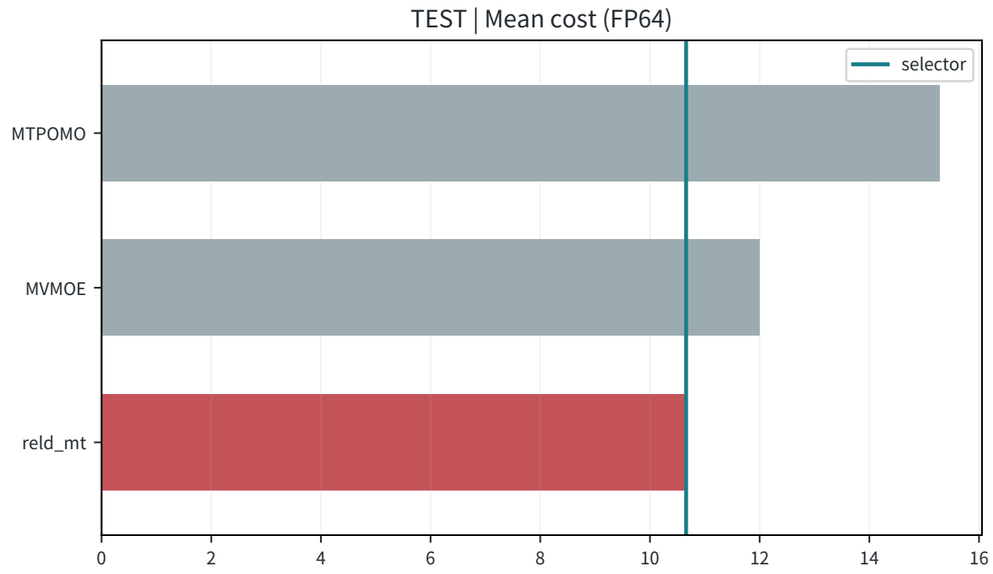
R101/B | OVRPBTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



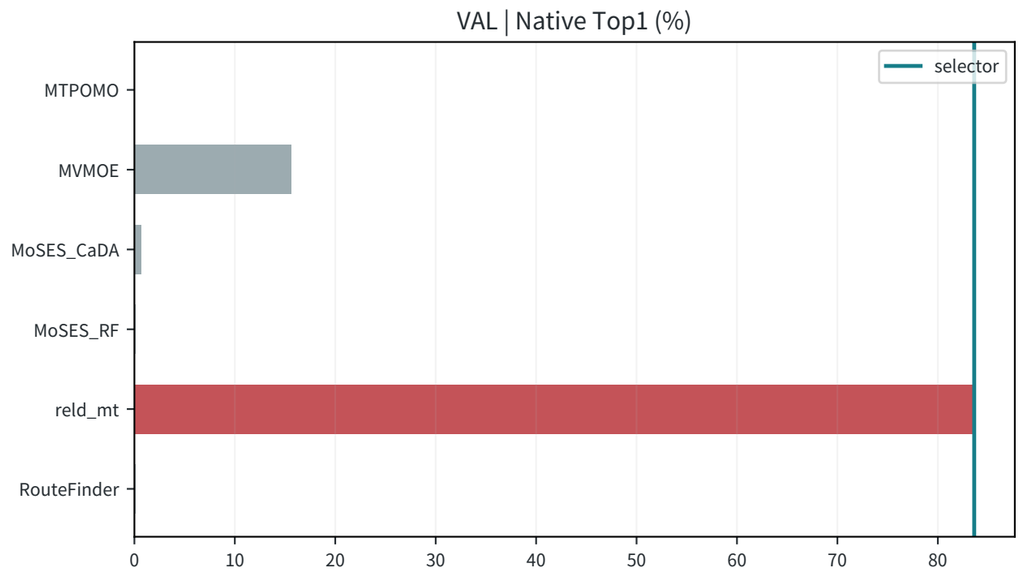


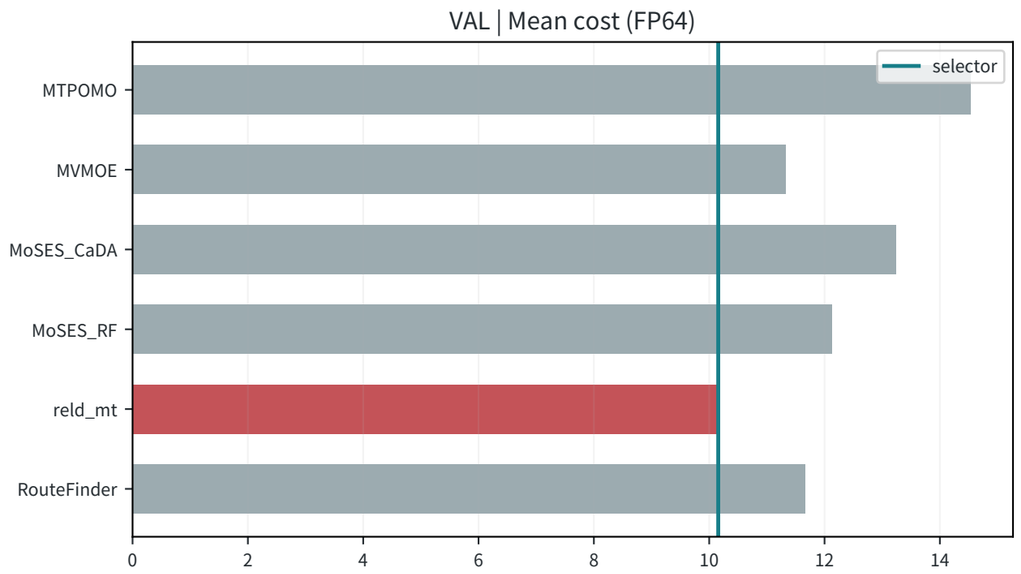


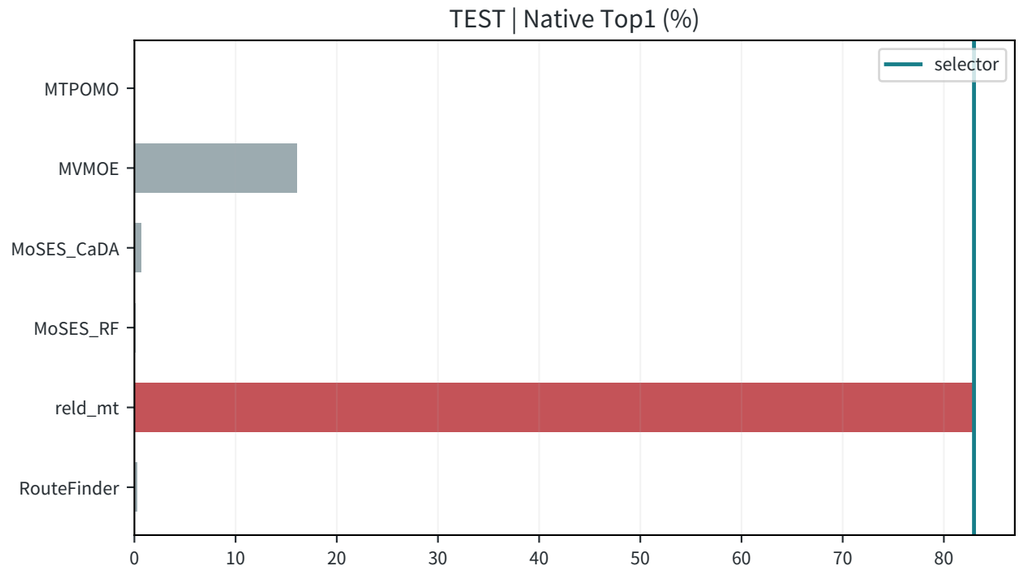


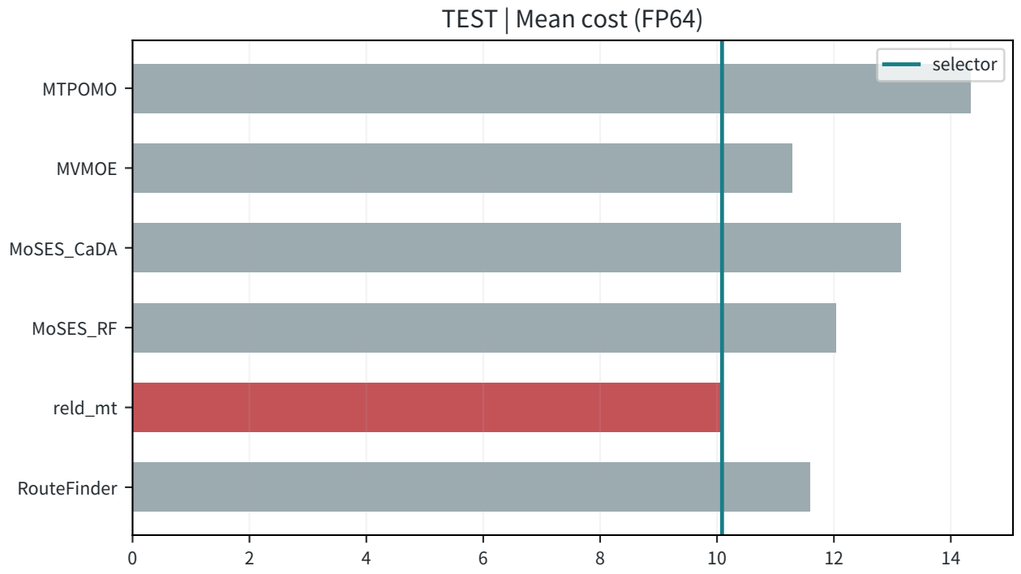
R101/B | OVRPLTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.



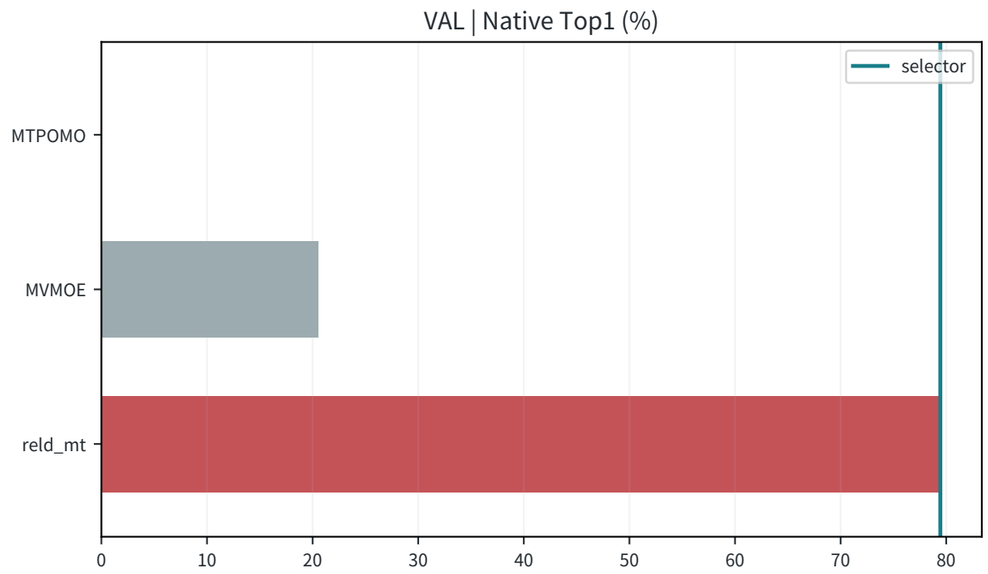


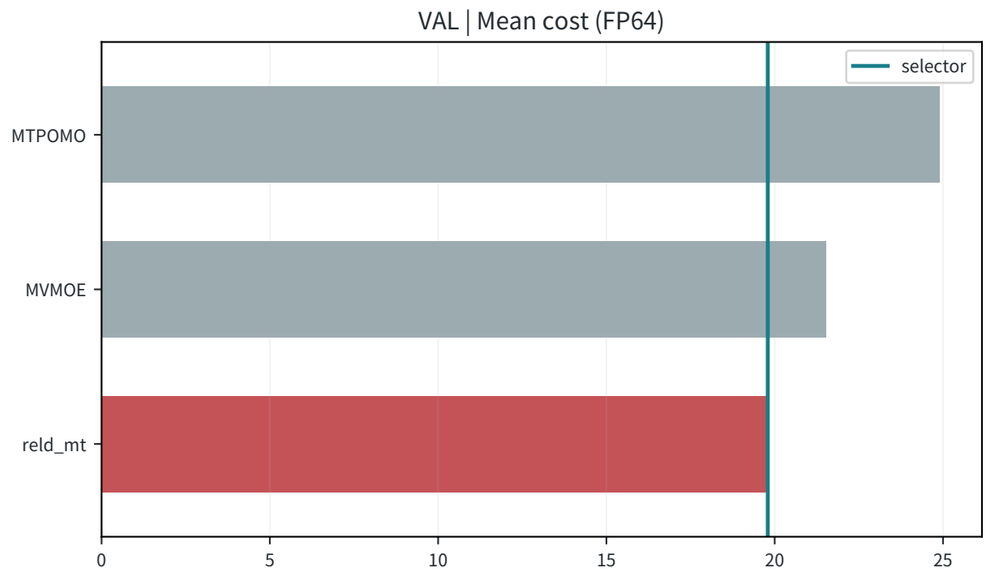


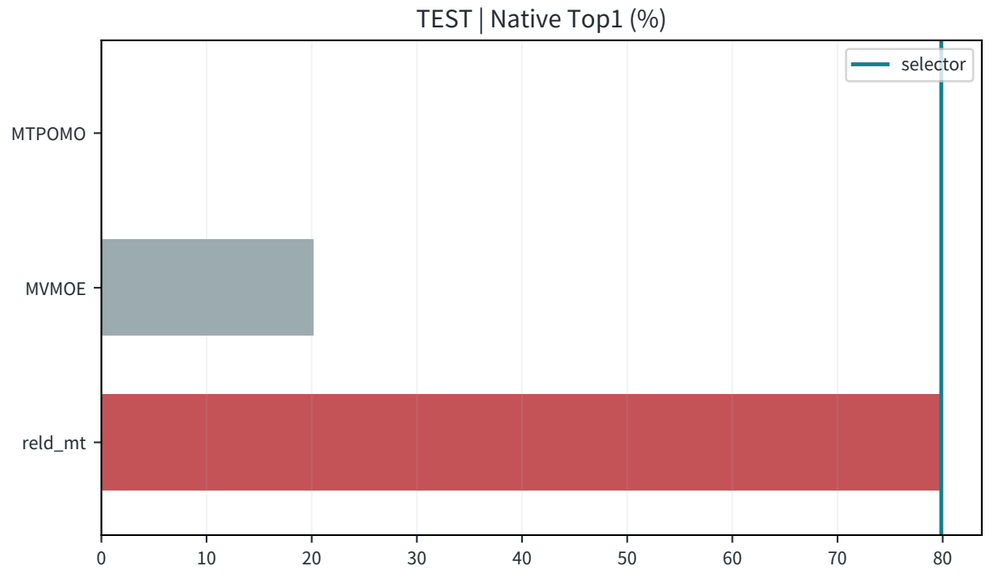


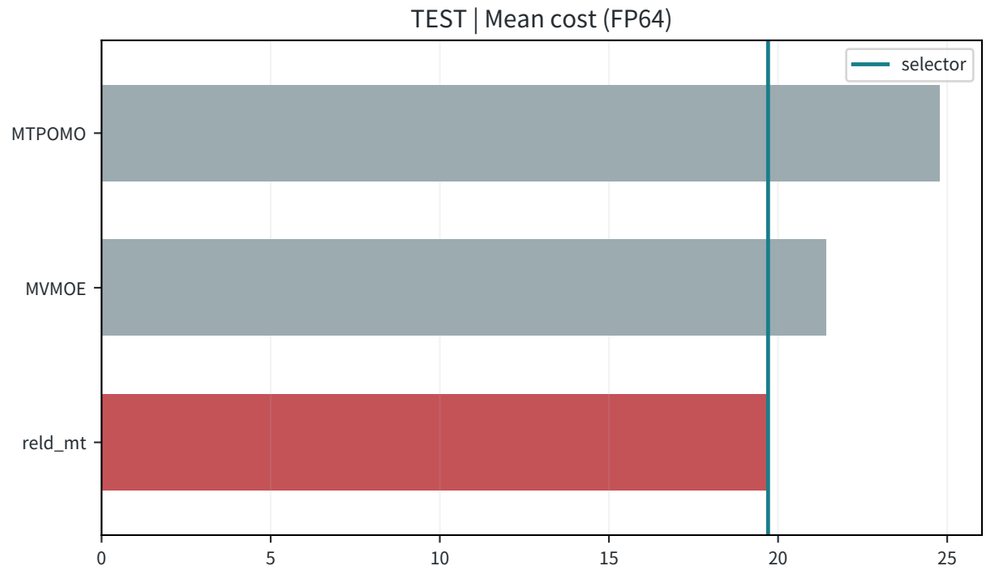
R101/B | VRPBLTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.









R101/B | OVRPBLTW: Selector vs Single Methods

Native winner Top1 and FP64 mean cost; every selection is from the same locked model. Lower cost is better.

